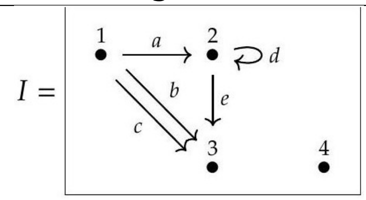
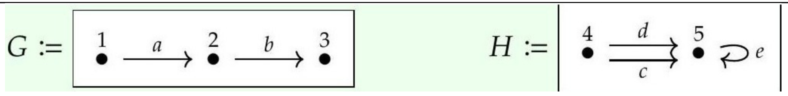
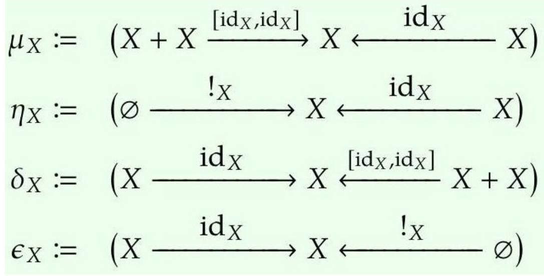
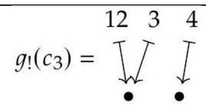
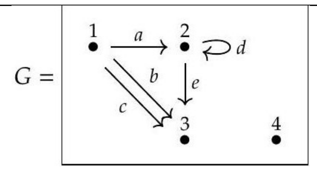
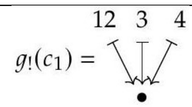
fong-spivak-seven-sketches — formula report

4029 inline formulas (section omitted; 1 did not render), 264 display equations, 63 tables, 503 unrecovered image regions.

Display equations

<table>
<tr><th>Identifier</th><th>Page</th><th>Conf.</th><th>LaTeX source</th><th>Rendered</th><th>Scan image</th></tr>
<tr><td>fong-spivak-seven-sketche EQ0061 (3.61) [conf 0.008]</td><td>110</td><td>0.008 C|+6</td><td><code>I=\left[\begin{array}{ccc} 1 & \underset{\bullet}{a} & \stackrel{2}{\bullet}{ }^{b} d \\ & \downarrow & \\ & \downarrow & \\ & \downarrow & \bullet \end{array}\right.</code></td><td>$I=\left[\begin{array}{ccc} 1 & \underset{\bullet}{a} & \stackrel{2}{\bullet}{ }^{b} d \\ & \downarrow & \\ & \downarrow & \\ & \downarrow & \bullet \end{array}\right.$</td><td></td></tr>
<tr><td>fong-spivak-seven-sketche EQ0063 [conf 0.014]</td><td>111</td><td>0.014 C|+4</td><td><code>G:=\stackrel{1}{\bullet} \stackrel{a}{\longrightarrow} \stackrel{b}{\longrightarrow} \stackrel{3}{\bullet} \quad H:=\stackrel{4}{\leftrightarrow} \xrightarrow[c]{d} \bullet \stackrel{5}{\hookrightarrow} \text { Pe }</code></td><td>$G:=\stackrel{1}{\bullet} \stackrel{a}{\longrightarrow} \stackrel{b}{\longrightarrow} \stackrel{3}{\bullet} \quad H:=\stackrel{4}{\leftrightarrow} \xrightarrow[c]{d} \bullet \stackrel{5}{\hookrightarrow} \text { Pe }$</td><td></td></tr>
<tr><td>fong-spivak-seven-sketche EQ0133 [conf 0.025]</td><td>214</td><td>0.025 S|+1</td><td><code>\begin{aligned} \mu_{X} & :=\left(X+X \xrightarrow{ \left[\mathrm{id}_{X}, \mathrm{id}_{X}\right]} X \longleftrightarrow \stackrel{i d}{\longrightarrow}_{X} X\right) \\ \eta_{X} & :=\left(\varnothing \xrightarrow{!_{X}} X \stackrel{\mathrm{id}_{X}}{ \longleftrightarrow} X\right) \\ \delta_{X} & := \left(X \xrightarrow{\mathrm{id}_{X}} X \stackrel{\left[\mathrm{id}_{X}, \mathrm{id}_{X}\right]}{-} X+X\right) \\ \epsilon_{X} & :=\left(X \xrightarrow{\mathrm{id}_{X}} X \longleftrightarrow!_{X} \varnothing\right) \end{aligned}</code></td><td>$\begin{aligned} \mu_{X} & :=\left(X+X \xrightarrow{ \left[\mathrm{id}_{X}, \mathrm{id}_{X}\right]} X \longleftrightarrow \stackrel{i d}{\longrightarrow}_{X} X\right) \\ \eta_{X} & :=\left(\varnothing \xrightarrow{!_{X}} X \stackrel{\mathrm{id}_{X}}{ \longleftrightarrow} X\right) \\ \delta_{X} & := \left(X \xrightarrow{\mathrm{id}_{X}} X \stackrel{\left[\mathrm{id}_{X}, \mathrm{id}_{X}\right]}{-} X+X\right) \\ \epsilon_{X} & :=\left(X \xrightarrow{\mathrm{id}_{X}} X \longleftrightarrow!_{X} \varnothing\right) \end{aligned}$</td><td></td></tr>
<tr><td>fong-spivak-seven-sketche EQ0188 [conf 0.038]</td><td>279</td><td>0.038 S|+2</td><td><code>g_{!}\left(c_{3}\right)={ }^{12}{ }_{\downarrow}{ }^{3} \stackrel{4}{\bullet}</code></td><td>$g_{!}\left(c_{3}\right)={ }^{12}{ }_{\downarrow}{ }^{3} \stackrel{4}{\bullet}$</td><td></td></tr>
<tr><td>fong-spivak-seven-sketche EQ0181 [conf 0.044]</td><td>274</td><td>0.044 C|+6</td><td><code>G=\left|\begin{array}{ccc} 1 & \underset{\bullet}{a} & \stackrel{2}{\bullet} \\ & \downarrow & \downarrow \\ & \downarrow & \\ & \downarrow & \\ & \downarrow & \end{array}\right|</code></td><td>$G=\left|\begin{array}{ccc} 1 & \underset{\bullet}{a} & \stackrel{2}{\bullet} \\ & \downarrow & \downarrow \\ & \downarrow & \\ & \downarrow & \\ & \downarrow & \end{array}\right|$</td><td></td></tr>
<tr><td>fong-spivak-seven-sketche EQ0186 [conf 0.050]</td><td>279</td><td>0.050 S|+2</td><td><code>g!\left(c_{1}\right)={ }^{12} 3{ }^{4}</code></td><td>$g!\left(c_{1}\right)={ }^{12} 3{ }^{4}$</td><td></td></tr>
<tr><td colspan="6">Conf. MathPix confidence: ≥0.9 0.5–0.9 <0.5 — no value
Residual render vs scan (inkdrill): C component W weak S stable N noise K clean not measured</td></tr>
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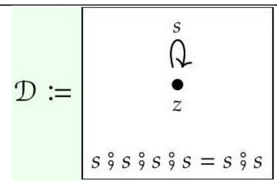
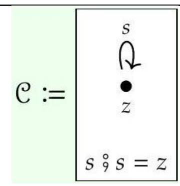
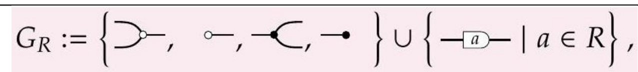
<tr><td>fong-spivak-seven-sketche EQ0189</td><td>279</td><td>0.108 W|+0</td><td>g_{!}\left(c_{4}\right)=\amalg_{\bullet}^{12} I^{3} \stackrel{4}{\downarrow}</td><td>$g_{!}\left(c_{4}\right)=\amalg_{\bullet}^{12} I^{3} \stackrel{4}{\downarrow}$</td><td>12 3 4
$g_!(c_4) =$</td></tr>
<tr><td>fong-spivak-seven-sketche EQ0073 (3.77)</td><td>118</td><td>0.114 U|+40</td><td>\mathcal{G}:=\left|\stackrel{\text { Email }}{\underset{\text { received_by }}{\stackrel{\text { sent_by }}{\rightleftharpoons}}}{ }^{\text {Address }} \bullet \right|</td><td>(not rendered)</td><td>$\mathcal{G} :=$ Email • $\underset{\text{received\_by}}{\overset{\text{sent\_by}}{\rightrightarrows}}$ Address •
no equations</td></tr>
<tr><td>fong-spivak-seven-sketche EQ0245</td><td>311</td><td>0.129 C|-7</td><td>a_{m}:=(1+\cdots+1+\cdots) ; \cdots ;(1+1+\supset-) ;(1+\cdots) ; \cdots: m \rightarrow 1</td><td>$a_{m}:=(1+\cdots+1+\cdots) ; \cdots ;(1+1+\supset-) ;(1+\cdots) ; \cdots: m \rightarrow 1$</td><td>$a_m := (1+\cdots+1+\supset\!-)\fatsemi\cdots\fatsemi(1+1+\supset\!-)\fatsemi(1+\supset\!-)\fatsemi\supset\!-: m \to 1$</td></tr>
<tr><td>fong-spivak-seven-sketche EQ0083</td><td>136</td><td>0.147 N|-1</td><td>\text { Chassis: }(\text { load } \times velocity }) \rightarrow \text { (torque } \text { × speed } \times \$ \text {) }</td><td>Chassis: (load × velocity) → (torque × speed × $)</td><td>Chassis: (load × velocity) ⇸ (torque × speed × $)</td></tr>
<tr><td>fong-spivak-seven-sketche EQ0205</td><td>290</td><td>0.147 C|-7</td><td>A, \quad A \div f, \quad A \div g, \quad A \div f \div h, \quad A \div g \text { i } i, \quad B, \quad B \doteqdot h, \quad C, \quad C: i, \quad D</td><td>$A, \quad A \div f, \quad A \div g, \quad A \div f \div h, \quad A \div g \text { i } i, \quad B, \quad B \doteqdot h, \quad C, \quad C: i, \quad D$</td><td>$A, \quad A\fatsemi f, \quad A\fatsemi g, \quad A\fatsemi f\fatsemi h, \quad A\fatsemi g\fatsemi i, \quad B, \quad B\fatsemi h, \quad C, \quad C\fatsemi i, \quad D$</td></tr>
<tr><td>fong-spivak-seven-sketche EQ0064 (3.65)</td><td>112</td><td>0.147 C|+31</td><td>\mathbf{G r}:=\stackrel{\text { Arrow }}{\rightarrow} \underset{\text { target }}{\stackrel{\text { source }}{\rightrightarrows}} \stackrel{\text { Vertex }}{\text { no equations }}</td><td>$\mathbf{G r}:=\stackrel{\text { Arrow }}{\rightarrow} \underset{\text { target }}{\stackrel{\text { source }}{\rightrightarrows}} \stackrel{\text { Vertex }}{\text { no equations }}$</td><td>$\mathbf{Gr} :=$ Arrow • $\underset{\text{target}}{\overset{\text{source}}{\rightrightarrows}}$ Vertex •
no equations
$\mathbf{DDS} :=$ State • next
no equations</td></tr>
<tr><td>fong-spivak-seven-sketche EQ0214</td><td>294</td><td>0.168 N|+0</td><td>H:=\stackrel{4}{\longrightarrow} \underset{c}{\stackrel{d}{\longrightarrow}} \stackrel{5}{\bullet} \gtrdot e</td><td>$H:=\stackrel{4}{\longrightarrow} \underset{c}{\stackrel{d}{\longrightarrow}} \stackrel{5}{\bullet} \gtrdot e$</td><td>$H :=$ $\stackrel{4}{\bullet}\ \underset{c}{\overset{d}{\rightrightarrows}}\ \stackrel{5}{\bullet}$ e</td></tr>
<tr><td>fong-spivak-seven-sketche EQ0115</td><td>182</td><td>0.203 C|+4</td><td>G_{R}:=\{\supset-, \circ-, \subset, \longrightarrow\} \cup\{-\mathrm{a}-\mid a \in R\},</td><td>$G_{R}:=\{\supset-, \circ-, \subset, \longrightarrow\} \cup\{-\mathrm{a}-\mid a \in R\},$</td><td>$G_R := \left\{\supset\!-,\ \circ\!-,\ -\!\subset,\ -\bullet\right\} \cup \left\{-\boxed{a}- \mid a \in R\right\},$</td></tr>
<tr><td>fong-spivak-seven-sketche EQ0187</td><td>279</td><td>0.215 S|+0</td><td>g_{!}\left(c_{2}\right)=\Psi_{\bullet}^{12} \mathrm{~J}^{3} 4^{4}</td><td>$g_{!}\left(c_{2}\right)=\Psi_{\bullet}^{12} \mathrm{~J}^{3} 4^{4}$</td><td>12 3 4
$g_!(c_2) =$</td></tr>
<tr><td>fong-spivak-seven-sketche EQ0094</td><td>148</td><td>0.259 S|+0</td><td>\begin{array}{ll} & S \times(T \times U):=\{(s,(t, u)) \mid s \in S, t \in T, u \in U\} \\ =? & (S \times T) \times U:=\{((s, t), u) \mid s \in S, t \in T, u \in U\} . \end{array}</td><td>$\begin{array}{ll} & S \times(T \times U):=\{(s,(t, u)) \mid s \in S, t \in T, u \in U\} \\ =? & (S \times T) \times U:=\{((s, t), u) \mid s \in S, t \in T, u \in U\} . \end{array}$</td><td>$\begin{array}{ll} & S\times(T\times U) := \left\{(s,(t,u)) \,\middle|\, s\in S, t\in T, u\in U\right\} \\ =^? & (S\times T)\times U := \left\{((s,t),u) \,\middle|\, s\in S, t\in T, u\in U\right\}. \end{array}$</td></tr>
<tr><td>fong-spivak-seven-sketche EQ0072</td><td>118</td><td>0.263 K|+0</td><td>\Sigma_{!}: \mathcal{C - I n s t} \rightarrow \text { Set } \quad \text { and } \quad \Pi_{!}: \mathcal{C}-\text { Inst → Set. }</td><td>$\Sigma_{!}: \mathcal{C}-\mathcal{I}\backslash\int\sqcup \rightarrow$ Set and $\Pi_{!}: \mathcal{C}-$ Inst → Set.</td><td>$\Sigma_!: \mathcal{C}\text{-}\mathbf{Inst} \to \mathbf{Set}$ and $\Pi_!: \mathcal{C}\text{-}\mathbf{Inst} \to \mathbf{Set}$.</td></tr>
</table>

Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value
Residual render vs scan (inkdrill): • C component • W weak • S stable • N noise • K clean • not measured

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
fong-spivak-seven-sketche EQ0020 (2.10)	055	0.266 W|+2	`\begin{array}{ll} h_{1}:=\{2 \diamond, 3 \diamond, 10 \bullet, J \bullet, Q \bullet\} & i_{1}:=\{4 \diamond, 4 \diamond, 6 \diamond, 6 \diamond, 10 \diamond\} \\ h_{2}:=\{2 \diamond, 3 \diamond, 4 \diamond, K \star, A \diamond\} & i_{2}:=\{5 \diamond, 5 \diamond, 7 \diamond, J \diamond, Q \diamond\} . \end{array}`	$h_1 := \{2\diamond, 3\diamond, 10\bullet, J\bullet, Q\bullet\} \quad i_1 := \{4\diamond, 4\diamond, 6\diamond, 6\diamond, 10\diamond\}$ $h_2 := \{2\diamond, 3\diamond, 4\diamond, K\star, A\diamond\} \quad i_2 := \{5\diamond, 5\diamond, 7\diamond, J\diamond, Q\diamond\}.$	$h_1 := \{2\heartsuit, 3\heartsuit, 10\spadesuit, J\spadesuit, Q\spadesuit\} \quad i_1 := \{4\clubsuit, 4\spadesuit, 6\heartsuit, 6\diamondsuit, 10\diamondsuit\}$ $h_2 := \{2\diamondsuit, 3\diamondsuit, 4\diamondsuit, K\spadesuit, A\spadesuit\} \quad i_2 := \{5\spadesuit, 5\heartsuit, 7\heartsuit, J\diamondsuit, Q\diamondsuit\}.$
fong-spivak-seven-sketche EQ0033	073	0.268 N|+0	`X:=\operatorname{Ob}(\mathfrak{X}) \quad d(x, y):= \mathfrak{X}(x, y) .`	$X := \operatorname{Ob}(\mathfrak{X}) \quad d(x,y) := \mathfrak{X}(x,y).$	$X := \operatorname{Ob}(\mathcal{X}) \qquad d(x,y) := \mathcal{X}(x,y).$
fong-spivak-seven-sketche EQ0175 (7.82)	268	0.278 C|+3	`\begin{aligned} & \forall(t: \mathbb{R}) \cdot @_{t} \left(\operatorname{bad} _ \text {pos }(D)\right) \Rightarrow \\ & \quad \exists(r: \mathbb{R}) \cdot(0<r<1) \wedge \forall\left(r^{\prime}: \mathbb{R} \right) \cdot 0 \leq r^{\prime} \leq 5 \Rightarrow @_{t+r+r^{\prime}}(\operatorname{engaged}(T)) . \end{aligned}`	$\forall(t:\mathbb{R})\cdot @_t(\operatorname{bad}\_\text{pos }(D)) \Rightarrow$ $\quad \exists(r:\mathbb{R})\cdot(0<r<1)\wedge\forall(r':\mathbb{R})\cdot 0\leq r'\leq 5 \Rightarrow @_{t+r+r'}(\operatorname{engaged}(T)).$	$\forall(t:\mathbb{R}).\,@_t\big(\text{bad\_pos}(D)\big) \Rightarrow$ $\quad \exists(r:\mathbb{R}).\,(0<r<1)\wedge\forall(r':\mathbb{R}).\,0\leq r'\leq 5 \Rightarrow @_{t+r+r'}\big(\text{engaged}(T)\big).$
fong-spivak-seven-sketche EQ0057	097	0.293 C|-3	`\mathcal{D}:=\left[\begin{array}{c} s \\ \hat{L} \\ \underset{z}{b} \\ s i s j s \text { is } s=s \text { is } \end{array}\right]`	$\mathcal{D} := \left[\begin{array}{c} s \\ \hat{L} \\ \underset{z}{b} \\ sisjs \text{ is } s=s \text{ is } \end{array}\right]$	
fong-spivak-seven-sketche EQ0184	274	0.304 C|+18	`\begin{array}{rlrl} (\bullet)(\circ)(*) & \leq(\bullet)(\circ)(*) & (\bullet)(\circ)(*) & \leq(\bullet \circ)(*) \\ (\bullet)(\circ)(*) & \leq(\bullet)(\circ *) & (\bullet)(\circ)(*) & \leq(\bullet \circ *) \\ (\bullet \circ)(*) & \leq(\bullet \circ *) & (\bullet *)(\circ) & \leq(\bullet *)(\circ) \\ (\bullet)(\circ *) & \leq(\bullet)(\circ *) & (\bullet)(\circ *) & \leq(\bullet \circ *) \\ (\bullet \circ)(*) & & \leq(\bullet \circ)(*) \\ (\bullet *)(\circ) & \leq(\bullet \circ *) \\ (\bullet \circ *) & \leq(\bullet \circ *) \end{array}`	$\begin{array}{rlrl} (\bullet)(\circ)(*) & \leq(\bullet)(\circ)(*) & (\bullet)(\circ)(*) & \leq(\bullet \circ)(*) \\ (\bullet)(\circ)(*) & \leq(\bullet)(\circ *) & (\bullet)(\circ)(*) & \leq(\bullet \circ *) \\ (\bullet \circ)(*) & \leq(\bullet \circ *) & (\bullet *)(\circ) & \leq(\bullet *)(\circ) \\ (\bullet)(\circ *) & \leq(\bullet)(\circ *) & (\bullet)(\circ *) & \leq(\bullet \circ *) \\ (\bullet \circ)(*) & & \leq(\bullet \circ)(*) \\ (\bullet *)(\circ) & \leq(\bullet \circ *) \\ (\bullet \circ *) & \leq(\bullet \circ *) \end{array}$	$\begin{array}{lll} (\bullet)(\circ)(*) \leq (\bullet)(\circ)(*) & (\bullet)(\circ)(*) \leq (\bullet\circ)(*) & (\bullet)(\circ)(*) \leq (\bullet*)(\circ) \\ (\bullet)(\circ)(*) \leq (\bullet)(\circ*) & (\bullet)(\circ)(*) \leq (\bullet\circ*) & (\bullet\circ)(*) \leq (\bullet\circ)(*) \\ (\bullet\circ)(*) \leq (\bullet\circ*) & (\bullet*)(\circ) \leq (\bullet*)(\circ) & (\bullet*)(\circ) \leq (\bullet\circ*) \\ (\bullet)(\circ*) \leq (\bullet)(\circ*) & (\bullet)(\circ*) \leq (\bullet\circ*) & (\bullet\circ*) \leq (\bullet\circ*) \end{array}$
fong-spivak-seven-sketche EQ0056	097	0.330 N|+1	`e:=\begin{array}{|c|} \hline s \\ \mathcal{Q} \\ z \\ s ; s=z \\ \hline \end{array}`	$e := \begin{array}{\|c\|} \hline s \\ \mathcal{Q} \\ z \\ s;s=z \\ \hline \end{array}$	
fong-spivak-seven-sketche EQ0109 (5.47)	175	0.331 N|-1	`G_{R}:=\{\supset-, \circ-, \subset, \longrightarrow\} \cup\{-a-\mid a \in R\},`	$G_R := \{\supset -, \circ -, \subset, \longrightarrow\} \cup \{-a- \mid a \in R\},$	

Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value
Residual render vs scan (inkdrill): • C component • W weak • S stable • N noise • K clean • not measured

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EQ0059 (3.47)</td><td>106</td><td>■0.333
● N|+1</td><td><code>C:=\begin{array}{|c|} \hline s \\ \mathcal{Q} \\ b \\ s ; s=s \\ \hline \end{array}</code></td><td>$C:=\begin{array}{|c|} \hline s \\ \mathcal{Q} \\ b \\ s ; s=s \\ \hline \end{array}$</td><td>$\mathcal{C} := \begin{array}{|c|} \hline s \\ \circlearrowleft \\ \bullet \\ z \\ \\ s ⨾ s = s \\ \hline \end{array}$</td></tr>
<tr><td>fong-spivak-seven-sketche
EQ0113</td><td>178</td><td>■0.335
● N|-1</td><td><code>-\mapsto(1) \quad \text { and } \quad X \mapsto\left(\begin{array}{ll} 0 & 1 \\ 1 & 0 \end{array}\right) \quad \text { and } \quad-a-\mapsto(a)</code></td><td>$-\mapsto(1) \quad \text { and } \quad X \mapsto\left(\begin{array}{ll} 0 & 1 \\ 1 & 0 \end{array}\right) \quad \text { and } \quad-a-\mapsto(a)$</td><td>$— \mapsto (1) \quad \text{and} \quad \times \mapsto \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \quad \text{and} \quad -\boxed{a}- \mapsto (a)$</td></tr>
<tr><td>fong-spivak-seven-sketche
EQ0226</td><td>303</td><td>■0.335
● N|+1</td><td><code>\mathcal{X}(x, z)=\mathcal{X}(x, z) \otimes I \leq \mathfrak{X}(x, z) \otimes \mathcal{X}(z, z) \leq \bigvee_{y \in \mathcal{X}} \mathcal{X}(x, y) \otimes \mathcal{X}(y, z),</code></td><td>$\mathcal{X}(x, z)=\mathcal{X}(x, z) \otimes I \leq \mathfrak{X}(x, z) \otimes \mathcal{X}(z, z) \leq \bigvee_{y \in \mathcal{X}} \mathcal{X}(x, y) \otimes \mathcal{X}(y, z),$</td><td>$\mathcal{X}(x,z) = \mathcal{X}(x,z) \otimes I \leq \mathcal{X}(x,z) \otimes \mathcal{X}(z,z) \leq \bigvee_{y \in \mathcal{X}} \mathcal{X}(x,y) \otimes \mathcal{X}(y,z),$</td></tr>
<tr><td>fong-spivak-seven-sketche
EQ0068</td><td>115</td><td>■0.352
● S|+2</td><td><code>\mathcal{C} \underset{R}{\stackrel{L}{ \Longleftrightarrow}} \mathcal{D}</code></td><td>$\mathcal{C} \underset{R}{\stackrel{L}{ \Longleftrightarrow}} \mathcal{D}$</td><td>$\mathcal{C} \underset{R}{\overset{L}{\rightleftarrows}} \mathcal{D}$</td></tr>
<tr><td>fong-spivak-seven-sketche
EQ0206</td><td>290</td><td>■0.362
● S|+0</td><td><code>A, \quad A ; f, \quad A ; g, \quad A ; j, \quad B, \quad B ; h, \quad C, \quad C ; i, \quad D</code></td><td>$A, \quad A ; f, \quad A ; g, \quad A ; j, \quad B, \quad B ; h, \quad C, \quad C ; i, \quad D$</td><td>$A, \quad A ⨾ f, \quad A ⨾ g, \quad A ⨾ j, \quad B, \quad B ⨾ h, \quad C, \quad C ⨾ i, \quad D$</td></tr>
<tr><td>fong-spivak-seven-sketche
EQ0098</td><td>156</td><td>■0.366
● N|+0</td><td><code>(x \times y)\left((x, y),\left(x^{\prime}, y^{\prime} \right)\right):=x\left(x, x^{\prime}\right) \otimes y \left(y, y^{\prime}\right) .</code></td><td>$(x \times y)\left((x, y),\left(x^{\prime}, y^{\prime} \right)\right):=x\left(x, x^{\prime}\right) \otimes y \left(y, y^{\prime}\right) .$</td><td>$(\mathcal{X} \times \mathcal{Y})((x,y),(x',y')) := \mathcal{X}(x,x') \otimes \mathcal{Y}(y,y').$</td></tr>
<tr><td>fong-spivak-seven-sketche
EQ0036</td><td>077</td><td>■0.376
● W|-2</td><td><code>\begin{array}{rlr} \mathcal{C}_{f}(c, d) \otimes_{W} \mathcal{C}_{f}(d, e) & =f(\mathcal{C}(c, d)) \otimes_{W} f(\mathcal{C}(d, e)) & \left(\text { definition of } \mathcal{C}_{f}\right) \\ & \leq f\left(\mathcal{C}(c, d) \otimes_{V} \mathcal{C}(d, e)\right) & (f \text { is monoidal monotone }) \\ & \leq f(\mathcal{C}(c, e)) & (\mathcal{C} \text { is } \mathcal{V} \text {-category }) \\ & =\mathcal{C}_{f}(c, e) & \left(\text { definition of } \mathscr{C}_{f}\right) \end{array}</code></td><td>$\begin{array}{rlr} \mathcal{C}_{f}(c, d) \otimes_{W} \mathcal{C}_{f}(d, e) & =f(\mathcal{C}(c, d)) \otimes_{W} f(\mathcal{C}(d, e)) & \left(\text { definition of } \mathcal{C}_{f}\right) \\ & \leq f\left(\mathcal{C}(c, d) \otimes_{V} \mathcal{C}(d, e)\right) & (f \text { is monoidal monotone }) \\ & \leq f(\mathcal{C}(c, e)) & (\mathcal{C} \text { is } \mathcal{V} \text {-category }) \\ & =\mathcal{C}_{f}(c, e) & \left(\text { definition of } \mathscr{C}_{f}\right) \end{array}$</td><td>$\begin{array}{rlr} \mathcal{C}_f(c,d) \otimes_W \mathcal{C}_f(d,e) & = f(\mathcal{C}(c,d)) \otimes_W f(\mathcal{C}(d,e)) & (\text{definition of } \mathcal{C}_f) \\ & \leq f\big(\mathcal{C}(c,d) \otimes_V \mathcal{C}(d,e)\big) & (f \text{ is monoidal monotone}) \\ & \leq f(\mathcal{C}(c,e)) & (\mathcal{C} \text{ is } \mathcal{V}\text{-category}) \\ & = \mathcal{C}_f(c,e) & (\text{definition of } \mathcal{C}_f) \end{array}$</td></tr>
<tr><td>fong-spivak-seven-sketche
EQ0142</td><td>227</td><td>■0.424
● N|+2</td><td><code>\begin{aligned} (g \circ f)\left(x_{1}, \ldots, x_{i-1}, w_{1}, \ldots, w_{m}\right. & \left., x_{i+1}, \ldots, x_{n}\right) \\ & =g\left(x_{1}, \ldots, x_{i-1}, f \left(w_{1}, \ldots, w_{m}\right), x_{i+1}, \ldots, x_{n} \right) \end{aligned}</code></td><td>$\begin{aligned} (g \circ f)\left(x_{1}, \ldots, x_{i-1}, w_{1}, \ldots, w_{m}\right. & \left., x_{i+1}, \ldots, x_{n}\right) \\ & =g\left(x_{1}, \ldots, x_{i-1}, f \left(w_{1}, \ldots, w_{m}\right), x_{i+1}, \ldots, x_{n} \right) \end{aligned}$</td><td>$\begin{aligned} (g \circ_i f)(x_1, \ldots, x_{i-1}, w_1, \ldots, w_m, x_{i+1}, \ldots, x_n) & \\ = g\big(x_1, \ldots, x_{i-1}, f(w_1, \ldots, w_m), x_{i+1}, \ldots, x_n\big) & \end{aligned}$</td></tr>
<tr><td>fong-spivak-seven-sketche
EQ0228</td><td>303</td><td>■0.461
● N|+0</td><td><code>\left(\alpha^{-1} ; \alpha\right)(x, z)=\bigvee_{(y, 1) \in \mathcal{X} \times \mathbf{1}} \alpha(x,(y, 1)) \otimes \alpha^{-1}((y, 1), z)=\bigvee_{y \in \mathcal{X}} x(x, y) \otimes x(y, z)=x(x, z) .</code></td><td>$\left(\alpha^{-1} ; \alpha\right)(x, z)=\bigvee_{(y, 1) \in \mathcal{X} \times \mathbf{1}} \alpha(x,(y, 1)) \otimes \alpha^{-1}((y, 1), z)=\bigvee_{y \in \mathcal{X}} x(x, y) \otimes x(y, z)=x(x, z) .$</td><td>$(\alpha^{-1} ⨾ \alpha)(x,z) = \bigvee_{(y,1) \in \mathcal{X} \times \mathbf{1}} \alpha(x,(y,1)) \otimes \alpha^{-1}((y,1),z) = \bigvee_{y \in \mathcal{X}} \mathcal{X}(x,y) \otimes \mathcal{X}(y,z) = \mathcal{X}(x,z).$</td></tr>
</table>

Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value
Residual render vs scan (inkdrill): • C component • W weak • S stable • N noise • K clean • not measured

<table>
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<tr><td>fong-spivak-seven-sketches_
EQ0227</td><td>303</td><td>■ 0.463
● N|+0</td><td><code>\bigvee_{y \in \mathcal{X}} x(x, y) \otimes \mathcal{X}(y, z) \leq \bigvee_{y \in \mathcal{X}} x(x, z)=\mathscr{X}(x, z) .</code></td><td>$\bigvee_{y \in \mathcal{X}} x(x, y) \otimes \mathcal{X}(y, z) \leq \bigvee_{y \in \mathcal{X}} x(x, z)=\mathscr{X}(x, z) .$</td><td>$\bigvee_{y \in \mathcal{X}} \mathcal{X}(x, y) \otimes \mathcal{X}(y, z) \leq \bigvee_{y \in \mathcal{X}} \mathcal{X}(x, z) = \mathcal{X}(x, z).$</td></tr>
<tr><td>fong-spivak-seven-sketches_
EQ0071 (3.75)</td><td>118</td><td>■ 0.470
● K|+0</td><td><code>!: \mathcal{C} \boldsymbol{\rightarrow} \mathbf{1} .</code></td><td>$!: \mathcal{C} \boldsymbol{\rightarrow} \mathbf{1} .$</td><td>$!\colon \mathcal{C} \to \mathbf{1}.$</td></tr>
<tr><td>fong-spivak-seven-sketches_
EQ0039 (2.77)</td><td>079</td><td>■ 0.472
● N|+0</td><td><code>(\mathcal{X} \times \mathcal{Y})\left(\left(x_{1}, y_{1}\right),\left(x_{2}, y_{2}\right)\right) \otimes(\mathcal{X} \times \mathcal{Y})\left(\left(x_{2}, y_{2}\right),\left(x_{3}, y_{3}\right)\right) \leq(\mathfrak{X} \times \mathcal{Y})\left(\left(x_{1}, y_{1} \right),\left(x_{3}, y_{3}\right)\right) .</code></td><td>$(\mathcal{X} \times \mathcal{Y})\left(\left(x_{1}, y_{1}\right),\left(x_{2}, y_{2}\right)\right) \otimes(\mathcal{X} \times \mathcal{Y})\left(\left(x_{2}, y_{2}\right),\left(x_{3}, y_{3}\right)\right) \leq(\mathfrak{X} \times \mathcal{Y})\left(\left(x_{1}, y_{1} \right),\left(x_{3}, y_{3}\right)\right) .$</td><td>$(\mathcal{X} \times \mathcal{Y})\big((x_1, y_1), (x_2, y_2)\big) \otimes (\mathcal{X} \times \mathcal{Y})\big((x_2, y_2), (x_3, y_3)\big) \leq (\mathcal{X} \times \mathcal{Y})\big((x_1, y_1), (x_3, y_3)\big).$</td></tr>
<tr><td>fong-spivak-seven-sketches_
EQ0163</td><td>257</td><td>■ 0.487
● W|+2</td><td><code>\Omega_{\text {Grph }}=(0,0 ; 0) \subset 0 \underset{(V, 0 ; 0)}{\stackrel{(0, V ; 0)}{\longleftarrow}}{ \underset{(V}{V}}_{(V, V ; A)}^{(V, V ; 0)}</code></td><td>$\Omega_{\text {Grph }}=(0,0 ; 0) \subset 0 \underset{(V, 0 ; 0)}{\stackrel{(0, V ; 0)}{\longleftarrow}}{ \underset{(V}{V}}_{(V, V ; A)}^{(V, V ; 0)}$</td><td>$\Omega_{\mathbf{Grph}} = \begin{tikzcd} 0 \arrow[loop left, "{(0,0;0)}"] \arrow[r, bend left, "{(0,V;0)}"] & V \arrow[l, bend left, "{(V,0;0)}"] \arrow[loop above, "{(V,V;0)}"] \arrow[loop below, "{(V,V;A)}"] \end{tikzcd}$</td></tr>
<tr><td>fong-spivak-seven-sketches_
EQ0258</td><td>317</td><td>■ 0.494
● N|-1</td><td><code>\left[\mathrm{id}_{A},!_{A}\right] \stackrel{\circ}{9} \iota_{A}=\left[\mathrm{id}_{A} \stackrel{\circ}{\circ} \iota_{A},!_{A}^{\circ} \stackrel{\circ}{\circ}_{ \iota_{A}}\right]=\left[\iota_{A},!_{A+\varnothing} \right]=\left[\iota_{A}, \iota_{\varnothing}\right]= \mathrm{id}_{A+\varnothing} .</code></td><td>$\left[\mathrm{id}_{A},!_{A}\right] \stackrel{\circ}{9} \iota_{A}=\left[\mathrm{id}_{A} \stackrel{\circ}{\circ} \iota_{A},!_{A}^{\circ} \stackrel{\circ}{\circ}_{ \iota_{A}}\right]=\left[\iota_{A},!_{A+\varnothing} \right]=\left[\iota_{A}, \iota_{\varnothing}\right]= \mathrm{id}_{A+\varnothing} .$</td><td>$[\mathrm{id}_A, !_A] \fatsemi \iota_A = [\mathrm{id}_A \fatsemi \iota_A, !_A \fatsemi \iota_A] = [\iota_A, !_{A+\varnothing}] = [\iota_A, \iota_\varnothing] = \mathrm{id}_{A+\varnothing}.$</td></tr>
<tr><td>fong-spivak-seven-sketches_
EQ0074</td><td>119</td><td>■ 0.500
● W|-1</td><td><code>\begin{array}{c|c} 1 \\ \hline \text { Bob-Grace-Pat-Emmy } \\ \text { Sue-Doug } \end{array}</code></td><td>$\begin{array}{c|c} 1 \\ \hline \text { Bob-Grace-Pat-Emmy } \\ \text { Sue-Doug } \end{array}$</td><td>$\begin{array}{c|} \mathbf{1} \\ \hline \text{Bob-Grace-Pat-Emmy} \\ \text{Sue-Doug} \end{array}$</td></tr>
<tr><td colspan="6">Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value
Residual render vs scan (inkdrill): • C component • W weak • S stable • N noise • K clean • not measured</td></tr>
</table>

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
fong-spivak-seven-sketche EQ0223	299	0.512 W|+0	`\begin{aligned} \Phi=M_{X}^{3} * M_{\Phi} * M_{Y}^{2} & = \left(\begin{array}{cccc} 0 & 6 & 3 & 11 \\ 2 & 0 & 5 & 5 \\ 5 & 3 & 0 & 8 \\ 11 & 9 & 6 & 0 \end{array}\right) \left(\begin{array}{ccc} \infty & \infty & \infty \\ 11 & \infty & \infty \\ \infty & \infty & \infty \\ \infty & 9 & \infty \end{array}\right)\left(\begin{array}{lll} 0 & 4 & 3 \\ 3 & 0 & 6 \\ 7 & 4 & 0 \end{array}\right) \\ & =\left(\begin{array}{ccc} 17 & 20 & \infty \\ 11 & 14 & \infty \\ 14 & 17 & \infty \\ 20 & 9 & \infty \end{array}\right)\left(\begin{array}{lll} 0 & 4 & 3 \\ 3 & 0 & 6 \\ 7 & 4 & 0 \end{array}\right) \\ & =\left(\begin{array}{ccc} 17 & 20 & 20 \\ 11 & 14 & 14 \\ 14 & 17 & 17 \\ 12 & 9 & 15 \end{array}\right) \end{aligned}`	$\begin{aligned} \Phi=M_{X}^{3} * M_{\Phi} * M_{Y}^{2} & = \left(\begin{array}{cccc} 0 & 6 & 3 & 11 \\ 2 & 0 & 5 & 5 \\ 5 & 3 & 0 & 8 \\ 11 & 9 & 6 & 0 \end{array}\right) \left(\begin{array}{ccc} \infty & \infty & \infty \\ 11 & \infty & \infty \\ \infty & \infty & \infty \\ \infty & 9 & \infty \end{array}\right)\left(\begin{array}{lll} 0 & 4 & 3 \\ 3 & 0 & 6 \\ 7 & 4 & 0 \end{array}\right) \\ & =\left(\begin{array}{ccc} 17 & 20 & \infty \\ 11 & 14 & \infty \\ 14 & 17 & \infty \\ 20 & 9 & \infty \end{array}\right)\left(\begin{array}{lll} 0 & 4 & 3 \\ 3 & 0 & 6 \\ 7 & 4 & 0 \end{array}\right) \\ & =\left(\begin{array}{ccc} 17 & 20 & 20 \\ 11 & 14 & 14 \\ 14 & 17 & 17 \\ 12 & 9 & 15 \end{array}\right) \end{aligned}$	$\begin{aligned} \Phi=M_{X}^{3} * M_{\Phi} * M_{Y}^{2} & = \begin{pmatrix} 0 & 6 & 3 & 11 \\ 2 & 0 & 5 & 5 \\ 5 & 3 & 0 & 8 \\ 11 & 9 & 6 & 0 \end{pmatrix} \begin{pmatrix} \infty & \infty & \infty \\ 11 & \infty & \infty \\ \infty & \infty & \infty \\ \infty & 9 & \infty \end{pmatrix}\begin{pmatrix} 0 & 4 & 3 \\ 3 & 0 & 6 \\ 7 & 4 & 0 \end{pmatrix} \\ & =\begin{pmatrix} 17 & 20 & \infty \\ 11 & 14 & \infty \\ 14 & 17 & \infty \\ 20 & 9 & \infty \end{pmatrix}\begin{pmatrix} 0 & 4 & 3 \\ 3 & 0 & 6 \\ 7 & 4 & 0 \end{pmatrix} \\ & =\begin{pmatrix} 17 & 20 & 20 \\ 11 & 14 & 14 \\ 14 & 17 & 17 \\ 12 & 9 & 15 \end{pmatrix} \end{aligned}$
fong-spivak-seven-sketche EQ0021	058	0.515 C|+3	`\left.\left.\frac{\bar{x}}{y}\right\}=\frac{x \otimes y}{z}\right\}=\left\{\frac{x}{y \otimes z}=\left\{\frac{y}{z}\right.\right.`	$\left.\left.\frac{\bar{x}}{y}\right\}=\frac{x \otimes y}{z}\right\}=\left\{\frac{x}{y \otimes z}=\left\{\frac{y}{z}\right.\right.$	$\frac{\left.\frac{x}{y}\right\}}{z} = \left.\frac{x \otimes y}{z}\right\} = \left\{\frac{x}{y \otimes z = \left\{\frac{y}{z}\right.}\right.$
fong-spivak-seven-sketche EQ0069	115	0.520 W|+2	`\text { Set } \underset{(-)^{B}}{\stackrel{-\times B}{\Rightarrow}} \text { Set } \quad \operatorname{Set}(A \times B, C) \cong \operatorname{Set}\left(A, C^{B}\right)`	$\text { Set } \underset{(-)^{B}}{\stackrel{-\times B}{\Rightarrow}} \text { Set } \quad \operatorname{Set}(A \times B, C) \cong \operatorname{Set}\left(A, C^{B}\right)$	$\mathbf{Set} \underset{(-)^{B}}{\overset{-\times B}{\rightleftarrows}} \mathbf{Set} \qquad \mathbf{Set}(A \times B, C) \cong \mathbf{Set}(A, C^{B})$
fong-spivak-seven-sketche EQ0116	186	0.520 N|-1	`\mathrm{B}(-\mathrm{C})=\{(x,(x, x)) \mid x \in R\} .`	$\mathrm{B}(-\mathrm{C})=\{(x,(x, x)) \mid x \in R\} .$	$\mathrm{B}(\multimap C) = \{(x,(x,x)) \mid x \in R\}.$
fong-spivak-seven-sketche EQ0257	317	0.522 N|+0	`\mathrm{id}_{A}+\mathrm{id}_{B}=\left[\mathrm{id}_{A} \stackrel{\circ}{\circ} \iota_{A}, \mathrm{id}_{B} \stackrel{\circ}{\circ} \iota_{B}\right]=\left[\iota_{A}, \iota_{B}\right]=\mathrm{id}_{A+B} .`	$\mathrm{id}_{A}+\mathrm{id}_{B}=\left[\mathrm{id}_{A} \stackrel{\circ}{\circ} \iota_{A}, \mathrm{id}_{B} \stackrel{\circ}{\circ} \iota_{B}\right]=\left[\iota_{A}, \iota_{B}\right]=\mathrm{id}_{A+B} .$	$\mathrm{id}_A + \mathrm{id}_B = [\mathrm{id}_A \fatsemi \iota_A, \mathrm{id}_B \fatsemi \iota_B] = [\iota_A, \iota_B] = \mathrm{id}_{A+B}.$
fong-spivak-seven-sketche EQ0054 (3.11)	095	0.526 C|+43	`\mathbf{3}:=\operatorname{Free}\left(\stackrel{v_{1}}{\bullet} \xrightarrow{f_{1}} \stackrel{v_{2}}{\longrightarrow} \xrightarrow{f_{2}} \stackrel{v_{3}}{\bullet}\right.`	$\mathbf{3}:=\operatorname{Free}\left(\stackrel{v_{1}}{\bullet} \xrightarrow{f_{1}} \stackrel{v_{2}}{\longrightarrow} \xrightarrow{f_{2}} \stackrel{v_{3}}{\bullet}\right.$	$\mathbf{3} := \mathrm{Free}\left(\boxed{\overset{v_1}{\bullet} \xrightarrow{f_1} \overset{v_2}{\bullet} \xrightarrow{f_2} \overset{v_3}{\bullet}}\right)$
fong-spivak-seven-sketche EQ0005	033	0.539 C|+4	`p=g(f(p)) \quad \text { and } \quad q=f(g(q)) .`	$p=g(f(p)) \quad \text { and } \quad q=f(g(q)) .$	$p = g(f(p)) \quad \text{and} \quad q = f(g(q)).$
fong-spivak-seven-sketche EQ0078 (4.3)	132	0.549 N|+2	`\Phi: x^{\mathrm{op}} \times y \rightarrow \text { Bool. }`	$\Phi: x^{\mathrm{op}} \times y \rightarrow \text { Bool. }$	$\Phi\colon \mathcal{X}^{\mathrm{op}} \times \mathcal{Y} \to \mathbf{Bool}.$

Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value

Residual render vs scan (inkdrill): • C component • W weak • S stable • N noise • K clean • not measured

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fong-spivak-seven-sketche EQ0091	142	0.550 C|+3	`\widehat{F}(p, q):=Q(F(p), q) \text { and } \check{F}(q, p):=Q(q, F(p))`	$\widehat{F}(p, q):=Q(F(p), q) \text { and } \check{F}(q, p):=Q(q, F(p))$	$\widehat{F}(p,q) \coloneqq \mathcal{Q}(F(p),q) \quad \text{and} \quad \check{F}(q,p) \coloneqq \mathcal{Q}(q,F(p))$
fong-spivak-seven-sketche EQ0104 (5.17)	165	0.566 N|+0	`G+G^{\prime}:=\left(\left(V \sqcup V^{\prime}\right),[\text { in }, \text { in' }],[\text { out }, \text { out' }],\left(\iota \sqcup \iota^{\prime}\right)\right) .`	$G+G^{\prime}:=\left(\left(V \sqcup V^{\prime}\right),[\text { in }, \text { in' }],[\text { out }, \text { out' }],\left(\iota \sqcup \iota^{\prime}\right)\right) .$	$G+G' \coloneqq \big((V \sqcup V'), [in, in'], [out, out'], (\iota \sqcup \iota')\big).$
fong-spivak-seven-sketche EQ0221	298	0.568 N|+0	`x\left(x^{\prime}, x\right) \otimes y\left(y, y^{\prime}\right) \leq \Phi(x, y) \multimap \Phi\left(x^{\prime}, y^{\prime}\right)`	$x\left(x^{\prime}, x\right) \otimes y\left(y, y^{\prime}\right) \leq \Phi(x, y) \multimap \Phi\left(x^{\prime}, y^{\prime}\right)$	$\mathcal{X}(x',x) \otimes \mathcal{Y}(y,y') \leq \Phi(x,y) \multimap \Phi(x',y')$
fong-spivak-seven-sketche EQ0253	313	0.570 N|+0	`\left\{(y, z) \mid \text { there exists } x \in R^{m} \text { such that } y=S(g)(x) \text { and } S(h)(x)=z \right\} \text {. }`	$\left\{(y, z) \mid \text { there exists } x \in R^{m} \text { such that } y=S(g)(x) \text { and } S(h)(x)=z \right\} \text {. }$	$\{(y,z) \mid \text{ there exists } x \in R^m \text{ such that } y = S(g)(x) \text{ and } S(h)(x) = z\}.$
fong-spivak-seven-sketche EQ0043	082	0.593 N|-1	`2 \mathrm{H}_{2} \mathrm{O} \rightarrow\left(2 \mathrm{Na} \rightarrow\left(2 \mathrm{NaOH}+ \mathrm{H}_{2}\right)\right)`	$2 \mathrm{H}_{2} \mathrm{O} \rightarrow\left(2 \mathrm{Na} \rightarrow\left(2 \mathrm{NaOH}+ \mathrm{H}_{2}\right)\right)$	$2\mathrm{H_2O} \longrightarrow (2\mathrm{Na} \multimap (2\mathrm{NaOH} + \mathrm{H_2}))$
fong-spivak-seven-sketche EQ0179	272	0.598 N|+0	`h, \quad 1, \quad 2, \quad 3 .`	$h, \quad 1, \quad 2, \quad 3 .$	$h, \quad 1, \quad 2, \quad 3.$
fong-spivak-seven-sketche EQ0002 (1.15)	021	0.658 N|+2	`A=\bigcup_{p \in P} A_{p} \quad\text { and } \quad\text { if } p \neq q \text { then } A_{p} \cap A_{q}= \varnothing .`	$A=\bigcup_{p \in P} A_{p} \quad\text { and } \quad\text { if } p \neq q \text { then } A_{p} \cap A_{q}= \varnothing .$	$A = \bigcup_{p \in P} A_p \qquad \text{and} \qquad \text{if } p \neq q \text{ then } A_p \cap A_q = \varnothing.$
fong-spivak-seven-sketche EQ0158 (7.37)	250	0.658 N|+0	`\operatorname{Sec}_{f}(U):=\{s: U \rightarrow X \mid(s ; f)(u)=u \text { for all } u \in U\} .`	$\operatorname{Sec}_{f}(U):=\{s: U \rightarrow X \mid(s ; f)(u)=u \text { for all } u \in U\} .$	$\mathsf{Sec}_f(U) \coloneqq \{s \colon U \to X \mid (s \mathbin{⨟} f)(u) = u \text{ for all } u \in U\}.$
fong-spivak-seven-sketche EQ0147 (7.2)	235	0.661 N|+1	`((A=\mathrm{go}) \Rightarrow \ddot{S}>1) \wedge((A=\text { stop }) \Rightarrow \ddot{S}=0) .`	$((A=\mathrm{go}) \Rightarrow \ddot{S}>1) \wedge((A=\text { stop }) \Rightarrow \ddot{S}=0) .$	$\big((A = \mathtt{go}) \Rightarrow \ddot{S} > 1\big) \wedge \big((A = \mathtt{stop}) \Rightarrow \ddot{S} = 0\big).$
fong-spivak-seven-sketche EQ0191	282	0.677 C|-8	`\begin{array}{rlr} t+u & \leq(v+w)+u & (\text { monotonicity, } t \leq v+w, u \leq u) \\ & =v+(w+u) & (\text { associativity }) \\ & \leq v+(x+z) & (\text { monotonicity, } v \leq v, w+u \leq x+v) \\ & =(v+x)+z & (\text { associativitiy }) \\ & \leq y+z . & (\text { monotonicity, } v+x \leq y, z \leq z) \end{array}`	$\begin{array}{rlr} t+u & \leq(v+w)+u & (\text { monotonicity, } t \leq v+w, u \leq u) \\ & =v+(w+u) & (\text { associativity }) \\ & \leq v+(x+z) & (\text { monotonicity, } v \leq v, w+u \leq x+v) \\ & =(v+x)+z & (\text { associativitiy }) \\ & \leq y+z . & (\text { monotonicity, } v+x \leq y, z \leq z) \end{array}$	$\begin{array}{llr} t+u & \leq (v+w)+u & (\text{monotonicity, } t \leq v+w, u \leq u) \\ & = v+(w+u) & (\text{associativity}) \\ & \leq v+(x+z) & (\text{monotonicity, } v \leq v, w+u \leq x+v) \\ & = (v+x)+z & (\text{associativitiy}) \\ & \leq y+z. & (\text{monotonicity, } v+x \leq y, z \leq z) \end{array}$

Conf. MathPix confidence: ≥0.9 0.5–0.9 <0.5 — no value
Residual render vs scan (inkdrill): C component W weak S stable N noise K clean not measured

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
fong-spivak-seven-sketches EQ0224	300	■ 0.683 ● W|+0	`\begin{aligned} \Phi: \Psi & =\left(M_{X}^{3} * M_{\Phi} * M_{Y}^{2}\right) *\left(M_{Y}^{2} * M_{\Psi} * M_{Z}^{3}\right) \\ & =M_{X}^{3} * M_{\Phi} * M_{Y}^{4} * M_{\Psi} * M_{Z}^{3} \\ & =\left(M_{X}^{3} * M_{\Phi} * M_{Y}^{2}\right) * M_{\Psi} * M_{Z}^{3} \\ & =\Phi * M_{\Psi} * M_{Z}^{3} \\ & =\left(\begin{array}{ccc} 17 & 20 & 20 \\ 11 & 14 & 14 \\ 14 & 17 & 17 \\ 12 & 9 & 15 \end{array}\right)\left(\begin{array}{cccc} \infty & \infty & \infty & \infty \\ \infty & \infty & 0 & \infty \\ 4 & \infty & \infty & 4 \end{array}\right) \left(\begin{array}{cccc} 0 & 2 & 4 & 5 \\ 4 & 0 & 2 & 3 \\ 2 & 4 & 0 & 1 \\ 1 & 3 & 5 & 0 \end{array}\right) \\ & =\left(\begin{array}{ccc} 17 & 20 & 20 \\ 11 & 14 & 14 \\ 14 & 17 & 17 \\ 12 & 9 & 15 \end{array}\right)\left(\begin{array}{cccc} \infty & \infty & \infty & \infty \\ 2 & 4 & 0 & 1 \\ 4 & 6 & 8 & 4 \end{array}\right) \\ & =\left(\begin{array}{cccc} 22 & 24 & 20 & 21 \\ 16 & 18 & 14 & 15 \\ 19 & 21 & 17 & 18 \\ 11 & 13 & 9 & 10 \end{array}\right) \end{aligned}`	$\begin{aligned} \Phi: \Psi & =\left(M_{X}^{3} * M_{\Phi} * M_{Y}^{2}\right) *\left(M_{Y}^{2} * M_{\Psi} * M_{Z}^{3}\right) \\ & =M_{X}^{3} * M_{\Phi} * M_{Y}^{4} * M_{\Psi} * M_{Z}^{3} \\ & =\left(M_{X}^{3} * M_{\Phi} * M_{Y}^{2}\right) * M_{\Psi} * M_{Z}^{3} \\ & =\Phi * M_{\Psi} * M_{Z}^{3} \\ & =\left(\begin{array}{ccc} 17 & 20 & 20 \\ 11 & 14 & 14 \\ 14 & 17 & 17 \\ 12 & 9 & 15 \end{array}\right)\left(\begin{array}{cccc} \infty & \infty & \infty & \infty \\ \infty & \infty & 0 & \infty \\ 4 & \infty & \infty & 4 \end{array}\right) \left(\begin{array}{cccc} 0 & 2 & 4 & 5 \\ 4 & 0 & 2 & 3 \\ 2 & 4 & 0 & 1 \\ 1 & 3 & 5 & 0 \end{array}\right) \\ & =\left(\begin{array}{ccc} 17 & 20 & 20 \\ 11 & 14 & 14 \\ 14 & 17 & 17 \\ 12 & 9 & 15 \end{array}\right)\left(\begin{array}{cccc} \infty & \infty & \infty & \infty \\ 2 & 4 & 0 & 1 \\ 4 & 6 & 8 & 4 \end{array}\right) \\ & =\left(\begin{array}{cccc} 22 & 24 & 20 & 21 \\ 16 & 18 & 14 & 15 \\ 19 & 21 & 17 & 18 \\ 11 & 13 & 9 & 10 \end{array}\right) \end{aligned}$	$\begin{aligned} \Phi \fatsemi \Psi & = (M_{X}^{3} * M_{\Phi} * M_{Y}^{2}) * (M_{Y}^{2} * M_{\Psi} * M_{Z}^{3}) \\ & = M_{X}^{3} * M_{\Phi} * M_{Y}^{4} * M_{\Psi} * M_{Z}^{3} \\ & = (M_{X}^{3} * M_{\Phi} * M_{Y}^{2}) * M_{\Psi} * M_{Z}^{3} \\ & = \Phi * M_{\Psi} * M_{Z}^{3} \\ & = \begin{pmatrix} 17 & 20 & 20 \\ 11 & 14 & 14 \\ 14 & 17 & 17 \\ 12 & 9 & 15 \end{pmatrix} \begin{pmatrix} \infty & \infty & \infty & \infty \\ \infty & \infty & 0 & \infty \\ 4 & \infty & \infty & 4 \end{pmatrix} \begin{pmatrix} 0 & 2 & 4 & 5 \\ 4 & 0 & 2 & 3 \\ 2 & 4 & 0 & 1 \\ 1 & 3 & 5 & 0 \end{pmatrix} \\ & = \begin{pmatrix} 17 & 20 & 20 \\ 11 & 14 & 14 \\ 14 & 17 & 17 \\ 12 & 9 & 15 \end{pmatrix} \begin{pmatrix} \infty & \infty & \infty & \infty \\ 2 & 4 & 0 & 1 \\ 4 & 6 & 8 & 4 \end{pmatrix} \\ & = \begin{pmatrix} 22 & 24 & 20 & 21 \\ 16 & 18 & 14 & 15 \\ 19 & 21 & 17 & 18 \\ 11 & 13 & 9 & 10 \end{pmatrix} \end{aligned}$
fong-spivak-seven-sketches EQ0120 (5.81)	187	■ 0.690 ● N|+0	`\mathbf{S F G}_{R}^{+}:=\text {Free }\left(G_{R} \sqcup G_{R}^{\mathrm{op}}\right) .`	$\mathbf{S F G}_{R}^{+}:=\text {Free }\left(G_{R} \sqcup G_{R}^{\mathrm{op}}\right) .$	$\mathbf{SFG}_{R}^{+} \coloneqq \mathbf{Free}\left(G_{R} \sqcup G_{R}^{\mathrm{op}}\right).$
fong-spivak-seven-sketches EQ0081 (◇)	133	■ 0.690 ● N|+0	`x\left(x^{\prime}, x\right) \otimes \Phi(x, y) \otimes y \left(y, y^{\prime}\right) \leq \Phi\left(x^{\prime}, y^{\prime}\right) .`	$x\left(x^{\prime}, x\right) \otimes \Phi(x, y) \otimes y \left(y, y^{\prime}\right) \leq \Phi\left(x^{\prime}, y^{\prime}\right) .$	$\mathcal{X}(x', x) \otimes \Phi(x, y) \otimes \mathcal{Y}(y, y') \leq \Phi(x', y').$
fong-spivak-seven-sketches EQ0140 (6.81)	220	■ 0.725 ● W|+0	`\begin{aligned} \psi_{V, V^{\prime}}: \operatorname{Circ}(V) \times \operatorname{Circ} \left(V^{\prime}\right) & \longrightarrow \operatorname{Circ}\left(V+V^{\prime}\right) ; \\ \left((V, A, s, t, \ell),\left(V^{\prime}, A^{\prime}, s^{\prime}, t^{\prime}, \ell^{\prime}\right)\right) & \longmapsto\left(V+V^{\prime}, A+A^{\prime}, s+s^{\prime}, t+t^{\prime},\left[\ell, \ell^{\prime}\right] \right) . \end{aligned}`	$\begin{aligned} \psi_{V, V^{\prime}}: \operatorname{Circ}(V) \times \operatorname{Circ} \left(V^{\prime}\right) & \longrightarrow \operatorname{Circ}\left(V+V^{\prime}\right) ; \\ \left((V, A, s, t, \ell),\left(V^{\prime}, A^{\prime}, s^{\prime}, t^{\prime}, \ell^{\prime}\right)\right) & \longmapsto\left(V+V^{\prime}, A+A^{\prime}, s+s^{\prime}, t+t^{\prime},\left[\ell, \ell^{\prime}\right] \right) . \end{aligned}$	$\begin{aligned} \psi_{V,V'} \colon \mathsf{Circ}(V) \times \mathsf{Circ}(V') & \longrightarrow \mathsf{Circ}(V + V'); \\ \big((V, A, s, t, \ell), (V', A', s', t', \ell')\big) & \longmapsto (V + V', A + A', s + s', t + t', [\ell, \ell']). \end{aligned}$
fong-spivak-seven-sketches EQ0222	298	■ 0.731 ● N|+0	`\mathcal{X}\left(x^{\prime}, x\right) \otimes \Phi(x, y) \otimes y\left(y, y^{\prime}\right) \leq \Phi\left(x^{\prime}, y^{\prime}\right) .`	$\mathcal{X}\left(x^{\prime}, x\right) \otimes \Phi(x, y) \otimes y\left(y, y^{\prime}\right) \leq \Phi\left(x^{\prime}, y^{\prime}\right) .$	$\mathcal{X}(x', x) \otimes \Phi(x, y) \otimes \mathcal{Y}(y, y') \leq \Phi(x', y').$

Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value
Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
fong-spivak-seven-sketches EQ0132	212	■0.742 ● C|+7	`\begin{array}{rrrr} \operatorname{in}(\mu):=2, & \operatorname{in}(\eta):=0, & \operatorname{in}(\delta):=1, & \operatorname{in}(\epsilon):=1, \\ \operatorname{out}(\mu):=1, & \operatorname{out}(\eta):=1, & \operatorname{out}(\delta):=2, & \operatorname{out}(\epsilon):=0 . \end{array}`	$\begin{array}{rrrr} \operatorname{in}(\mu):=2, & \operatorname{in}(\eta):=0, & \operatorname{in}(\delta):=1, & \operatorname{in}(\epsilon):=1, \\ \operatorname{out}(\mu):=1, & \operatorname{out}(\eta):=1, & \operatorname{out}(\delta):=2, & \operatorname{out}(\epsilon):=0 . \end{array}$	$\begin{array}{llll} in(\mu) := 2, & in(\eta) := 0, & in(\delta) := 1, & in(\epsilon) := 1, \\ out(\mu) := 1, & out(\eta) := 1, & out(\delta) := 2, & out(\epsilon) := 0. \end{array}$
fong-spivak-seven-sketches EQ0097	155	■0.755 ● U|+26	`\[ \alpha: \beta:=\iota_{A \sqcup C}^{*}\left(\left( \iota_{A \sqcup B}\right)!(\alpha) \vee\left(\iota_{B \sqcup C}\right)!(\beta)\right) . \] }`	*(not rendered)*	$\alpha \mathbin{;} \beta := \iota^*_{A \sqcup C}((\iota_{A \sqcup B})_!(\alpha) \vee (\iota_{B \sqcup C})_!(\beta)).$
fong-spivak-seven-sketches EQ0230	304	■0.759 ● W|+0	`\begin{aligned} & \quad \bigvee_{a, b, c, d, e \in X} \alpha^{-1}(x,(a, 1)) \otimes\left(\mathrm{U}_{X} \times \eta_{X}\right)((a, 1),(b, c, d)) \\ & = \bigvee_{a, b, c, d, e \in X} \alpha^{-1}(x,(a, 1)) \otimes \mathrm{U}_{X}(a, b) \otimes \eta_{X}(1, c, d) \quad \otimes\left(\epsilon_{X} \times \mathrm{U}_{X} \right)((b, c, d),(1, e)) \otimes \alpha((1, e), y) \\ & =\bigvee_{a, b, c, d, e \in X} x(x, a) \otimes X(a, b) \otimes X(c, d) \otimes X(b, c) \otimes X(d, e) \otimes X(e, y) \\ & =X(x, y) \end{aligned} \quad \otimes \epsilon_{X}(b, c, 1) \otimes \mathrm{U}_{X}(d, e) \otimes \alpha((1, e), y)`	$\begin{aligned} & \quad \bigvee_{a, b, c, d, e \in X} \alpha^{-1}(x,(a, 1)) \otimes\left(\mathrm{U}_{X} \times \eta_{X}\right)((a, 1),(b, c, d)) \\ & = \bigvee_{a, b, c, d, e \in X} \alpha^{-1}(x,(a, 1)) \otimes \mathrm{U}_{X}(a, b) \otimes \eta_{X}(1, c, d) \quad \otimes\left(\epsilon_{X} \times \mathrm{U}_{X} \right)((b, c, d),(1, e)) \otimes \alpha((1, e), y) \\ & =\bigvee_{a, b, c, d, e \in X} x(x, a) \otimes X(a, b) \otimes X(c, d) \otimes X(b, c) \otimes X(d, e) \otimes X(e, y) \\ & =X(x, y) \end{aligned} \quad \otimes \epsilon_{X}(b, c, 1) \otimes \mathrm{U}_{X}(d, e) \otimes \alpha((1, e), y)$	$\begin{aligned} & \bigvee_{a,b,c,d,e \in \mathcal{X}} \alpha^{-1}(x,(a,1)) \otimes (\mathrm{U}_{\mathcal{X}} \times \eta_{\mathcal{X}})((a,1),(b,c,d)) \\ & \qquad \otimes (\epsilon_{\mathcal{X}} \times \mathrm{U}_{\mathcal{X}})((b,c,d),(1,e)) \otimes \alpha((1,e),y) \\ = & \bigvee_{a,b,c,d,e \in \mathcal{X}} \alpha^{-1}(x,(a,1)) \otimes \mathrm{U}_{\mathcal{X}}(a,b) \otimes \eta_{\mathcal{X}}(1,c,d) \\ & \qquad \otimes \epsilon_{\mathcal{X}}(b,c,1) \otimes \mathrm{U}_{\mathcal{X}}(d,e) \otimes \alpha((1,e),y) \\ = & \bigvee_{a,b,c,d,e \in \mathcal{X}} \mathcal{X}(x,a) \otimes \mathcal{X}(a,b) \otimes \mathcal{X}(c,d) \otimes \mathcal{X}(b,c) \otimes \mathcal{X}(d,e) \otimes \mathcal{X}(e,y) \\ = & \ \mathcal{X}(x,y) \end{aligned}$
fong-spivak-seven-sketches EQ0242	310	■0.775 ● C|-4	`c_{n}:=-c:(1+-c):(1+1+-c): \cdots:(1+\cdots+1+-c): 1 \rightarrow n`	$c_{n}:=-c:(1+-c):(1+1+-c): \cdots:(1+\cdots+1+-c): 1 \rightarrow n$	$c_n := \multimap c \mathbin{;} (1 + \multimap c) \mathbin{;} (1+1+\multimap c) \mathbin{;} \cdots \mathbin{;} (1+\cdots+1+\multimap c) \colon 1 \to n$
fong-spivak-seven-sketches EQ0156	248	■0.780 ● N|+0	`\mathbf{O p}_{1}:=\{\varnothing,\{1\}, X\} \quad \text { and } \quad \mathbf{O p}_{2}:=\{\varnothing,\{2\}, X\}`	$\mathbf{O p}_{1}:=\{\varnothing,\{1\}, X\} \quad \text { and } \quad \mathbf{O p}_{2}:=\{\varnothing,\{2\}, X\}$	$\mathbf{Op}_1 := \{\varnothing, \{1\}, X\} \qquad \text{and} \qquad \mathbf{Op}_2 := \{\varnothing, \{2\}, X\}$
fong-spivak-seven-sketches EQ0219	298	■0.795 ● C|-7	`\begin{aligned} \Lambda(\text { monoid }, \text { nothing }) & =\Lambda(\text { monoid }, \text { this book }) \ \ & =\Lambda(\text { preorder, this book }) \\ \Lambda(\text { preorder, nothing }) & =\Lambda(\text { category, nothing })=\text { false. } \end{aligned} \quad=\Lambda(\text { category, this book })=\text { true }`	$\begin{aligned} \Lambda(\text { monoid }, \text { nothing }) & =\Lambda(\text { monoid }, \text { this book }) \\ & =\Lambda(\text { preorder, this book }) \\ \Lambda(\text { preorder, nothing }) & =\Lambda(\text { category, nothing })=\text { false. } \end{aligned} \quad=\Lambda(\text { category, this book })=\text { true }$	$\begin{aligned} \Lambda(\text{monoid, nothing}) &= \Lambda(\text{monoid, this book}) \\ &= \Lambda(\text{preorder, this book}) \quad = \Lambda(\text{category, this book}) = \texttt{true} \\ \Lambda(\text{preorder, nothing}) &= \Lambda(\text{category, nothing}) = \texttt{false}. \end{aligned}$
fong-spivak-seven-sketches EQ0077	124	■0.797 ● C|+3	`\begin{aligned} & \lim _{\mathcal{J}} D:=\left\{ \left(d_{1}, \ldots, d_{n}\right) \mid d_{i} \in D \left(v_{i}\right) \text { for all } 1 \leq i \leq n \right. \text { and } \\ & \left.\qquad \text { for all } a: v_{i} \rightarrow v_{j} \in A, \text { we have } D(a) \left(d_{i}\right)=d_{j}\right\} . \end{aligned}`	$\begin{aligned} & \lim _{\mathcal{J}} D:=\left\{ \left(d_{1}, \ldots, d_{n}\right) \mid d_{i} \in D \left(v_{i}\right) \text { for all } 1 \leq i \leq n \right. \text { and } \\ & \left.\qquad \text { for all } a: v_{i} \rightarrow v_{j} \in A, \text { we have } D(a) \left(d_{i}\right)=d_{j}\right\} . \end{aligned}$	$\begin{aligned} \lim_{\mathcal{J}} D := \Big\{ & (d_1, \ldots, d_n) \mid d_i \in D(v_i) \text{ for all } 1 \le i \le n \text{ and} \\ & \text{for all } a \colon v_i \to v_j \in A, \text{ we have } D(a)(d_i) = d_j \Big\}. \end{aligned}$
fong-spivak-seven-sketches EQ0131	211	■0.808 ● S|+1	`m \text { legs }\{\lessapprox \in\} n \text { legs }`	$m \text { legs }\{\lessapprox \in\} n \text { legs }$	m legs $\Big\{$  $\Big\}$ n legs

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fong-spivak-seven-sketches_ EQ0193	286	0.825 W|+0	`\begin{aligned} (x & \times y)\left(\left(x_{1}, y_{1} \right),\left(x_{2}, y_{2}\right)\right) \otimes(x \times y)\left(\left(x_{2}, y_{2}\right),\left(x_{3}, y_{3}\right)\right) \\ & =x\left(x_{1}, x_{2}\right) \otimes y\left(y_{1}, y_{2}\right) \otimes x\left(x_{2}, x_{3}\right) \otimes y\left(y_{2}, y_{3}\right) \\ & =x \left(x_{1}, x_{2}\right) \otimes x\left(x_{2}, x_{3} \right) \otimes y\left(y_{1}, y_{2}\right) \otimes y \left(y_{2}, y_{3}\right) \\ & \leq x\left(x_{1}, x_{3} \right) \otimes y\left(y_{1}, y_{3}\right) \\ & =(x \times y)\left(\left(x_{1}, y_{1}\right),\left(x_{3}, y_{3}\right)\right) . \end{aligned}`	$\begin{aligned} (x & \times y)\left(\left(x_{1}, y_{1} \right),\left(x_{2}, y_{2}\right)\right) \otimes(x \times y)\left(\left(x_{2}, y_{2}\right),\left(x_{3}, y_{3}\right)\right) \\ & =x\left(x_{1}, x_{2}\right) \otimes y\left(y_{1}, y_{2}\right) \otimes x\left(x_{2}, x_{3}\right) \otimes y\left(y_{2}, y_{3}\right) \\ & =x \left(x_{1}, x_{2}\right) \otimes x\left(x_{2}, x_{3} \right) \otimes y\left(y_{1}, y_{2}\right) \otimes y \left(y_{2}, y_{3}\right) \\ & \leq x\left(x_{1}, x_{3} \right) \otimes y\left(y_{1}, y_{3}\right) \\ & =(x \times y)\left(\left(x_{1}, y_{1}\right),\left(x_{3}, y_{3}\right)\right) . \end{aligned}$	$\begin{aligned} &(\mathcal{X} \times \mathcal{Y})((x_1,y_1),(x_2,y_2)) \otimes (\mathcal{X} \times \mathcal{Y})((x_2,y_2),(x_3,y_3)) \\ &\quad = \mathcal{X}(x_1,x_2) \otimes \mathcal{Y}(y_1,y_2) \otimes \mathcal{X}(x_2,x_3) \otimes \mathcal{Y}(y_2,y_3) \\ &\quad = \mathcal{X}(x_1,x_2) \otimes \mathcal{X}(x_2,x_3) \otimes \mathcal{Y}(y_1,y_2) \otimes \mathcal{Y}(y_2,y_3) \\ &\quad \leq \mathcal{X}(x_1,x_3) \otimes \mathcal{Y}(y_1,y_3) \\ &\quad = (\mathcal{X} \times \mathcal{Y})((x_1,y_1),(x_3,y_3)). \end{aligned}$
fong-spivak-seven-sketches_ EQ0172	264	0.827 N|+0	`\mathbb{I} \mathbb{R}:=\{[d, u] \subseteq \mathbb{R} \mid d \leq u\} .`	$\mathbb{I} \mathbb{R}:=\{[d, u] \subseteq \mathbb{R} \mid d \leq u\} .$	$\mathbb{IR} := \{[d,u] \subseteq \mathbb{R} \mid d \leq u\}.$
fong-spivak-seven-sketches_ EQ0093	143	0.828 C|+9	`\operatorname{Col}(\Phi)(a, b):= \begin{cases}x(a, b) & \text { if } a, b \in X \\ \Phi(a, b) & \text { if } a \in X, b \in y \\ \varnothing & \text { if } a \in y, b \in X \\ y(a, b) & \text { if } a, b \in y\end{cases}`	$\operatorname{Col}(\Phi)(a, b):= \begin{cases}x(a, b) & \text { if } a, b \in X \\ \Phi(a, b) & \text { if } a \in X, b \in y \\ \varnothing & \text { if } a \in y, b \in X \\ y(a, b) & \text { if } a, b \in y\end{cases}$	$\mathbf{Col}(\Phi)(a,b) := \begin{cases} \mathcal{X}(a,b) & \text{if } a,b \in \mathcal{X} \\ \Phi(a,b) & \text{if } a \in \mathcal{X}, b \in \mathcal{Y} \\ \varnothing & \text{if } a \in \mathcal{Y}, b \in \mathcal{X} \\ \mathcal{Y}(a,b) & \text{if } a,b \in \mathcal{Y} \end{cases}$
fong-spivak-seven-sketches_ EQ0022	058	0.830 W|+0	`\frac{\frac{x}{y}}{z} \quad=\quad \frac{\frac{x}{y}}{ \frac{z}{z}}`	$\frac{\frac{x}{y}}{z} \quad=\quad \frac{\frac{x}{y}}{ \frac{z}{z}}$	$\frac{\frac{x}{y}}{z} \quad = \quad \frac{x}{\frac{y}{z}}$
fong-spivak-seven-sketches_ EQ0146	229	0.839 K|+0	`\operatorname{Circ}(\varphi): \operatorname{Circ}(2) \times \operatorname{Circ}(2) \times \operatorname{Circ}(2) \rightarrow \operatorname{Circ}(0) .`	$\operatorname{Circ}(\varphi): \operatorname{Circ}(2) \times \operatorname{Circ}(2) \times \operatorname{Circ}(2) \rightarrow \operatorname{Circ}(0) .$	$\mathsf{Circ}(\varphi) \colon \mathsf{Circ}(2) \times \mathsf{Circ}(2) \times \mathsf{Circ}(2) \longrightarrow \mathsf{Circ}(0).$
fong-spivak-seven-sketches_ EQ0086 (4.25)	140	0.842 K|+0	`\mathrm{U}_{x}(x, y):=\mathcal{X}(x, y) .`	$\mathrm{U}_{x}(x, y):=\mathcal{X}(x, y) .$	$\mathrm{U}_{\mathcal{X}}(x,y) := \mathcal{X}(x,y).$
fong-spivak-seven-sketches_ EQ0136	219	0.843 N|+1	`C:=\{\text { light }, \text { switch, battery }\} \sqcup \left\{x \Omega \mid x \in \mathbb{R}^{+}\right\} .`	$C:=\{\text { light }, \text { switch, battery }\} \sqcup \left\{x \Omega \mid x \in \mathbb{R}^{+}\right\} .$	$C := \{\texttt{light}, \texttt{switch}, \texttt{battery}\} \sqcup \{x\Omega \mid x \in \mathbb{R}^+\}.$
fong-spivak-seven-sketches_ EQ0125	197	0.843 N|+1	`\mathcal{C}:=\stackrel{A}{\bullet} \underset{g}{ \stackrel{f}{\rightrightarrows}} \stackrel{B}{\bullet}`	$\mathcal{C}:=\stackrel{A}{\bullet} \underset{g}{ \stackrel{f}{\rightrightarrows}} \stackrel{B}{\bullet}$	$\mathcal{C} := \; \stackrel{A}{\bullet} \underset{g}{\stackrel{f}{\rightrightarrows}} \stackrel{B}{\bullet}$
fong-spivak-seven-sketches_ EQ0003	024	0.849 C|+7	`S:=\begin{array}{|ccc|} \hline 11 & 12 & 13 \\ \bullet & & \\ 21 & 22 & 23 \\ \bullet & \bullet & \bullet \end{array}`	$S:=\begin{array}{\|ccc\|} \hline 11 & 12 & 13 \\ \bullet & & \\ 21 & 22 & 23 \\ \bullet & \bullet & \bullet \end{array}$	$S := \begin{array}{ccc} \stackrel{11}{\bullet} & \stackrel{12}{\bullet} & \stackrel{13}{\bullet} \\ \stackrel{21}{\bullet} & \stackrel{22}{\bullet} & \stackrel{23}{\bullet} \end{array}$

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fong-spivak-seven-sketche EQ0121	188	0.868 N|+1	`\mathrm{B}\left(g \circ\left(h^{\mathrm{op}}\right) \right)=\{(x, y) \mid S(g)(x)=S(h)(y)\} .`	$\mathrm{B}\left(g \circ\left(h^{\mathrm{op}}\right)\right)=\{(x, y) \mid S(g)(x)=S(h)(y)\}.$	$\mathsf{B}(g \mathbin{⨟} (h^{\mathrm{op}})) = \{(x,y) \mid S(g)(x) = S(h)(y)\}.$
fong-spivak-seven-sketche EQ0107	174	0.873 W|+1	`\begin{array}{r} \dot{x}+3 \ddot{y}-2 z=0 \\ \ddot{y}+5 \dot{z}=0 \end{array}`	$\begin{array}{r} \dot{x}+3 \ddot{y}-2 z=0 \\ \ddot{y}+5 \dot{z}=0 \end{array}$	$\begin{array}{r} \dot{x}+3 \ddot{y}-2 z=0 \\ \ddot{y}+5 \dot{z}=0 \end{array}$
fong-spivak-seven-sketche EQ0010 (1.108)	042	0.877 C|+4	`p \leq g(f(p)) \quad \text { and } \quad f(g(q)) \leq q \text {. }`	$p \leq g(f(p)) \quad \text { and } \quad f(g(q)) \leq q \text {. }$	$p \leq g(f(p)) \quad \text{and} \quad f(g(q)) \leq q.$
fong-spivak-seven-sketche EQ0079 (4.5)	132	0.881 K|+0	`b \wedge c \leq d \quad \text { iff } \quad b \leq(c \Rightarrow d) .`	$b \wedge c \leq d \quad \text { iff } \quad b \leq(c \Rightarrow d) .$	$b \wedge c \leq d \quad \text{iff} \quad b \leq (c \Rightarrow d).$
fong-spivak-seven-sketche EQ0217	296	0.890 C|-4	`\begin{aligned} \lim _{\mathfrak{J}} D:=\left\{ \left(d_{1}, \ldots, d_{n}\right) \mid d_{i} \in D \left(v_{i}\right) \text { for all } 1 \leq i \leq n \right. \text { and } & \\ & \text { for all } \left.a: v_{i} \rightarrow v_{j} \in A, \text { we have } D(a) \left(d_{i}\right)=d_{j}\right\} . \end{aligned}`	$\begin{aligned} \lim _{\mathfrak{J}} D:=\left\{ \left(d_{1}, \ldots, d_{n}\right) \mid d_{i} \in D \left(v_{i}\right) \text { for all } 1 \leq i \leq n \right. \text { and } & \\ & \text { for all } \left.a: v_{i} \rightarrow v_{j} \in A, \text { we have } D(a) \left(d_{i}\right)=d_{j}\right\} . \end{aligned}$	$\begin{aligned} \lim_{\mathcal{J}} D := \{(d_1, \ldots, d_n) \mid d_i \in D(v_i) \text{ for all } 1 \leq i \leq n \text{ and} & \\ & \text{for all } a\colon v_i \to v_j \in A, \text{ we have } D(a)(d_i) = d_j\}. \end{aligned}$
fong-spivak-seven-sketche EQ0153 (7.18)	242	0.894 C|+3	`X=\{(\text { true }, \text { true }),(\text { true }, \text { false }),(\text { false }, \text { true })\} \subseteq \mathbb{B} \times \mathbb{B} .`	$X=\{(\text { true }, \text { true }),(\text { true }, \text { false }),(\text { false }, \text { true })\} \subseteq \mathbb{B} \times \mathbb{B} .$	$X = \{(\mathtt{true},\mathtt{true}),(\mathtt{true},\mathtt{false}),(\mathtt{false},\mathtt{true})\} \subseteq \mathbb{B} \times \mathbb{B}.$
fong-spivak-seven-sketche EQ0105	167	0.897 N|+0	`a \subset \bullet>b`	$a \subset \bullet>b$	$a \circlearrowleft \bullet \circlearrowright b$
fong-spivak-seven-sketche EQ0143 (6.95)	227	0.902 K|+0	`\left(g \circ{ }_{i} f\right): X_{1} \times \cdots \times X_{i-1} \times W_{1} \times \cdots \times W_{m} \times X_{i+1} \times \cdots \times X_{n} \longrightarrow Y`	$\left(g \circ{ }_{i} f\right): X_{1} \times \cdots \times X_{i-1} \times W_{1} \times \cdots \times W_{m} \times X_{i+1} \times \cdots \times X_{n} \longrightarrow Y$	$(g \circ_i f)\colon X_1 \times \cdots \times X_{i-1} \times W_1 \times \cdots \times W_m \times X_{i+1} \times \cdots \times X_n \longrightarrow Y$
fong-spivak-seven-sketche EQ0129	205	0.913 N|+0	`\left\{(v, d) \mid d \in D(v), \text { where } v=v_{1} \text { or } v=v_{2}\right\} .`	$\left\{(v, d) \mid d \in D(v), \text { where } v=v_{1} \text { or } v=v_{2}\right\} .$	$\{(v,d) \mid d \in D(v), \text{ where } v = v_1 \text{ or } v = v_2\}.$
fong-spivak-seven-sketche EQ0164	257	0.917 C|+22	`\stackrel{A}{\longrightarrow} \stackrel{B}{\bullet} \quad \stackrel{C}{\bullet}`	$\stackrel{A}{\longrightarrow} \stackrel{B}{\bullet} \quad \stackrel{C}{\bullet}$	$\boxed{\overset{A}{\bullet} \longrightarrow \overset{B}{\bullet} \qquad \overset{C}{\bullet}} \subseteq \overset{A}{\bullet} \underset{g}{\overset{f}{\rightleftarrows}} \overset{B}{\bullet} \xrightarrow{h} \overset{C}{\bullet} \xrightarrow{i} \overset{D}{\bullet}$
fong-spivak-seven-sketche EQ0256	315	0.919 C|-10	`\begin{aligned} f(m) *_{R} f(n) & =(\underbrace{1_{R}+\cdots+1_{R}}_{m \text { summands }}) * \underbrace{1_{R}+\cdots+1_{R}}_{n \text { summands }}) \\ & =\underbrace{1_{R} *(\underbrace{1_{R}+\cdots+1_{R}}_{n \text { summands }})+\cdots+1_{R} *(\underbrace{1_{R}+\cdots+1_{R}}_{n \text { summands }})}_{m \text { summands }} \\ & =\underbrace{1_{R}+\cdots+1_{R}}_{m n \text { summands }}=f(m * n) . \end{aligned}`	$\begin{aligned} f(m) *_{R} f(n) & =(\underbrace{1_{R}+\cdots+1_{R}}_{m \text { summands }}) * \underbrace{1_{R}+\cdots+1_{R}}_{n \text { summands }}) \\ & =\underbrace{1_{R} *(\underbrace{1_{R}+\cdots+1_{R}}_{n \text { summands }})+\cdots+1_{R} *(\underbrace{1_{R}+\cdots+1_{R}}_{n \text { summands }})}_{m \text { summands }} \\ & =\underbrace{1_{R}+\cdots+1_{R}}_{m n \text { summands }}=f(m * n) . \end{aligned}$	$\begin{aligned} f(m) *_R f(n) &= (\underbrace{1_R + \cdots + 1_R}_{m \text{ summands}}) *_R (\underbrace{1_R + \cdots + 1_R}_{n \text{ summands}}) \\ &= \underbrace{1_R * (\underbrace{1_R + \cdots + 1_R}_{n \text{ summands}}) + \cdots + 1_R * (\underbrace{1_R + \cdots + 1_R}_{n \text{ summands}})}_{m \text{ summands}} \\ &= \underbrace{1_R + \cdots + 1_R}_{mn \text{ summands}} = f(m * n). \end{aligned}$
fong-spivak-seven-sketche EQ0096	149	0.919 S|+0	`(f \times g) ;\left(f^{\prime} \times g^{\prime} \right)=(f ; g) \times\left(f^{\prime} ; g^{\prime} \right) .`	$(f \times g) ;\left(f^{\prime} \times g^{\prime}\right)=(f ; g) \times\left(f^{\prime} ; g^{\prime}\right) .$	$(f \times g) \mathbin{⨟} (f' \times g') = (f \mathbin{⨟} g) \times (f' \mathbin{⨟} g').$

Conf. MathPix confidence: ≥0.9 0.5–0.9 <0.5 — no value

Residual render vs scan (inkdrill): • C component • W weak • S stable • N noise • K clean • not measured

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
fong-spivak-seven-sketche EQ0259	321	0.923 C|-3	`\begin{aligned} \varphi_{S^{\prime}, T^{\prime}}\left(\left(\operatorname{im}_{f} \times \operatorname{im}_{g} \right)(A \times B)\right) & =\varphi_{S^{\prime}, T^{ \prime}}(\{f(a) \mid a \in A\} \times\{g(b) \mid b \in B\}) \\ & =\{(f(a), g(b)) \mid a \in A, b \in B\} \\ & =\operatorname{im}_{f \times g}(A \times B) \\ & = \operatorname{im}_{f \times g}\left(\varphi_{S, T}(A, B) \right) . \end{aligned}`	$\begin{aligned} \varphi_{S^{\prime}, T^{\prime}}\left(\left(\operatorname{im}_{f} \times \operatorname{im}_{g} \right)(A \times B)\right) & =\varphi_{S^{\prime}, T^{ \prime}}(\{f(a) \mid a \in A\} \times\{g(b) \mid b \in B\}) \\ & =\{(f(a), g(b)) \mid a \in A, b \in B\} \\ & =\operatorname{im}_{f \times g}(A \times B) \\ & = \operatorname{im}_{f \times g}\left(\varphi_{S, T}(A, B) \right) . \end{aligned}$	$\begin{aligned} \varphi_{S', T'}\Big((\operatorname{im}_f \times \operatorname{im}_g)(A \times B)\Big) &= \varphi_{S',T'}(\{f(a) \mid a \in A\} \times \{g(b) \mid b \in B\}) \\ &= \{(f(a), g(b)) \mid a \in A,\, b \in B\} \\ &= \operatorname{im}_{f\times g}(A \times B) \\ &= \operatorname{im}_{f \times g}(\varphi_{S,T}(A,B)). \end{aligned}$
fong-spivak-seven-sketche EQ0087	140	0.928 N|+0	`\mathrm{U}_{\mathcal{P}} ; \Phi=\Phi=\Phi ; \mathrm{U}_{ \mathcal{Q}}`	$\mathrm{U}_{\mathcal{P}} ; \Phi=\Phi=\Phi ; \mathrm{U}_{ \mathcal{Q}}$	$\mathrm{U}_{\mathcal{P}} \fatsemi \Phi = \Phi = \Phi \fatsemi \mathrm{U}_{\mathcal{Q}}$
fong-spivak-seven-sketche EQ0199	288	0.928 W|+0	`\begin{aligned} ((M * N) * P)(w, z) & =\bigvee_{y \in Y} \left(\bigvee_{x \in X} M(w, x) \otimes N(x, y)\right) \otimes P(y, z) \\ & \cong \bigvee_{y \in Y, x \in X} M(w, x) \otimes N(x, y) \otimes P(y, z) \\ & \cong \bigvee_{x \in X} M(w, x) \otimes\left(\bigvee_{y \in Y} N(x, y) \otimes P(y, z)\right) \\ & =(M *(N * P))(w, z) . \end{aligned}`	$\begin{aligned} ((M * N) * P)(w, z) & =\bigvee_{y \in Y} \left(\bigvee_{x \in X} M(w, x) \otimes N(x, y)\right) \otimes P(y, z) \\ & \cong \bigvee_{y \in Y, x \in X} M(w, x) \otimes N(x, y) \otimes P(y, z) \\ & \cong \bigvee_{x \in X} M(w, x) \otimes\left(\bigvee_{y \in Y} N(x, y) \otimes P(y, z)\right) \\ & =(M *(N * P))(w, z) . \end{aligned}$	$\begin{aligned} ((M * N) * P)(w, z) &= \bigvee_{y \in Y}\Big(\bigvee_{x \in X} M(w,x) \otimes N(x,y)\Big) \otimes P(y,z) \\ &\cong \bigvee_{y \in Y, x \in X} M(w,x) \otimes N(x,y) \otimes P(y,z) \\ &\cong \bigvee_{x \in X} M(w,x) \otimes \Big(\bigvee_{y \in Y} N(x,y) \otimes P(y,z)\Big) \\ &= (M * (N * P))(w,z). \end{aligned}$
fong-spivak-seven-sketche EQ0251	313	0.930 N|+0	`\mathrm{B}\left(g ;\left(h^{\mathrm{op}}\right)\right)= \{(x, y) \mid S(g)(x)=S(h)(y)\} .`	$\mathrm{B}\left(g ;\left(h^{\mathrm{op}}\right)\right)= \{(x, y) \mid S(g)(x)=S(h)(y)\} .$	$\mathsf{B}(g \fatsemi (h^{\mathrm{op}})) = \{(x, y) \mid S(g)(x) = S(h)(y)\}.$
fong-spivak-seven-sketche EQ0034	073	0.932 N|-1	`d_{L}(U, V):=\sup _{u \in U} \inf _{v \in V} d(u, v),`	$d_{L}(U, V):=\sup _{u \in U} \inf _{v \in V} d(u, v),$	$d_L(U, V) := \sup_{u \in U} \inf_{v \in V} d(u, v),$
fong-spivak-seven-sketche EQ0128	205	0.939 C|+3	`\operatorname{colim}_{\mathcal{J}} D:=\{(v, d) \mid v \in V \text { and } d \in D(v)\} / \sim`	$\operatorname{colim}_{\mathcal{J}} D:=\{(v, d) \mid v \in V \text { and } d \in D(v)\} / \sim$	$\operatorname{colim}_{\mathcal{J}} D := \big\{(v, d) \mid v \in V \text{ and } d \in D(v)\big\}/\sim$
fong-spivak-seven-sketche EQ0053 (3.8)	094	0.941 S|+1	`\mathbf{2}:=\operatorname{Free}\left(\begin{array}{|ccc|} \hline v_{1} & f_{1} & v_{2} \\ \bullet & & \bullet \\ \hline \end{array}\right)`	$\mathbf{2}:=\operatorname{Free}\left(\begin{array}{\|ccc\|} \hline v_{1} & f_{1} & v_{2} \\ \bullet & & \bullet \\ \hline \end{array}\right)$	$\mathbf{2} := \mathbf{Free}\left(\boxed{\overset{v_1}{\bullet} \xrightarrow{f_1} \overset{v_2}{\bullet}}\right)$
fong-spivak-seven-sketche EQ0075	119	0.943 U|-7	`\frac{1}{\text { Em_6 }}`	*(not rendered)*	$\frac{\mathbf{1}}{\text{Em\_6}}$
fong-spivak-seven-sketche EQ0026	060	0.948 N|-1	`\mathrm{H}_{2} \mathrm{O}, \quad \mathrm{NaCl}, \quad 2 \mathrm{NaOH}, \quad \mathrm{CH}_{4}+3 \mathrm{O}_{2}`	$\mathrm{H}_{2} \mathrm{O}, \quad \mathrm{NaCl}, \quad 2 \mathrm{NaOH}, \quad \mathrm{CH}_{4}+3 \mathrm{O}_{2}$	$\mathrm{H_2O}, \quad \mathrm{NaCl}, \quad 2\mathrm{NaOH}, \quad \mathrm{CH_4} + 3\mathrm{O_2}$
fong-spivak-seven-sketche EQ0111	176	0.950 N|+1	`\left(\begin{array}{cc} A & 0 \\ 0 & B \end{array}\right)`	$\left(\begin{array}{cc} A & 0 \\ 0 & B \end{array}\right)$	$\begin{pmatrix} A & 0 \\ 0 & B \end{pmatrix}$

Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value

Residual render vs scan (inkdrill): • C component • W weak • S stable • N noise • K clean • not measured

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
fong-spivak-seven-sketche… EQ0159 (7.43)	252	■0.951 ● N|+0	`\operatorname{Sec}_{f}(U):=\{g: U \rightarrow X \mid g \text { is continuous and }(g \circ f)(u)=u \text { for all } u \in U\},`	$\operatorname{Sec}_{f}(U):=\{g: U \rightarrow X \mid g \text { is continuous and }(g \circ f)(u)=u \text { for all } u \in U\},$	$\operatorname{Sec}_{f}(U):=\{g: U \rightarrow X \mid g \text { is continuous and }(g \fatsemi f)(u)=u \text { for all } u \in U\},$
fong-spivak-seven-sketche… EQ0007	038	■0.952 ● N|+1	`f(a) \vee f(b) \not \equiv f(a \vee b) .`	$f(a) \vee f(b) \not \equiv f(a \vee b) .$	$f(a) \vee f(b) \not \equiv f(a \vee b) .$
fong-spivak-seven-sketche… EQ0030 (2.25)	062	■0.954 ● N|+0	`\mathrm{H}_{2} \mathrm{O}+\mathrm{NaCl} \rightarrow^{?} \mathrm{H}_{2} \mathrm{O}`	$\mathrm{H}_{2} \mathrm{O}+\mathrm{NaCl} \rightarrow^{?} \mathrm{H}_{2} \mathrm{O}$	$\mathrm{H}_{2} \mathrm{O}+\mathrm{NaCl} \rightarrow^{?} \mathrm{H}_{2} \mathrm{O}$
fong-spivak-seven-sketche… EQ0194	287	■0.955 ● K|+0	`A=(A \cap \bar{B}) \cup(A \cap B) \subseteq \bar{B} \cup C .`	$A=(A \cap \bar{B}) \cup(A \cap B) \subseteq \bar{B} \cup C .$	$A=(A \cap \bar{B}) \cup(A \cap B) \subseteq \bar{B} \cup C .$
fong-spivak-seven-sketche… EQ0171 (7.73)	263	■0.958 ● K|+0	`\forall(t: T) . \exists(s: S) . f(s)=t`	$\forall(t: T) . \exists(s: S) . f(s)=t$	$\forall(t: T) . \exists(s: S) . f(s)=t$
fong-spivak-seven-sketche… EQ0040 (2.80)	081	■0.960 ● K|+0	`(a \otimes v) \leq w \quad \text { iff } \quad a \leq(v \multimap w) .`	$(a \otimes v) \leq w \quad \text { iff } \quad a \leq(v \multimap w) .$	$(a \otimes v) \leq w \quad \text { iff } \quad a \leq(v \multimap w) .$
fong-spivak-seven-sketche… EQ0262	328	■0.963 ● K|+0	`\bigcup_{i \in I} A_{i}=\bigcup_{i \in I}\left(B_{i} \cap Y\right)=\left(\bigcup_{i \in i} B_{i}\right) \cap Y \in \mathbf{O p}_{? \cap Y} .`	$\bigcup_{i \in I} A_{i}=\bigcup_{i \in I}\left(B_{i} \cap Y\right)=\left(\bigcup_{i \in i} B_{i}\right) \cap Y \in \mathbf{O p}_{? \cap Y} .$	$\bigcup_{i \in I} A_{i}=\bigcup_{i \in I}\left(B_{i} \cap Y\right)=\left(\bigcup_{i \in i} B_{i}\right) \cap Y \in \mathbf{O p}_{? \cap Y} .$
fong-spivak-seven-sketche… EQ0092 (4.40)	143	■0.967 ● N|+0	`\mathcal{P}(p, G(q)) \cong \mathcal{Q}(F(p), q)`	$\mathcal{P}(p, G(q)) \cong \mathcal{Q}(F(p), q)$	$\mathcal{P}(p, G(q)) \cong \mathcal{Q}(F(p), q)$
fong-spivak-seven-sketche… EQ0218	297	■0.967 ● C|-3	`\lim _{\mathcal{J}} D:=\left\{\left(d_{1}, d_{2}\right) \mid d_{1} \in A \text { and } d_{2} \in B\right\} .`	$\lim _{\mathcal{J}} D:=\left\{\left(d_{1}, d_{2}\right) \mid d_{1} \in A \text { and } d_{2} \in B\right\} .$	$\lim _{\mathcal{J}} D:=\left\{\left(d_{1}, d_{2}\right) \mid d_{1} \in A \text { and } d_{2} \in B\right\} .$
fong-spivak-seven-sketche… EQ0168	260	■0.967 ● K|+0	`s: S \mid p(s) \vdash q(s) \quad \text { or } \quad p(s) \vdash_{s: S} q(s) \text {. }`	$s: S \mid p(s) \vdash q(s) \quad \text { or } \quad p(s) \vdash_{s: S} q(s) \text {. }$	$s: S \mid p(s) \vdash q(s) \quad \text { or } \quad p(s) \vdash_{s: S} q(s) .$
fong-spivak-seven-sketche… EQ0066 (3.69)	114	■0.968 ● N|+0	`f(p) \leq q \text { if and only if } p \leq g(q) .`	$f(p) \leq q \text { if and only if } p \leq g(q) .$	$f(p) \leq q \text { if and only if } p \leq g(q) .$
fong-spivak-seven-sketche… EQ0119 (5.78)	187	■0.969 ● N|+0	`B_{1} ; B_{2}:=\left\{(x, z) \mid \text { there exists } y \in R^{n} \text { such that }(x, y) \in B_{1} \text { and }(y, z) \in B_{2}\right\} .`	$B_{1} ; B_{2}:=\left\{(x, z) \mid \text { there exists } y \in R^{n} \text { such that }(x, y) \in B_{1} \text { and }(y, z) \in B_{2}\right\} .$	$B_{1} \fatsemi B_{2}:=\left\{(x, z) \mid \text { there exists } y \in R^{n} \text { such that }(x, y) \in B_{1} \text { and }(y, z) \in B_{2}\right\} .$
fong-spivak-seven-sketche… EQ0157	249	■0.969 ● N|+2	`\left.s_{i}\right|_{U_{i} \cap U_{j}}=\left.s_{j}\right|_{U_{i} \cap U_{j}} .`	$\left.s_{i}\right\vert_{U_{i} \cap U_{j}}=\left.s_{j}\right\vert_{U_{i} \cap U_{j}} .$	$\left.s_{i}\right\vert_{U_{i} \cap U_{j}}=\left.s_{j}\right\vert_{U_{i} \cap U_{j}} .$
fong-spivak-seven-sketche… EQ0100	157	■0.973 ● N|+0	`\eta_{x}: \mathbf{1} \times \mathcal{X}^{\mathrm{op}} \times \mathcal{X} \rightarrow \mathcal{V} ;`	$\eta_{x}: \mathbf{1} \times \mathcal{X}^{\mathrm{op}} \times \mathcal{X} \rightarrow \mathcal{V} ;$	$\eta_{\mathcal{X}}: \mathbf{1} \times \mathcal{X}^{\mathrm{op}} \times \mathcal{X} \rightarrow \mathcal{V} ;$
fong-spivak-seven-sketche… EQ0045	085	■0.974 ● W|-1	`d(U, V):=\sup _{u \in U} \inf _{v \in V} d(u, v) \quad X(U, V):=\bigwedge_{u \in U} \bigvee_{v \in V} x(u, v) .`	$d(U, V):=\sup _{u \in U} \inf _{v \in V} d(u, v) \quad X(U, V):=\bigwedge_{u \in U} \bigvee_{v \in V} x(u, v) .$	$d(U, V):=\sup _{u \in U} \inf _{v \in V} d(u, v) \qquad \mathcal{X}(U, V):=\bigwedge_{u \in U} \bigvee_{v \in V} \mathcal{X}(u, v) .$
fong-spivak-seven-sketche… EQ0046	085	■0.974 ● N|+1	`v \rightarrow w:=\bigvee_{\{a \in V \mid a \otimes v \leq w\}} a .`	$v \rightarrow w:=\bigvee_{\{a \in V \mid a \otimes v \leq w\}} a .$	$v \multimap w:=\bigvee_{\{a \in V \mid a \otimes v \leq w\}} a .$
fong-spivak-seven-sketche… EQ0085	139	■0.976 ● C|+6	`(\Phi ; \Psi)(p, r)=\bigvee_{q \in Q}(\Phi(p, q) \otimes \Psi(q, r)) .`	$(\Phi ; \Psi)(p, r)=\bigvee_{q \in Q}(\Phi(p, q) \otimes \Psi(q, r)) .$	$(\Phi \fatsemi \Psi)(p, r)=\bigvee_{q \in Q}(\Phi(p, q) \otimes \Psi(q, r)) .$
fong-spivak-seven-sketche… EQ0195	287	■0.979 ● K|+0	`A \cap B \subseteq(\bar{B} \cup C) \cap B=(\bar{B} \cap B) \cup(C \cap B)=C \cap B \subseteq C .`	$A \cap B \subseteq(\bar{B} \cup C) \cap B=(\bar{B} \cap B) \cup(C \cap B)=C \cap B \subseteq C .$	$A \cap B \subseteq(\bar{B} \cup C) \cap B=(\bar{B} \cap B) \cup(C \cap B)=C \cap B \subseteq C .$
fong-spivak-seven-sketche… EQ0173	265	■0.979 ● W|+0	`\mathbb{R} \cong\{[d, u] \in \mathbb{I} \mathbb{R} \mid d=u\} .`	$\mathbb{R} \cong\{[d, u] \in \mathbb{I} \mathbb{R} \mid d=u\} .$	$\mathbb{R} \cong\{[d, u] \in \mathbb{I} \mathbb{R} \mid d=u\} .$

Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value
Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
fong-spivak-seven-sketche EQ0137	220	0.982 K|+0	`\text { Circ: }(\text { FinSet },+) \longrightarrow(\text { Set }, \times)`	$\text { Circ: }(\text { FinSet },+) \longrightarrow(\text { Set }, \times)$	$\mathrm{Circ}\colon (\mathbf{FinSet},+) \longrightarrow (\mathbf{Set},\times)$
fong-spivak-seven-sketche EQ0169	260	0.983 N|+1	`\forall(t: T) . p(s, t) .`	$\forall(t: T) . p(s, t) .$	$\forall(t:T).\, p(s,t).$
fong-spivak-seven-sketche EQ0018	046	0.985 N|-1	`\operatorname{fix}_{j}:=\{p \in P \mid j(p) \cong p\} .`	$\operatorname{fix}_{j}:=\{p \in P \mid j(p) \cong p\} .$	$\operatorname{fix}_j := \{p \in P \mid j(p) \cong p\}.$
fong-spivak-seven-sketche EQ0103	165	0.985 N|+0	`\iota^{\prime \prime}(x)= \begin{cases}\iota(x) & \text { if } \iota(x) \in I \\ \iota^{\prime}(\iota(x)) & \text { if } \iota(x) \in \underline{n} \\ \iota^{\prime}(x) & \text { if } x \in O .^{\prime}\end{cases}`	$\iota^{\prime \prime}(x)= \begin{cases}\iota(x) & \text { if } \iota(x) \in I \\ \iota^{\prime}(\iota(x)) & \text { if } \iota(x) \in \underline{n} \\ \iota^{\prime}(x) & \text { if } x \in O .^{\prime}\end{cases}$	$\iota''(x) = \begin{cases}\iota(x) & \text{if } \iota(x) \in I \\ \iota'(\iota(x)) & \text{if } \iota(x) \in \underline{n} \\ \iota'(x) & \text{if } x \in O.'\end{cases}$
fong-spivak-seven-sketche EQ0210	292	0.986 N|+1	`F(a):=a^{\prime}, \quad G(a):=a^{\prime}, \quad F(b):=b^{\prime}, \quad G(b):=b^{\prime}, \quad F(f):=f_{1}, \text { and } G(f):=f_{2} .`	$F(a):=a^{\prime}, \quad G(a):=a^{\prime}, \quad F(b):=b^{\prime}, \quad G(b):=b^{\prime}, \quad F(f):=f_{1}, \text { and } G(f):=f_{2} .$	$F(a) := a', \quad G(a) := a', \quad F(b) := b', \quad G(b) := b', \quad F(f) := f_1, \text{ and } G(f) := f_2.$
fong-spivak-seven-sketche EQ0117 (5.75)	186	0.988 N|+0	`\mathrm{B}(g)=\left\{(x, S(g)(x)) \mid x \in R^{m}\right\} \subseteq R^{m} \times R^{n} .`	$\mathrm{B}(g)=\left\{(x, S(g)(x)) \mid x \in R^{m}\right\} \subseteq R^{m} \times R^{n} .$	$\mathrm{B}(g) = \left\{(x, S(g)(x)) \mid x \in R^m\right\} \subseteq R^m \times R^n.$
fong-spivak-seven-sketche EQ0031	067	0.989 K|+0	`\text { Cost }:=([0, \infty], \geq, 0,+),`	$\text { Cost }:=([0, \infty], \geq, 0,+),$	$\mathbf{Cost} := ([0,\infty], \geq, 0, +),$
fong-spivak-seven-sketche EQ0204	289	0.989 N|+0	`\stackrel{v_{1}}{\bullet} \xrightarrow{f_{1}} \stackrel{v_{2}}{\bullet} \xrightarrow{f_{2}} \cdots \xrightarrow{f_{n-1}} \stackrel{v_{n}}{\bullet}`	$\stackrel{v_{1}}{\bullet} \xrightarrow{f_{1}} \stackrel{v_{2}}{\bullet} \xrightarrow{f_{2}} \cdots \xrightarrow{f_{n-1}} \stackrel{v_{n}}{\bullet}$	$\stackrel{v_1}{\bullet} \xrightarrow{f_1} \stackrel{v_2}{\bullet} \xrightarrow{f_2} \cdots \xrightarrow{f_{n-1}} \stackrel{v_n}{\bullet}$
fong-spivak-seven-sketche EQ0213	294	0.989 N|+1	`G:=\stackrel{1}{\bullet} \xrightarrow{a} \stackrel{2}{\bullet} \xrightarrow{b} \stackrel{3}{\bullet}`	$G:=\stackrel{1}{\bullet} \xrightarrow{a} \stackrel{2}{\bullet} \xrightarrow{b} \stackrel{3}{\bullet}$	$G := \boxed{\stackrel{1}{\bullet} \xrightarrow{a} \stackrel{2}{\bullet} \xrightarrow{b} \stackrel{3}{\bullet}}$
fong-spivak-seven-sketche EQ0250	313	0.990 N|-1	`\left\{(x, y) \mid \text { there exists } z \in R^{n} \text { such that } S(g)(x)=z \text { and } z=S(h)(y)\right\} .`	$\left\{(x, y) \mid \text { there exists } z \in R^{n} \text { such that } S(g)(x)=z \text { and } z=S(h)(y)\right\} .$	$\{(x,y) \mid \text{ there exists } z \in R^n \text{ such that } S(g)(x) = z \text{ and } z = S(h)(y)\}.$
fong-spivak-seven-sketche EQ0038	078	0.990 N|+1	`\mathcal{X}\left(x_{1}, x_{2}\right) \leq y\left(F\left(x_{1}\right), F\left(x_{2}\right)\right),`	$\mathcal{X}\left(x_{1}, x_{2}\right) \leq y\left(F\left(x_{1}\right), F\left(x_{2}\right)\right),$	$\mathcal{X}(x_1, x_2) \leq \mathcal{Y}(F(x_1), F(x_2)),$
fong-spivak-seven-sketche EQ0102	162	0.991 N|+0	`R ; S:=\{(i, k) \in \underline{m} \times \underline{p} \mid \exists(j \in \underline{n}) .(i, j) \in R \text { and }(j, k) \in S\} .`	$R ; S:=\{(i, k) \in \underline{m} \times \underline{p} \mid \exists(j \in \underline{n}) .(i, j) \in R \text { and }(j, k) \in S\} .$	$R \mathbin{⨟} S := \{(i,k) \in \underline{m} \times \underline{p} \mid \exists(j \in \underline{n}).\,(i,j) \in R \text{ and } (j,k) \in S\}.$
fong-spivak-seven-sketche EQ0080 (4.6)	133	0.991 N|+1	`\Phi: x^{\mathrm{op}} \times y \rightarrow \mathcal{V} .`	$\Phi: x^{\mathrm{op}} \times y \rightarrow \mathcal{V} .$	$\Phi\colon \mathcal{X}^{\mathrm{op}} \times \mathcal{Y} \to \mathcal{V}.$
fong-spivak-seven-sketche EQ0134	216	0.992 N|+0	`\varphi_{c_{1}, c_{2}}: F\left(c_{1}\right) \otimes_{\mathcal{D}} F\left(c_{2}\right) \rightarrow F\left(c_{1} \otimes_{\mathcal{C}} c_{2}\right),`	$\varphi_{c_{1}, c_{2}}: F\left(c_{1}\right) \otimes_{\mathcal{D}} F\left(c_{2}\right) \rightarrow F\left(c_{1} \otimes_{\mathcal{C}} c_{2}\right),$	$\varphi_{c_1,c_2}\colon F(c_1) \otimes_{\mathcal{D}} F(c_2) \to F(c_1 \otimes_{\mathcal{C}} c_2),$
fong-spivak-seven-sketche EQ0148	236	0.992 U|+8	`\{p \in P \mid \text { likes_cats }(p)\} .`	*(not rendered)*	$\{p \in P \mid \mathtt{likes\_cats}(p)\}.$
fong-spivak-seven-sketche EQ0144	228	0.992 N|+1	`(f \circ i g):=f \circ(\mathrm{id}, \ldots, \mathrm{id}, g, \mathrm{id}, \ldots, \mathrm{id})`	$(f \circ i g):=f \circ(\mathrm{id}, \ldots, \mathrm{id}, g, \mathrm{id}, \ldots, \mathrm{id})$	$(f \circ_i g) := f \circ (\mathrm{id}, \ldots, \mathrm{id}, g, \mathrm{id}, \ldots, \mathrm{id})$
fong-spivak-seven-sketche EQ0123 (5.86)	189	0.993 N|-1	`\{(0,(x, x)) \mid x \in R\} \subseteq R^{0} \times R^{2} \quad \text { and } \quad\{((x, x), 0) \mid x \in R\} \subseteq R^{2} \times R^{0} .`	$\{(0,(x, x)) \mid x \in R\} \subseteq R^{0} \times R^{2} \quad \text { and } \quad\{((x, x), 0) \mid x \in R\} \subseteq R^{2} \times R^{0} .$	$\{(0,(x,x)) \mid x \in R\} \subseteq R^0 \times R^2 \quad \text{and} \quad \{((x,x),0) \mid x \in R\} \subseteq R^2 \times R^0.$

Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value

Residual render vs scan (inkdrill): • C component • W weak • S stable • N noise • K clean • not measured

<table>
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<tr><td>fong-spivak-seven-sketches_ EQ0229</td><td>304</td><td>■ 0.993 ● W|+0</td><td><code>x \xrightarrow{\alpha^{-1}} x \times \mathbf{1} \xrightarrow{\mathrm{U}_{x} \times \eta_{x}} x \times x^{\mathrm{op}} \times x \xrightarrow{\epsilon_{x} \times \mathrm{U}_{x}} \mathbf{1} \times x \xrightarrow{\alpha} x</code></td><td>$x \xrightarrow{\alpha^{-1}} x \times \mathbf{1} \xrightarrow{\mathrm{U}_{x} \times \eta_{x}} x \times x^{\mathrm{op}} \times x \xrightarrow{\epsilon_{x} \times \mathrm{U}_{x}} \mathbf{1} \times x \xrightarrow{\alpha} x$</td><td>$\mathcal{X} \xrightarrow{\alpha^{-1}} \mathcal{X} \times \mathbf{1} \xrightarrow{\mathrm{U}_{\mathcal{X}} \times \eta_{\mathcal{X}}} \mathcal{X} \times \mathcal{X}^{\mathrm{op}} \times \mathcal{X} \xrightarrow{\epsilon_{\mathcal{X}} \times \mathrm{U}_{\mathcal{X}}} \mathbf{1} \times \mathcal{X} \xrightarrow{\alpha} \mathcal{X}$</td></tr>
<tr><td>fong-spivak-seven-sketches_ EQ0225</td><td>301</td><td>■ 0.993 ● W|+0</td><td><code>\begin{aligned} ((\Phi ; \Psi) ; \Upsilon)(p, s) & =\bigvee_{r \in \mathcal{R}}\left(\bigvee_{q \in \mathcal{Q}} \Phi(p, q) \otimes \Psi(q, r)\right) \otimes \Upsilon(r, s) \\ & =\bigvee_{r \in \mathcal{R}, q \in \mathcal{Q}} \Phi(p, q) \otimes \Psi(q, r) \otimes \Upsilon(r, s) \\ & =\bigvee_{q \in \mathcal{Q}} \Phi(p, q) \otimes \bigvee_{r \in \mathcal{R}}(\Psi(q, r) \otimes \Upsilon(r, s)) \\ & =(\Phi ;(\Psi ; \Upsilon))(p, s) \end{aligned}</code></td><td>$\begin{aligned} ((\Phi ; \Psi) ; \Upsilon)(p, s) & =\bigvee_{r \in \mathcal{R}}\left(\bigvee_{q \in \mathcal{Q}} \Phi(p, q) \otimes \Psi(q, r)\right) \otimes \Upsilon(r, s) \\ & =\bigvee_{r \in \mathcal{R}, q \in \mathcal{Q}} \Phi(p, q) \otimes \Psi(q, r) \otimes \Upsilon(r, s) \\ & =\bigvee_{q \in \mathcal{Q}} \Phi(p, q) \otimes \bigvee_{r \in \mathcal{R}}(\Psi(q, r) \otimes \Upsilon(r, s)) \\ & =(\Phi ;(\Psi ; \Upsilon))(p, s) \end{aligned}$</td><td>$\begin{aligned} ((\Phi \fatsemi \Psi) \fatsemi \Upsilon)(p, s) & =\bigvee_{r \in \mathcal{R}}\left(\bigvee_{q \in \mathcal{Q}} \Phi(p, q) \otimes \Psi(q, r)\right) \otimes \Upsilon(r, s) \\ & =\bigvee_{r \in \mathcal{R}, q \in \mathcal{Q}} \Phi(p, q) \otimes \Psi(q, r) \otimes \Upsilon(r, s) \\ & =\bigvee_{q \in \mathcal{Q}} \Phi(p, q) \otimes \bigvee_{r \in \mathcal{R}}\Big(\Psi(q, r) \otimes \Upsilon(r, s)\Big) \\ & =(\Phi \fatsemi (\Psi \fatsemi \Upsilon))(p, s) \end{aligned}$</td></tr>
<tr><td>fong-spivak-seven-sketches_ EQ0089 (4.29)</td><td>140</td><td>■ 0.994 ● N|+0</td><td><code>\mathcal{P}\left(p, p_{1}\right) \otimes \Phi\left(p_{1}, q\right)=\mathcal{P}\left(p, p_{1}\right) \otimes \Phi \left(p_{1}, q\right) \otimes I \leq \mathcal{P}\left(p, p_{1}\right) \otimes \Phi\left(p_{1}, q\right) \otimes \mathcal{Q}(q, q) \leq \Phi(p, q) .</code></td><td>$\mathcal{P}\left(p, p_{1}\right) \otimes \Phi\left(p_{1}, q\right)=\mathcal{P}\left(p, p_{1}\right) \otimes \Phi \left(p_{1}, q\right) \otimes I \leq \mathcal{P}\left(p, p_{1}\right) \otimes \Phi\left(p_{1}, q\right) \otimes \mathcal{Q}(q, q) \leq \Phi(p, q) .$</td><td>$\mathcal{P}(p, p_{1}) \otimes \Phi(p_{1}, q)=\mathcal{P}(p, p_{1}) \otimes \Phi(p_{1}, q) \otimes I \leq \mathcal{P}(p, p_{1}) \otimes \Phi(p_{1}, q) \otimes \mathcal{Q}(q, q) \leq \Phi(p, q).$</td></tr>
<tr><td>fong-spivak-seven-sketches_ EQ0165 (7.56)</td><td>257</td><td>■ 0.995 ● K|+0</td><td><code>(U \wedge V):=U \cap V \quad \text { and } \quad(U \vee V):=U \cup V .</code></td><td>$(U \wedge V):=U \cap V \quad \text { and } \quad(U \vee V):=U \cup V .$</td><td>$(U \wedge V) := U \cap V \qquad \text{and} \qquad (U \vee V) := U \cup V.$</td></tr>
<tr><td>fong-spivak-seven-sketches_ EQ0150 (7.14)</td><td>241</td><td>■ 0.995 ● N|+0</td><td><code>\Omega_{\text {Set }}:=\mathbb{B}=\{\text { true }, \text { false }\} .</code></td><td>$\Omega_{\text {Set }}:=\mathbb{B}=\{\text { true }, \text { false }\} .$</td><td>$\Omega_{\mathbf{Set}} := \mathbb{B} = \{\mathtt{true}, \mathtt{false}\}.$</td></tr>
<tr><td>fong-spivak-seven-sketches_ EQ0160 (7.50)</td><td>255</td><td>■ 0.996 ● N|+0</td><td><code>\Omega(U):=\left\{U^{\prime} \in \mathbf{O p} \mid U^{\prime} \subseteq U\right\} .</code></td><td>$\Omega(U):=\left\{U^{\prime} \in \mathbf{O p} \mid U^{\prime} \subseteq U\right\} .$</td><td>$\Omega(U) := \{U' \in \mathbf{Op} \mid U' \subseteq U\}.$</td></tr>
<tr><td>fong-spivak-seven-sketches_ EQ0260</td><td>324</td><td>■ 0.996 ● S|+0</td><td><code>p ; h_{3}=r ; g ; h_{3}=r ; h_{2} ; g^{\prime}=q ; f^{\prime} ; g^{\prime}</code></td><td>$p ; h_{3}=r ; g ; h_{3}=r ; h_{2} ; g^{\prime}=q ; f^{\prime} ; g^{\prime}$</td><td>$p \fatsemi h_3 = r \fatsemi g \fatsemi h_3 = r \fatsemi h_2 \fatsemi g' = q \fatsemi f' \fatsemi g'$</td></tr>
<tr><td>fong-spivak-seven-sketches_ EQ0088 (4.28)</td><td>140</td><td>■ 0.996 ● W|+1</td><td><code>\Phi(p, q)=I \otimes \Phi(p, q) \leq \mathcal{P}(p, p) \otimes \Phi(p, q) \leq \bigvee_{p_{1} \in P}\left(\mathcal{P}\left(p, p_{1}\right) \otimes \Phi\left(p_{1}, q\right)\right)=\left(\mathrm{U}_{\mathcal{P}} ; \Phi \right)(p, q) .</code></td><td>$\Phi(p, q)=I \otimes \Phi(p, q) \leq \mathcal{P}(p, p) \otimes \Phi(p, q) \leq \bigvee_{p_{1} \in P}\left(\mathcal{P}\left(p, p_{1}\right) \otimes \Phi\left(p_{1}, q\right)\right)=\left(\mathrm{U}_{\mathcal{P}} ; \Phi \right)(p, q) .$</td><td>$\Phi(p, q) = I \otimes \Phi(p, q) \leq \mathcal{P}(p, p) \otimes \Phi(p, q) \leq \bigvee_{p_1 \in P} \big(\mathcal{P}(p, p_1) \otimes \Phi(p_1, q)\big) = (\mathrm{U}_{\mathcal{P}} \fatsemi \Phi)(p, q).$</td></tr>
<tr><td>fong-spivak-seven-sketches_ EQ0185</td><td>276</td><td>■ 0.996 ● N|-1</td><td><code>f^{*}(P)=\{\{x \in \mathbb{Z} \mid x \leq 0\},\{x \in \mathbb{Z} \mid x \geq 1\}\} \quad \text { and } \quad f^{*}(Q)=\{\{0\},\{x \in \mathbb{Z} \mid x \neq 0\}\} .</code></td><td>$f^{*}(P)=\{\{x \in \mathbb{Z} \mid x \leq 0\},\{x \in \mathbb{Z} \mid x \geq 1\}\} \quad \text { and } \quad f^{*}(Q)=\{\{0\},\{x \in \mathbb{Z} \mid x \neq 0\}\} .$</td><td>$f^*(P) = \{\{x \in \mathbb{Z} \mid x \leq 0\}, \{x \in \mathbb{Z} \mid x \geq 1\}\} \quad \text{and} \quad f^*(Q) = \{\{0\}, \{x \in \mathbb{Z} \mid x \neq 0\}\}.$</td></tr>
<tr><td>fong-spivak-seven-sketches_ EQ0166 (7.57)</td><td>257</td><td>■ 0.996 ● N|+0</td><td><code>(U \Rightarrow V):=\bigcup_{\{R \in \mathbf{O} \mathbf{p} \mid R \cap U \subseteq V\}} R .</code></td><td>$(U \Rightarrow V):=\bigcup_{\{R \in \mathbf{O} \mathbf{p} \mid R \cap U \subseteq V\}} R .$</td><td>$(U \Rightarrow V) := \bigcup_{\{R \in \mathbf{Op} \mid R \cap U \subseteq V\}} R.$</td></tr>
<tr><td>fong-spivak-seven-sketches_ EQ0013 (1.116)</td><td>044</td><td>■ 0.997 ● K|+0</td><td><code>f(p):=\bigwedge\{q \in Q \mid p \leq g(q)\} ;</code></td><td>$f(p):=\bigwedge\{q \in Q \mid p \leq g(q)\} ;$</td><td>$f(p) := \bigwedge \{q \in Q \mid p \leq g(q)\};$</td></tr>
</table>

Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value

Residual render vs scan (inkdrill): • C component • W weak • S stable • N noise • K clean • not measured

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<tr><td>fong-spivak-seven-sketche EQ0154</td><td>243</td><td>■0.997 ● W|+2</td><td><code>\begin{array}{cc||c|c} P & Q & P \wedge Q & P=(P \wedge Q) \\ \text { true } & \text { true } & ? & ? \\ \text { true } & \text { false } & ? & ? \\ \text { false } & \text { true } & ? & ? \\ \text { false } & \text { false } & ? & ? \end{array}</code></td><td>$$\begin{array}{cc||c|c} P & Q & P \wedge Q & P=(P \wedge Q) \\ \text { true } & \text { true } & ? & ? \\ \text { true } & \text { false } & ? & ? \\ \text { false } & \text { true } & ? & ? \\ \text { false } & \text { false } & ? & ? \end{array}$$</td><td>$$\begin{array}{cc||c|c} P & Q & P \wedge Q & P=(P \wedge Q) \\ \texttt{true} & \texttt{true} & ? & ? \\ \texttt{true} & \texttt{false} & ? & ? \\ \texttt{false} & \texttt{true} & ? & ? \\ \texttt{false} & \texttt{false} & ? & ? \end{array}$$</td></tr>
<tr><td>fong-spivak-seven-sketche EQ0041</td><td>081</td><td>■0.998 ● N|-1</td><td><code>a+x \geq y \quad \text { iff } \quad a \geq y-x \quad \text { iff } \quad \max (0, a) \geq \max (0, y-x) \quad \text { iff } \quad a \geq(x \multimap y)</code></td><td>$a+x \geq y \quad \text { iff } \quad a \geq y-x \quad \text { iff } \quad \max (0, a) \geq \max (0, y-x) \quad \text { iff } \quad a \geq(x \multimap y)$</td><td>$a+x \geq y \quad \text { iff } \quad a \geq y-x \quad \text { iff } \quad \max (0, a) \geq \max (0, y-x) \quad \text { iff } \quad a \geq(x \multimap y)$</td></tr>
<tr><td>fong-spivak-seven-sketche EQ0082 (4.14)</td><td>135</td><td>■0.998 ● N|+0</td><td><code>\Phi(B, x)=11, \quad \Phi(A, z)=20, \quad \Phi(C, y)=17 .</code></td><td>$\Phi(B, x)=11, \quad \Phi(A, z)=20, \quad \Phi(C, y)=17 .$</td><td>$\Phi(B, x)=11, \quad \Phi(A, z)=20, \quad \Phi(C, y)=17 .$</td></tr>
<tr><td>fong-spivak-seven-sketche EQ0124</td><td>190</td><td>■0.998 ● S|+0</td><td><code>v=\int \frac{1}{m} F(t) d t+u(t)+p \int a u(t)+b v(t) d t .</code></td><td>$v=\int \frac{1}{m} F(t) d t+u(t)+p \int a u(t)+b v(t) d t .$</td><td>$v=\int \frac{1}{m} F(t) d t+u(t)+p \int a u(t)+b v(t) d t .$</td></tr>
<tr><td>fong-spivak-seven-sketche EQ0198</td><td>288</td><td>■0.998 ● N|+0</td><td><code>\begin{aligned} I_{X} * M(x, y) & =\bigvee_{x^{\prime} \in X} I_{X}\left(x, x^{\prime}\right) \otimes M\left(x^{\prime}, y\right) \\ & =\left(I_{X}(x, x) \otimes M(x, y)\right) \vee\left(\bigvee_{x^{\prime} \in X, x^{\prime} \neq x} I_{X}\left(x, x^{\prime}\right) \otimes M \left(x^{\prime}, y\right)\right) \\ & =(I \otimes M(x, y)) \vee\left(\bigvee_{x^{\prime} \in X, x^{\prime} \neq x} 0 \otimes M\left(x^{\prime}, y\right)\right) \\ & =M(x, y) \vee 0=M(x, y) . \end{aligned}</code></td><td>$$\begin{aligned} I_{X} * M(x, y) & =\bigvee_{x^{\prime} \in X} I_{X}\left(x, x^{\prime}\right) \otimes M\left(x^{\prime}, y\right) \\ & =\left(I_{X}(x, x) \otimes M(x, y)\right) \vee\left(\bigvee_{x^{\prime} \in X, x^{\prime} \neq x} I_{X}\left(x, x^{\prime}\right) \otimes M \left(x^{\prime}, y\right)\right) \\ & =(I \otimes M(x, y)) \vee\left(\bigvee_{x^{\prime} \in X, x^{\prime} \neq x} 0 \otimes M\left(x^{\prime}, y\right)\right) \\ & =M(x, y) \vee 0=M(x, y) . \end{aligned}$$</td><td>$$\begin{aligned} I_{X} * M(x, y) & =\bigvee_{x^{\prime} \in X} I_{X}\left(x, x^{\prime}\right) \otimes M\left(x^{\prime}, y\right) \\ & =\left(I_{X}(x, x) \otimes M(x, y)\right) \vee\left(\bigvee_{x^{\prime} \in X, x^{\prime} \neq x} I_{X}\left(x, x^{\prime}\right) \otimes M \left(x^{\prime}, y\right)\right) \\ & =(I \otimes M(x, y)) \vee\left(\bigvee_{x^{\prime} \in X, x^{\prime} \neq x} 0 \otimes M\left(x^{\prime}, y\right)\right) \\ & =M(x, y) \vee 0=M(x, y) . \end{aligned}$$</td></tr>
<tr><td>fong-spivak-seven-sketche EQ0070</td><td>117</td><td>■0.999 ● N|-2</td><td><code>\Sigma_{F}(I)(\text { Airline Seat })=I(\text { Economy }) \sqcup I(\text { First Class }) .</code></td><td>$\Sigma_{F}(I)(\text { Airline Seat })=I(\text { Economy }) \sqcup I(\text { First Class }) .$</td><td>$\Sigma_{F}(I)(\text{Airline Seat})=I(\text{Economy}) \sqcup I(\text{First Class}) .$</td></tr>
<tr><td>fong-spivak-seven-sketche EQ0084 (4.20)</td><td>138</td><td>■0.999 ● N|+0</td><td><code>(\Phi ; \Psi)(p, r):=\bigvee_{q \in Q} \Phi(p, q) \wedge \Psi(q, r) .</code></td><td>$(\Phi ; \Psi)(p, r):=\bigvee_{q \in Q} \Phi(p, q) \wedge \Psi(q, r) .$</td><td>$(\Phi \fatsemi \Psi)(p, r):=\bigvee_{q \in \mathrm{Q}} \Phi(p, q) \wedge \Psi(q, r) .$</td></tr>
<tr><td>fong-spivak-seven-sketche EQ0008 (1.96)</td><td>039</td><td>■0.999 ● N|+0</td><td><code>f(p) \leq q \quad \text { if and only if } \quad p \leq g(q) .</code></td><td>$f(p) \leq q \quad \text { if and only if } \quad p \leq g(q) .$</td><td>$f(p) \leq q \quad \text { if and only if } \quad p \leq g(q) .$</td></tr>
<tr><td>fong-spivak-seven-sketche EQ0035</td><td>076</td><td>■0.999 ● W|+1</td><td><code>\begin{array}{rlr} I_{W} & \leq f\left(I_{V}\right) & (f \text { is monoidal monotone }) \\ & \leq f(\mathcal{C}(c, c)) & (\mathcal{C} \text { is } \mathcal{V} \text {-category }) \\ & =\mathcal{C}_{f}(c, c) & \left(\text { definition of } \mathcal{C}_{f}\right) \end{array}</code></td><td>$$\begin{array}{rlr} I_{W} & \leq f\left(I_{V}\right) & (f \text { is monoidal monotone }) \\ & \leq f(\mathcal{C}(c, c)) & (\mathcal{C} \text { is } \mathcal{V} \text {-category }) \\ & =\mathcal{C}_{f}(c, c) & \left(\text { definition of } \mathcal{C}_{f}\right) \end{array}$$</td><td>$$\begin{array}{rlr} I_{W} & \leq f\left(I_{V}\right) & (f \text { is monoidal monotone}) \\ & \leq f(\mathcal{C}(c, c)) & (\mathcal{C} \text { is } \mathcal{V} \text {-category}) \\ & =\mathcal{C}_{f}(c, c) & (\text{definition of } \mathcal{C}_{f}) \end{array}$$</td></tr>
</table>

Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value

Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
fong-spivak-seven-sketches EQ0042	082	■0.999 ● N|-1	`a+c \rightarrow d \quad \text { if and only if } \quad a \rightarrow(c \rightarrow d) .`	$a+c \rightarrow d \quad \text { if and only if } \quad a \rightarrow(c \rightarrow d) .$	$a+c \rightarrow d \quad \text{if and only if} \quad a \rightarrow (c \multimap d).$
fong-spivak-seven-sketches EQ0065 (3.66)	113	■0.999 ● N|+0	`\text { Gr } \xrightarrow{F} \text { DDS } \xrightarrow{I} \text { Set. }`	$\text { Gr } \xrightarrow{F} \text { DDS } \xrightarrow{I} \text { Set. }$	$\mathbf{Gr} \xrightarrow{F} \mathbf{DDS} \xrightarrow{I} \mathbf{Set}.$
fong-spivak-seven-sketches EQ0126 (6.15)	199	■0.999 ● S|+0	`[f, g](x)= \begin{cases}f(x) & \text { if } x= \iota_{A}(a) \text { for some } a \in A ; \\ g(x) & \text { if } x=\iota_{B}(b) \text { for some } b \in B . \end{cases}`	$[f, g](x)= \begin{cases}f(x) & \text { if } x= \iota_{A}(a) \text { for some } a \in A ; \\ g(x) & \text { if } x=\iota_{B}(b) \text { for some } b \in B . \end{cases}$	$[f, g](x) = \begin{cases} f(x) & \text{if } x = \iota_A(a) \text{ for some } a \in A; \\ g(x) & \text{if } x = \iota_B(b) \text{ for some } b \in B. \end{cases}$
fong-spivak-seven-sketches EQ0139	220	■0.999 ● W|+0	`\begin{aligned} \operatorname{Circ}(f): \operatorname{Circ}(V) & \longrightarrow \operatorname{Circ}\left(V^{\prime}\right) ; \\ (V, A, s, t, \ell) & \longmapsto\left(V^{\prime}, A,(s ; f),(t ; f), \ell\right) . \end{aligned}`	$\begin{aligned} \operatorname{Circ}(f): \operatorname{Circ}(V) & \longrightarrow \operatorname{Circ}\left(V^{\prime}\right) ; \\ (V, A, s, t, \ell) & \longmapsto\left(V^{\prime}, A,(s ; f),(t ; f), \ell\right) . \end{aligned}$	$\begin{aligned} \mathrm{Circ}(f)\colon \mathrm{Circ}(V) & \longrightarrow \mathrm{Circ}(V'); \\ (V, A, s, t, \ell) & \longmapsto \big(V', A, (s \mathbin{⨟} f), (t \mathbin{⨟} f), \ell\big). \end{aligned}$
fong-spivak-seven-sketches EQ0032	068	■1.000 ● S|-1	`d(x):=\left\{\begin{array}{ll} \text { false } & \text { if } x>0 \\ \text { true } & \text { if } x=0 \end{array} \quad u(x):= \begin{cases}\text { false } & \text { if } x=\infty \\ \text { true } & \text { if } x<\infty \end{cases}\right.`	$d(x):=\left\{\begin{array}{ll} \text { false } & \text { if } x>0 \\ \text { true } & \text { if } x=0 \end{array} \quad u(x):= \begin{cases}\text { false } & \text { if } x=\infty \\ \text { true } & \text { if } x<\infty \end{cases}\right.$	$d(x) := \begin{cases} \texttt{false} & \text{if } x > 0 \\ \texttt{true} & \text{if } x = 0 \end{cases} \qquad u(x) := \begin{cases} \texttt{false} & \text{if } x = \infty \\ \texttt{true} & \text{if } x < \infty \end{cases}$
fong-spivak-seven-sketches EQ0176	272	■1.000 ● K|+0	`\varnothing, \quad\{1\}, \quad\{2\}, \quad\{3\}, \quad \{1,2\}, \quad\{1,3\}, \quad\{2,3\}, \quad\{1,2,3\} .`	$\varnothing, \quad\{1\}, \quad\{2\}, \quad\{3\}, \quad \{1,2\}, \quad\{1,3\}, \quad\{2,3\}, \quad\{1,2,3\} .$	$\varnothing, \quad \{1\}, \quad \{2\}, \quad \{3\}, \quad \{1,2\}, \quad \{1,3\}, \quad \{2,3\}, \quad \{1,2,3\}.$
fong-spivak-seven-sketches EQ0252	313	■1.000 ● W|+0	`\begin{aligned} \mathrm{B}\left(g^{\mathrm{op}}\right) & =\left\{(y, x) \in R^{n} \times R^{m} \mid y=S(g)(x) \right\} \\ \mathrm{B}(h) & =\left\{(x, z) \in R^{m} \times R^{p} \mid S(h)(x)=z\right\} \end{aligned}`	$\begin{aligned} \mathrm{B}\left(g^{\mathrm{op}}\right) & =\left\{(y, x) \in R^{n} \times R^{m} \mid y=S(g)(x) \right\} \\ \mathrm{B}(h) & =\left\{(x, z) \in R^{m} \times R^{p} \mid S(h)(x)=z\right\} \end{aligned}$	$\begin{aligned} \mathrm{B}(g^{\mathrm{op}}) &= \{(y,x) \in R^n \times R^m \mid y = S(g)(x)\} \\ \mathrm{B}(h) &= \{(x,z) \in R^m \times R^p \mid S(h)(x) = z\} \end{aligned}$
fong-spivak-seven-sketches EQ0001 (1.8)	018	■1.000 ● K|+0	`\Phi(A) \vee \Phi(B) \leq \Phi(A \vee B) .`	$\Phi(A) \vee \Phi(B) \leq \Phi(A \vee B) .$	$\Phi(A) \vee \Phi(B) \leq \Phi(A \vee B).$
fong-spivak-seven-sketches EQ0004	024	■1.000 ● N|+0	`f(11)=a, \quad f(12)=a, \quad f(13)=b, \quad f(21)=c, \quad f(22)=d, \quad f(23)=d`	$f(11)=a, \quad f(12)=a, \quad f(13)=b, \quad f(21)=c, \quad f(22)=d, \quad f(23)=d$	$f(11)=a, \quad f(12)=a, \quad f(13)=b, \quad f(21)=c, \quad f(22)=d, \quad f(23)=d$
fong-spivak-seven-sketches EQ0006	035	■1.000 ● N|+0	`\left\{\left.\frac{1}{n+1} \right\rvert\, n \in \mathbb{N}\right\}=\left\{1, \frac{1}{2}, \frac{1}{3}, \frac{1}{4}, \ldots\right\} \subseteq \mathbb{R}`	$\left\{\left.\frac{1}{n+1} \right\rvert\, n \in \mathbb{N}\right\}=\left\{1, \frac{1}{2}, \frac{1}{3}, \frac{1}{4}, \ldots\right\} \subseteq \mathbb{R}$	$\left\{ \frac{1}{n+1} \;\middle\vert\; n \in \mathbb{N} \right\} = \left\{1, \frac{1}{2}, \frac{1}{3}, \frac{1}{4}, \ldots \right\} \subseteq \mathbb{R}$
fong-spivak-seven-sketches EQ0009	039	■1.000 ● K|+0	`\lceil x / 3\rceil \leq y \text { if and only if } x \leq 3 y .`	$\lceil x / 3\rceil \leq y \text { if and only if } x \leq 3 y .$	$\lceil x/3 \rceil \leq y \text{ if and only if } x \leq 3y.$
fong-spivak-seven-sketches EQ0011	043	■1.000 ● C|+4	`g(\bigwedge A) \cong \bigwedge g(A) .`	$g(\bigwedge A) \cong \bigwedge g(A) .$	$g\left(\bigwedge A\right) \cong \bigwedge g(A).$
fong-spivak-seven-sketches EQ0012	043	■1.000 ● C|+3	`f(\bigvee A) \cong \bigvee f(A)`	$f(\bigvee A) \cong \bigvee f(A)$	$f\left(\bigvee A\right) \cong \bigvee f(A).$

Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value

Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
fong-spivak-seven-sketche EQ0014	044	■1.000 ● N|+0	`p_{0} \leq \bigwedge\left\{g(q) \in P \mid p_{0} \leq g(q)\right\} \cong g\left(\bigwedge\left\{q \in Q \mid p_{0} \leq g(q)\right\}\right)=g\left(f\left(p_{0}\right)\right),`	$p_{0} \leq \bigwedge\left\{g(q) \in P \mid p_{0} \leq g(q)\right\} \cong g\left(\bigwedge\left\{q \in Q \mid p_{0} \leq g(q)\right\}\right)=g\left(f\left(p_{0}\right)\right),$	$p_0 \leq \bigwedge\{g(q) \in P \mid p_0 \leq g(q)\} \cong g\left(\bigwedge\{q \in Q \mid p_0 \leq g(q)\}\right) = g(f(p_0)),$
fong-spivak-seven-sketche EQ0015	044	■1.000 ● N|+0	`f\left(g\left(q_{0}\right)\right)=\bigwedge\left\{q \in Q \mid g\left(q_{0}\right) \leq g(q)\right\} \leq \bigwedge \left\{q_{0}\right\}=q_{0},`	$f\left(g\left(q_{0}\right)\right)=\bigwedge\left\{q \in Q \mid g\left(q_{0}\right) \leq g(q)\right\} \leq \bigwedge \left\{q_{0}\right\}=q_{0},$	$f(g(q_0)) = \bigwedge\{q \in Q \mid g(q_0) \leq g(q)\} \leq \bigwedge\{q_0\} = q_0,$
fong-spivak-seven-sketche EQ0016	045	■1.000 ● N|-1	`f_{!}\left(A^{\prime}\right):=\left\{b \in B \mid \text { there exists } a \in A^{\prime} \text { such that } f(a)=b\right\}`	$f_{!}\left(A^{\prime}\right):=\left\{b \in B \mid \text { there exists } a \in A^{\prime} \text { such that } f(a)=b\right\}$	$f_!(A') := \{b \in B \mid \text{there exists } a \in A' \text{ such that } f(a) = b\}$
fong-spivak-seven-sketche EQ0017	045	■1.000 ● N|-1	`f_{*}\left(A^{\prime}\right):=\left\{b \in B \mid \text { for all } a \text { such that } f(a)=b, \text { we have } a \in A^{\prime}\right\}`	$f_{*}\left(A^{\prime}\right):=\left\{b \in B \mid \text { for all } a \text { such that } f(a)=b, \text { we have } a \in A^{\prime}\right\}$	$f_*(A') := \{b \in B \mid \text{for all } a \text{ such that } f(a) = b, \text{ we have } a \in A'\}$
fong-spivak-seven-sketche EQ0019 (2.7)	054	■1.000 ● N|+0	`m * e=m, \quad e * m=m, \quad(m * n) * p=m *(n * p)`	$m * e=m, \quad e * m=m, \quad(m * n) * p=m *(n * p)$	$m * e = m, \qquad e * m = m, \qquad (m * n) * p = m * (n * p)$
fong-spivak-seven-sketche EQ0023 (2.16)	059	■1.000 ● K|+0	`t \leq v+w \quad w+u \leq x+z \quad v+x \leq y`	$t \leq v+w \quad w+u \leq x+z \quad v+x \leq y$	$t \leq v + w \qquad w + u \leq x + z \qquad v + x \leq y$
fong-spivak-seven-sketche EQ0024 (2.17)	059	■1.000 ● K|+0	`t+u \leq y+z .`	$t+u \leq y+z .$	$t + u \leq y + z.$
fong-spivak-seven-sketche EQ0025 (2.18)	059	■1.000 ● N|+0	`t+u \leq v+w+u \leq v+x+z \leq y+z .`	$t+u \leq v+w+u \leq v+x+z \leq y+z .$	$t + u \;\leq\; v + w + u \;\leq\; v + x + z \;\leq\; y + z.$
fong-spivak-seven-sketche EQ0027 (2.24)	061	■1.000 ● N|+0	`2 \mathrm{H}_{2} \mathrm{O}+2 \mathrm{Na} \rightarrow 2 \mathrm{NaOH}+\mathrm{H}_{2} .`	$2 \mathrm{H}_{2} \mathrm{O}+2 \mathrm{Na} \rightarrow 2 \mathrm{NaOH}+\mathrm{H}_{2} .$	$2\mathrm{H_2O} + 2\mathrm{Na} \longrightarrow 2\mathrm{NaOH} + \mathrm{H_2}.$
fong-spivak-seven-sketche EQ0028 (2.22)	061	■1.000 ● N|+0	`y+k \rightarrow y^{\prime}+k^{\prime} \quad x+y^{\prime} \rightarrow z^{\prime} \quad z^{\prime}+k^{\prime} \rightarrow z+k`	$y+k \rightarrow y^{\prime}+k^{\prime} \quad x+y^{\prime} \rightarrow z^{\prime} \quad z^{\prime}+k^{\prime} \rightarrow z+k$	$y + k \longrightarrow y' + k' \qquad x + y' \longrightarrow z' \qquad z' + k' \longrightarrow z + k$
fong-spivak-seven-sketche EQ0029 (2.23)	061	■1.000 ● N|+0	`x+y+k \rightarrow x+y^{\prime}+k^{\prime} \rightarrow z^{\prime}+k^{\prime} \rightarrow z+k .`	$x+y+k \rightarrow x+y^{\prime}+k^{\prime} \rightarrow z^{\prime}+k^{\prime} \rightarrow z+k .$	$x + y + k \longrightarrow x + y' + k' \longrightarrow z' + k' \longrightarrow z + k.$
fong-spivak-seven-sketche EQ0037 (2.66)	077	■1.000 ● S|+1	`f(x):= \begin{cases}\text { true } & \text { if } x=0 \\ \text { false } & \text { if } x>0\end{cases}`	$f(x):= \begin{cases}\text { true } & \text { if } x=0 \\ \text { false } & \text { if } x>0\end{cases}$	$f(x) := \begin{cases} \texttt{true} & \text{if } x = 0 \\ \texttt{false} & \text{if } x > 0 \end{cases}$
fong-spivak-seven-sketche EQ0044 (2.88)	082	■1.000 ● C|+8	`\left(v \otimes \bigvee_{a \in A} a\right) \cong \bigvee_{a \in A}(v \otimes a) .`	$\left(v \otimes \bigvee_{a \in A} a\right) \cong \bigvee_{a \in A}(v \otimes a) .$	$\left(v \otimes \bigvee_{a \in A} a\right) \cong \bigvee_{a \in A} (v \otimes a).$
fong-spivak-seven-sketche EQ0047 (2.99)	085	■1.000 ● N|+0	`(M * N)(i, k)=\sum_{j} M(i, j) * N(j, k) .`	$(M * N)(i, k)=\sum_{j} M(i, j) * N(j, k) .$	$(M * N)(i, k) = \sum_j M(i, j) * N(j, k).$
fong-spivak-seven-sketche EQ0048 (2.101)	086	■1.000 ● N|+1	`(M * N)(x, z):=\bigvee_{y \in Y} M(x, y) \otimes N(y, z) .`	$(M * N)(x, z):=\bigvee_{y \in Y} M(x, y) \otimes N(y, z) .$	$(M * N)(x, z) := \bigvee_{y \in Y} M(x, y) \otimes N(y, z).$

Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value
Residual render vs scan (inkdrill): • C component • W weak • S stable • N noise • K clean • not measured

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
fong-spivak-seven-sketche EQ0049	086	■1.000 ● C|+6	`\left(\begin{array}{cc} \text { false } & \text { false } \\ \text { false } & \text { true } \\ \text { true } & \text { true } \end{array}\right) *\left(\begin{array}{ccc} \text { true } & \text { true } & \text { false } \\ \text { true } & \text { false } & \text { true } \end{array}\right)=\left(\begin{array}{ccc} \text { false } & \text { false } & \text { false } \\ \text { true } & \text { false } & \text { true } \\ \text { true } & \text { true } & \text { true } \end{array}\right)`	$\left(\begin{array}{cc} \text{false} & \text{false} \\ \text{false} & \text{true} \\ \text{true} & \text{true} \end{array}\right) * \left(\begin{array}{ccc} \text{true} & \text{true} & \text{false} \\ \text{true} & \text{false} & \text{true} \end{array}\right) = \left(\begin{array}{ccc} \text{false} & \text{false} & \text{false} \\ \text{true} & \text{false} & \text{true} \\ \text{true} & \text{true} & \text{true} \end{array}\right)$	$\left(\begin{array}{cc} \texttt{false} & \texttt{false} \\ \texttt{false} & \texttt{true} \\ \texttt{true} & \texttt{true} \end{array}\right) * \left(\begin{array}{ccc} \texttt{true} & \texttt{true} & \texttt{false} \\ \texttt{true} & \texttt{false} & \texttt{true} \end{array}\right) = \left(\begin{array}{ccc} \texttt{false} & \texttt{false} & \texttt{false} \\ \texttt{true} & \texttt{false} & \texttt{true} \\ \texttt{true} & \texttt{true} & \texttt{true} \end{array}\right)$
fong-spivak-seven-sketche EQ0050	086	■1.000 ● N|+0	`I_{X}(x, y):= \begin{cases}I & \text { if } x=y \\ 0 & \text { if } x \neq y .\end{cases}`	$I_X(x,y) := \begin{cases} I & \text{if } x=y \\ 0 & \text{if } x \neq y. \end{cases}$	$I_X(x,y) := \begin{cases} I & \text{if } x=y \\ 0 & \text{if } x \neq y. \end{cases}$
fong-spivak-seven-sketche EQ0051	087	■1.000 ● N|+0	`M_{Y}^{2}=\begin{array}{c|ccc} \nearrow & x & y & z \\ \hline x & 0 & 4 & 3 \\ y & 3 & 0 & 6 \\ z & 7 & 4 & 0 \end{array}`	$M_Y^2 = \begin{array}{c\|ccc} \nearrow & x & y & z \\ \hline x & 0 & 4 & 3 \\ y & 3 & 0 & 6 \\ z & 7 & 4 & 0 \end{array}$	$M_Y^2 = \begin{array}{c\|ccc} \nearrow & x & y & z \\ \hline x & 0 & 4 & 3 \\ y & 3 & 0 & 6 \\ z & 7 & 4 & 0 \end{array}$
fong-spivak-seven-sketche EQ0052	087	■1.000 ● N|+0	`M_{Y}^{3}=\begin{array}{c|ccc} \nearrow & x & y & z \\ \hline x & 0 & 4 & 3 \\ y & 3 & 0 & 6 \\ z & 7 & 4 & 0 \end{array}`	$M_Y^3 = \begin{array}{c\|ccc} \nearrow & x & y & z \\ \hline x & 0 & 4 & 3 \\ y & 3 & 0 & 6 \\ z & 7 & 4 & 0 \end{array}$	$M_Y^3 = \begin{array}{c\|ccc} \nearrow & x & y & z \\ \hline x & 0 & 4 & 3 \\ y & 3 & 0 & 6 \\ z & 7 & 4 & 0 \end{array}$
fong-spivak-seven-sketche EQ0055	096	■1.000 ● N|+0	`\{A, B, C, D, f, g, h, i, f ; h\}`	$\{A,B,C,D,f,g,h,i,f;h\}$	$\{A,B,C,D,f,g,h,i,f \fatsemi h\}$
fong-spivak-seven-sketche EQ0058	104	■1.000 ● S|+0	`f^{\prime} ; h^{\prime}=F(f) ; F(h)=F(f ; h)=F(g ; i)=F(g) ; F(i)=g^{\prime} ; i^{\prime}`	$f';h' = F(f);F(h) = F(f;h) = F(g;i) = F(g);F(i) = g';i'$	$f' \fatsemi h' = F(f) \fatsemi F(h) = F(f \fatsemi h) = F(g \fatsemi i) = F(g) \fatsemi F(i) = g' \fatsemi i'$
fong-spivak-seven-sketche EQ0060 (3.50)	107	■1.000 ● N|+1	`F(f) ; \alpha_{d}=\alpha_{c} ; G(f) .`	$F(f);\alpha_d = \alpha_c;G(f).$	$F(f) \fatsemi \alpha_d = \alpha_c \fatsemi G(f).$
fong-spivak-seven-sketche EQ0062	110	■1.000 ● C|+4	`\alpha_{\text {Vertex }}: G(\text { Vertex }) \rightarrow H(\text { Vertex }) \quad \text { and } \quad \alpha_{ \text {Arrow }}: G(\text { Arrow }) \rightarrow H(\text { Arrow }),`	$\alpha_{\text{Vertex}} : G(\text{ Vertex }) \rightarrow H(\text{ Vertex }) \quad \text{and} \quad \alpha_{\text{Arrow}} : G(\text{ Arrow }) \rightarrow H(\text{ Arrow }),$	$\alpha_{\text{Vertex}} \colon G(\text{Vertex}) \to H(\text{Vertex}) \qquad \text{and} \qquad \alpha_{\text{Arrow}} \colon G(\text{Arrow}) \to H(\text{Arrow}),$
fong-spivak-seven-sketche EQ0067	114	■1.000 ● C|+4	`\alpha_{c, d}: \mathcal{C}(c, R(d)) \xrightarrow{\cong} \mathcal{D}(L(c), d)`	$\alpha_{c,d} : \mathcal{C}(c,R(d)) \xrightarrow{\cong} \mathcal{D}(L(c),d)$	$\alpha_{c,d} \colon \mathcal{C}(c,R(d)) \xrightarrow{\cong} \mathcal{D}(L(c),d)$
fong-spivak-seven-sketche EQ0076	121	■1.000 ● K|+0	`X \times Y:=\{(x, y) \mid x \in X, y \in Y\},`	$X \times Y := \{(x,y) \mid x \in X, y \in Y\},$	$X \times Y := \{(x,y) \mid x \in X, y \in Y\},$
fong-spivak-seven-sketche EQ0090	141	■1.000 ● C|+3	`(\Phi ; \Psi) ; \Upsilon=\Phi ;(\Psi ; \Upsilon) .`	$(\Phi;\Psi);\Upsilon = \Phi;(\Psi;\Upsilon).$	$(\Phi \fatsemi \Psi) \fatsemi \Upsilon = \Phi \fatsemi (\Psi \fatsemi \Upsilon).$
fong-spivak-seven-sketche EQ0095	148	■1.000 ● N|+0	`\alpha_{s, t, u}:\{(s,(t, u)) \mid s \in S, t \in T, u \in U\} \xrightarrow{\cong}\{((s, t), u) \mid s \in S, t \in T, u \in U\} .`	$\alpha_{s,t,u} : \{(s,(t,u)) \mid s \in S, t \in T, u \in U\} \xrightarrow{\cong} \{((s,t),u) \mid s \in S, t \in T, u \in U\}.$	$\alpha_{s,t,u} \colon \{(s,(t,u)) \mid s \in S, t \in T, u \in U\} \xrightarrow{\cong} \{((s,t),u) \mid s \in S, t \in T, u \in U\}.$

Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value
Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
fong-spivak-seven-sketche EQ0099	156	1.000 N|+1	`(\Phi \times \Psi)\left(\left(x_{1}, y_{1}\right), \left(x_{2}, y_{2}\right)\right):=\Phi\left(x_{1}, x_{2} \right) \otimes \Psi\left(y_{1}, y_{2}\right) .`	$(\Phi \times \Psi)\left(\left(x_{1}, y_{1}\right), \left(x_{2}, y_{2}\right)\right):=\Phi\left(x_{1}, x_{2} \right) \otimes \Psi\left(y_{1}, y_{2}\right) .$	$(\Phi \times \Psi)((x_1, y_1), (x_2, y_2)) \coloneqq \Phi(x_1, x_2) \otimes \Psi(y_1, y_2).$
fong-spivak-seven-sketche EQ0101 (5.4)	162	1.000 W|-1	`i \longmapsto \begin{cases}f(i) & \text { if } 1 \leq i \leq m ; \\ m^{\prime}+g(i) & \text { if } m+1 \leq i \leq m+n .\end{cases}`	$i \longmapsto \begin{cases}f(i) & \text { if } 1 \leq i \leq m ; \\ m^{\prime}+g(i) & \text { if } m+1 \leq i \leq m+n .\end{cases}$	$i \longmapsto \begin{cases}f(i) & \text{if } 1 \leq i \leq m; \\ m'+g(i) & \text{if } m+1 \leq i \leq m+n.\end{cases}$
fong-spivak-seven-sketche EQ0106 (5.26)	168	1.000 S|+0	`G:=\left\{\rho_{m, n} \mid m, n \in \mathbb{N}\right\}, \quad s\left(\rho_{m, n}\right):=m, \quad t\left(\rho_{m, n}\right):=n,`	$G:=\left\{\rho_{m, n} \mid m, n \in \mathbb{N}\right\}, \quad s\left(\rho_{m, n}\right):=m, \quad t\left(\rho_{m, n}\right):=n,$	$G \coloneqq \{\rho_{m,n} \mid m, n \in \mathbb{N}\}, \quad s(\rho_{m,n}) \coloneqq m, \quad t(\rho_{m,n}) \coloneqq n,$
fong-spivak-seven-sketche EQ0108	174	1.000 W|+0	`\begin{array}{r} D x+3 D^{2} y-2 z=0 \\ D^{2} y+5 D z=0 \end{array}`	$\begin{array}{r} D x+3 D^{2} y-2 z=0 \\ D^{2} y+5 D z=0 \end{array}$	$\begin{array}{r} Dx+3D^2y-2z=0 \\ D^2y+5Dz=0 \end{array}$
fong-spivak-seven-sketche EQ0110 (5.48)	176	1.000 N|+1	`M ; N(a, c):=\sum_{b \in \underline{n}} M(a, b) \times N(b, c),`	$M ; N(a, c):=\sum_{b \in \underline{n}} M(a, b) \times N(b, c),$	$M \mathbin{;} N(a, c) \coloneqq \sum_{b \in \underline{n}} M(a, b) \times N(b, c),$
fong-spivak-seven-sketche EQ0112	176	1.000 K|+0	`A=\left(\begin{array}{lll} 3 & 3 & 1 \\ 2 & 0 & 4 \end{array}\right) \quad B=\left(\begin{array}{llll} 2 & 5 & 6 & 1 \end{array}\right) .`	$A=\left(\begin{array}{lll} 3 & 3 & 1 \\ 2 & 0 & 4 \end{array}\right) \quad B=\left(\begin{array}{llll} 2 & 5 & 6 & 1 \end{array}\right) .$	$A = \begin{pmatrix} 3 & 3 & 1 \\ 2 & 0 & 4 \end{pmatrix} \qquad B = \begin{pmatrix} 2 & 5 & 6 & 1 \end{pmatrix}.$
fong-spivak-seven-sketche EQ0114	179	1.000 N|+0	`\sum_{j \in \underline{n}} \alpha(i, j) * \beta(j, k) .`	$\sum_{j \in \underline{n}} \alpha(i, j) * \beta(j, k) .$	$\sum_{j \in \underline{n}} \alpha(i, j) * \beta(j, k).$
fong-spivak-seven-sketche EQ0118 (5.76)	186	1.000 K|+0	`\mathrm{B}\left(g^{\mathrm{op}}\right):=\left\{(S(g)(x), x) \mid x \in R^{m}\right\} \subseteq R^{n} \times R^{m} .`	$\mathrm{B}\left(g^{\mathrm{op}}\right):=\left\{(S(g)(x), x) \mid x \in R^{m}\right\} \subseteq R^{n} \times R^{m} .$	$\mathsf{B}(g^{\mathrm{op}}) \coloneqq \big\{\big(S(g)(x), x\big) \mid x \in R^m\big\} \subseteq R^n \times R^m.$
fong-spivak-seven-sketche EQ0122	188	1.000 N|+0	`\mathrm{B}\left(\left(g^{\mathrm{op}}\right) ; h\right)=\left\{(S(g)(x), S(h)(x)) \mid x \in R^{m}\right\} .`	$\mathrm{B}\left(\left(g^{\mathrm{op}}\right) ; h\right)=\left\{(S(g)(x), S(h)(x)) \mid x \in R^{m}\right\} .$	$\mathsf{B}((g^{\mathrm{op}}) \mathbin{;} h) = \{(S(g)(x), S(h)(x)) \mid x \in R^m\}.$
fong-spivak-seven-sketche EQ0127	203	1.000 N|+1	`y(0)=y(g(0))=x(f(0))=x(0) .`	$y(0)=y(g(0))=x(f(0))=x(0).$	$y(0) = y(g(0)) = x(f(0)) = x(0).$
fong-spivak-seven-sketche EQ0130	206	1.000 K|+0	`\{(v, d) \mid d \in D(v)\}`	$\{(v, d) \mid d \in D(v)\}$	$\{(v, d) \mid d \in D(v)\}$
fong-spivak-seven-sketche EQ0135 (6.76)	219	1.000 N|+0	`F\left(\left[\iota_{N}, \iota_{P}\right]\right)\left(\varphi_{N, P}(s, t)\right)`	$F\left(\left[\iota_{N}, \iota_{P}\right]\right)\left(\varphi_{N, P}(s, t)\right)$	$F([\iota_N, \iota_P])(\varphi_{N,P}(s, t))$
fong-spivak-seven-sketche EQ0138	220	1.000 N|+0	`\operatorname{Circ}(V):=\{(V, A, s, t, \ell) \mid \text { where } s, t: A \rightarrow V, \ell: E \rightarrow C\} .`	$\operatorname{Circ}(V):=\{(V, A, s, t, \ell) \mid \text { where } s, t: A \rightarrow V, \ell: E \rightarrow C\}.$	$\mathsf{Circ}(V) \coloneqq \{(V, A, s, t, \ell) \mid \text{ where } s, t\colon A \to V, \ell\colon E \to C\}.$
fong-spivak-seven-sketche EQ0141	225	1.000 C|+7	`\circ_{i}: \mathcal{O}\left(s_{1}, \ldots, s_{m} ; t_{i}\right) \times \mathcal{O}\left(t_{1}, \ldots, t_{n} ; t\right) \rightarrow \mathcal{O}\left(t_{1}, \ldots, t_{i-1}, s_{1}, \ldots, s_{m}, t_{i+1}, \ldots, t_{n} ; t\right) ;`	$\circ_{i}: \mathcal{O}\left(s_{1}, \ldots, s_{m} ; t_{i}\right) \times \mathcal{O}\left(t_{1}, \ldots, t_{n} ; t\right) \rightarrow \mathcal{O}\left(t_{1}, \ldots, t_{i-1}, s_{1}, \ldots, s_{m}, t_{i+1}, \ldots, t_{n} ; t\right) ;$	$\circ_i\colon \mathcal{O}(s_1, \ldots, s_m; t_i) \times \mathcal{O}(t_1, \ldots, t_n; t) \to \mathcal{O}(t_1, \ldots, t_{i-1}, s_1, \ldots, s_m, t_{i+1}, \ldots, t_n; t);$
fong-spivak-seven-sketche EQ0145	229	1.000 N|+2	`F: \mathcal{O}\left(t_{1}, \ldots, t_{n} ; t\right) \rightarrow \mathcal{P}\left(f\left(t_{1}\right), \ldots, f\left(t_{n}\right) ; f(t)\right)`	$F: \mathcal{O}\left(t_{1}, \ldots, t_{n} ; t\right) \rightarrow \mathcal{P}\left(f\left(t_{1}\right), \ldots, f\left(t_{n}\right) ; f(t)\right)$	$F\colon \mathcal{O}(t_1, \ldots, t_n; t) \to \mathcal{P}(f(t_1), \ldots, f(t_n); f(t))$
fong-spivak-seven-sketche EQ0149 (7.10)	239	1.000 N|+0	`\mathcal{C}(A \times C, D) \cong \mathcal{C}\left(A, D^{C}\right)`	$\mathcal{C}(A \times C, D) \cong \mathcal{C}\left(A, D^{C}\right)$	$\mathcal{C}(A \times C, D) \cong \mathcal{C}(A, D^C)$

Conf. MathPix confidence: ■ ≥ 0.9 ■ 0.5–0.9 ■ <0.5 — no value
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fong-spivak-seven-sketche EQ0151	241	1.000 W|+1	`\ulcorner m\urcorner(y):= \begin{cases}\text { true } & \text { if } m(x)=y \text { for some } x \in X \\ \text { false } & \text { otherwise }\end{cases}`	$\ulcorner m\urcorner(y):= \begin{cases}\text { true } & \text { if } m(x)=y \text { for some } x \in X \\ \text { false } & \text { otherwise }\end{cases}$	$\ulcorner m \urcorner(y) := \begin{cases}\texttt{true} & \text{if } m(x)=y \text{ for some } x \in X \\ \texttt{false} & \text{otherwise}\end{cases}$
fong-spivak-seven-sketche EQ0152 (7.15)	241	1.000 N|+0	`X=\{y \in Y \mid p(y)=\text { true }\} .`	$X=\{y \in Y \mid p(y)=\text { true }\} .$	$X = \{y \in Y \mid p(y) = \texttt{true}\}.$
fong-spivak-seven-sketche EQ0155	246	1.000 N|+0	`B(p ; \epsilon):=\left\{p^{\prime} \in \mathbb{R}^{2} \mid d\left(p, p^{\prime}\right)<\epsilon\right\}^{7}`	$B(p ; \epsilon):=\left\{p^{\prime} \in \mathbb{R}^{2} \mid d\left(p, p^{\prime}\right)<\epsilon\right\}^{7}$	$B(p;\epsilon) := \{p' \in \mathbb{R}^2 \mid d(p,p') < \epsilon\}^7$
fong-spivak-seven-sketche EQ0161 (7.51)	255	1.000 N|+0	`\begin{aligned} \Omega(U) & \rightarrow \Omega(V) \\ U^{\prime} & \mapsto U^{\prime} \cap V . \end{aligned}`	$\begin{aligned} \Omega(U) & \rightarrow \Omega(V) \\ U^{\prime} & \mapsto U^{\prime} \cap V . \end{aligned}$	$\begin{aligned} \Omega(U) & \to \Omega(V) \\ U' & \mapsto U' \cap V. \end{aligned}$
fong-spivak-seven-sketche EQ0162	256	1.000 W|+0	`V \cap U_{j}=\left(\bigcup_{i \in I} V_{i}\right) \cap U_{j}=\bigcup_{i \in I}\left(V_{i} \cap U_{j}\right)=\bigcup_{i \in I}\left(V_{j} \cap U_{i}\right)=\left(\bigcup_{i \in I} U_{i}\right) \cap V_{j}=V_{j}`	$V \cap U_{j}=\left(\bigcup_{i \in I} V_{i}\right) \cap U_{j}=\bigcup_{i \in I}\left(V_{i} \cap U_{j}\right)=\bigcup_{i \in I}\left(V_{j} \cap U_{i}\right)=\left(\bigcup_{i \in I} U_{i}\right) \cap V_{j}=V_{j}$	$V \cap U_j = \left(\bigcup_{i\in I} V_i\right) \cap U_j = \bigcup_{i\in I}(V_i \cap U_j) = \bigcup_{i\in I}(V_j \cap U_i) = \left(\bigcup_{i\in I} U_i\right) \cap V_j = V_j$
fong-spivak-seven-sketche EQ0167 (7.63)	259	1.000 K|+0	`\left(\left|\Omega^{S}\right|, \leq^{S}\right)`	$\left(\left\vert\Omega^{S}\right\vert, \leq^{S}\right)$	$(\vert\Omega^S\vert, \leq^S)$
fong-spivak-seven-sketche EQ0170	261	1.000 W|+2	`p(n, z)= \begin{cases}\text { true } & \text { if } n \leq|z| \\ \text { false } & \text { if } n>|z| .\end{cases}`	$p(n, z)= \begin{cases}\text { true } & \text { if } n \leq\vert z\vert \\ \text { false } & \text { if } n>\vert z\vert .\end{cases}$	$p(n,z) = \begin{cases}\texttt{true} & \text{if } n \leq \vert z\vert \\ \texttt{false} & \text{if } n > \vert z\vert.\end{cases}$
fong-spivak-seven-sketche EQ0174	265	1.000 N|-2	`S\left(o_{[a, b]}\right) \cong \lim _{\epsilon>0} S \left(o_{[a-\epsilon, b+\epsilon]}\right) .`	$S\left(o_{[a, b]}\right) \cong \lim _{\epsilon>0} S \left(o_{[a-\epsilon, b+\epsilon]}\right) .$	$S(o_{[a,b]}) \cong \lim_{\epsilon>0} S(o_{[a-\epsilon,b+\epsilon]}).$
fong-spivak-seven-sketche EQ0177	272	1.000 N|+0	`(h, 1), \quad(h, 2), \quad(h, 3), \quad(1,1), \quad(1,2), \quad(1,3) .`	$(h, 1), \quad(h, 2), \quad(h, 3), \quad(1,1), \quad(1,2), \quad(1,3) .$	$(h,1), \quad (h,2), \quad (h,3), \quad (1,1), \quad (1,2), \quad (1,3).$
fong-spivak-seven-sketche EQ0178	272	1.000 N|+0	`(h, 1), \quad(1,1), \quad(1,2), \quad(2,2), \quad(3,2) .`	$(h, 1), \quad(1,1), \quad(1,2), \quad(2,2), \quad(3,2) .$	$(h,1), \quad (1,1), \quad (1,2), \quad (2,2), \quad (3,2).$
fong-spivak-seven-sketche EQ0180	273	1.000 S|+0	`\begin{array}{lllll} (11,11) & (11,12) & (12,11) & (12,12) & (13,13) \\ (21,21) & (22,22) & (12,23) & (23,22) & (23,23) \end{array}`	$\begin{array}{lllll} (11,11) & (11,12) & (12,11) & (12,12) & (13,13) \\ (21,21) & (22,22) & (12,23) & (23,22) & (23,23) \end{array}$	$\begin{array}{lllll} (11,11) & (11,12) & (12,11) & (12,12) & (13,13) \\ (21,21) & (22,22) & (12,23) & (23,22) & (23,23) \end{array}$
fong-spivak-seven-sketche EQ0182	274	1.000 K|+0	`1 \leq 1, \quad 1 \leq 2, \quad 1 \leq 3, \quad 2 \leq 2, \quad 2 \leq 3, \quad 3 \leq 3, \quad 4 \leq 4`	$1 \leq 1, \quad 1 \leq 2, \quad 1 \leq 3, \quad 2 \leq 2, \quad 2 \leq 3, \quad 3 \leq 3, \quad 4 \leq 4$	$1 \leq 1, \quad 1 \leq 2, \quad 1 \leq 3, \quad 2 \leq 2, \quad 2 \leq 3, \quad 3 \leq 3, \quad 4 \leq 4$
fong-spivak-seven-sketche EQ0183	274	1.000 S|+0	`(\bullet)(\circ)(*), \quad(\bullet \circ)(*), \quad(\bullet *)(\circ), \quad(\bullet)(\circ *), \quad(\bullet \circ *)`	$(\bullet)(\circ)(*), \quad(\bullet \circ)(*), \quad(\bullet *)(\circ), \quad(\bullet)(\circ *), \quad(\bullet \circ *)$	$(\bullet)(\circ)(*), \quad (\bullet\circ)(*), \quad (\bullet *)(\circ), \quad (\bullet)(\circ *), \quad (\bullet \circ *)$
fong-spivak-seven-sketche EQ0190	280	1.000 N|+0	`g(q) \leq g^{\prime}(f(g(q))) \leq g^{\prime}(q) .`	$g(q) \leq g^{\prime}(f(g(q))) \leq g^{\prime}(q) .$	$g(q) \leq g'(f(g(q))) \leq g'(q).$

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fong-spivak-seven-sketche EQ0192	283	■1.000 ● C|+4	`\begin{gathered} g(\text { false })+g(\text { false })=\infty+\infty \geq \infty=g(\text { false } \wedge \text { false }) \\ g(\text { false })+g(\text { true })=\infty+0 \geq \infty=g(\text { false } \wedge \text { true }) \\ g(\text { true })+g(\text { false })=0+\infty \geq \infty=g(\text { true } \wedge \text { false }) \\ g(\text { true })+g(\text { true })=0+0 \geq 0=g(\text { true } \wedge \text { true }) \end{gathered}`	$\begin{gathered} g(\text { false })+g(\text { false })=\infty+\infty \geq \infty=g(\text { false } \wedge \text { false }) \\ g(\text { false })+g(\text { true })=\infty+0 \geq \infty=g(\text { false } \wedge \text { true }) \\ g(\text { true })+g(\text { false })=0+\infty \geq \infty=g(\text { true } \wedge \text { false }) \\ g(\text { true })+g(\text { true })=0+0 \geq 0=g(\text { true } \wedge \text { true }) \end{gathered}$	$\begin{gathered} g(\texttt{false})+g(\texttt{false})=\infty+\infty \geq \infty=g(\texttt{false} \wedge \texttt{false}) \\ g(\texttt{false})+g(\texttt{true})=\infty+0 \geq \infty=g(\texttt{false} \wedge \texttt{true}) \\ g(\texttt{true})+g(\texttt{false})=0+\infty \geq \infty=g(\texttt{true} \wedge \texttt{false}) \\ g(\texttt{true})+g(\texttt{true})=0+0 \geq 0=g(\texttt{true} \wedge \texttt{true}). \end{gathered}$
fong-spivak-seven-sketche EQ0196	287	■1.000 ● W|+1	`\left(\begin{array}{ll} 1 & 0 \\ 0 & 1 \end{array}\right), \quad\left(\begin{array}{cc} \text { true } & \text { false } \\ \text { false } & \text { true } \end{array}\right), \quad \text { and } \quad\left(\begin{array}{cc} 0 & \infty \\ \infty & 0 \end{array}\right) .`	$\left(\begin{array}{ll} 1 & 0 \\ 0 & 1 \end{array}\right), \quad\left(\begin{array}{cc} \text { true } & \text { false } \\ \text { false } & \text { true } \end{array}\right), \quad \text { and } \quad\left(\begin{array}{cc} 0 & \infty \\ \infty & 0 \end{array}\right) .$	$\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \quad \begin{pmatrix} \texttt{true} & \texttt{false} \\ \texttt{false} & \texttt{true} \end{pmatrix}, \quad \text{and} \quad \begin{pmatrix} 0 & \infty \\ \infty & 0 \end{pmatrix}.$
fong-spivak-seven-sketche EQ0197	287	■1.000 ● N|+0	`0 \otimes v \cong v \otimes 0 \cong\left(v \otimes \bigvee_{a \in \varnothing} a\right)=\bigvee_{a \in \varnothing}(v \otimes a)=0 .`	$0 \otimes v \cong v \otimes 0 \cong\left(v \otimes \bigvee_{a \in \varnothing} a\right)=\bigvee_{a \in \varnothing}(v \otimes a)=0 .$	$0 \otimes v \cong v \otimes 0 \cong \left(v \otimes \bigvee_{a \in \varnothing} a\right) = \bigvee_{a \in \varnothing} (v \otimes a) = 0.$
fong-spivak-seven-sketche EQ0200	288	■1.000 ● C|-6	`M_{X}=\left(\begin{array}{cccc} 0 & \infty & 3 & \infty \\ 2 & 0 & \infty & 5 \\ \infty & 3 & 0 & \infty \\ \infty & \infty & 6 & 0 \end{array}\right) \quad M_{X}^{2}=\left(\begin{array}{cccc} 0 & 6 & 3 & \infty \\ 2 & 0 & 5 & 5 \\ 5 & 3 & 0 & 8 \\ \infty & 9 & 6 & 0 \end{array}\right) \quad M_{X}^{3}=M_{X}^{4}=\left(\begin{array}{cccc} 0 & 6 & 3 & 11 \\ 2 & 0 & 5 & 5 \\ 5 & 3 & 0 & 8 \\ 11 & 9 & 6 & 0 \end{array}\right)`	$M_{X}=\left(\begin{array}{cccc} 0 & \infty & 3 & \infty \\ 2 & 0 & \infty & 5 \\ \infty & 3 & 0 & \infty \\ \infty & \infty & 6 & 0 \end{array}\right) \quad M_{X}^{2}=\left(\begin{array}{cccc} 0 & 6 & 3 & \infty \\ 2 & 0 & 5 & 5 \\ 5 & 3 & 0 & 8 \\ \infty & 9 & 6 & 0 \end{array}\right) \quad M_{X}^{3}=M_{X}^{4}=\left(\begin{array}{cccc} 0 & 6 & 3 & 11 \\ 2 & 0 & 5 & 5 \\ 5 & 3 & 0 & 8 \\ 11 & 9 & 6 & 0 \end{array}\right)$	$M_X = \begin{pmatrix} 0 & \infty & 3 & \infty \\ 2 & 0 & \infty & 5 \\ \infty & 3 & 0 & \infty \\ \infty & \infty & 6 & 0 \end{pmatrix} \quad M_X^2 = \begin{pmatrix} 0 & 6 & 3 & \infty \\ 2 & 0 & 5 & 5 \\ 5 & 3 & 0 & 8 \\ \infty & 9 & 6 & 0 \end{pmatrix} \quad M_X^3 = M_X^4 = \begin{pmatrix} 0 & 6 & 3 & 11 \\ 2 & 0 & 5 & 5 \\ 5 & 3 & 0 & 8 \\ 11 & 9 & 6 & 0 \end{pmatrix}$
fong-spivak-seven-sketche EQ0201	289	■1.000 ● W|-1	`t(p):=\left\{\begin{array}{l} v \text { if } n=0 \\ t \left(a_{n}\right) \text { if } n \geq 1 \end{array}\right.`	$t(p):=\left\{\begin{array}{l} v \text { if } n=0 \\ t \left(a_{n}\right) \text { if } n \geq 1 \end{array}\right.$	$t(p) := \begin{cases} v \text{ if } n = 0 \\ t(a_n) \text{ if } n \geq 1 \end{cases}$
fong-spivak-seven-sketche EQ0202	289	■1.000 ● N|+0	`(p ; q):=\left(v, a_{1}, \ldots, a_{m}, b_{1}, \ldots, b_{n}\right) .`	$(p ; q):=\left(v, a_{1}, \ldots, a_{m}, b_{1}, \ldots, b_{n}\right) .$	$(p \mathbin{⨾} q) := (v, a_1, \ldots, a_m, b_1, \ldots, b_n).$
fong-spivak-seven-sketche EQ0203	289	■1.000 ● N|+0	`\left(v, a_{1}, \ldots, a_{m}, b_{1}, \ldots, b_{n}, c_{1}, \ldots, c_{o}\right) .`	$\left(v, a_{1}, \ldots, a_{m}, b_{1}, \ldots, b_{n}, c_{1}, \ldots, c_{o}\right) .$	$(v, a_1, \ldots, a_m, b_1, \ldots, b_n, c_1, \ldots, c_o).$
fong-spivak-seven-sketche EQ0207	291	■1.000 ● K|+0	`\begin{array}{lll} (1,1) & (1,2) & (1,3) \\ (2,1) & (2,2) & (2,3) \\ (3,1) & (3,2) & (3,3) \end{array}`	$\begin{array}{lll} (1,1) & (1,2) & (1,3) \\ (2,1) & (2,2) & (2,3) \\ (3,1) & (3,2) & (3,3) \end{array}$	$\begin{array}{lll} (1,1) & (1,2) & (1,3) \\ (2,1) & (2,2) & (2,3) \\ (3,1) & (3,2) & (3,3) \end{array}$
fong-spivak-seven-sketche EQ0208	291	■1.000 ● N|+0	`f^{-1}(1)=b, \quad f^{-1}(2)=a, \quad f^{-1}(3)=c .`	$f^{-1}(1)=b, \quad f^{-1}(2)=a, \quad f^{-1}(3)=c .$	$f^{-1}(1) = b, \quad f^{-1}(2) = a, \quad f^{-1}(3) = c.$

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Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
fong-spivak-seven-sketche EQ0209	291	1.000 W|+0	`\begin{gathered} A^{\prime} \mapsto A, \quad f^{\prime} \mapsto f, \quad g^{\prime} \mapsto g, \quad f^{\prime} ; h^{\prime} \mapsto f ; h, \quad g^{\prime} ; i^{\prime} \mapsto f ; h, \\ B^{\prime} \mapsto B, \quad h^{\prime} \mapsto h, \quad C^{\prime} \mapsto C, \quad i^{\prime} \mapsto i, \quad D^{\prime} \mapsto D . \end{gathered}`	$\begin{gathered} A^{\prime} \mapsto A, \quad f^{\prime} \mapsto f, \quad g^{\prime} \mapsto g, \quad f^{\prime} ; h^{\prime} \mapsto f ; h, \quad g^{\prime} ; i^{\prime} \mapsto f ; h, \\ B^{\prime} \mapsto B, \quad h^{\prime} \mapsto h, \quad C^{\prime} \mapsto C, \quad i^{\prime} \mapsto i, \quad D^{\prime} \mapsto D . \end{gathered}$	$\begin{gathered} A' \mapsto A, \quad f' \mapsto f, \quad g' \mapsto g, \quad f' \fatsemi h' \mapsto f \fatsemi h, \quad g' \fatsemi i' \mapsto f \fatsemi h, \\ B' \mapsto B, \quad h' \mapsto h, \quad C' \mapsto C, \quad i' \mapsto i, \quad D' \mapsto D. \end{gathered}$
fong-spivak-seven-sketche EQ0211	292	1.000 S|+0	`(F ; G)(f ; g)=G(F(f) ; F(g))=G(F(f)) ; G(F(g))`	$(F ; G)(f ; g)=G(F(f) ; F(g))=G(F(f)) ; G(F(g))$	$(F \fatsemi G)(f \fatsemi g) = G(F(f) \fatsemi F(g)) = G(F(f)) \fatsemi G(F(g))$
fong-spivak-seven-sketche EQ0212	292	1.000 S|+0	`((F ; G) ; H)(c)=H((F ; G)(c))=H(G(F(c)))=(G ; H)(F(c))=(F ;(G ; H))(c) .`	$((F ; G) ; H)(c)=H((F ; G)(c))=H(G(F(c)))=(G ; H)(F(c))=(F ;(G ; H))(c) .$	$((F \fatsemi G) \fatsemi H)(c) = H((F \fatsemi G)(c)) = H(G(F(c))) = (G \fatsemi H)(F(c)) = (F \fatsemi (G \fatsemi H))(c).$
fong-spivak-seven-sketche EQ0215	295	1.000 W|+0	`\left(f_{1} ; f_{2}\right)^{B}(g)=g ;\left(f_{1} ; f_{2}\right)=\left(g ; f_{1}\right) ; f_{2}=\left(f_{1}^{B} ; f_{2}^{B}\right)(g)`	$\left(f_{1} ; f_{2}\right)^{B}(g)=g ;\left(f_{1} ; f_{2}\right)=\left(g ; f_{1}\right) ; f_{2}=\left(f_{1}^{B} ; f_{2}^{B}\right)(g)$	$(f_1 \fatsemi f_2)^B(g) = g \fatsemi (f_1 \fatsemi f_2) = (g \fatsemi f_1) \fatsemi f_2 = (f_1^B \fatsemi f_2^B)(g)$
fong-spivak-seven-sketche EQ0216	296	1.000 W|+0	`\left(c_{1}, d_{1}\right) \xrightarrow{\left(f_{1}, g_{1}\right)}\left(c_{2}, d_{2}\right) \xrightarrow{ \left(f_{2}, g_{2}\right)}\left(c_{3}, d_{3}\right) \xrightarrow{\left(f_{3}, g_{3}\right)}\left(c_{4}, d_{4} \right) .`	$\left(c_{1}, d_{1}\right) \xrightarrow{\left(f_{1}, g_{1}\right)}\left(c_{2}, d_{2}\right) \xrightarrow{ \left(f_{2}, g_{2}\right)}\left(c_{3}, d_{3}\right) \xrightarrow{\left(f_{3}, g_{3}\right)}\left(c_{4}, d_{4} \right) .$	$(c_1, d_1) \xrightarrow{(f_1, g_1)} (c_2, d_2) \xrightarrow{(f_2, g_2)} (c_3, d_3) \xrightarrow{(f_3, g_3)} (c_4, d_4).$
fong-spivak-seven-sketche EQ0220	298	1.000 N|+1	`\left(\mathcal{X}^{\mathrm{op}} \times y\right)\left((x, y),\left(x^{\prime}, y^{\prime}\right)\right) \leq \mathcal{V}\left(\Phi(x, y), \Phi\left(x^{\prime}, y^{ \prime}\right)\right) .`	$\left(\mathcal{X}^{\mathrm{op}} \times y\right)\left((x, y),\left(x^{\prime}, y^{\prime}\right)\right) \leq \mathcal{V}\left(\Phi(x, y), \Phi\left(x^{\prime}, y^{ \prime}\right)\right) .$	$(\mathcal{X}^{\mathrm{op}} \times \mathcal{Y})\big((x, y), (x', y')\big) \leq \mathcal{V}\big(\Phi(x, y), \Phi(x', y')\big).$
fong-spivak-seven-sketche EQ0231	306	1.000 W|-2	`\sigma_{m, n}(i):= \begin{cases}i+n & \text { if } i \leq m \\ i-m & \text { if } m+1 \leq i\end{cases}`	$\sigma_{m, n}(i):= \begin{cases}i+n & \text { if } i \leq m \\ i-m & \text { if } m+1 \leq i\end{cases}$	$\sigma_{m,n}(i) := \begin{cases} i+n & \text{if } i \leq m \\ i-m & \text{if } m+1 \leq i \end{cases}$
fong-spivak-seven-sketche EQ0232	306	1.000 N|+0	`(f+g)(i):= \begin{cases}f(i) & \text { if } i \leq m \\ g(i-m) & \text { if } m+1 \leq i\end{cases}`	$(f+g)(i):= \begin{cases}f(i) & \text { if } i \leq m \\ g(i-m) & \text { if } m+1 \leq i\end{cases}$	$(f+g)(i) := \begin{cases} f(i) & \text{if } i \leq m \\ g(i-m) & \text{if } m+1 \leq i \end{cases}$
fong-spivak-seven-sketche EQ0233	308	1.000 N|+0	`a^{0}, a^{1}, a^{2}, a^{3}, \ldots, a^{2019}, \ldots`	$a^{0}, a^{1}, a^{2}, a^{3}, \ldots, a^{2019}, \ldots$	$a^0, a^1, a^2, a^3, \ldots, a^{2019}, \ldots$
fong-spivak-seven-sketche EQ0234	308	1.000 N|+0	`[], \quad[a], \quad[b], \quad[a, a], \quad[a, b], \quad \ldots, \quad[b, a, b, b, a, b, a, a, a, a], \quad \ldots`	$[], \quad[a], \quad[b], \quad[a, a], \quad[a, b], \quad \ldots, \quad[b, a, b, b, a, b, a, a, a, a], \quad \ldots$	$[\,], \quad [a], \quad [b], \quad [a, a], \quad [a, b], \quad \ldots, \quad [b, a, b, b, a, b, a, a, a, a], \quad \ldots$
fong-spivak-seven-sketche EQ0235	308	1.000 W|+0	`G:=\left\{\rho_{m, n}: m \rightarrow n \mid m, n \in \mathbb{N}\right\}, \quad s\left(\rho_{m, n}\right):=m, \quad t\left(\rho_{m, n}\right):=n ;`	$G:=\left\{\rho_{m, n}: m \rightarrow n \mid m, n \in \mathbb{N}\right\}, \quad s\left(\rho_{m, n}\right):=m, \quad t\left(\rho_{m, n}\right):=n ;$	$G := \{\rho_{m,n} \colon m \to n \mid m, n \in \mathbb{N}\}, \quad s(\rho_{m,n}) := m, \quad t(\rho_{m,n}) := n;$
fong-spivak-seven-sketche EQ0236	309	1.000 K|+0	`\left(\begin{array}{llll} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{array}\right) .`	$\left(\begin{array}{llll} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{array}\right) .$	$\begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}.$
fong-spivak-seven-sketche EQ0237	309	1.000 N|+0	`A * B=\left(\begin{array}{ll} 0 & 1 \\ 0 & 0 \end{array}\right) *\left(\begin{array}{ll} 0 & 1 \\ 1 & 0 \end{array}\right) \neq\left(\begin{array}{ll} 0 & 1 \\ 1 & 0 \end{array}\right) *\left(\begin{array}{ll} 0 & 1 \\ 0 & 0 \end{array}\right)=B * A`	$A * B=\left(\begin{array}{ll} 0 & 1 \\ 0 & 0 \end{array}\right) *\left(\begin{array}{ll} 0 & 1 \\ 1 & 0 \end{array}\right) \neq\left(\begin{array}{ll} 0 & 1 \\ 1 & 0 \end{array}\right) *\left(\begin{array}{ll} 0 & 1 \\ 0 & 0 \end{array}\right)=B * A$	$A * B = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} * \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \neq \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} * \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} = B * A$

Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value
Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
fong-spivak-seven-sketches EQ0238	309	■1.000 ● K|+0	`A+B=\left(\begin{array}{lllllll} 3 & 3 & 1 & 0 & 0 & 0 & 0 \\ 2 & 0 & 4 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 2 & 5 & 6 & 1 \end{array}\right)`	$A+B=\left(\begin{array}{lllllll} 3 & 3 & 1 & 0 & 0 & 0 & 0 \\ 2 & 0 & 4 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 2 & 5 & 6 & 1 \end{array}\right)$	$A+B=\left(\begin{array}{lllllll} 3 & 3 & 1 & 0 & 0 & 0 & 0 \\ 2 & 0 & 4 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 2 & 5 & 6 & 1 \end{array}\right)$
fong-spivak-seven-sketches EQ0239	310	■1.000 ● K|+0	`\left(\begin{array}{lll} 1 & 1 & 1 \end{array}\right)`	$\left(\begin{array}{lll} 1 & 1 & 1 \end{array}\right)$	$\left(\begin{array}{lll} 1 & 1 & 1 \end{array}\right)$
fong-spivak-seven-sketches EQ0240	310	■1.000 ● K|+0	`\left(\begin{array}{lll} 1 & 1 & 1 \end{array}\right)`	$\left(\begin{array}{lll} 1 & 1 & 1 \end{array}\right)$	$\left(\begin{array}{lll} 1 & 1 & 1 \end{array}\right)$
fong-spivak-seven-sketches EQ0241	310	■1.000 ● N|+0	`g_{1}:=c_{n}+\cdots+c_{n}: m \rightarrow(m \times n),`	$g_{1}:=c_{n}+\cdots+c_{n}: m \rightarrow(m \times n),$	$g_{1}:=c_{n}+\cdots+c_{n}: m \rightarrow(m \times n),$
fong-spivak-seven-sketches EQ0243	310	■1.000 ● N|+0	`\begin{aligned} g_{2}:= & s_{M(1,1)}+\cdots+s_{M(1, n)} \\ & +s_{M(2,1)}+\cdots+s_{M(2, n)} \\ & +\cdots \\ & +s_{M(m, 1)}+\cdots+s_{M(m, n)}:(m \times n) \rightarrow(m \times n), \end{aligned}`	$\begin{aligned} g_{2}:= & s_{M(1,1)}+\cdots+s_{M(1, n)} \\ & +s_{M(2,1)}+\cdots+s_{M(2, n)} \\ & +\cdots \\ & +s_{M(m, 1)}+\cdots+s_{M(m, n)}:(m \times n) \rightarrow(m \times n), \end{aligned}$	$\begin{aligned} g_{2}:= & s_{M(1,1)}+\cdots+s_{M(1, n)} \\ & +s_{M(2,1)}+\cdots+s_{M(2, n)} \\ & +\cdots \\ & +s_{M(m, 1)}+\cdots+s_{M(m, n)}:(m \times n) \rightarrow(m \times n), \end{aligned}$
fong-spivak-seven-sketches EQ0244	310	■1.000 ● N|+1	`g_{4}:=a_{m}+\cdots+a_{m}:(m \times n) \rightarrow n,`	$g_{4}:=a_{m}+\cdots+a_{m}:(m \times n) \rightarrow n,$	$g_{4}:=a_{m}+\cdots+a_{m}:(m \times n) \rightarrow n,$
fong-spivak-seven-sketches EQ0246	312	■1.000 ● K|+0	`\mu:=\left(\begin{array}{ll} 1 & 0 \\ 0 & 1 \\ 1 & 0 \\ 0 & 1 \end{array}\right)`	$\mu:=\left(\begin{array}{ll} 1 & 0 \\ 0 & 1 \\ 1 & 0 \\ 0 & 1 \end{array}\right)$	$\mu:=\left(\begin{array}{ll} 1 & 0 \\ 0 & 1 \\ 1 & 0 \\ 0 & 1 \end{array}\right)$
fong-spivak-seven-sketches EQ0247	313	■1.000 ● N|+0	`U(\mu)\left(\left(a_{1}, \ldots, a_{m}\right), \left(b_{1}, \ldots, b_{m}\right)\right):= \left(a_{1}+b_{1}, \ldots, a_{m}+b_{m}\right) .`	$U(\mu)\left(\left(a_{1}, \ldots, a_{m}\right), \left(b_{1}, \ldots, b_{m}\right)\right):= \left(a_{1}+b_{1}, \ldots, a_{m}+b_{m}\right) .$	$U(\mu)((a_{1}, \ldots, a_{m}), (b_{1}, \ldots, b_{m})):= (a_{1}+b_{1}, \ldots, a_{m}+b_{m}).$
fong-spivak-seven-sketches EQ0248	313	■1.000 ● N|-1	`B+C:=\left\{(w, y, x, z) \in R^{m+p} \times R^{n+q} \mid(w, x) \in B \text { and }(y, z) \in C\right\} .`	$B+C:=\left\{(w, y, x, z) \in R^{m+p} \times R^{n+q} \mid(w, x) \in B \text { and }(y, z) \in C\right\} .$	$B+C:=\{(w, y, x, z) \in R^{m+p} \times R^{n+q} \mid(w, x) \in B \text { and }(y, z) \in C\}.$
fong-spivak-seven-sketches EQ0249	313	■1.000 ● W|+0	`\begin{aligned} \mathrm{B}(g) & =\left\{(x, z) \in R^{m} \times R^{n} \mid S(g)(x)=z\right\} \\ \mathrm{B} \left(h^{\mathrm{op}}\right) & =\left\{(z, y) \in R^{n} \times R^{\ell} \mid z=S(h)(y)\right\} \end{aligned}`	$\begin{aligned} \mathrm{B}(g) & =\left\{(x, z) \in R^{m} \times R^{n} \mid S(g)(x)=z\right\} \\ \mathrm{B} \left(h^{\mathrm{op}}\right) & =\left\{(z, y) \in R^{n} \times R^{\ell} \mid z=S(h)(y)\right\} \end{aligned}$	$\begin{aligned} \mathrm{B}(g) & =\{(x, z) \in R^{m} \times R^{n} \mid S(g)(x)=z\} \\ \mathrm{B}(h^{\mathrm{op}}) & =\{(z, y) \in R^{n} \times R^{\ell} \mid z=S(h)(y)\} \end{aligned}$
fong-spivak-seven-sketches EQ0254	313	■1.000 ● N|+0	`\mathrm{B}\left(\left(g^{\mathrm{op}}\right) ; h\right)= \left\{(S(g)(x), S(h)(x)) \mid x \in R^{m}\right\} .`	$\mathrm{B}\left(\left(g^{\mathrm{op}}\right) ; h\right)= \left\{(S(g)(x), S(h)(x)) \mid x \in R^{m}\right\} .$	$\mathrm{B}((g^{\mathrm{op}}) \fatsemi h)= \{(S(g)(x), S(h)(x)) \mid x \in R^{m}\}.$
fong-spivak-seven-sketches EQ0255	315	■1.000 ● C|-14	`f(m)=f(\underbrace{1+1+\cdots+1}_{m \text { summands }})= \underbrace{f(1)+f(1)+\cdots+f(1)}_{m \text { summands }}=\underbrace{1_{R}+1_{R}+\cdots+1_{R}}_{m \text { summands }} .`	$f(m)=f(\underbrace{1+1+\cdots+1}_{m \text { summands }})= \underbrace{f(1)+f(1)+\cdots+f(1)}_{m \text { summands }}=\underbrace{1_{R}+1_{R}+\cdots+1_{R}}_{m \text { summands }} .$	$f(m)=f(\underbrace{1+1+\cdots+1}_{m \text { summands }})= \underbrace{f(1)+f(1)+\cdots+f(1)}_{m \text { summands }}=\underbrace{1_{R}+1_{R}+\cdots+1_{R}}_{m \text { summands }} .$

Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value

Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured

<table>
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<tr><td>fong-spivak-seven-sketches
EQ0261 (A.3)</td><td>327</td><td>■ 1.000
● C|+4</td><td>\begin{array}{cc||c|c} P & Q & P \wedge Q & P=(P \wedge Q) \\ \text { true } & \text { true } & \text { true } & \text { true } \\ \text { true } & \text { false } & \text { false } & \text { false } \\ \text { false } & \text { true } & \text { false } & \text { true } \\ \text { false } & \text { false } & \text { false } & \text { true } \end{array}</td><td>$\begin{array}{cc||c|c} P & Q & P \wedge Q & P=(P \wedge Q) \\ \text { true } & \text { true } & \text { true } & \text { true } \\ \text { true } & \text { false } & \text { false } & \text { false } \\ \text { false } & \text { true } & \text { false } & \text { true } \\ \text { false } & \text { false } & \text { false } & \text { true } \end{array}$</td><td>$\begin{array}{cc||c|c} P & Q & P \wedge Q & P=(P \wedge Q) \\ \texttt{true} & \texttt{true} & \texttt{true} & \texttt{true} \\ \texttt{true} & \texttt{false} & \texttt{false} & \texttt{false} \\ \texttt{false} & \texttt{true} & \texttt{false} & \texttt{true} \\ \texttt{false} & \texttt{false} & \texttt{false} & \texttt{true} \end{array}$</td></tr>
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EQ0263</td><td>333</td><td>■ 1.000
● N|+0</td><td>U=\bigcup_{i \in I}\left\{x \in \mathbb{R} \mid a_{i}<x<b_{i}\right\}</td><td>$U=\bigcup_{i \in I}\left\{x \in \mathbb{R} \mid a_{i}<x<b_{i}\right\}$</td><td>$U=\bigcup_{i \in I}\{x \in \mathbb{R} \mid a_{i}<x<b_{i}\}$</td></tr>
<tr><td>fong-spivak-seven-sketches
EQ0264</td><td>333</td><td>■ 1.000
● N|+1</td><td>U=\bigcup_{j \in J}\left\{x \in \mathbb{R} \mid a_{j}<x<b_{j}\right\}=\bigcup_{j \in J}\left(o_{ \left[a_{j}, b_{j}\right]} \cap \mathbb{R}\right)=\left(\bigcup_{j \in J} o_{\left[a_{j}, b_{j}\right]}\right) \cap \mathbb{R}</td><td>$U=\bigcup_{j \in J}\left\{x \in \mathbb{R} \mid a_{j}<x<b_{j}\right\}=\bigcup_{j \in J}\left(o_{\left[a_{j}, b_{j}\right]} \cap \mathbb{R}\right)=\left(\bigcup_{j \in J} o_{\left[a_{j}, b_{j}\right]}\right) \cap \mathbb{R}$</td><td>$U=\bigcup_{j \in J}\{x \in \mathbb{R} \mid a_{j}<x<b_{j}\}=\bigcup_{j \in J}(o_{[a_{j}, b_{j}]} \cap \mathbb{R})=\left(\bigcup_{j \in J} o_{[a_{j}, b_{j}]}\right) \cap \mathbb{R}$</td></tr>
<tr><td colspan="6">Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value
Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured</td></tr>
</table>

Tables

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
fong-spivak-seven-sketches_TAB_001	026	—	(no LaTeX source; 760 × 335 px region)	—	$\begin{array}{l\|\|l\|l} \textbf{arrow } a & \textbf{source } s(a)\in V & \textbf{target } t(a)\in V \\ \hline a & 1 & ? \\ b & 1 & 3 \\ c & ? & ? \\ d & ? & ? \\ e & ? & ? \end{array}$
fong-spivak-seven-sketches_TAB_002	064	—	(no LaTeX source; 824 × 172 px region)	—	$\begin{array}{c\|cc} \wedge & \texttt{false} & \texttt{true} \\ \hline \texttt{false} & \texttt{false} & \texttt{false} \\ \texttt{true} & \texttt{false} & \texttt{true} \end{array} \qquad \begin{array}{c\|cc} * & 0 & 1 \\ \hline 0 & 0 & 0 \\ 1 & 0 & 1 \end{array}$
fong-spivak-seven-sketches_TAB_003	065	—	(no LaTeX source; 391 × 172 px region)	—	$\begin{array}{c\|cc} \vee & \texttt{false} & \texttt{true} \\ \hline \texttt{false} & \texttt{false} & \texttt{true} \\ \texttt{true} & \texttt{true} & \texttt{true} \end{array}$
fong-spivak-seven-sketches_TAB_004	065	—	(no LaTeX source; 226 × 172 px region)	—	$\begin{array}{c\|cc} \max & 0 & 1 \\ \hline 0 & 0 & 1 \\ 1 & 1 & 1 \end{array}$
fong-spivak-seven-sketches_TAB_005	066	—	(no LaTeX source; 443 × 222 px region)	—	$\begin{array}{c\|ccc} \min & \texttt{no} & \texttt{maybe} & \texttt{yes} \\ \hline \texttt{no} & ? & ? & ? \\ \texttt{maybe} & ? & ? & ? \\ \texttt{yes} & ? & ? & ? \end{array}$
fong-spivak-seven-sketches_TAB_006	070	—	(no LaTeX source; 731 × 335 px region)	—	$\begin{array}{c\|ccccc} \Rightarrow & p & q & r & s & t \\ \hline p & \texttt{true} & \texttt{true} & \texttt{true} & \texttt{true} & \texttt{true} \\ q & \texttt{false} & \texttt{true} & \texttt{false} & \texttt{true} & \texttt{true} \\ r & \texttt{false} & \texttt{false} & \texttt{true} & \texttt{true} & \texttt{true} \\ s & \texttt{false} & \texttt{false} & \texttt{false} & \texttt{true} & \texttt{true} \\ t & \texttt{false} & \texttt{false} & \texttt{false} & \texttt{false} & \texttt{true} \end{array}$
fong-spivak-seven-sketches_TAB_007	074	—	(no LaTeX source; 305 × 229 px region)	—	$\begin{array}{c\|ccc} d(\nearrow) & x & y & z \\ \hline x & 0 & 4 & 3 \\ y & 3 & 0 & 6 \\ z & 7 & 4 & 0 \end{array}$
fong-spivak-seven-sketches_TAB_008	074	—	(no LaTeX source; 387 × 281 px region)	—	$\begin{array}{c\|cccc} d(\nearrow) & A & B & C & D \\ \hline A & 0 & ? & ? & ? \\ B & 2 & ? & 5 & ? \\ C & ? & ? & ? & ? \\ D & ? & ? & ? & ? \end{array}$
fong-spivak-seven-sketches_TAB_009	075	—	(no LaTeX source; 400 × 226 px region)	—	$M_Y := \begin{array}{c\|ccc} \nearrow & x & y & z \\ \hline x & 0 & 4 & 3 \\ y & 3 & 0 & \infty \\ z & \infty & 4 & 0 \end{array}$

Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value
Residual render vs scan (inkdrill): • C component • W weak • S stable • N noise • K clean • not measured

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
fong-spivak-seven-sketches_TAB_010	075	—	(no LaTeX source; 452 × 278 px region)	—	$M_X = \begin{array}{c\|cccc} \nearrow & A & B & C & D \\ \hline A & 0 & ? & ? & ? \\ B & 2 & 0 & \infty & ? \\ C & ? & ? & ? & ? \\ D & ? & ? & ? & ? \end{array}$
fong-spivak-seven-sketches_TAB_011	080	—	(no LaTeX source; 695 × 224 px region)	—	$\begin{array}{c\|ccc} \mathcal{X} & A & B & C \\ \hline A & 0 & 2 & 5 \\ B & \infty & 0 & 3 \\ C & \infty & \infty & 0 \end{array} \qquad \begin{array}{c\|cc} \mathcal{Y} & p & q \\ \hline p & 0 & 5 \\ q & 8 & 0 \end{array}$
fong-spivak-seven-sketches_TAB_012	080	—	(no LaTeX source; 905 × 400 px region)	—	$\begin{array}{c\|\|ccc\|ccc} \mathcal{X}\times\mathcal{Y} & (A,p) & (B,p) & (C,p) & (A,q) & (B,q) & (C,q) \\ \hline\hline (A,p) & 0 & 2 & 5 & 5 & 7 & 10 \\ (B,p) & \infty & 0 & 3 & \infty & 5 & 8 \\ (C,p) & \infty & \infty & 0 & \infty & \infty & 5 \\ \hline (A,q) & 8 & 10 & 13 & 0 & 2 & 5 \\ (B,q) & \infty & 8 & 11 & \infty & 0 & 3 \\ (C,q) & \infty & \infty & 8 & \infty & \infty & 0 \end{array}$
fong-spivak-seven-sketches_TAB_013	086	—	(no LaTeX source; 301 × 224 px region)	—	$\begin{array}{c\|ccc} M_Y & x & y & z \\ \hline x & 0 & 4 & 3 \\ y & 3 & 0 & \infty \\ z & \infty & 4 & 0 \end{array}$
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fong-spivak-seven-sketches_TAB_015	089	—	(no LaTeX source; 710 × 222 px region)	—	$\begin{array}{c\|ccc} \textbf{Employee} & \textbf{FName} & \textbf{WorksIn} & \textbf{Mngr} \\ \hline 1 & \text{Alan} & 101 & 2 \\ 2 & \text{Ruth} & 101 & 2 \\ 3 & \text{Kris} & 102 & 3 \end{array}$
fong-spivak-seven-sketches_TAB_016	089	—	(no LaTeX source; 548 × 222 px region)	—	$\begin{array}{c\|cc} \textbf{Department} & \textbf{DName} & \textbf{Secr} \\ \hline 101 & \text{Sales} & 1 \\ 102 & \text{IT} & 3 \end{array}$
fong-spivak-seven-sketches_TAB_017	101	—	(no LaTeX source; 1136 × 507 px region)	—	$\begin{array}{c\|} \textbf{Planet of Sol} \\ \hline \text{Mercury} \\ \text{Venus} \\ \text{Earth} \\ \text{Mars} \\ \text{Jupiter} \\ \text{Saturn} \\ \text{Uranus} \\ \text{Neptune} \end{array} \qquad \begin{array}{c\|} \textbf{Prime number} \\ \hline 2 \\ 3 \\ 5 \\ 7 \\ 11 \\ 13 \\ 17 \\ \vdots \end{array} \qquad \begin{array}{c\|} \textbf{Flying pig} \\ \hline \\ \end{array}$

Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value

Residual render vs scan (inkdrill): • C component • W weak • S stable • N noise • K clean • not measured

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Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value

Residual render vs scan (inkdrill): • C component • W weak • S stable • N noise • K clean • not measured

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Residual render vs scan (inkdrill): • C component • W weak • S stable • N noise • K clean • not measured</td></tr>
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Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
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fong-spivak-seven-sketches_TAB_032	135	—	(no LaTeX source; 380 × 278 px region)	—	$\begin{array}{c\|cccc} M_X & A & B & C & D \\ \hline A & 0 & \infty & 3 & \infty \\ B & 2 & 0 & \infty & 5 \\ C & \infty & 3 & 0 & \infty \\ D & \infty & \infty & 4 & 0 \end{array}$
fong-spivak-seven-sketches_TAB_033	135	—	(no LaTeX source; 314 × 276 px region)	—	$\begin{array}{c\|ccc} M_\Phi & x & y & z \\ \hline A & \infty & \infty & \infty \\ B & 11 & \infty & \infty \\ C & \infty & \infty & \infty \\ D & \infty & 9 & \infty \end{array}$
fong-spivak-seven-sketches_TAB_034	135	—	(no LaTeX source; 301 × 224 px region)	—	$\begin{array}{c\|ccc} M_Y & x & y & z \\ \hline x & 0 & 4 & 3 \\ y & 3 & 0 & \infty \\ z & \infty & 4 & 0 \end{array}$
fong-spivak-seven-sketches_TAB_035	138	—	(no LaTeX source; 715 × 281 px region)	—	$\begin{array}{c\|ccccc} \Phi & a & b & c & d & e \\ \hline N & \texttt{true} & \texttt{false} & \texttt{true} & \texttt{false} & \texttt{false} \\ E & \texttt{true} & \texttt{false} & \texttt{true} & \texttt{false} & \texttt{true} \\ W & \texttt{true} & \texttt{true} & \texttt{true} & \texttt{true} & \texttt{false} \\ S & \texttt{true} & \texttt{true} & \texttt{true} & \texttt{true} & \texttt{true} \end{array}$
fong-spivak-seven-sketches_TAB_036	138	—	(no LaTeX source; 344 × 335 px region)	—	$\begin{array}{c\|cc} \Psi & x & y \\ \hline a & \texttt{false} & \texttt{true} \\ b & \texttt{true} & \texttt{true} \\ c & \texttt{false} & \texttt{true} \\ d & \texttt{true} & \texttt{true} \\ e & \texttt{false} & \texttt{false} \end{array}$
fong-spivak-seven-sketches_TAB_037	138	—	(no LaTeX source; 382 × 283 px region)	—	$\begin{array}{c\|cc} \Phi \mathbin{⨟} \Psi & x & y \\ \hline N & \texttt{false} & \texttt{true} \\ E & \texttt{false} & \texttt{true} \\ W & \texttt{true} & \texttt{true} \\ S & \texttt{true} & \texttt{true} \end{array}$
fong-spivak-seven-sketches_TAB_038	139	—	(no LaTeX source; 369 × 278 px region)	—	$\begin{array}{c\|cccc} \Phi \mathbin{⨟} \Psi & p & q & r & s \\ \hline A & ? & 24 & ? & ? \\ B & ? & ? & ? & ? \\ C & ? & ? & ? & ? \\ D & ? & ? & 9 & ? \end{array}$

Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value

Residual render vs scan (inkdrill): • C component • W weak • S stable • N noise • K clean • not measured

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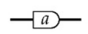
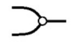
<tr><td>amplify by $a \in R$</td><td></td><td>(a)</td><td>$1 \to 1$</td></tr>
<tr><td>add</td><td></td><td>$\begin{pmatrix}1\\1\end{pmatrix}$</td><td>$2 \to 1$</td></tr>
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Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value

Residual render vs scan (inkdrill): • C component • W weak • S stable • N noise • K clean • not measured

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
fong-spivak-seven-sketches_TAB_046	284	—	(no LaTeX source; 362 × 224 px region)	—	$\begin{array}{ccccc} d(\nearrow) & A & B & C & D \\ \hline A & 0 & 6 & 3 & 11 \\ B & 2 & 0 & 5 & 5 \\ C & 5 & 3 & 0 & 8 \\ D & 11 & 9 & 6 & 0 \end{array}$
fong-spivak-seven-sketches_TAB_047	284	—	(no LaTeX source; 419 × 231 px region)	—	$M_X = \begin{array}{ccccc} \nearrow & A & B & C & D \\ \hline A & 0 & \infty & 3 & \infty \\ B & 2 & 0 & \infty & 5 \\ C & \infty & 3 & 0 & \infty \\ D & \infty & \infty & 6 & 0 \end{array}$
fong-spivak-seven-sketches_TAB_048	285	—	(no LaTeX source; 738 × 226 px region)	—	$\begin{array}{ccccc} \mathcal{X}(\nearrow) & A & B & C & D \\ \hline A & M & \{\text{boat}\} & \varnothing & \{\text{boat}\} \\ B & \varnothing & M & \varnothing & \{\text{boat}\} \\ C & \{\text{foot}, \text{boat}\} & \{\text{boat}\} & M & M \\ D & \{\text{foot}\} & \varnothing & \{\text{foot}, \text{car}\} & M \end{array}$
fong-spivak-seven-sketches_TAB_049	285	—	(no LaTeX source; 324 × 181 px region)	—	$\begin{array}{cccc} M(\nearrow) & A & B & C \\ \hline A & \infty & 6 & 10 \\ B & 10 & \infty & 10 \\ C & 10 & 6 & \infty \end{array}$
fong-spivak-seven-sketches_TAB_050	287	—	(no LaTeX source; 339 × 142 px region)	—	$\begin{array}{ccc} \Rightarrow & \texttt{false} & \texttt{true} \\ \hline \texttt{false} & \texttt{true} & \texttt{true} \\ \texttt{true} & \texttt{false} & \texttt{true} \end{array}$
fong-spivak-seven-sketches_TAB_051	289	—	(no LaTeX source; 676 × 317 px region)	—	$\begin{array}{ccccccc} \nearrow & v_1 & f_1 & f_1 \fatsemi f_2 & v_2 & f_2 & v_3 \\ \hline v_1 & v_1 & f_1 & f_1 \fatsemi f_2 & \boxtimes & \boxtimes & \boxtimes \\ f_1 & \boxtimes & \boxtimes & \boxtimes & f_1 & f_1 \fatsemi f_2 & \boxtimes \\ f_1 \fatsemi f_2 & \boxtimes & \boxtimes & \boxtimes & \boxtimes & \boxtimes & f_1 \fatsemi f_2 \\ v_2 & \boxtimes & \boxtimes & \boxtimes & v_2 & f_2 & \boxtimes \\ f_2 & \boxtimes & \boxtimes & \boxtimes & \boxtimes & \boxtimes & f_2 \\ v_3 & \boxtimes & \boxtimes & \boxtimes & \boxtimes & \boxtimes & v_3 \end{array}$
fong-spivak-seven-sketches_TAB_052	294	—	(no LaTeX source; 581 × 269 px region)	—	$\begin{array}{rll} \textbf{Arrow} & \textbf{source} & \textbf{target} \\ \hline \text{Mngr} & \text{Employee} & \text{Employee} \\ \text{WorksIn} & \text{Employee} & \text{Department} \\ \text{Secr} & \text{Department} & \text{Employee} \\ \text{FName} & \text{Employee} & \text{string} \\ \text{DName} & \text{Department} & \text{string} \end{array}$
fong-spivak-seven-sketches_TAB_053	294	—	(no LaTeX source; 217 × 269 px region)	—	$\begin{array}{r} \textbf{Vertex} \\ \hline \text{Department} \\ \text{Employee} \\ \text{string} \end{array}$
fong-spivak-seven-sketches_TAB_054	295	—	(no LaTeX source; 376 × 357 px region)	—	$\begin{array}{ccc} \textbf{Arrow} & \textbf{source} & \textbf{target} \\ \hline 1 & 4 & 1 \\ 2 & 4 & 2 \\ 3 & 5 & 3 \\ 4 & 5 & 4 \\ 5 & 5 & 5 \\ 6 & 7 & 6 \\ 7 & 6 & 7 \end{array}$

Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value

Residual render vs scan (inkdrill): • C component • W weak • S stable • N noise • K clean • not measured

<table>
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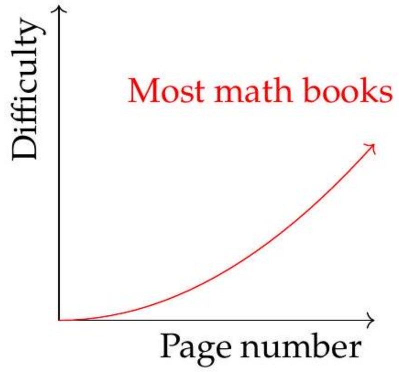
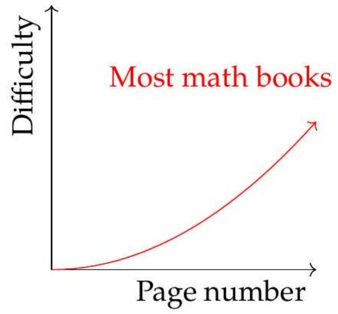
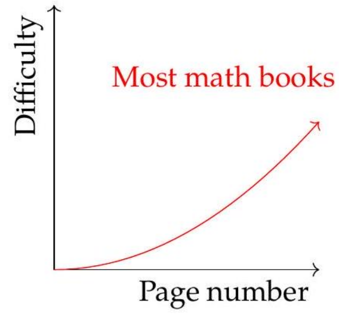
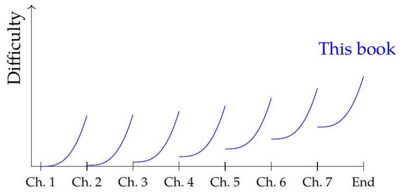
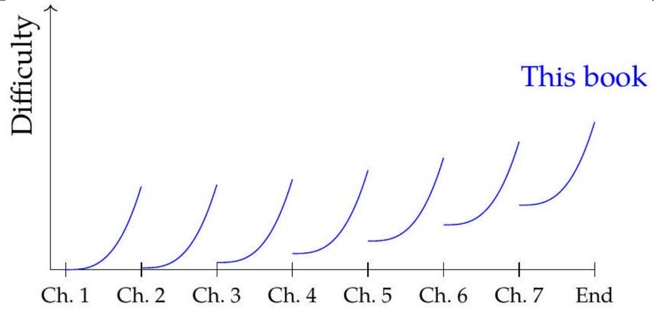
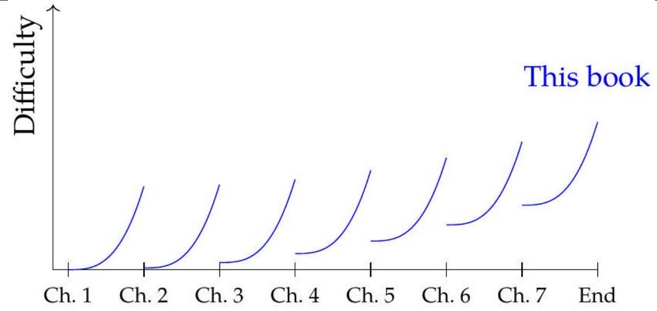
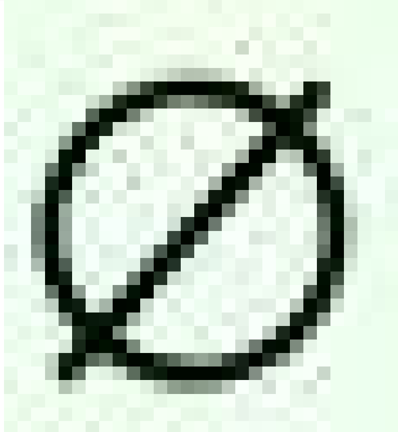
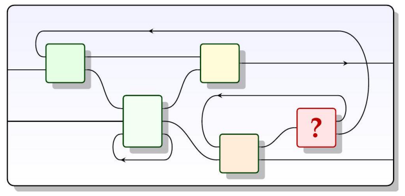
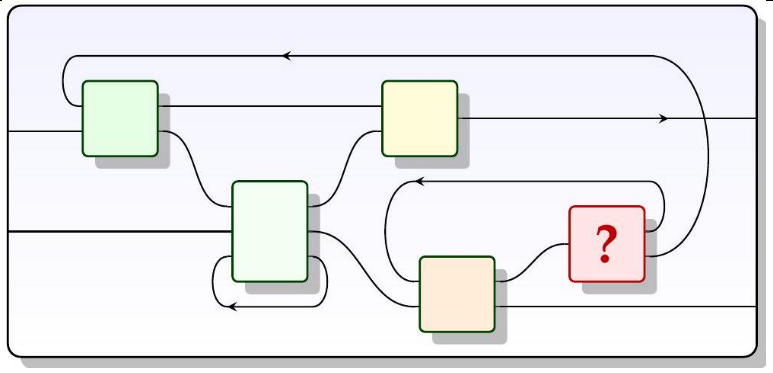
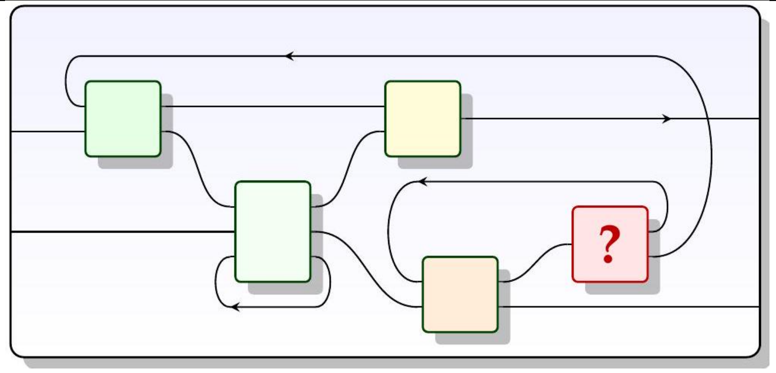
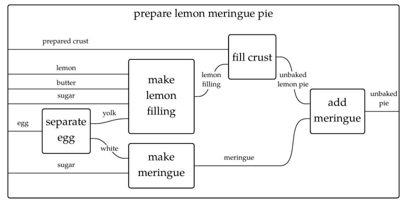
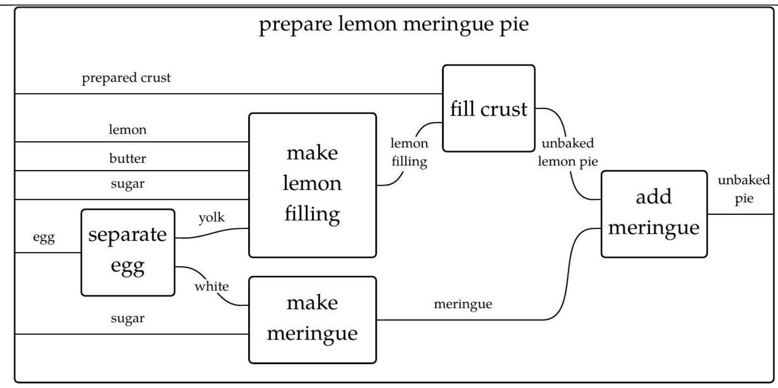
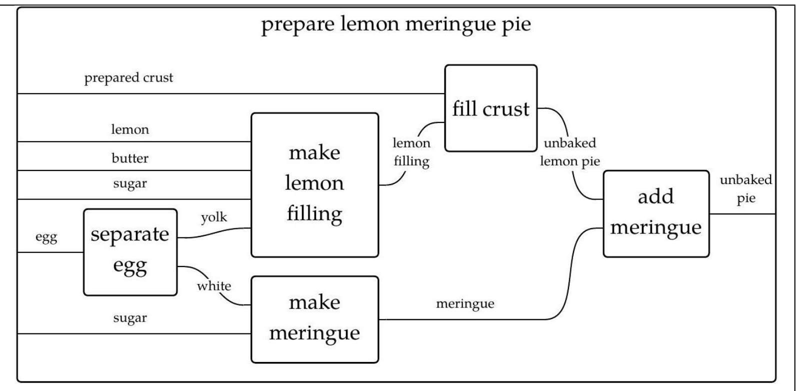
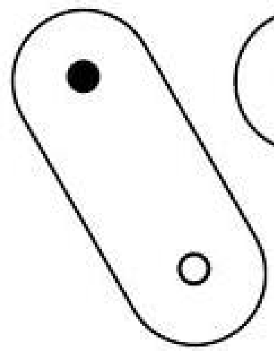
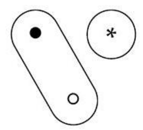
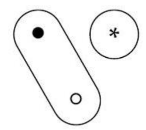
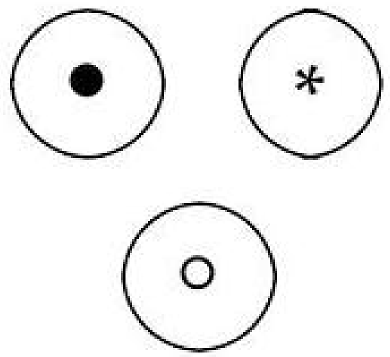
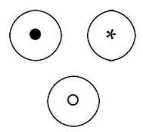
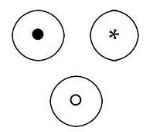
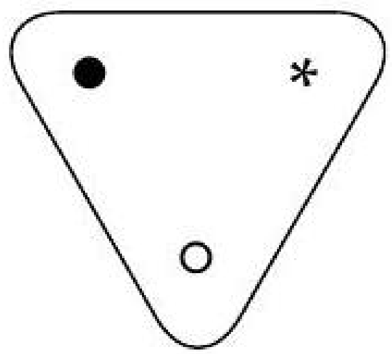
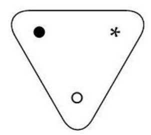
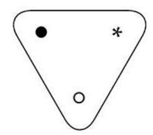
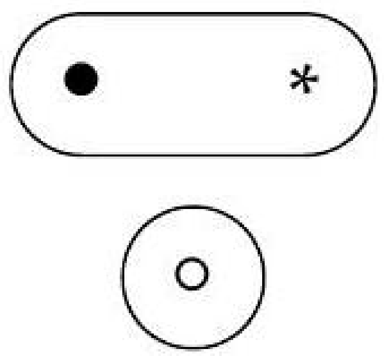
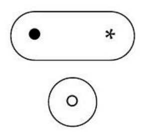
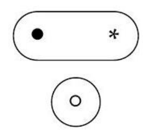
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value

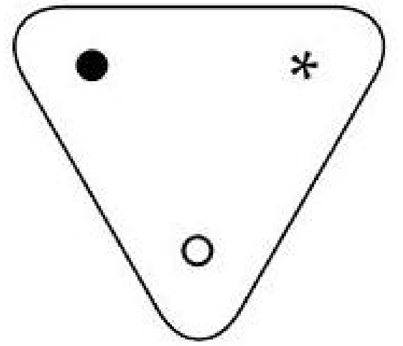
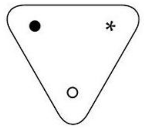
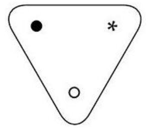
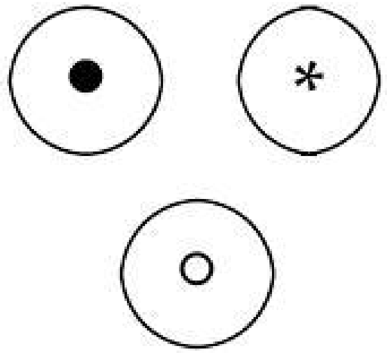
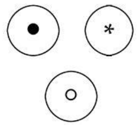
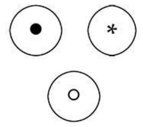
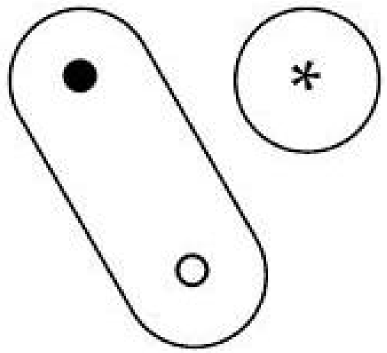
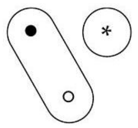
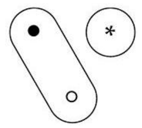
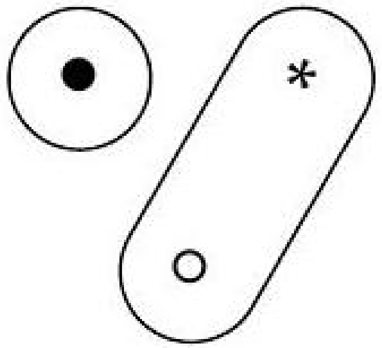
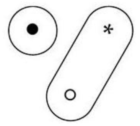
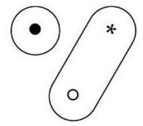
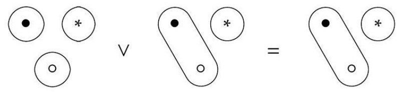
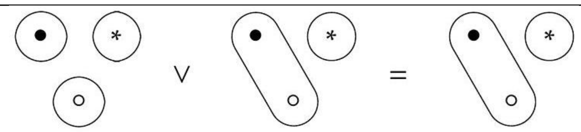
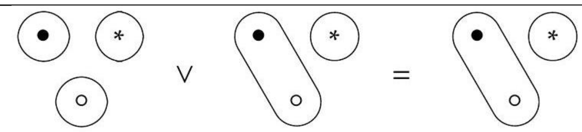
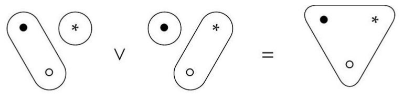
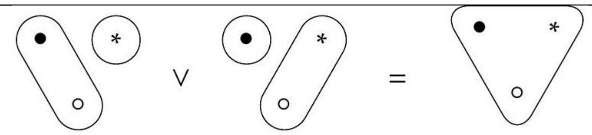
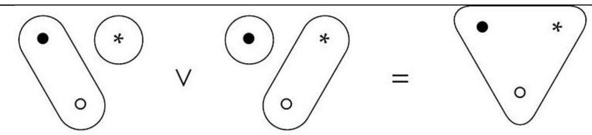
Residual render vs scan (inkdrill): • C component • W weak • S stable • N noise • K clean • not measured

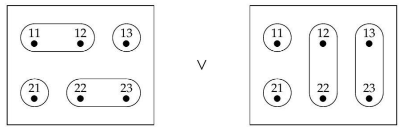
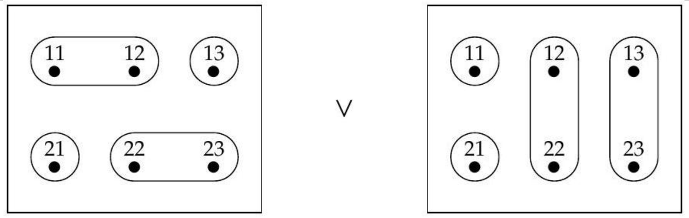
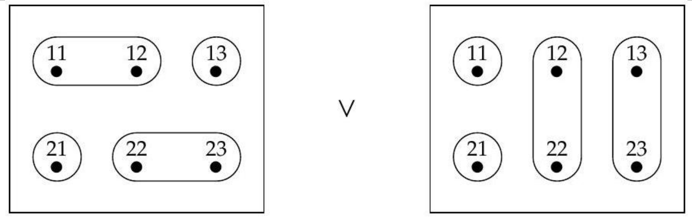
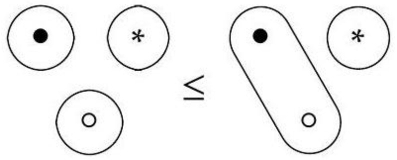
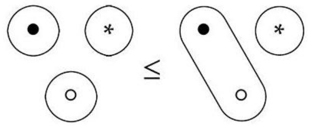
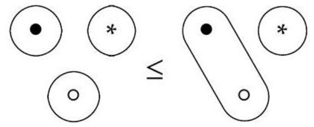
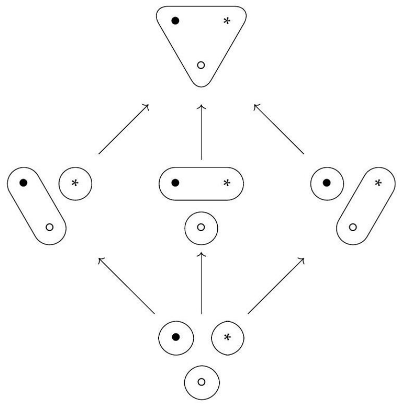
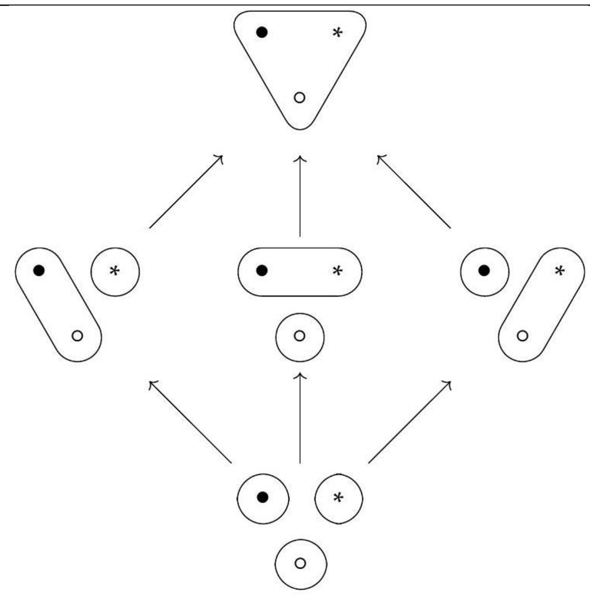
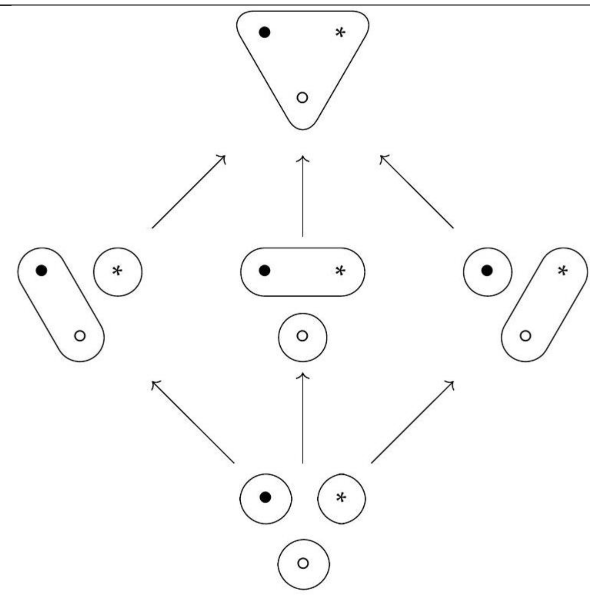
Image regions — rendered against scan

Two comparable cells per row. **Rendered**: the region's own LaTeX compiled as its OWN document, or — where no LaTeX exists — the scan again, so every row has two cells and the ink difference reads as a floor rather than as missing-versus-present. A duplicated row is marked *(dup)* and its distance is a SELF-comparison, not agreement between two sources. **Scan**: the `tex.zip` image whose filename region 5-tuple matches this row, else the CDN crop. The Class column carries the residual class beside MathPix's confidence, as the equation rows do. Setting a region's LaTeX does NOT say it renders to what is on the page — that is what the two columns are for, and 281 is the open question they exist to let a reader answer. A region with no LaTeX can be reconstructed with `pdfdrill vision`; verify any LLM result against the real ink with inkdrill.

Identifier	Page	Class	Source	Author source	Rendered	Scan
fong-spivak-seven-sketches_PIC_0001	005	— *(dup)*	7f1e85c6-4350-401f-a960-eb8a0176e786-005_407_430_1561_351.jpg			
fong-spivak-seven-sketches_PIC_0002	005	— *(dup)*	7f1e85c6-4350-401f-a960-eb8a0176e786-005_403_842_1561_923.jpg			
fong-spivak-seven-sketches_PIC_0003	029	— *(dup)*	7f1e85c6-4350-401f-a960-eb8a0176e786-029_32_29_1683_1045.jpg		$\varnothing$	$\varnothing$
fong-spivak-seven-sketches_PIC_0004	317	— *(dup)*	7f1e85c6-4350-401f-a960-eb8a0176e786-317_41_1294_1900_484.jpg	a) The map $[\mathrm{id}_A + \iota_B, \iota_C] = [[\iota_A, \iota_B \fatsemi \iota_{B+C}], \iota_C \fatsemi \iota_{B+C}]\colon (A+B)+C \to A+(B+C)$ is an	a) The map $[\mathrm{id}_A + \iota_B, \iota_C] = [[\iota_A, \iota_B \fatsemi \iota_{B+C}], \iota_C \fatsemi \iota_{B+C}]\colon (A+B)+C \to A+(B+C)$ is an	a) The map $[\mathrm{id}_A + \iota_B, \iota_C] = [[\iota_A, \iota_B \fatsemi \iota_{B+C}], \iota_C \fatsemi \iota_{B+C}]\colon (A+B)+C \to A+(B+C)$ is an
fong-spivak-seven-sketches_DIA_0001	001	— *(dup)*	7f1e85c6-4350-401f-a960-eb8a0176e786-001_483_1004_899_560.jpg			

fong-spivak-seven-sketches_DIA_0002	004	— (dup)	7f1e85c6-4350-401f-a960-eb8a0176e786-004_670_1353_919_380.jpg			
fong-spivak-seven-sketches_DIA_0003	015	— (dup)	7f1e85c6-4350-401f-a960-eb8a0176e786-015_161_174_344_977.jpg			
fong-spivak-seven-sketches_DIA_0004	015	— (dup)	7f1e85c6-4350-401f-a960-eb8a0176e786-015_167_174_977_742.jpg			
fong-spivak-seven-sketches_DIA_0005	015	— (dup)	7f1e85c6-4350-401f-a960-eb8a0176e786-015_165_181_984_1181.jpg			
fong-spivak-seven-sketches_DIA_0006	015	— (dup)	7f1e85c6-4350-401f-a960-eb8a0176e786-015_161_174_1633_742.jpg			

fong-spivak-seven-sketches_DIA_0007	015	—	(dup)	7f1e85c6-4350-401f-a960-eb8a0176e786-015_165_183_1638_1179.jpg			
fong-spivak-seven-sketches_DIA_0008	015	—	(dup)	7f1e85c6-4350-401f-a960-eb8a0176e786-015_165_174_1968_624.jpg			
fong-spivak-seven-sketches_DIA_0009	015	—	(dup)	7f1e85c6-4350-401f-a960-eb8a0176e786-015_158_172_1973_968.jpg			
fong-spivak-seven-sketches_DIA_0010	015	—	(dup)	7f1e85c6-4350-401f-a960-eb8a0176e786-015_158_174_1973_1308.jpg			
fong-spivak-seven-sketches_DIA_0011	016	—	(dup)	7f1e85c6-4350-401f-a960-eb8a0176e786-016_167_742_738_688.jpg			
fong-spivak-seven-sketches_DIA_0012	016	—	(dup)	7f1e85c6-4350-401f-a960-eb8a0176e786-016_170_751_971_683.jpg			

fong-spivak-seven-sketches_DIA_0013 — *(dup)*	016		7f1e85c6-4350-401f-a960-eb8a0176e786-016_281_887_1231_615.jpg			
fong-spivak-seven-sketches_DIA_0014 — *(dup)*	017		7f1e85c6-4350-401f-a960-eb8a0176e786-017_165_398_742_860.jpg			
fong-spivak-seven-sketches_DIA_0015 — *(dup)*	017		7f1e85c6-4350-401f-a960-eb8a0176e786-017_767_753_1050_681.jpg			
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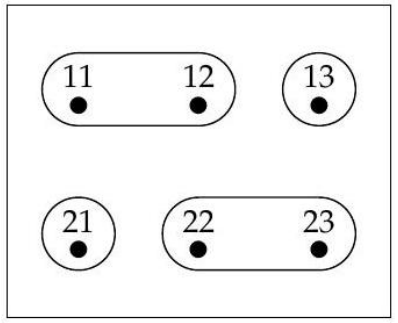
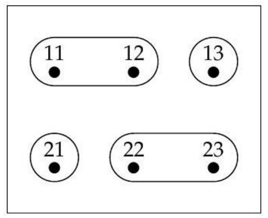
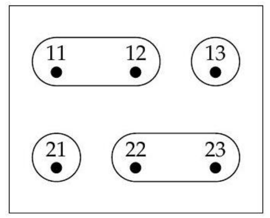
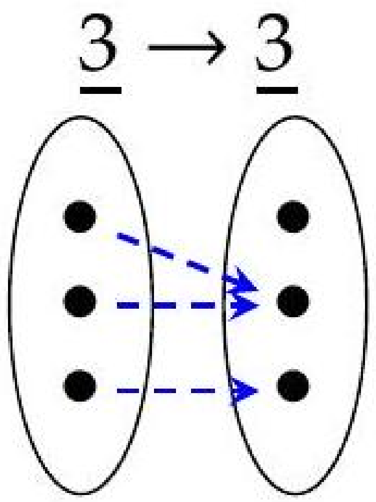
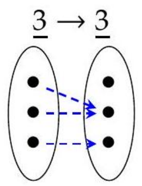
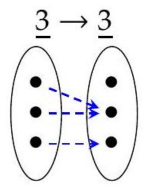
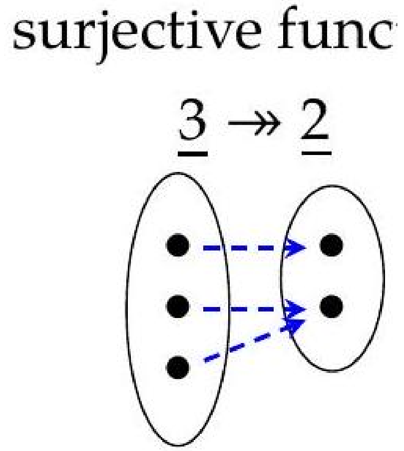
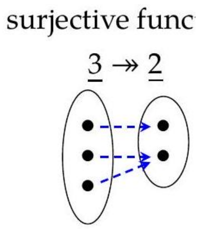
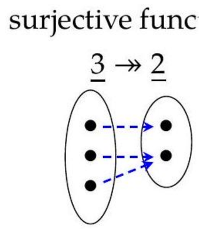
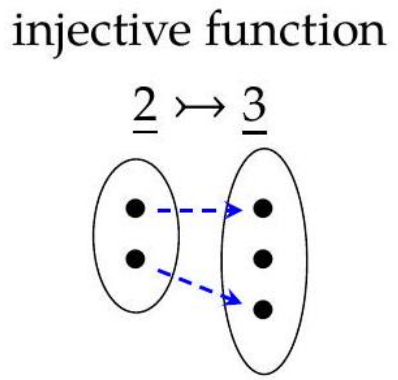
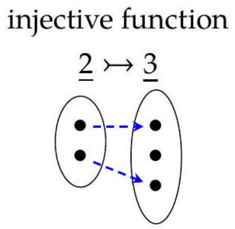
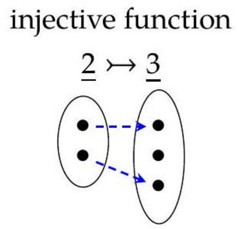
fong-spivak-seven-sketches_DIA_0017 — *(dup)*	021		7f1e85c6-4350-401f-a960-eb8a0176e786-021_279_339_1306_889.jpg			
fong-spivak-seven-sketches_DIA_0018 — *(dup)*	023		7f1e85c6-4350-401f-a960-eb8a0176e786-023_238_181_493_437.jpg			

fong-spivak-seven-sketches_DIA_0019	023	— *(dup)*	7f1e85c6-4350-401f-a960-eb8a0176e786-023_294_251_437_733.jpg			
fong-spivak-seven-sketches_DIA_0020	023	— *(dup)*	7f1e85c6-4350-401f-a960-eb8a0176e786-023_296_303_434_1109.jpg			
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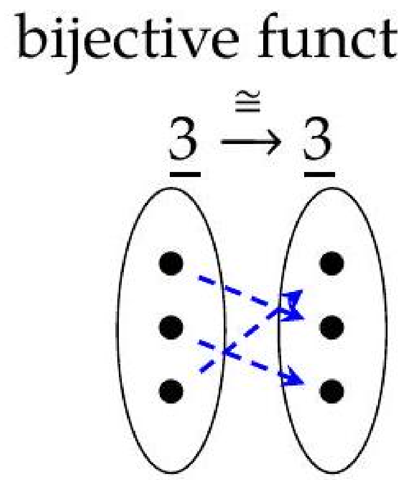
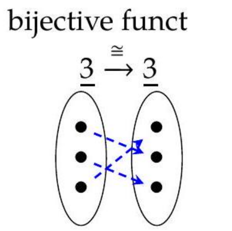
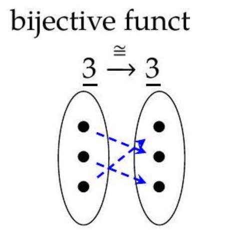
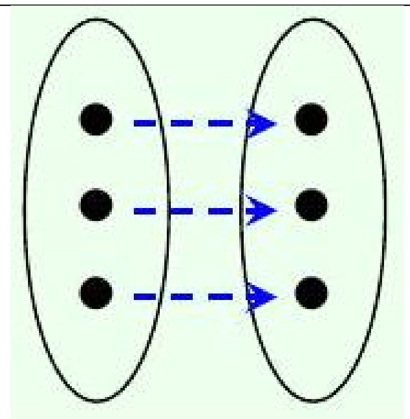
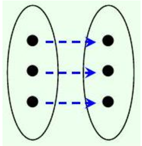
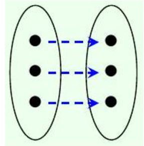
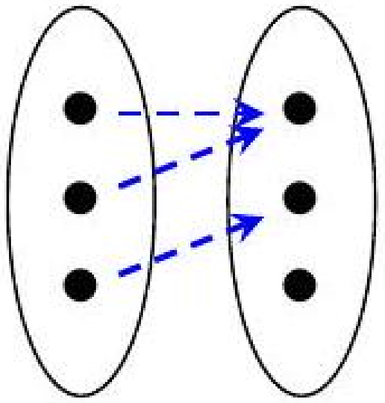
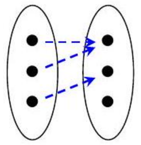
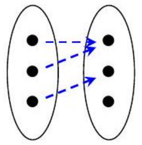
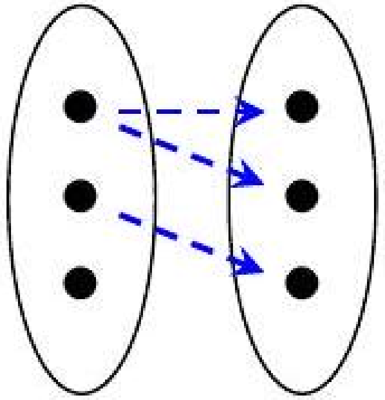
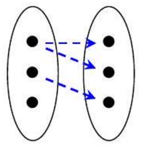
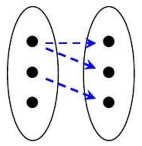
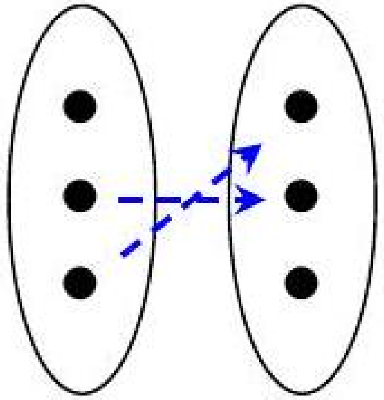
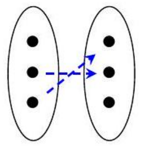
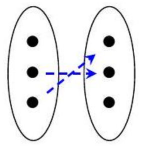
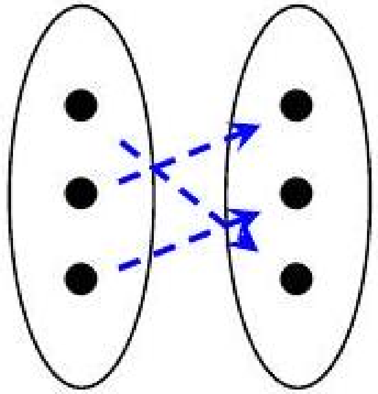
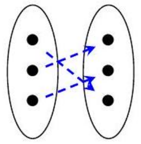
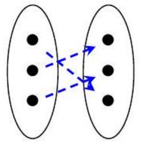
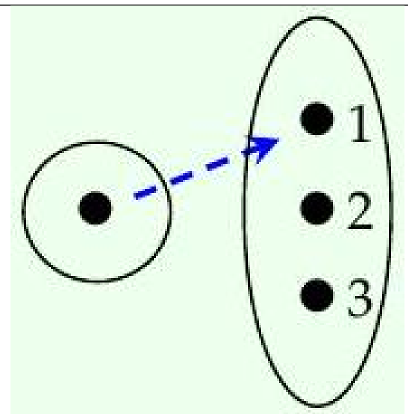
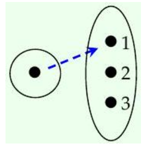
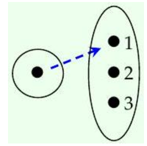
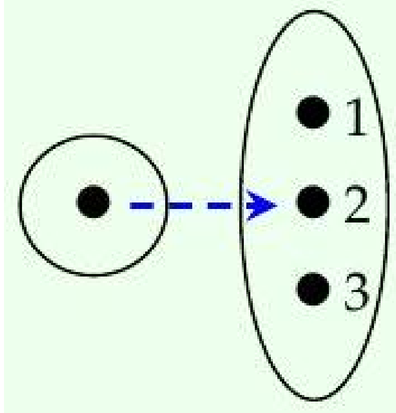
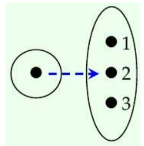
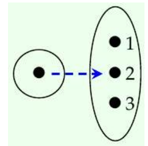
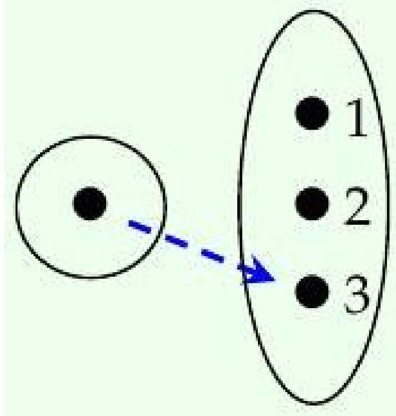
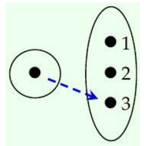
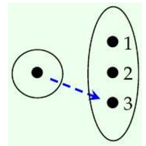
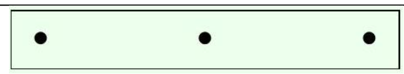
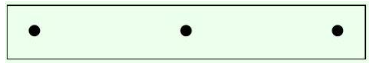
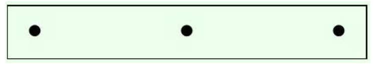
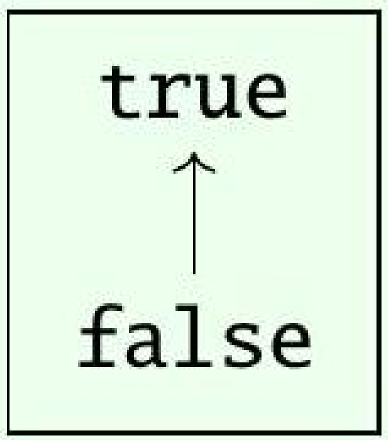
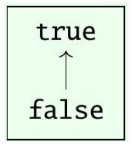
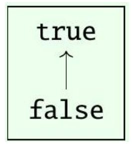
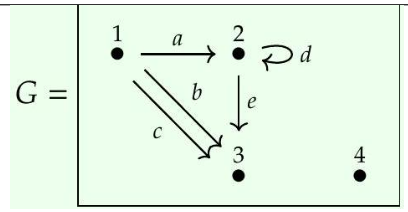
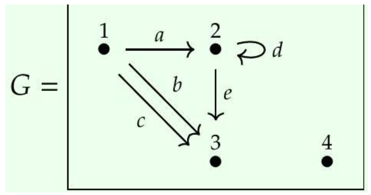
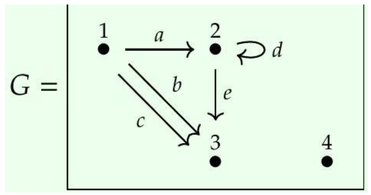
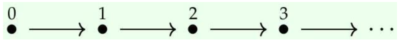
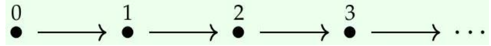
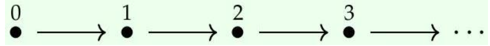
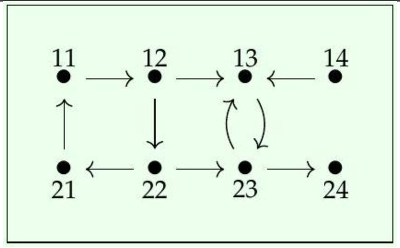
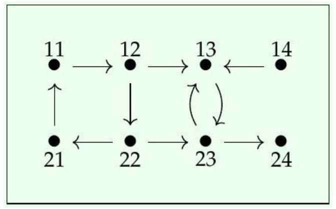
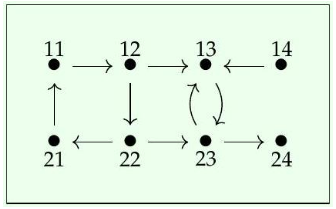
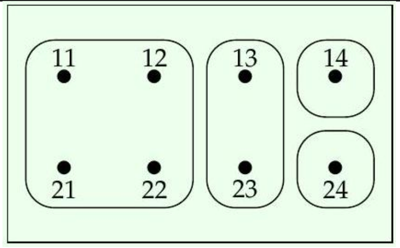
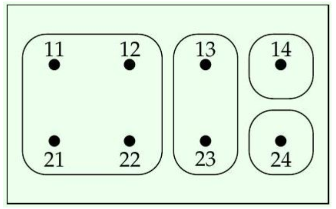
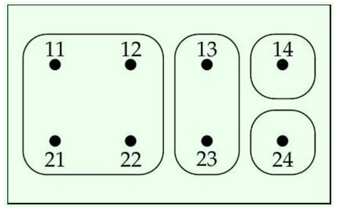
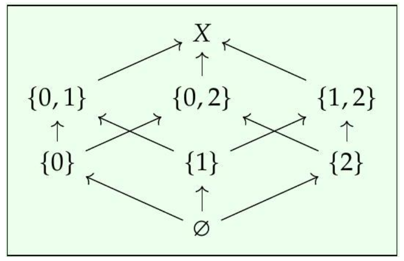
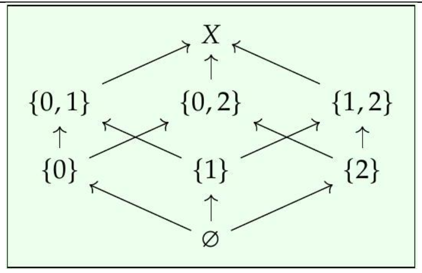
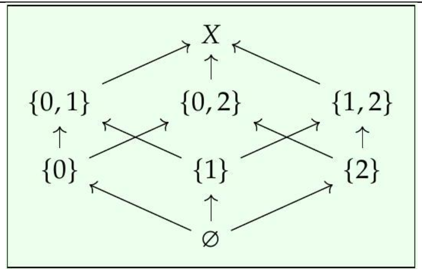
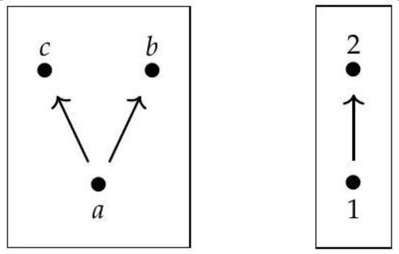
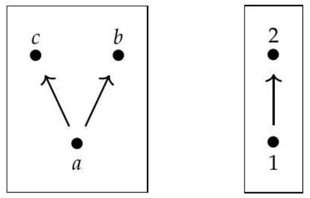
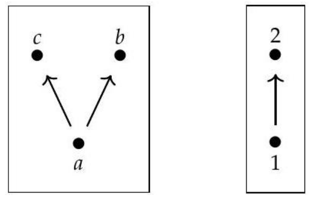
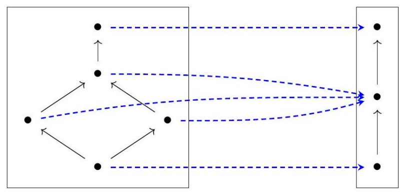
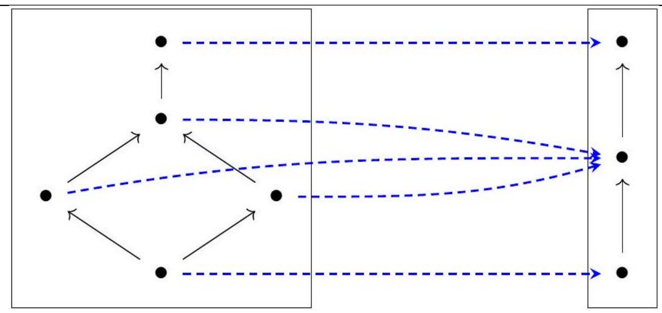
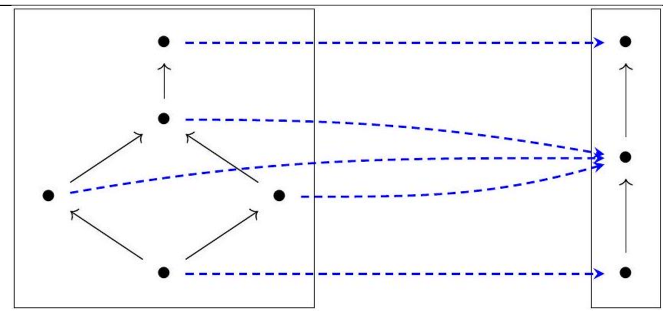
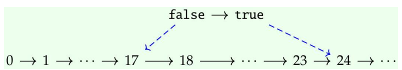
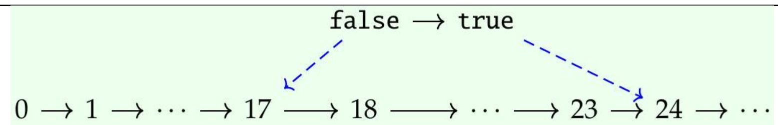
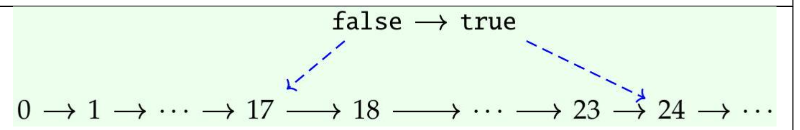
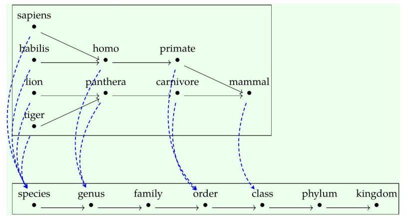
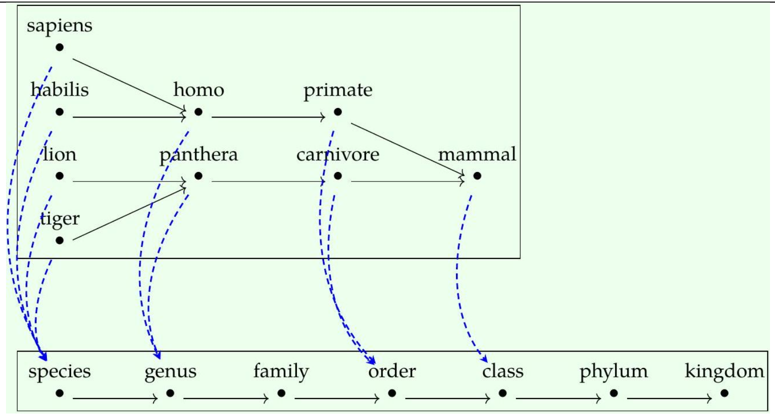
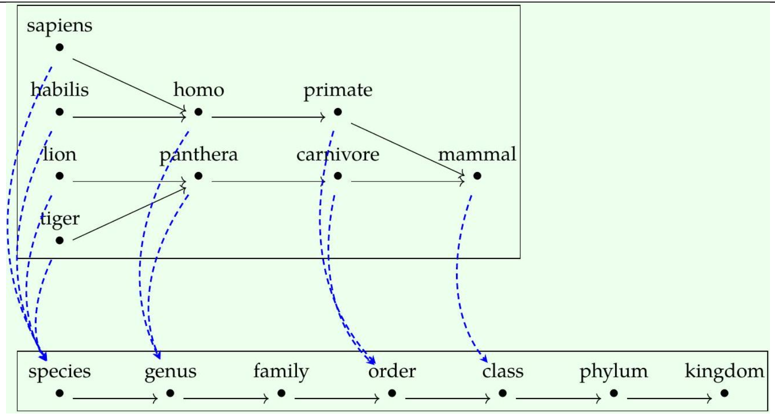
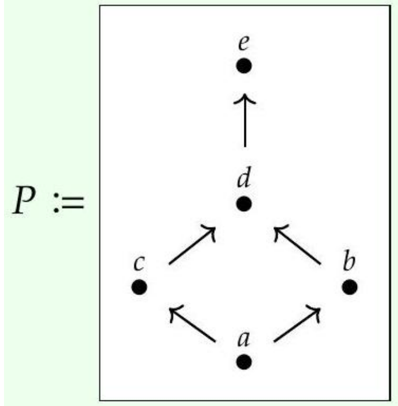
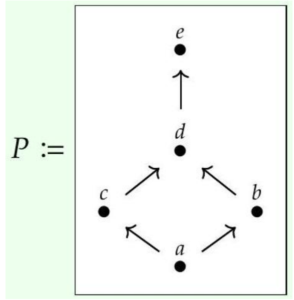
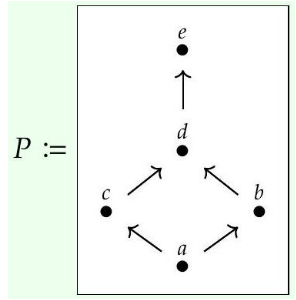
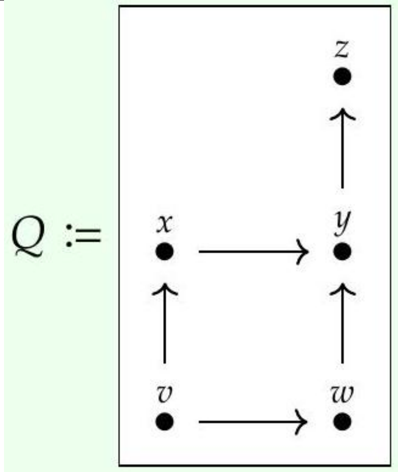
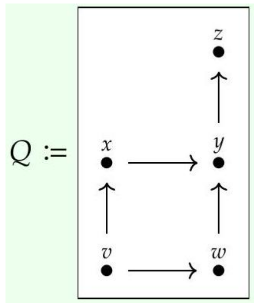
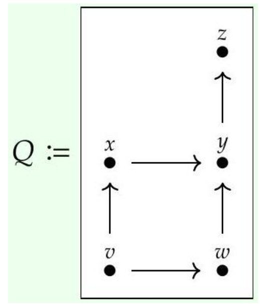
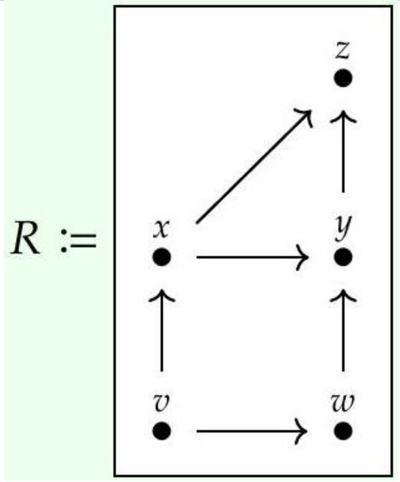
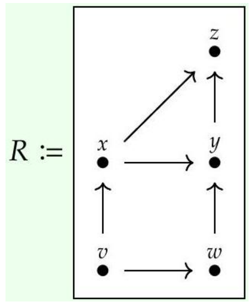
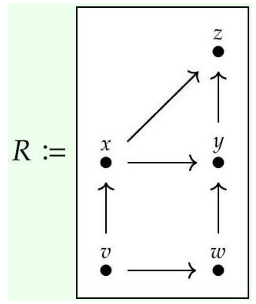
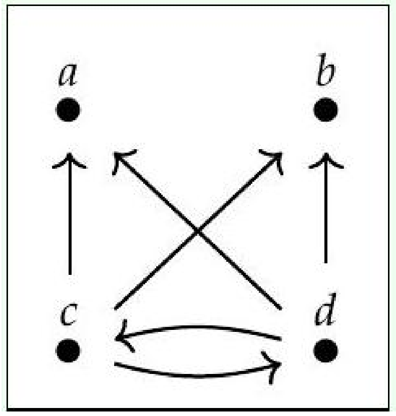
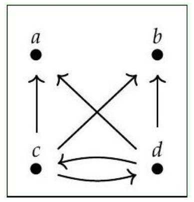
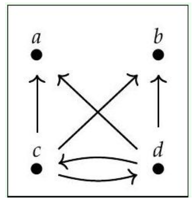
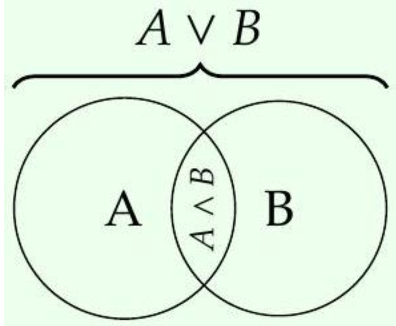
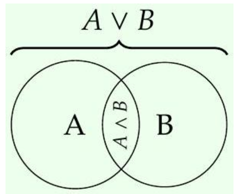
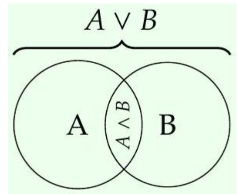
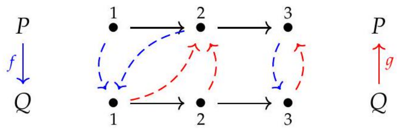
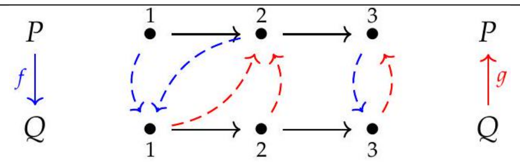
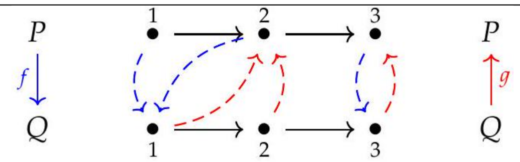
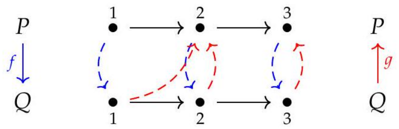
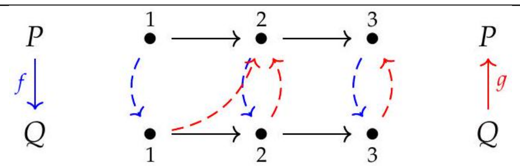
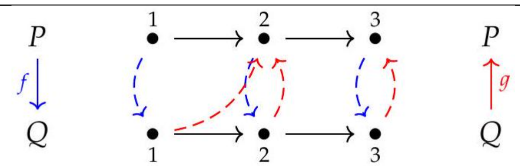
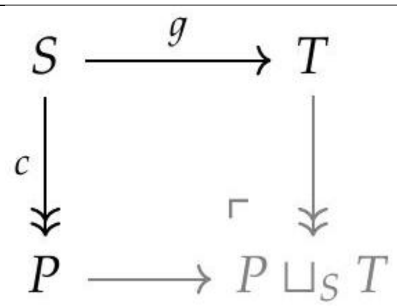
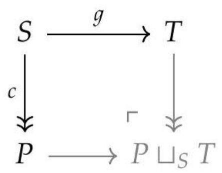
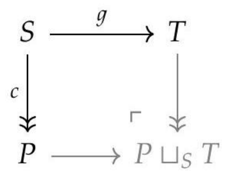
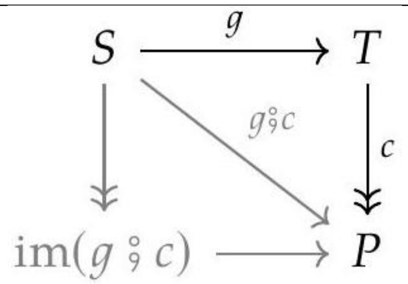
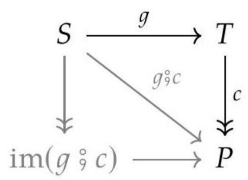
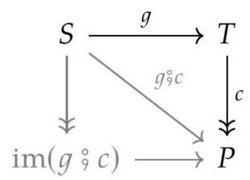
fong-spivak-seven-sketches_DIA_0022	023	— *(dup)*	7f1e85c6-4350-401f-a960-eb8a0176e786-023_299_240_432_1475.jpg			
fong-spivak-seven-sketches_DIA_0023	023	— *(dup)*	7f1e85c6-4350-401f-a960-eb8a0176e786-023_188_179_950_971.jpg			

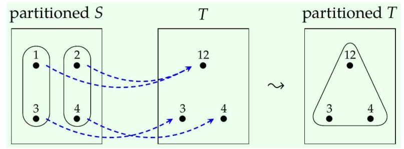
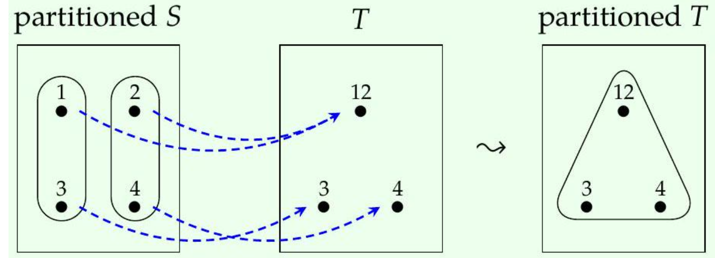
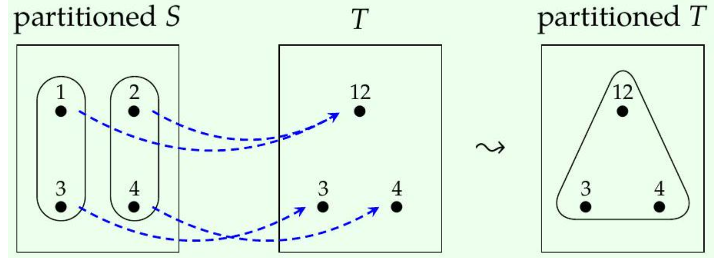
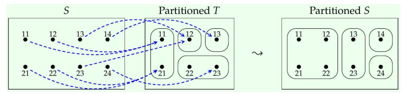
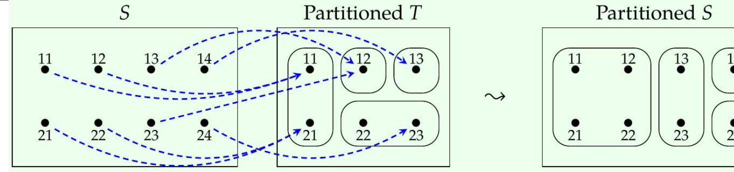
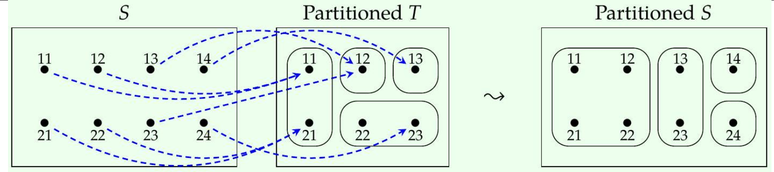
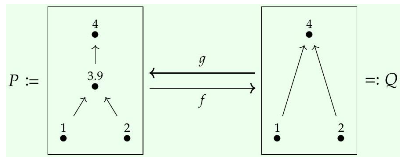
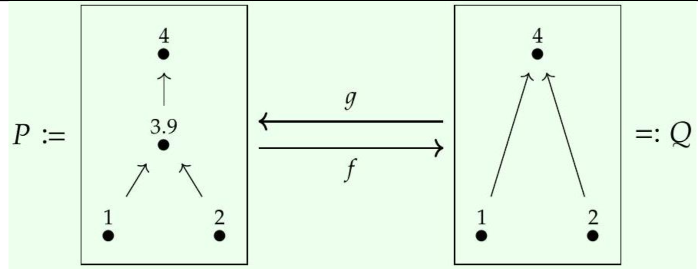
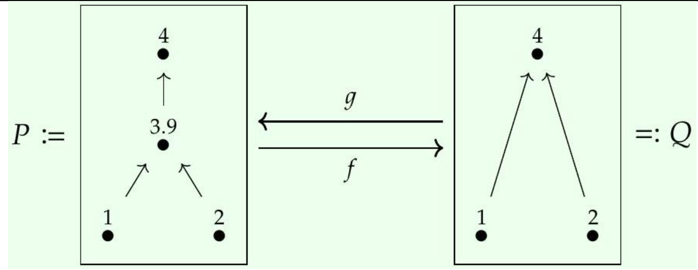
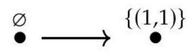
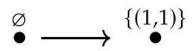
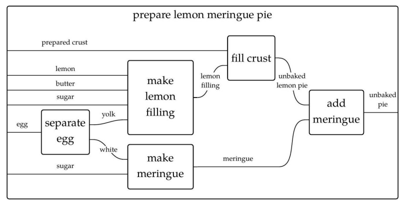
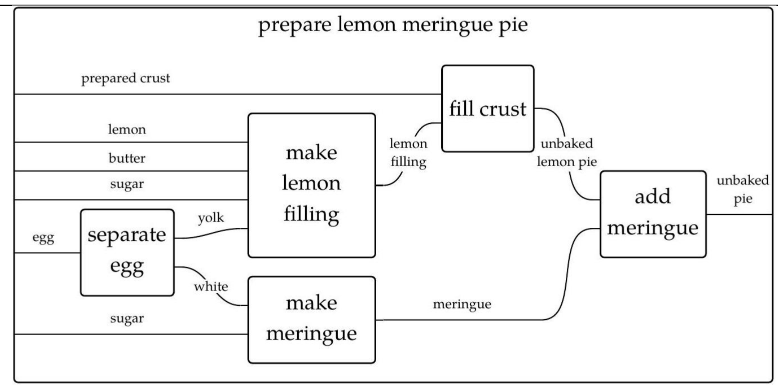
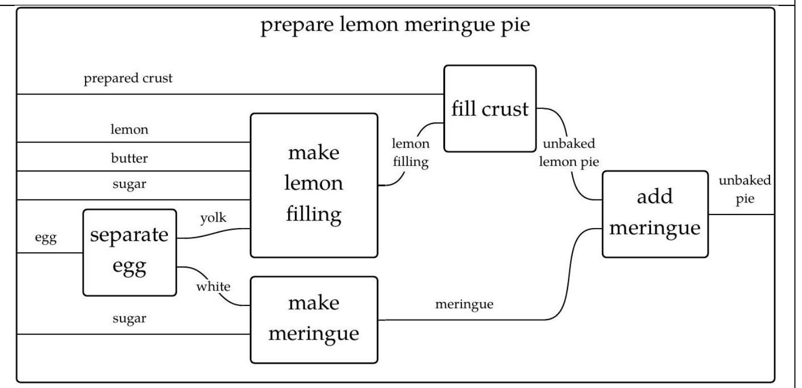
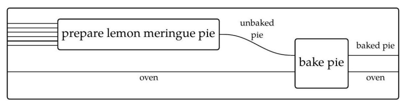
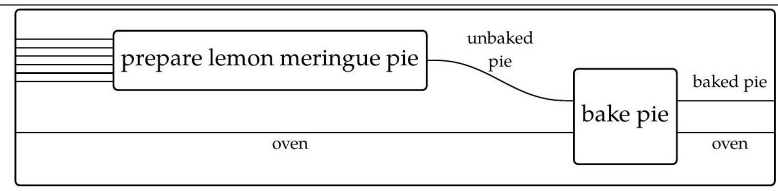
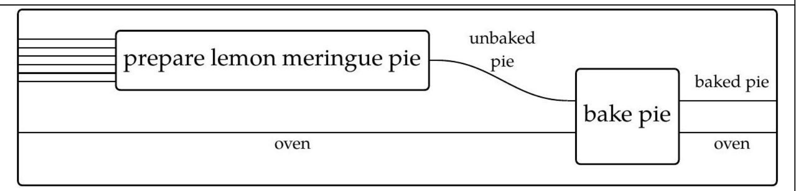
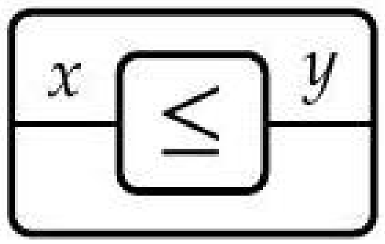
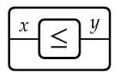
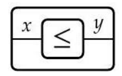
fong-spivak-seven-sketches_DIA_0024	023	— (dup)	7f1e85c6-4350-401f-a960-eb8a0176e786-023_190_176_1692_529.jpg			
fong-spivak-seven-sketches_DIA_0025	023	— (dup)	7f1e85c6-4350-401f-a960-eb8a0176e786-023_188_176_1695_824.jpg			
fong-spivak-seven-sketches_DIA_0026	023	— (dup)	7f1e85c6-4350-401f-a960-eb8a0176e786-023_188_176_1695_1118.jpg			
fong-spivak-seven-sketches_DIA_0027	023	— (dup)	7f1e85c6-4350-401f-a960-eb8a0176e786-023_188_179_1695_1414.jpg			

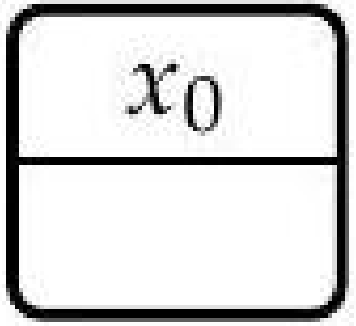
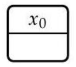
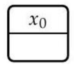
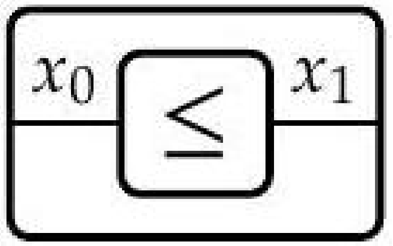
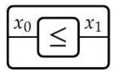
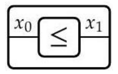
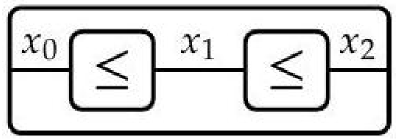
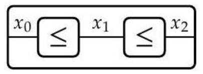
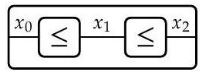
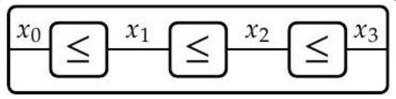
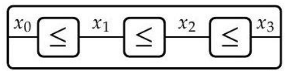
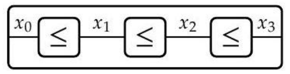
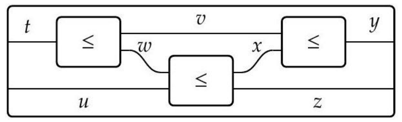
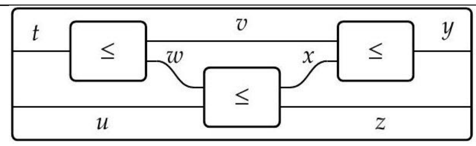
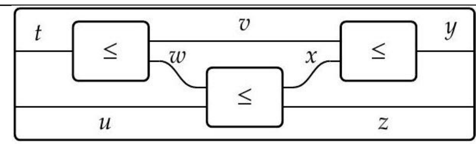
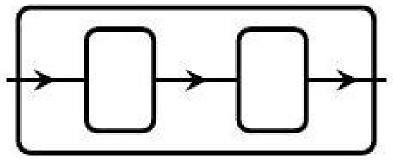
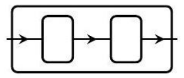
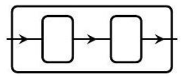
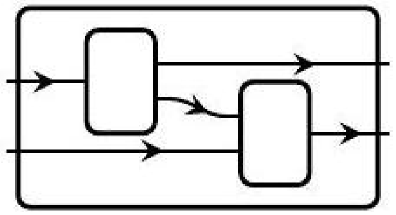
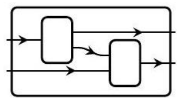
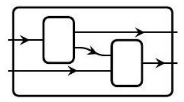
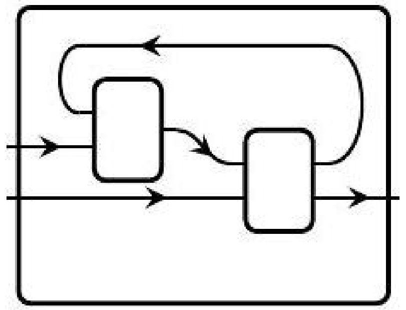
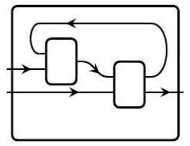
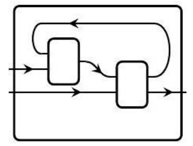
fong-spivak-seven-sketches_DIA_0028	024	— *(dup)*	7f1e85c6-4350-401f-a960-eb8a0176e786-024_183_176_1561_674.jpg			
fong-spivak-seven-sketches_DIA_0029	024	— *(dup)*	7f1e85c6-4350-401f-a960-eb8a0176e786-024_186_176_1561_968.jpg			
fong-spivak-seven-sketches_DIA_0030	024	— *(dup)*	7f1e85c6-4350-401f-a960-eb8a0176e786-024_186_174_1561_1265.jpg			
fong-spivak-seven-sketches_DIA_0031	025	— *(dup)*	7f1e85c6-4350-401f-a960-eb8a0176e786-025_81_473_826_824.jpg			
fong-spivak-seven-sketches_DIA_0032	025	— *(dup)*	7f1e85c6-4350-401f-a960-eb8a0176e786-025_181_161_1416_980.jpg			

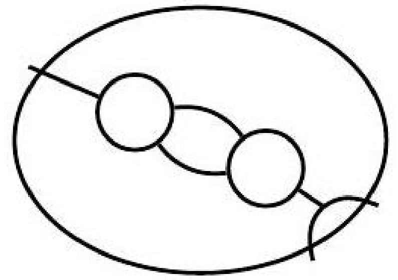
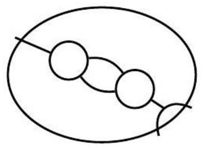
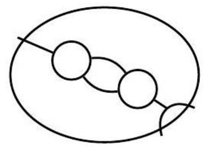
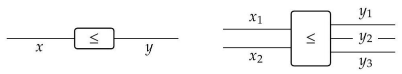
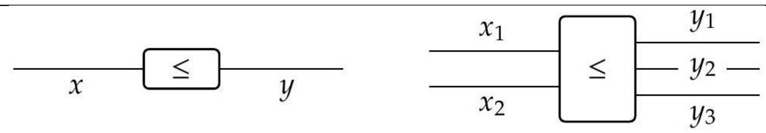
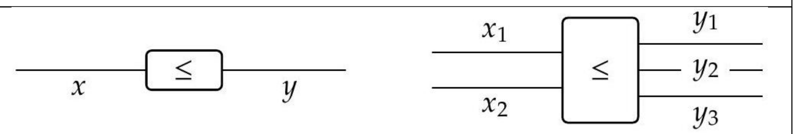
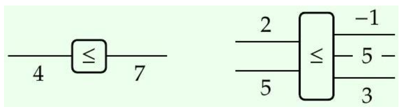
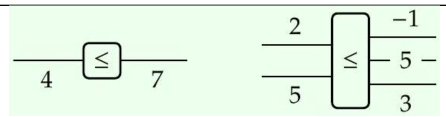
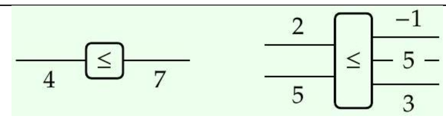
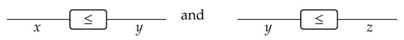
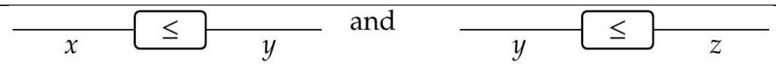
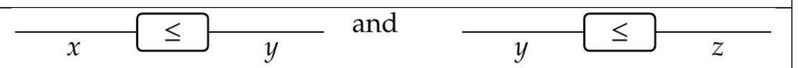
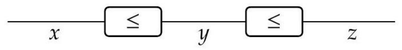
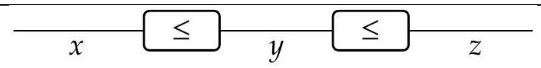
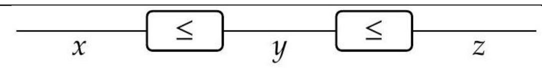
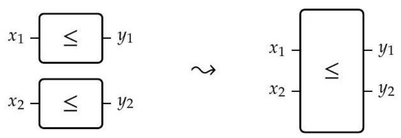
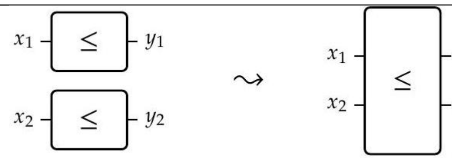
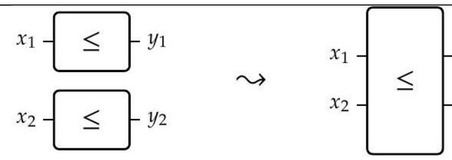
fong-spivak-seven-sketches_DIA_0033 — *(dup)*	026	7f1e85c6-4350-401f-a960-eb8a0176e786-026_244_471_919_824.jpg			
fong-spivak-seven-sketches_DIA_0034 — *(dup)*	027	7f1e85c6-4350-401f-a960-eb8a0176e786-027_59_629_1717_744.jpg			
fong-spivak-seven-sketches_DIA_0035 — *(dup)*	028	7f1e85c6-4350-401f-a960-eb8a0176e786-028_267_428_1054_550.jpg			
fong-spivak-seven-sketches_DIA_0036 — *(dup)*	028	7f1e85c6-4350-401f-a960-eb8a0176e786-028_267_428_1054_1140.jpg			
fong-spivak-seven-sketches_DIA_0037 — *(dup)*	028	7f1e85c6-4350-401f-a960-eb8a0176e786-028_346_536_1744_792.jpg			
fong-spivak-seven-sketches_DIA_0038 — *(dup)*	029	*no image for this region (498 in zip)*	*(no tex.zip crop for this row)*	*(image not on disk — pass –texzip or download the CDN crop)*	*(image not on disk — pass –texzip or download the CDN crop)*
fong-spivak-seven-sketches_DIA_0039 — *(dup)*	030	7f1e85c6-4350-401f-a960-eb8a0176e786-030_253_396_287_862.jpg			
fong-spivak-seven-sketches_DIA_0040 — *(dup)*	030	7f1e85c6-4350-401f-a960-eb8a0176e786-030_396_846_1989_636.jpg			
fong-spivak-seven-sketches_DIA_0041 — *(dup)*	031	7f1e85c6-4350-401f-a960-eb8a0176e786-031_167_1063_480_527.jpg			

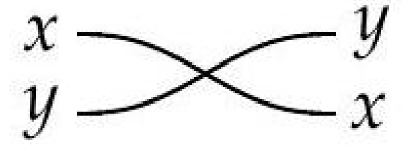
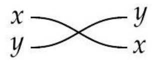
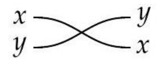
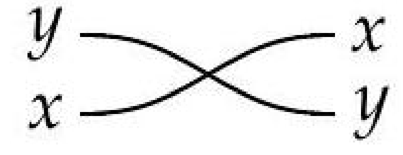
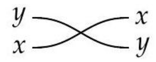
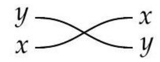
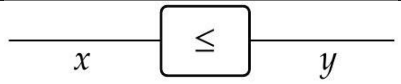
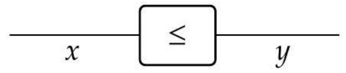
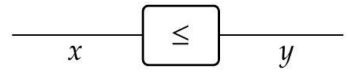
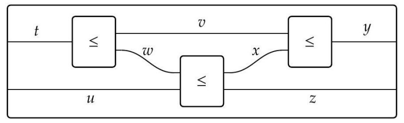
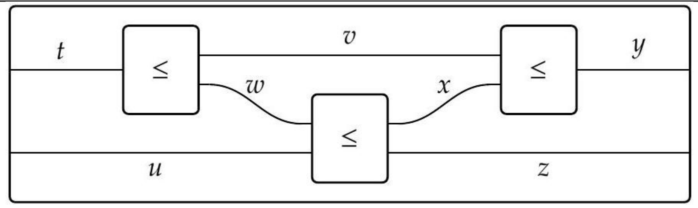
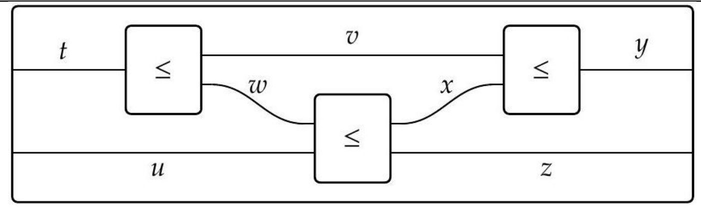
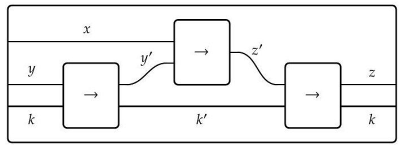
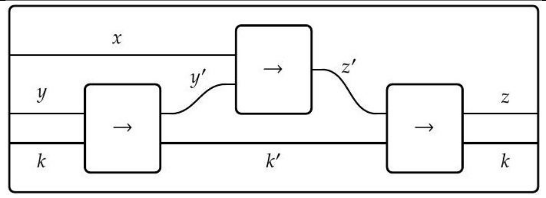
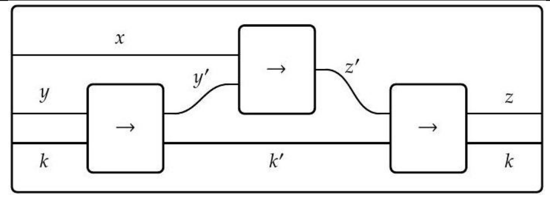
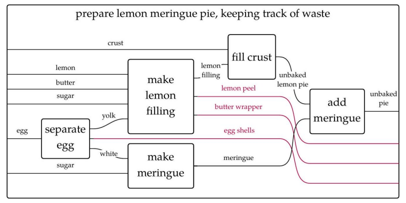
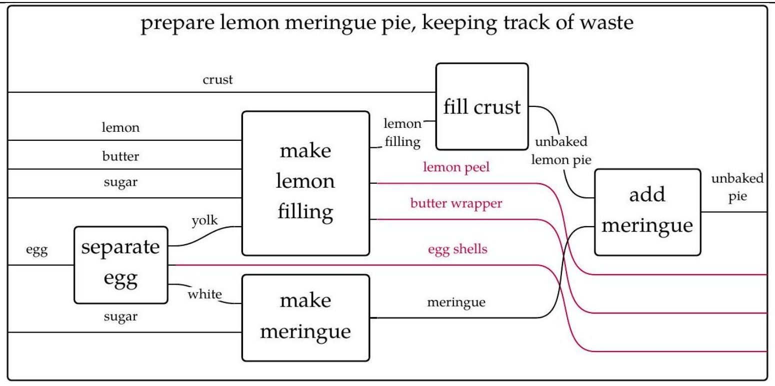
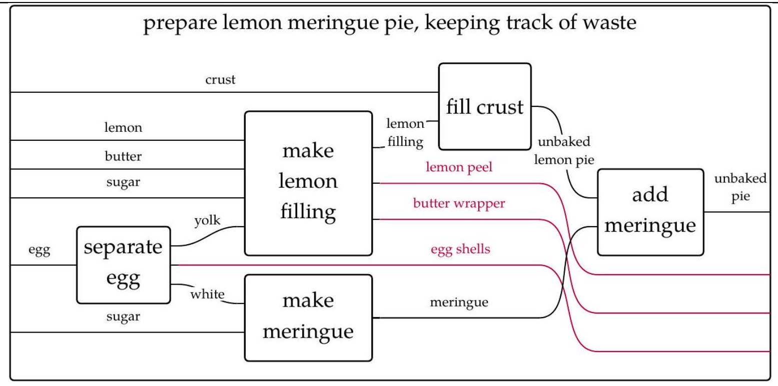
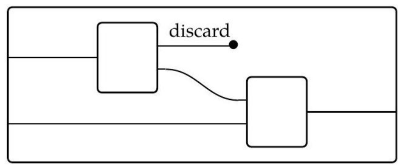
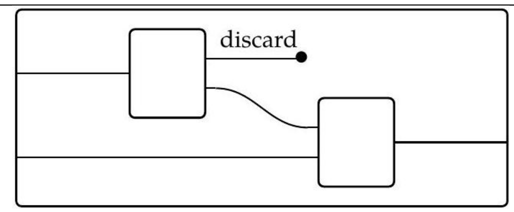
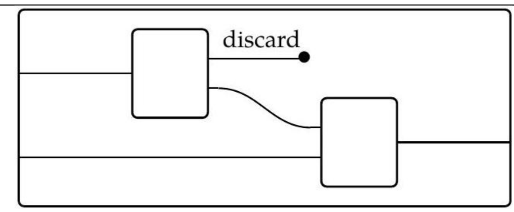
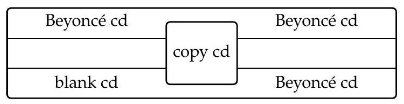
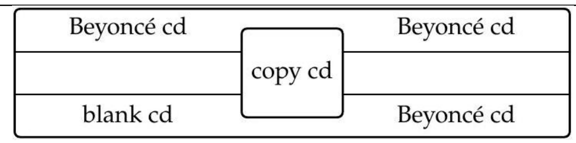
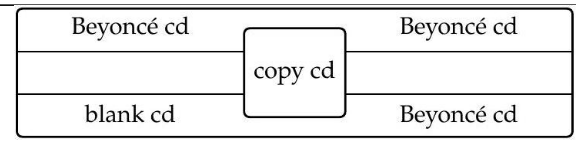
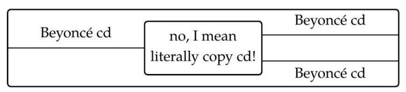
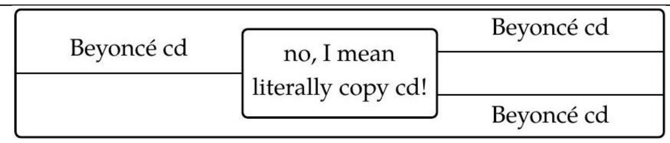
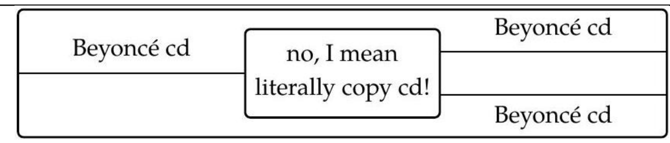
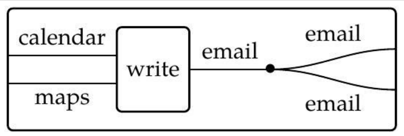
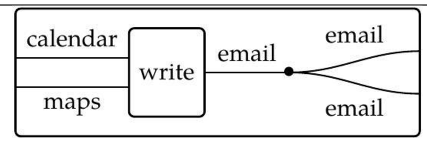
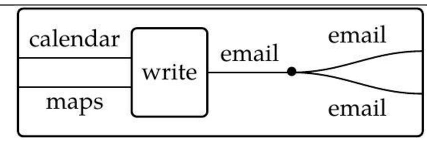
fong-spivak-seven-sketches_031_0042 — *(dup)*	031		7f1e85c6-4350-401f-a960-eb8a0176e786-031_713_1326_1052_407.jpg			
fong-spivak-seven-sketches_034_0043 — *(dup)*	034		7f1e85c6-4350-401f-a960-eb8a0176e786-034_387_373_414_387.jpg			
fong-spivak-seven-sketches_034_0044 — *(dup)*	034		7f1e85c6-4350-401f-a960-eb8a0176e786-034_389_324_414_923.jpg			
fong-spivak-seven-sketches_034_0045 — *(dup)*	034		7f1e85c6-4350-401f-a960-eb8a0176e786-034_387_317_414_1412.jpg			

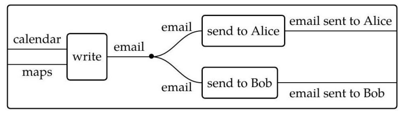
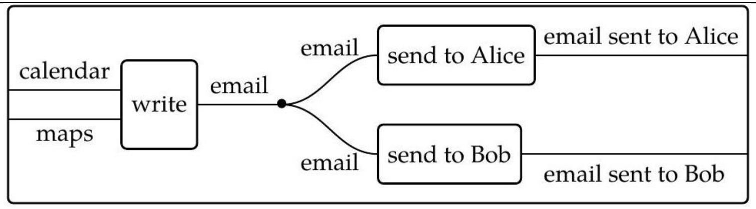
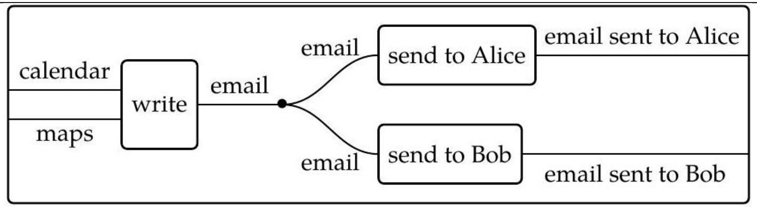
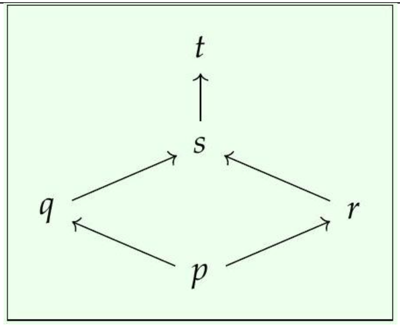
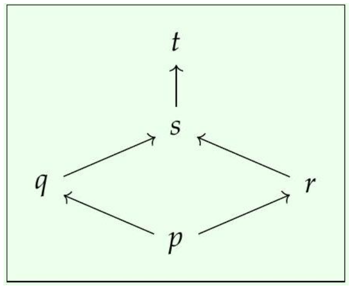
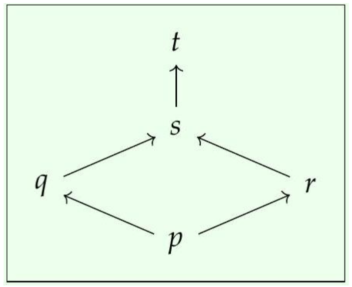
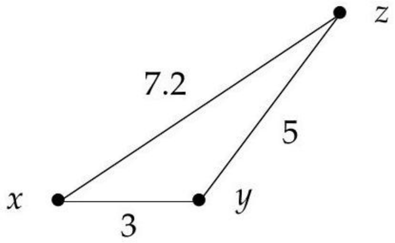
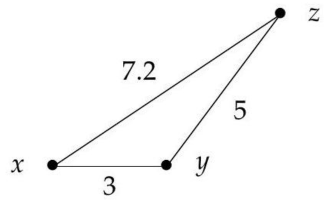
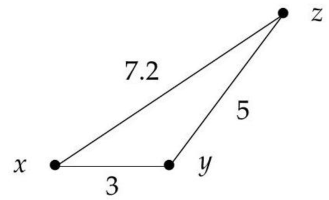
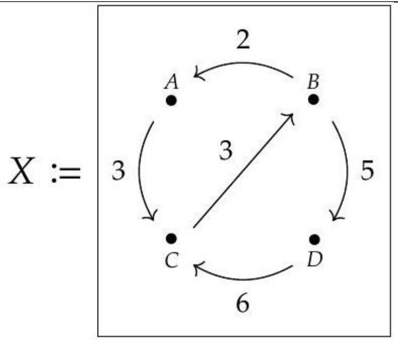
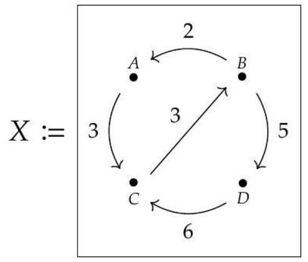
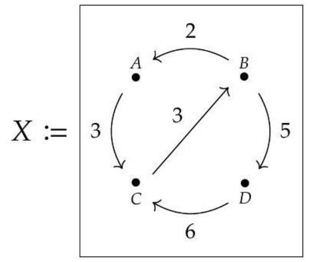
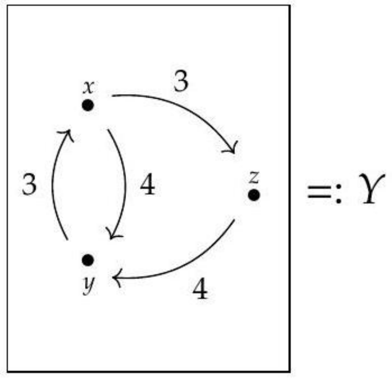
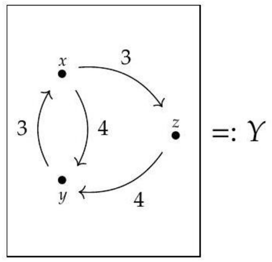
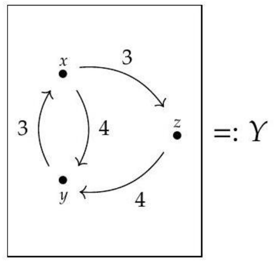
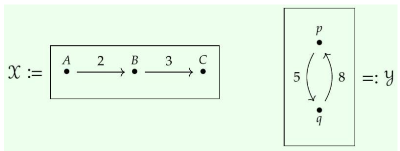
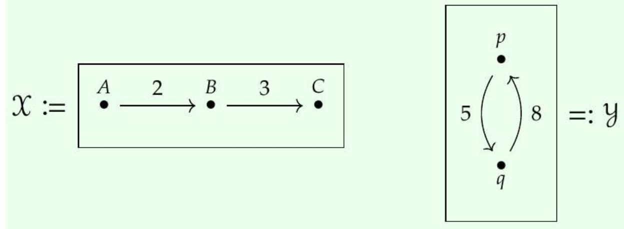
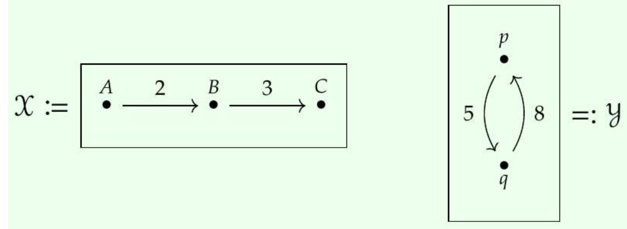
fong-spivak-seven-sketches_DIA_0046	036	— (dup)	7f1e85c6-4350-401f-a960-eb8a0176e786-036_253_242_1597_939.jpg			
fong-spivak-seven-sketches_DIA_0047	037	— (dup)	7f1e85c6-4350-401f-a960-eb8a0176e786-037_247_296_715_912.jpg			
fong-spivak-seven-sketches_DIA_0048	039	— (dup)	7f1e85c6-4350-401f-a960-eb8a0176e786-039_204_654_2158_778.jpg			
fong-spivak-seven-sketches_DIA_0049	040	— (dup)	7f1e85c6-4350-401f-a960-eb8a0176e786-040_213_656_348_778.jpg			
fong-spivak-seven-sketches_DIA_0050	040	— (dup)	7f1e85c6-4350-401f-a960-eb8a0176e786-040_217_285_1860_624.jpg			
fong-spivak-seven-sketches_DIA_0051	040	— (dup)	7f1e85c6-4350-401f-a960-eb8a0176e786-040_222_312_1860_1181.jpg			

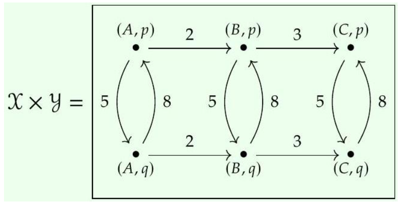
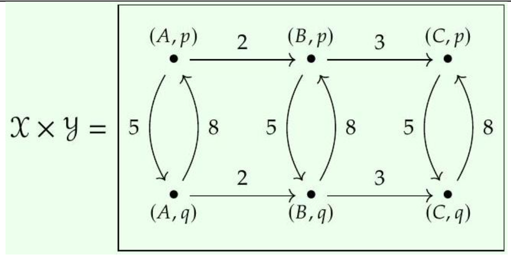
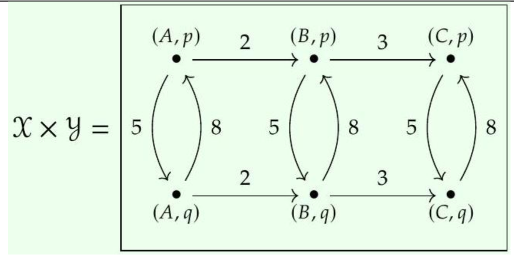
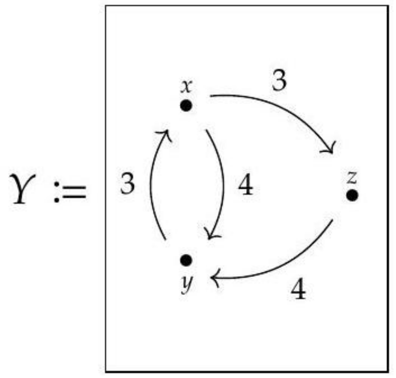
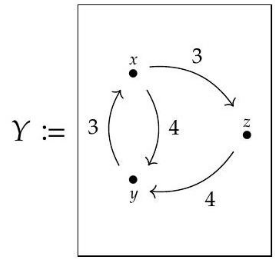
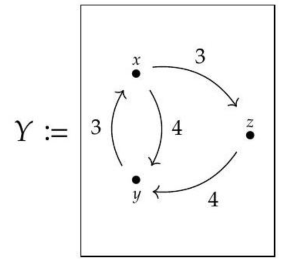
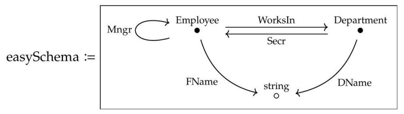
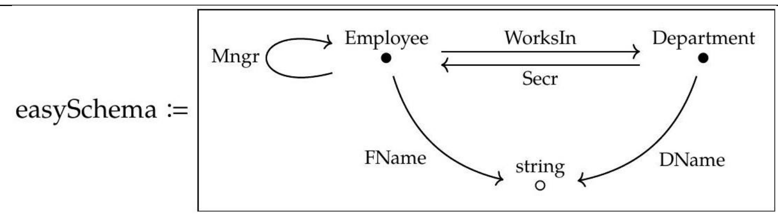
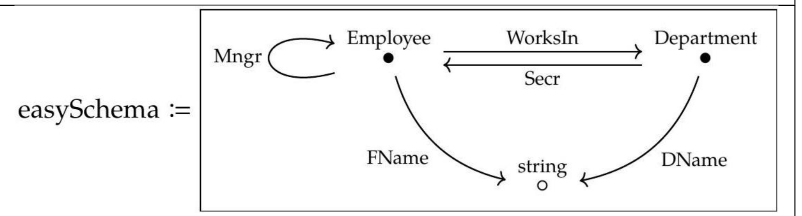
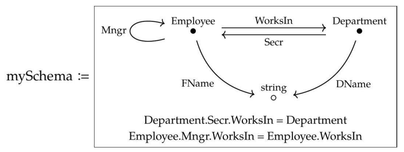
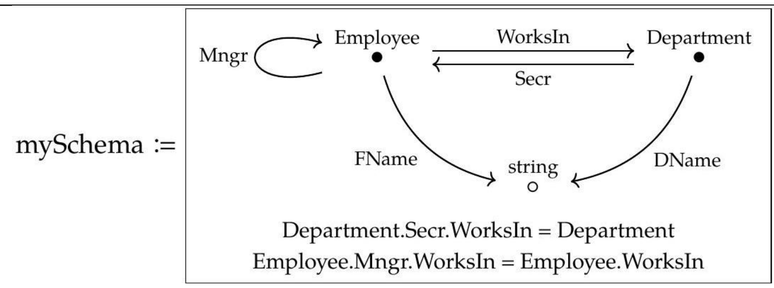
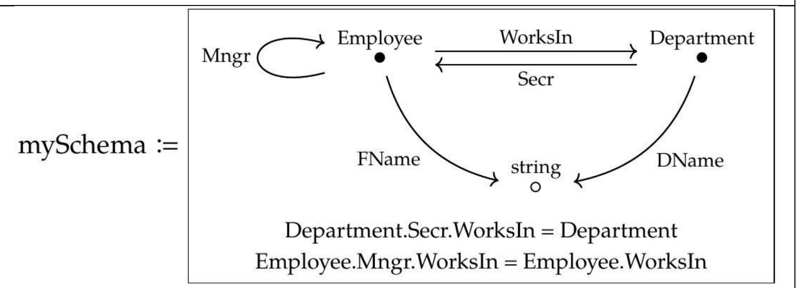
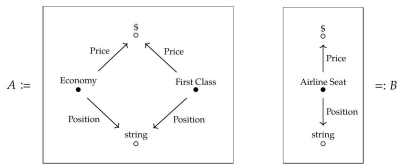
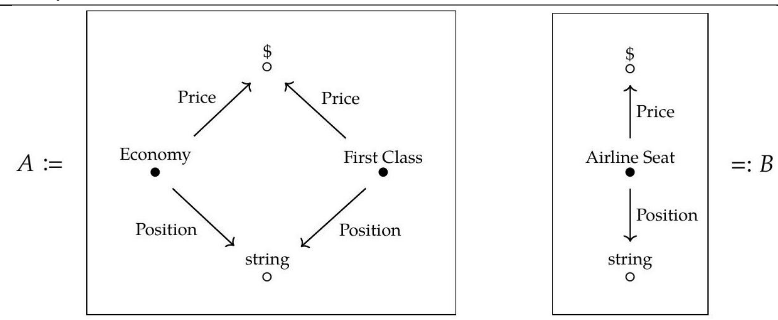
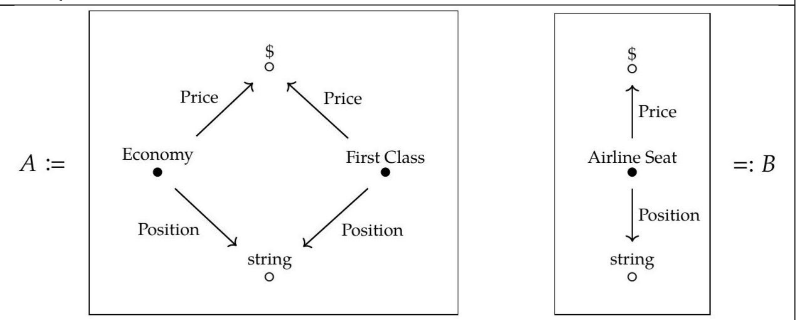
fong-spivak-seven-sketches_041_0052	041	— (dup)	7f1e85c6-4350-401f-a960-eb8a0176e786-041_339_919_828_600.jpg			
fong-spivak-seven-sketches_041_0053	041	— (dup)	7f1e85c6-4350-401f-a960-eb8a0176e786-041_319_1419_1600_351.jpg			
fong-spivak-seven-sketches_043_0054	043	— (dup)	7f1e85c6-4350-401f-a960-eb8a0176e786-043_348_896_1887_611.jpg			
fong-spivak-seven-sketches_047_0055	047	— (dup)	7f1e85c6-4350-401f-a960-eb8a0176e786-047_61_233_1466_941.jpg			
fong-spivak-seven-sketches_052_0056	052	— (dup)	7f1e85c6-4350-401f-a960-eb8a0176e786-052_671_1354_566_381.jpg			
fong-spivak-seven-sketches_052_0057	052	— (dup)	7f1e85c6-4350-401f-a960-eb8a0176e786-052_293_1228_1507_444.jpg			
fong-spivak-seven-sketches_055_0058	055	— (dup)	7f1e85c6-4350-401f-a960-eb8a0176e786-055_90_143_2072_989.jpg			

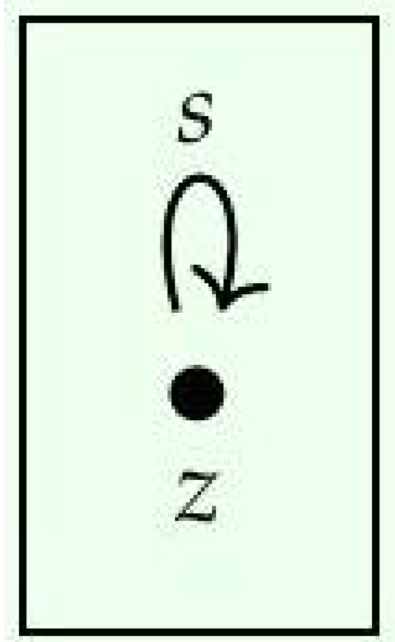
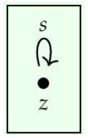
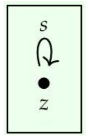
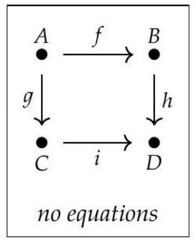
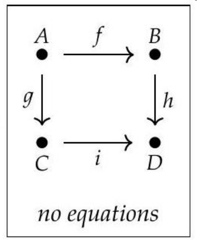
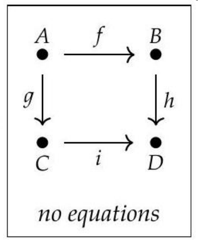
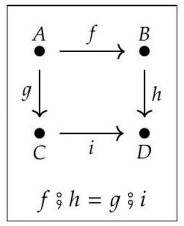
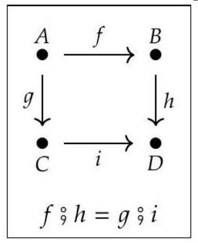
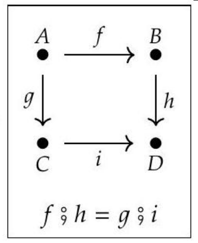
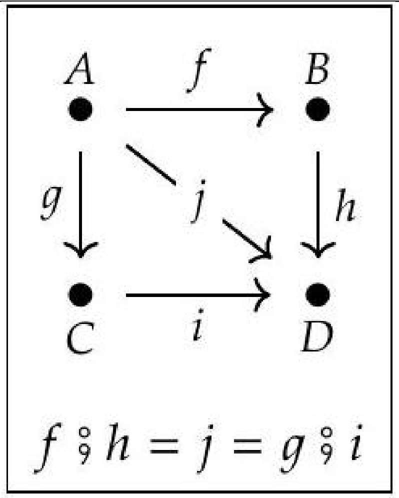
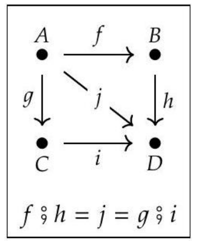
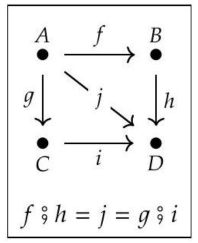
fong-spivak-seven-sketches_DIA_0059	055	— (dup)	7f1e85c6-4350-401f-a960-eb8a0176e786-055_86_93_2364_394.jpg			
fong-spivak-seven-sketches_DIA_0060	055	— (dup)	7f1e85c6-4350-401f-a960-eb8a0176e786-055_90_140_2364_593.jpg			
fong-spivak-seven-sketches_DIA_0061	055	— (dup)	7f1e85c6-4350-401f-a960-eb8a0176e786-055_90_253_2364_842.jpg			
fong-spivak-seven-sketches_DIA_0062	055	— (dup)	7f1e85c6-4350-401f-a960-eb8a0176e786-055_90_369_2364_1201.jpg			
fong-spivak-seven-sketches_DIA_0063	056	— (dup)	7f1e85c6-4350-401f-a960-eb8a0176e786-056_179_609_394_756.jpg			
fong-spivak-seven-sketches_DIA_0064	056	— (dup)	7f1e85c6-4350-401f-a960-eb8a0176e786-056_100_229_1041_394.jpg			
fong-spivak-seven-sketches_DIA_0065	056	— (dup)	7f1e85c6-4350-401f-a960-eb8a0176e786-056_124_226_1029_729.jpg			
fong-spivak-seven-sketches_DIA_0066	056	— (dup)	7f1e85c6-4350-401f-a960-eb8a0176e786-056_186_231_1000_1086.jpg			

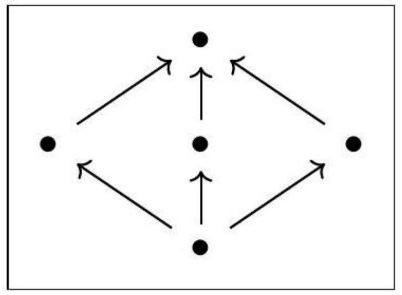
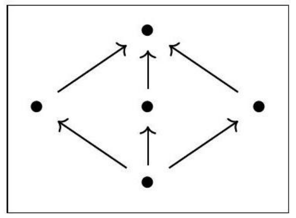
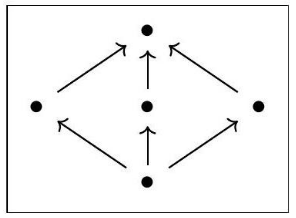
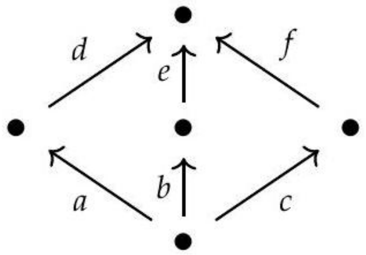
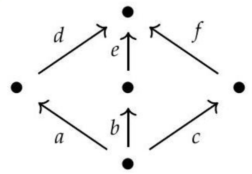
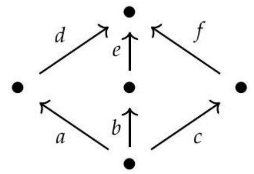
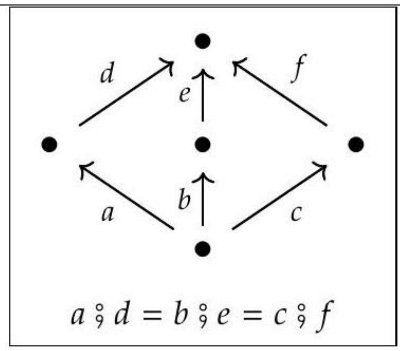
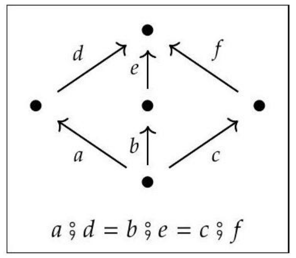
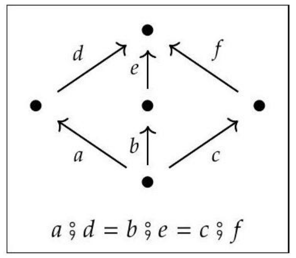
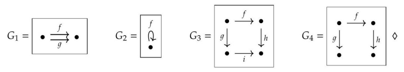
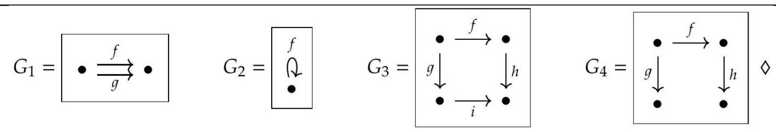
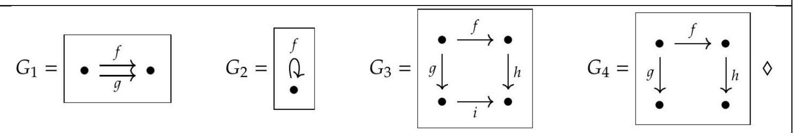
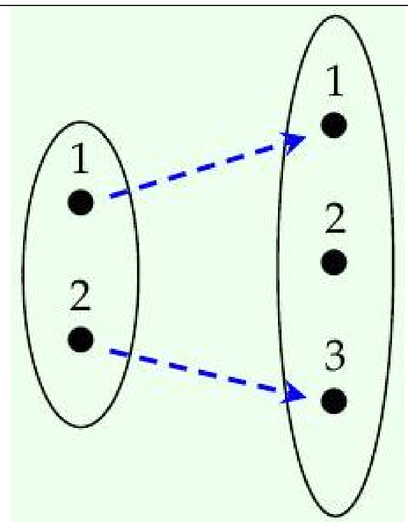
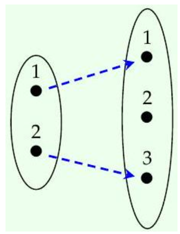
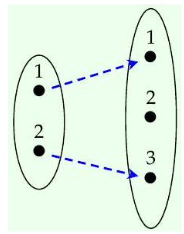
fong-spivak-seven-sketches_DIA_0067	056	—	(dup)	7f1e85c6-4350-401f-a960-eb8a0176e786-056_188_260_1002_1462.jpg			
fong-spivak-seven-sketches_DIA_0068	056	—	(dup)	7f1e85c6-4350-401f-a960-eb8a0176e786-056_120_109_2222_853.jpg	$\frac{x}{}$ $\frac{}{y}$	$\frac{x}{}$ $\frac{}{y}$	$\frac{x}{}$ $\frac{}{y}$
fong-spivak-seven-sketches_DIA_0069	056	—	(dup)	7f1e85c6-4350-401f-a960-eb8a0176e786-056_61_104_2272_1149.jpg	$\overline{x \otimes y}$	$\overline{x \otimes y}$	$\overline{x \otimes y}$
fong-spivak-seven-sketches_DIA_0070	057	—	(dup)	7f1e85c6-4350-401f-a960-eb8a0176e786-057_72_464_299_821.jpg	$\overline{I}$ *nothing*	$\overline{I}$ *nothing*	$\overline{I}$ *nothing*
fong-spivak-seven-sketches_DIA_0071	057	—	(dup)	7f1e85c6-4350-401f-a960-eb8a0176e786-057_163_984_814_593.jpg			
fong-spivak-seven-sketches_DIA_0072	057	—	(dup)	7f1e85c6-4350-401f-a960-eb8a0176e786-057_127_1330_1471_389.jpg	$\overline{3.14}$ $\overline{-1}$ $\left.\begin{array}{c}\overline{3.14}\\ \overline{-1}\end{array}\right\} = \overline{2.14}$ $\overline{0} =$ *nothing*	$\overline{3.14}$ $\overline{-1}$ $\left.\begin{array}{c}\overline{3.14}\\ \overline{-1}\end{array}\right\} = \overline{2.14}$ $\overline{0} =$ *nothing*	$\overline{3.14}$ $\overline{-1}$ $\left.\begin{array}{c}\overline{3.14}\\ \overline{-1}\end{array}\right\} = \overline{2.14}$ $\overline{0} =$ *nothing*
fong-spivak-seven-sketches_DIA_0073	057	—	(dup)	7f1e85c6-4350-401f-a960-eb8a0176e786-057_143_561_1762_792.jpg			
fong-spivak-seven-sketches_DIA_0074	057	—	(dup)	7f1e85c6-4350-401f-a960-eb8a0176e786-057_41_305_2204_907.jpg	$\frac{}{x}$	$\frac{}{x}$	$\frac{}{x}$
fong-spivak-seven-sketches_DIA_0075	057	—	(dup)	7f1e85c6-4350-401f-a960-eb8a0176e786-057_81_1061_2430_554.jpg			
fong-spivak-seven-sketches_DIA_0076	058	—	(dup)	7f1e85c6-4350-401f-a960-eb8a0176e786-058_81_688_346_740.jpg			
fong-spivak-seven-sketches_DIA_0077	058	—	(dup)	7f1e85c6-4350-401f-a960-eb8a0176e786-058_199_615_667_751.jpg			
fong-spivak-seven-sketches_DIA_0078	058	—	(dup)	7f1e85c6-4350-401f-a960-eb8a0176e786-058_113_403_1050_857.jpg			

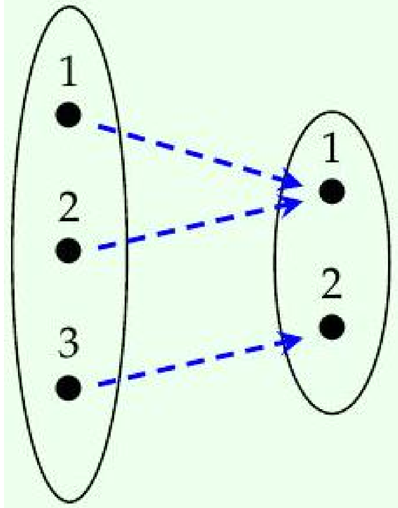
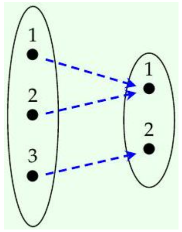
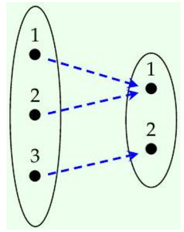
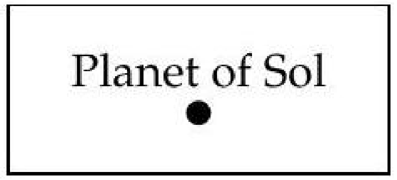
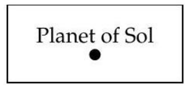
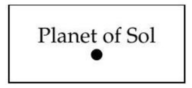
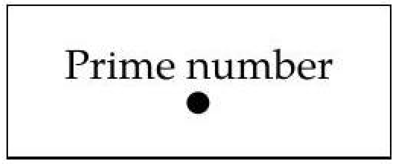
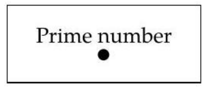
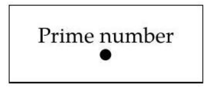
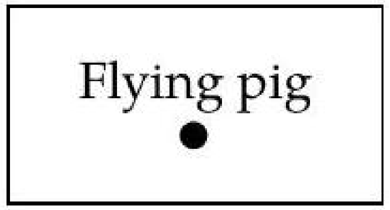
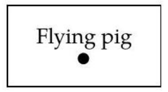
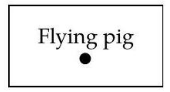
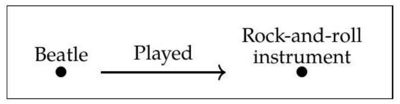
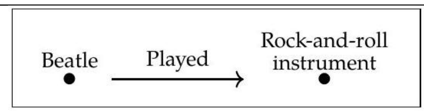
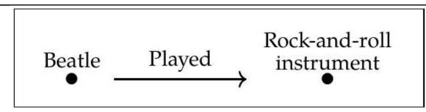
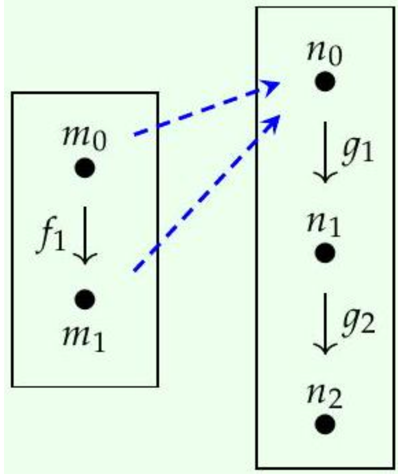
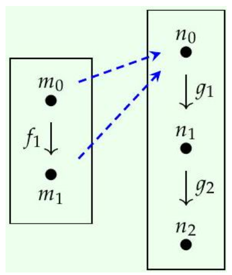
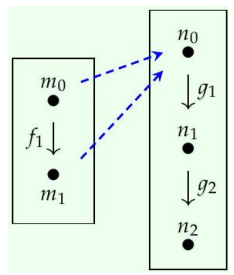
fong-spivak-seven-sketches_[illegible]_0079	058	— (dup)	7f1e85c6-4350-401f-a960-eb8a0176e786-058_75_192_2258_767.jpg			
fong-spivak-seven-sketches_[illegible]_0080	058	— (dup)	7f1e85c6-4350-401f-a960-eb8a0176e786-058_75_195_2258_1158.jpg			
fong-spivak-seven-sketches_[illegible]_0081	059	— (dup)	7f1e85c6-4350-401f-a960-eb8a0176e786-059_93_439_439_864.jpg			
fong-spivak-seven-sketches_[illegible]_0082	059	— (dup)	7f1e85c6-4350-401f-a960-eb8a0176e786-059_265_896_767_611.jpg			
fong-spivak-seven-sketches_[illegible]_0083	061	— (dup)	7f1e85c6-4350-401f-a960-eb8a0176e786-061_253_699_2002_710.jpg			
fong-spivak-seven-sketches_[illegible]_0084	062	— (dup)	7f1e85c6-4350-401f-a960-eb8a0176e786-062_671_1351_819_381.jpg			
fong-spivak-seven-sketches_[illegible]_0085	062	— (dup)	7f1e85c6-4350-401f-a960-eb8a0176e786-062_264_648_1723_734.jpg			
fong-spivak-seven-sketches_[illegible]_0086	063	— (dup)	7f1e85c6-4350-401f-a960-eb8a0176e786-063_174_731_1249_695.jpg			
fong-spivak-seven-sketches_[illegible]_0087	063	— (dup)	7f1e85c6-4350-401f-a960-eb8a0176e786-063_174_851_1588_633.jpg			
fong-spivak-seven-sketches_[illegible]_0088	063	— (dup)	7f1e85c6-4350-401f-a960-eb8a0176e786-063_172_534_2328_792.jpg			

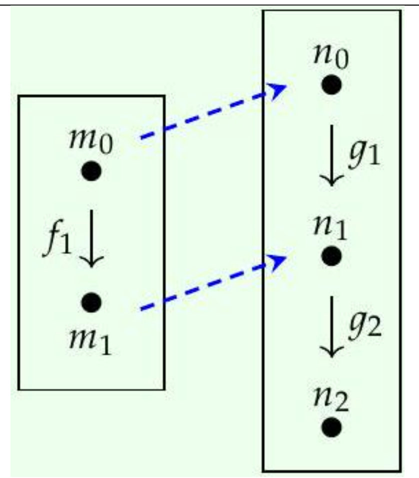
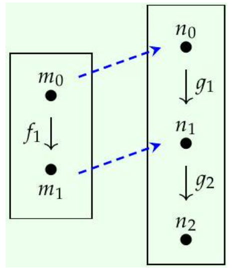
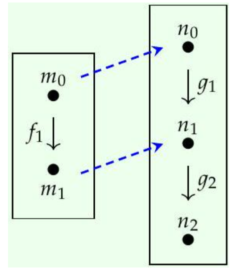
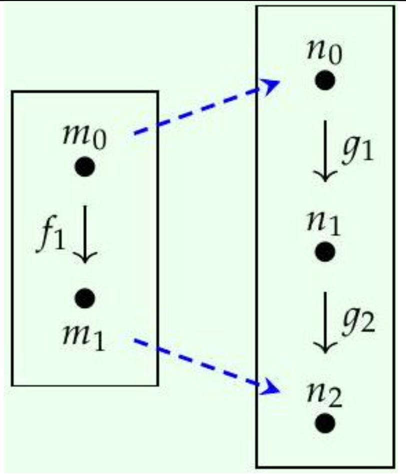
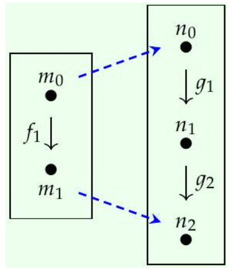
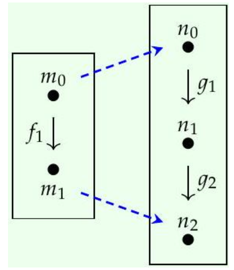
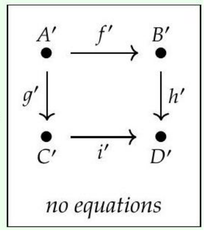
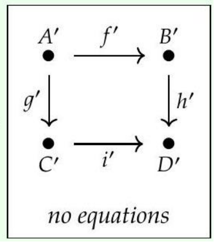
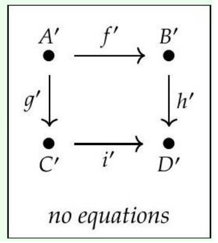
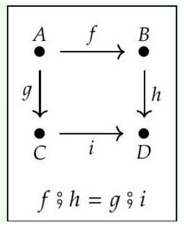
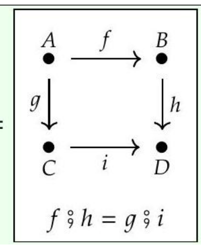
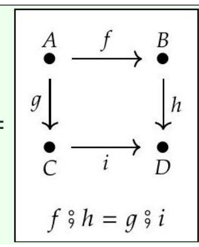
fong-spivak-seven-sketches_DIA_0089	064	—	*(dup)*	7f1e85c6-4350-401f-a960-eb8a0176e786-064_265_973_342_572.jpg			
fong-spivak-seven-sketches_DIA_0090	070	—	*(dup)*	7f1e85c6-4350-401f-a960-eb8a0176e786-070_367_450_335_835.jpg			
fong-spivak-seven-sketches_DIA_0091	072	—	*(dup)*	7f1e85c6-4350-401f-a960-eb8a0176e786-072_256_414_878_851.jpg			
fong-spivak-seven-sketches_DIA_0092	074	—	*(dup)*	7f1e85c6-4350-401f-a960-eb8a0176e786-074_335_391_799_550.jpg			
fong-spivak-seven-sketches_DIA_0093	074	—	*(dup)*	7f1e85c6-4350-401f-a960-eb8a0176e786-074_337_351_799_1217.jpg			
fong-spivak-seven-sketches_DIA_0094	079	—	*(dup)*	7f1e85c6-4350-401f-a960-eb8a0176e786-079_299_805_1534_656.jpg			

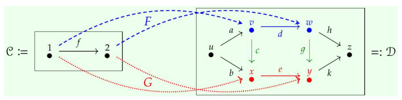
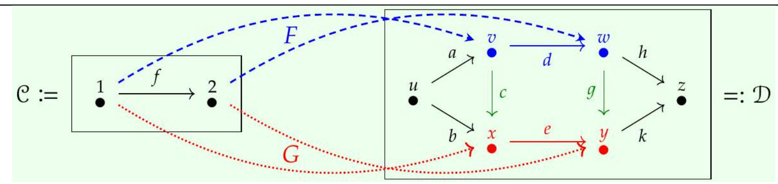
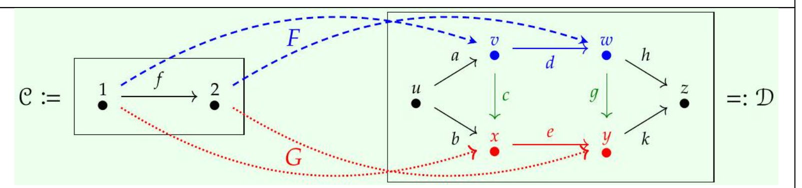
fong-spivak-seven-sketches_…_0095 — *(dup)*	080		7f1e85c6-4350-401f-a960-eb8a0176e786-080_330_654_333_738.jpg			
fong-spivak-seven-sketches_…_0096 — *(dup)*	086		7f1e85c6-4350-401f-a960-eb8a0176e786-086_339_351_1982_495.jpg			
fong-spivak-seven-sketches_…_0097 — *(dup)*	090		7f1e85c6-4350-401f-a960-eb8a0176e786-090_299_1097_1321_509.jpg			
fong-spivak-seven-sketches_…_0098 — *(dup)*	091		7f1e85c6-4350-401f-a960-eb8a0176e786-091_395_1079_559_520.jpg			
fong-spivak-seven-sketches_…_0099 — *(dup)*	091		*no image for this region (498 in zip)*	*(no tex.zip crop for this row)*	*(image not on disk — pass –texzip or download the CDN crop)*	*(image not on disk — pass –texzip or download the CDN crop)*
fong-spivak-seven-sketches_…_0100 — *(dup)*	092		7f1e85c6-4350-401f-a960-eb8a0176e786-092_500_1217_972_412.jpg			

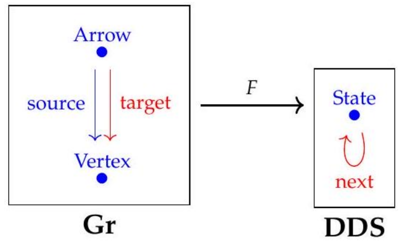
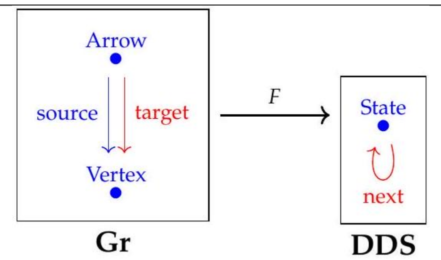
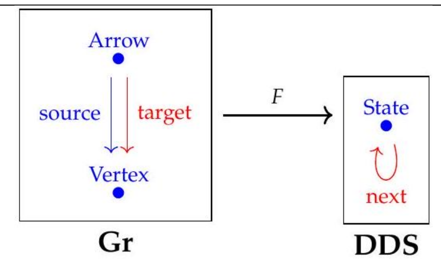
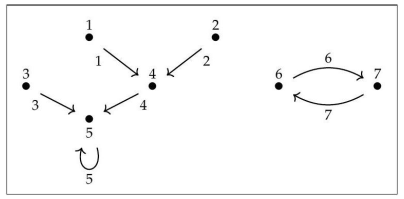
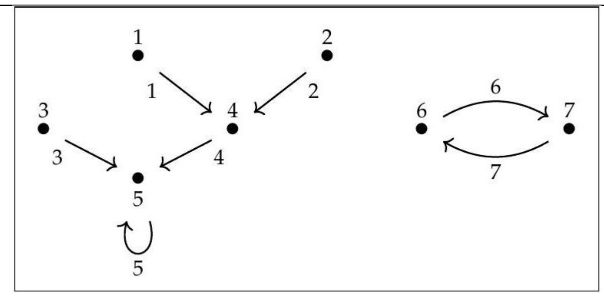
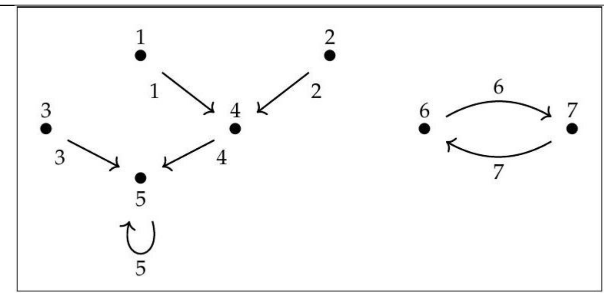
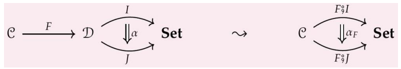
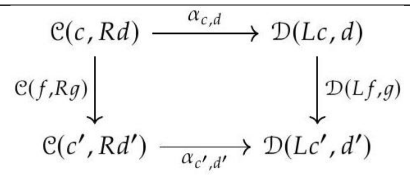
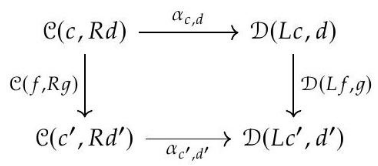
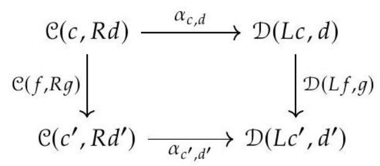
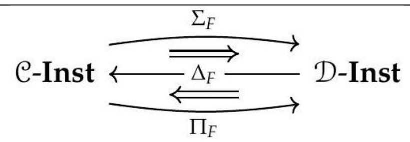
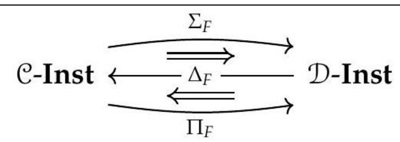
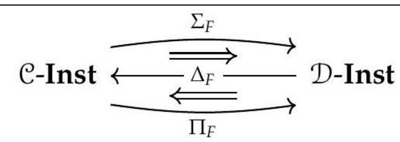
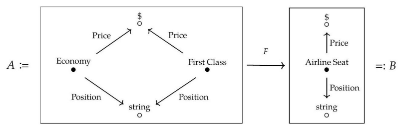
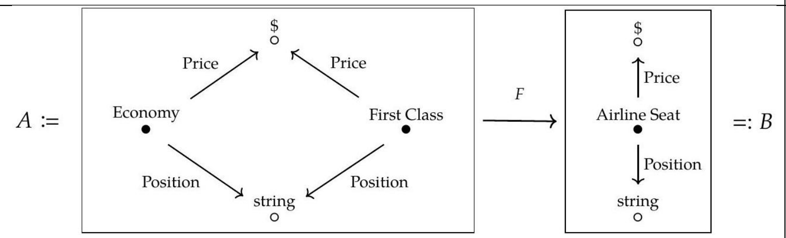
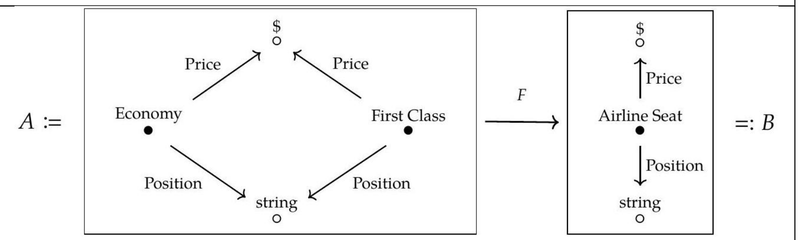
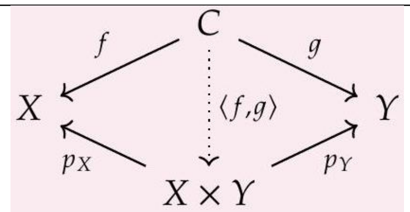
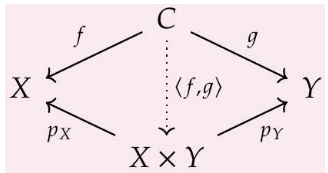
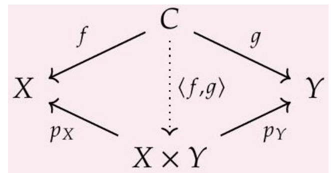
fong-spivak-seven-sketches_DIA_0101	095	— (dup)	7f1e85c6-4350-401f-a960-eb8a0176e786-095_174_106_1269_1007.jpg			
fong-spivak-seven-sketches_DIA_0102	096	— (dup)	7f1e85c6-4350-401f-a960-eb8a0176e786-096_312_514_683_430.jpg	Free_square := 	Free_square := 	Free_square := 
fong-spivak-seven-sketches_DIA_0103	096	— (dup)	7f1e85c6-4350-401f-a960-eb8a0176e786-096_312_552_683_1131.jpg	Comm_square := 	Comm_square := 	Comm_square := 
fong-spivak-seven-sketches_DIA_0104	096	— (dup)	7f1e85c6-4350-401f-a960-eb8a0176e786-096_308_244_1900_937.jpg			

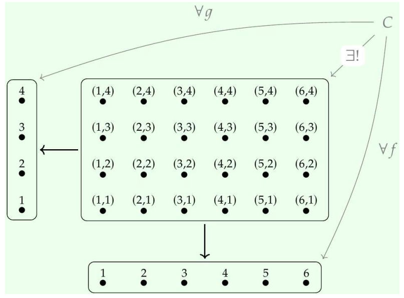
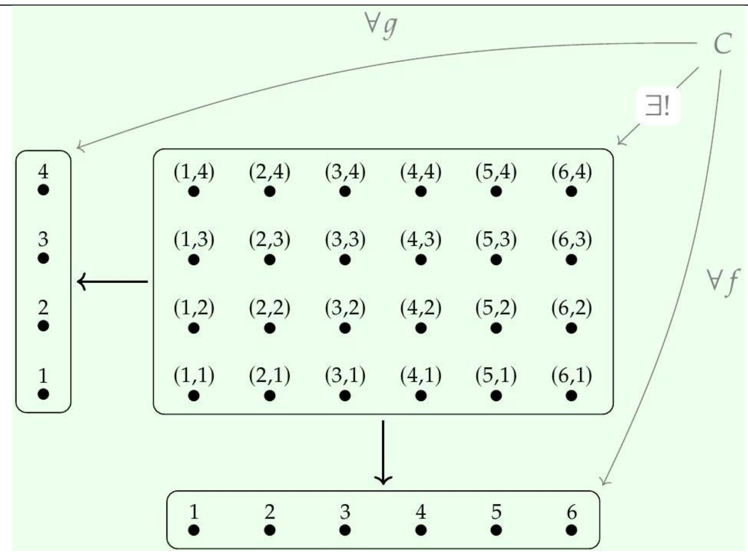
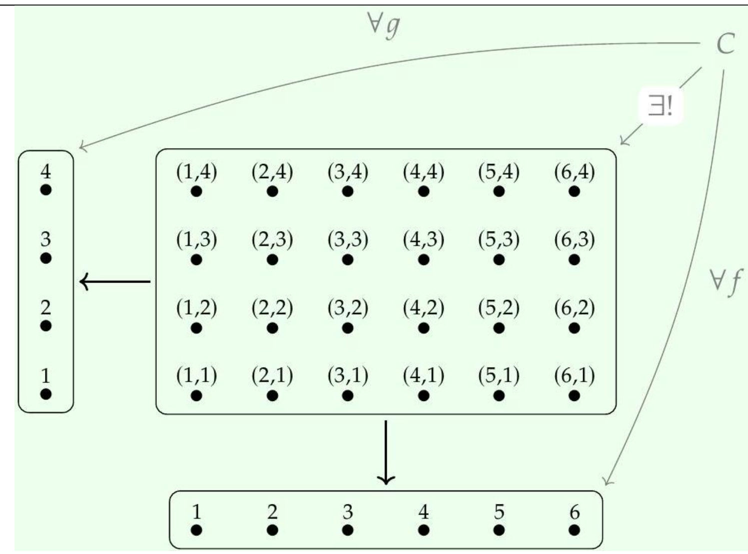
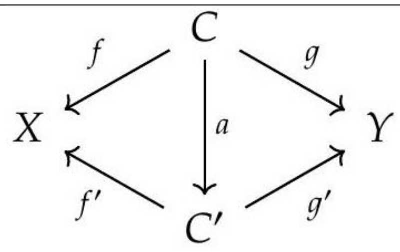
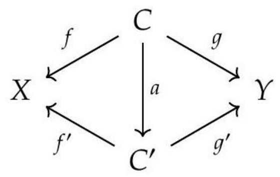
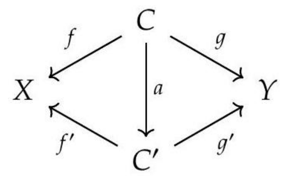
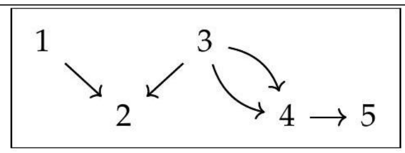
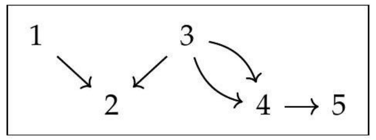
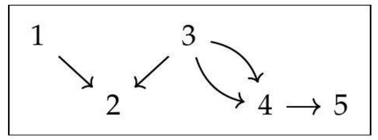
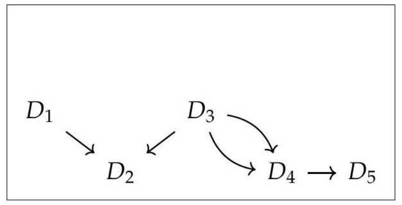
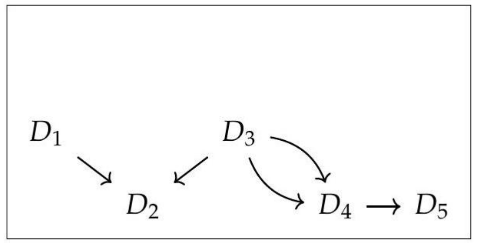
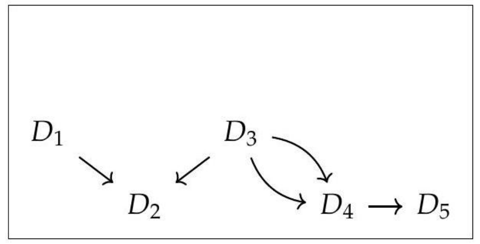
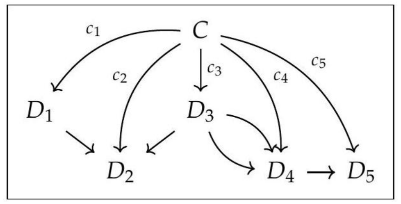
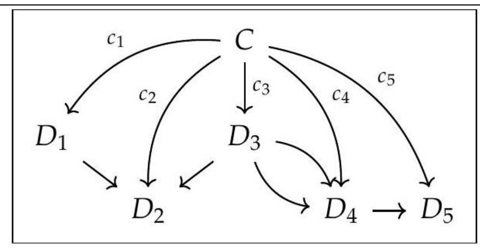
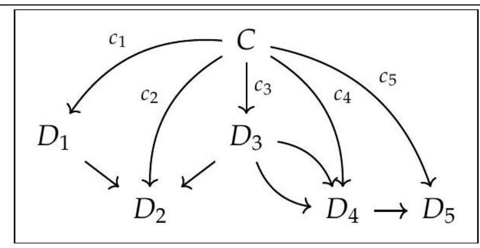
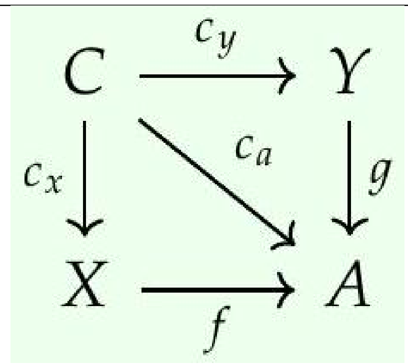
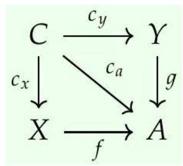
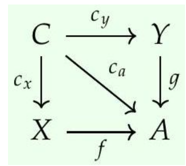
fong-spivak-seven-sketches_DIA_0105	097	— *(dup)*	7f1e85c6-4350-401f-a960-eb8a0176e786-097_278_373_2215_355.jpg			
fong-spivak-seven-sketches_DIA_0106	097	— *(dup)*	7f1e85c6-4350-401f-a960-eb8a0176e786-097_238_342_2199_878.jpg			
fong-spivak-seven-sketches_DIA_0107	097	— *(dup)*	7f1e85c6-4350-401f-a960-eb8a0176e786-097_326_373_2195_1389.jpg			
fong-spivak-seven-sketches_DIA_0108	098	— *(dup)*	7f1e85c6-4350-401f-a960-eb8a0176e786-098_235_1423_733_357.jpg			
fong-spivak-seven-sketches_DIA_0109	101	— *(dup)*	7f1e85c6-4350-401f-a960-eb8a0176e786-101_294_224_742_701.jpg			

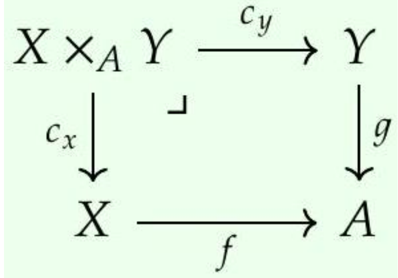
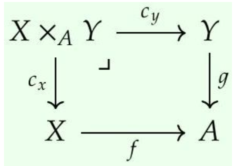
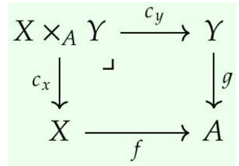
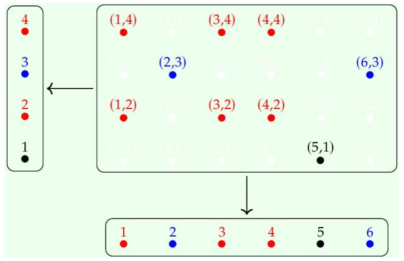
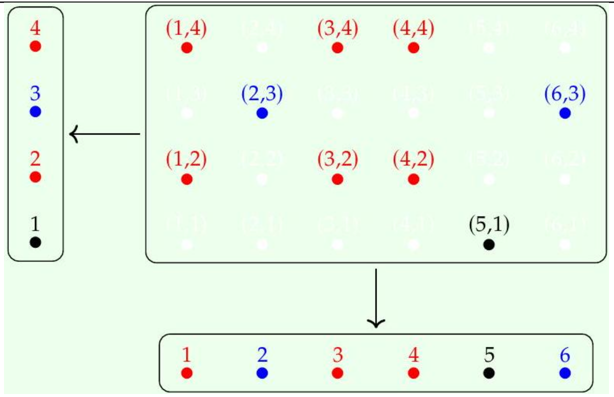
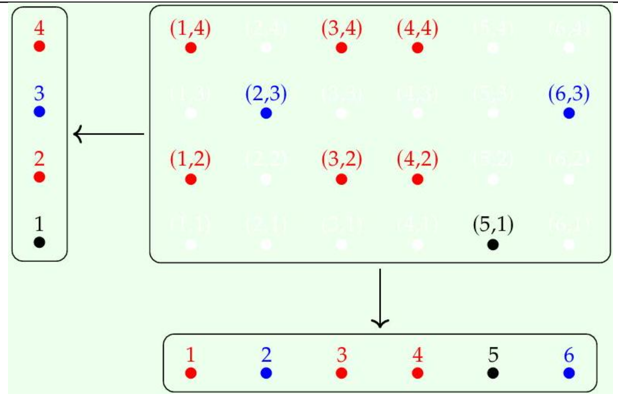
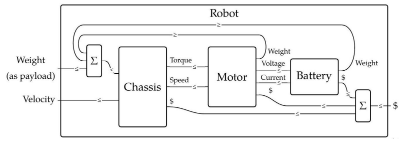
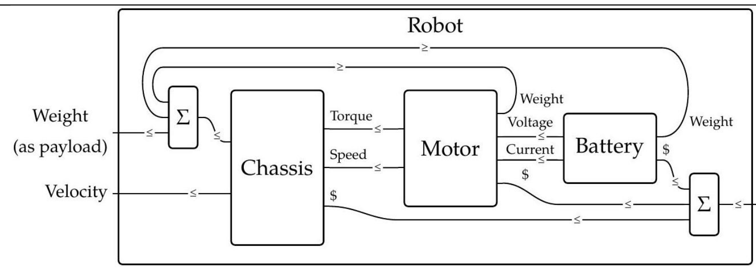
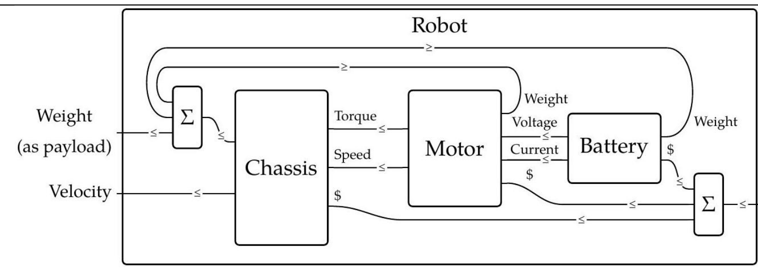
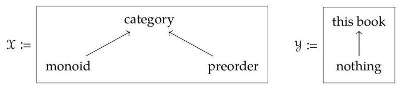
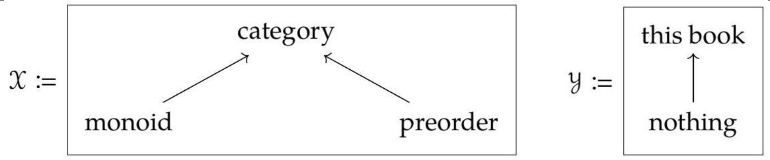
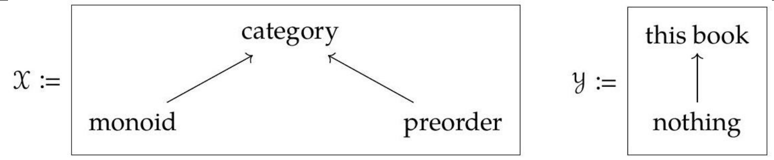
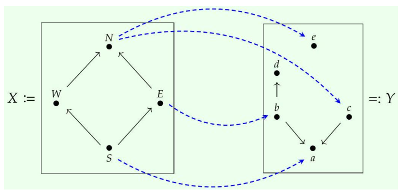
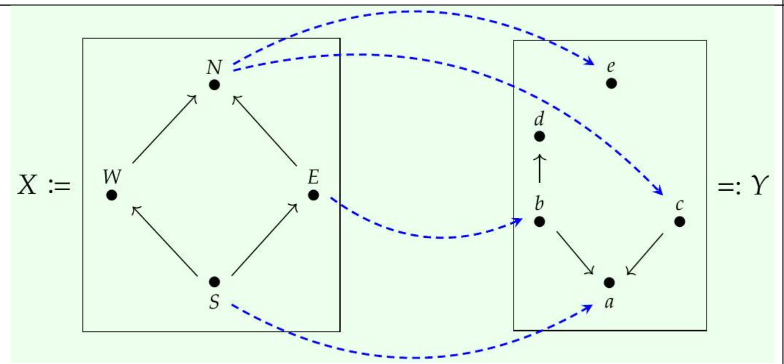
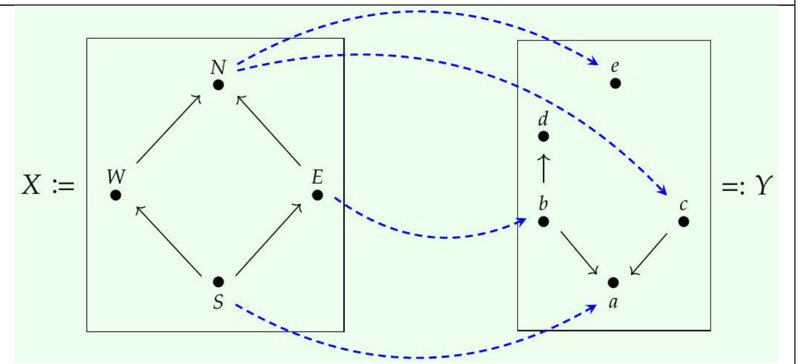
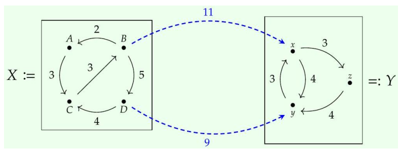
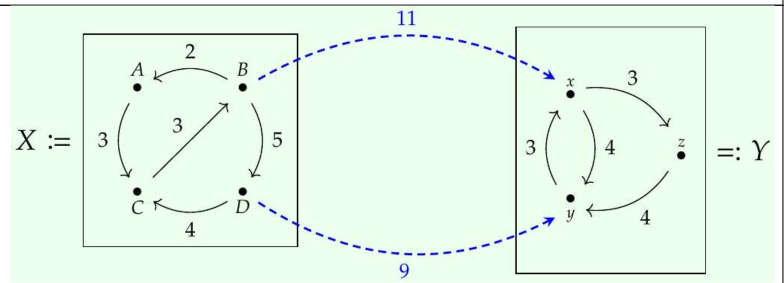
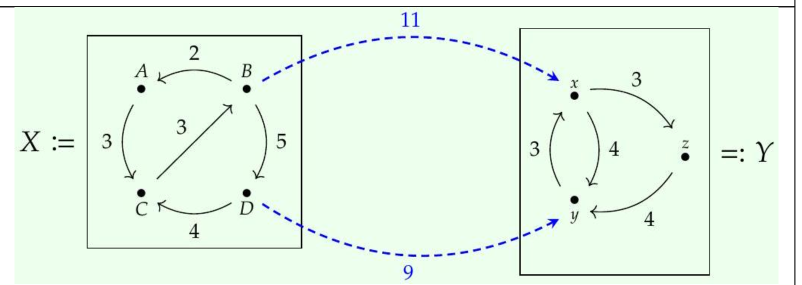
fong-spivak-seven-sketches_DIA_0110	101	— *(dup)*	7f1e85c6-4350-401f-a960-eb8a0176e786-101_294_226_742_1190.jpg			
fong-spivak-seven-sketches_DIA_0111	102	— *(dup)*	7f1e85c6-4350-401f-a960-eb8a0176e786-102_114_236_722_463.jpg			
fong-spivak-seven-sketches_DIA_0112	102	— *(dup)*	7f1e85c6-4350-401f-a960-eb8a0176e786-102_111_259_725_944.jpg			
fong-spivak-seven-sketches_DIA_0113	102	— *(dup)*	7f1e85c6-4350-401f-a960-eb8a0176e786-102_117_210_719_1444.jpg			
fong-spivak-seven-sketches_DIA_0114	102	— *(dup)*	7f1e85c6-4350-401f-a960-eb8a0176e786-102_136_543_2081_788.jpg			
fong-spivak-seven-sketches_DIA_0115	103	— *(dup)*	7f1e85c6-4350-401f-a960-eb8a0176e786-103_346_287_1188_387.jpg			

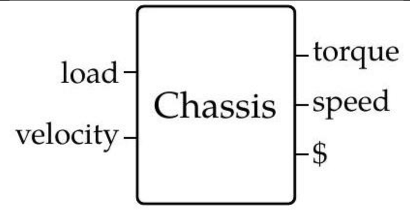
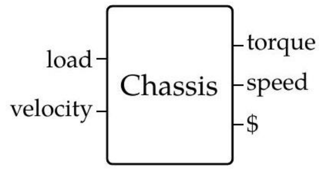
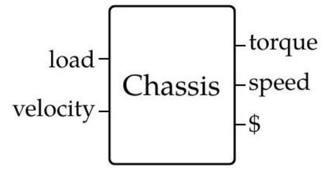
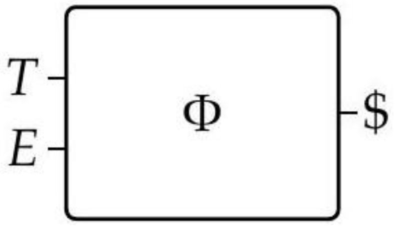
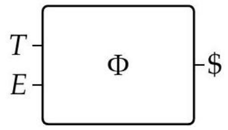
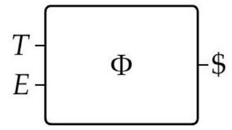
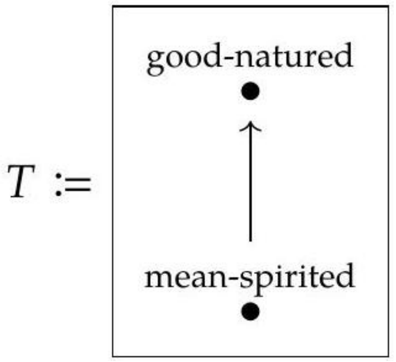
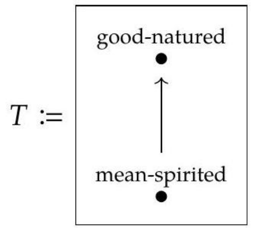
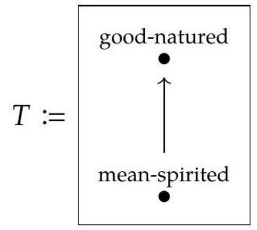
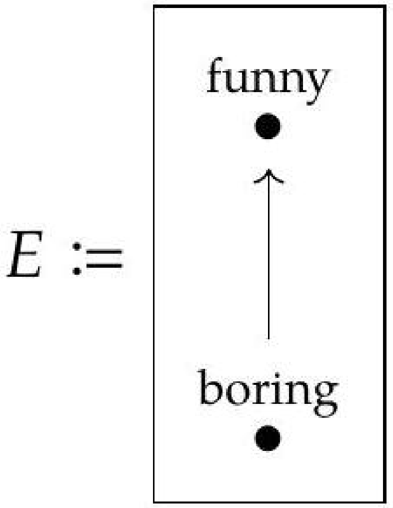
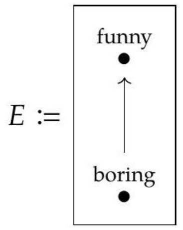
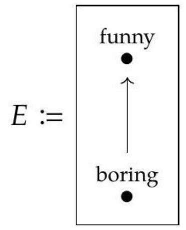
fong-spivak-seven-sketches_0116	103	— *(dup)*	7f1e85c6-4350-401f-a960-eb8a0176e786-103_344_287_1190_912.jpg			
fong-spivak-seven-sketches_0117	103	— *(dup)*	7f1e85c6-4350-401f-a960-eb8a0176e786-103_344_287_1190_1437.jpg			
fong-spivak-seven-sketches_0118	104	— *(dup)*	7f1e85c6-4350-401f-a960-eb8a0176e786-104_314_538_330_466.jpg	Free_square := 	Free_square := 	Free_square := 
fong-spivak-seven-sketches_0119	104	— *(dup)*	7f1e85c6-4350-401f-a960-eb8a0176e786-104_310_554_333_1093.jpg	Comm_square := 	Comm_square := 	Comm_square := 

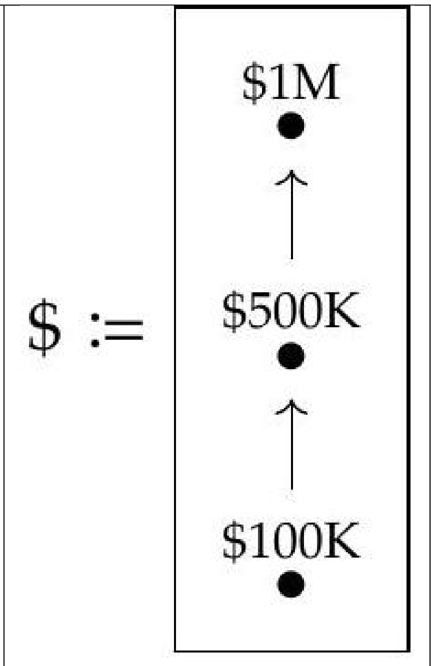
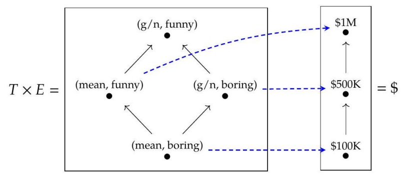
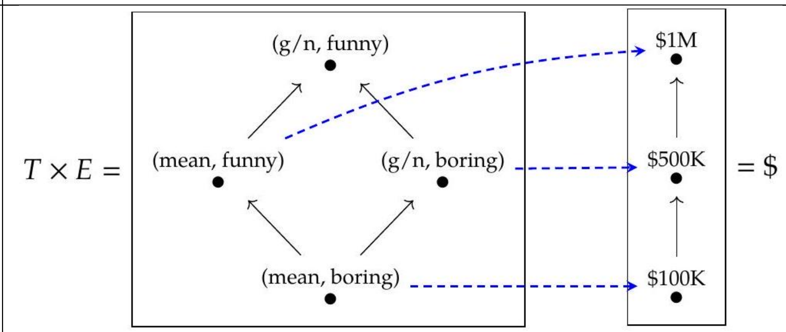
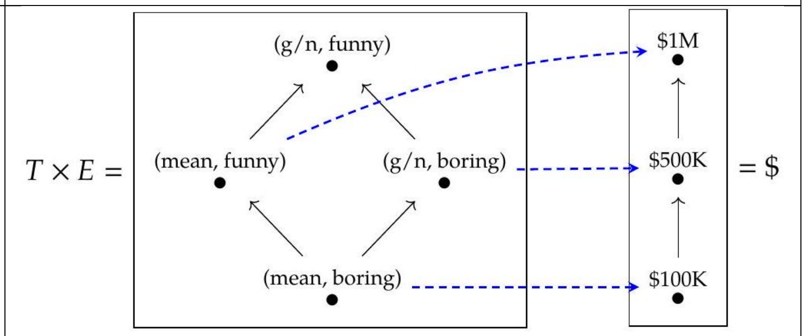
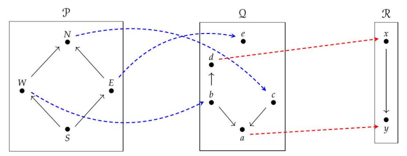
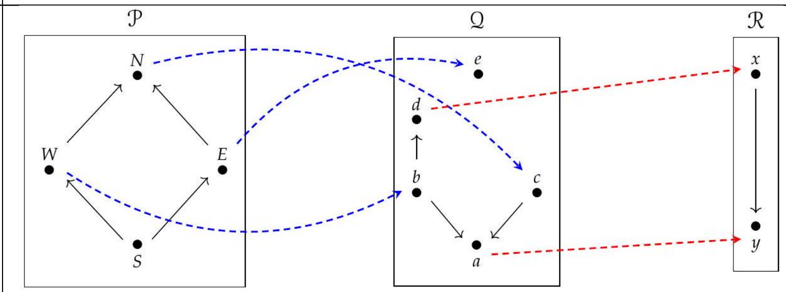
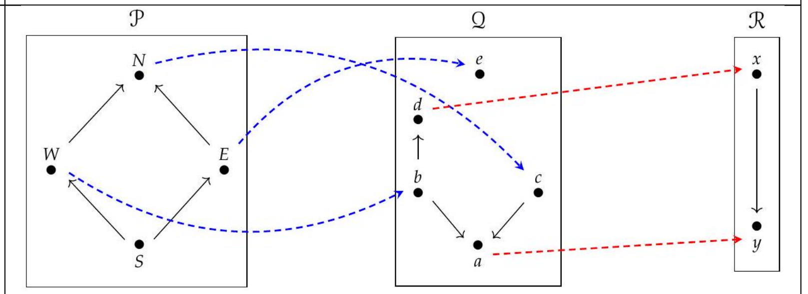
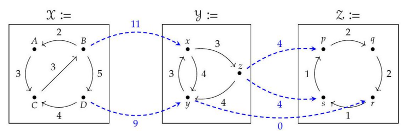
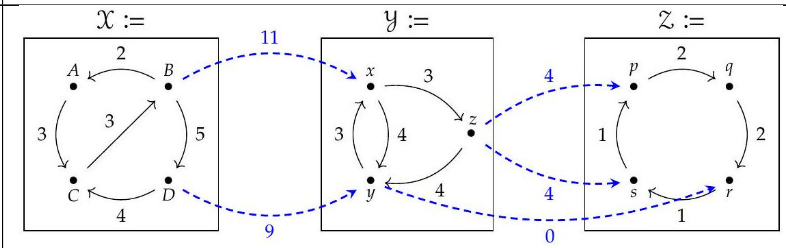
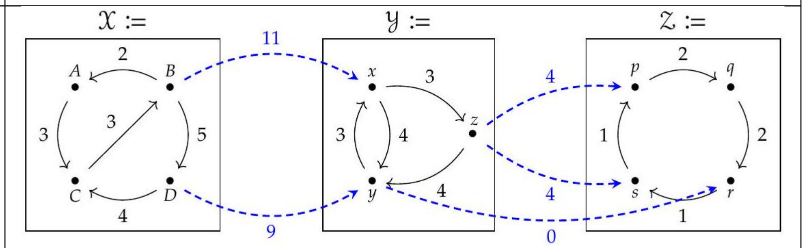
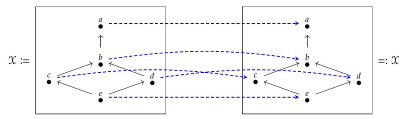
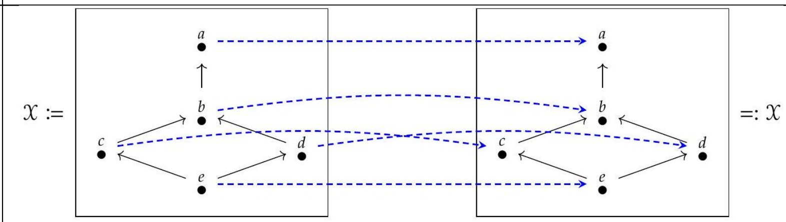
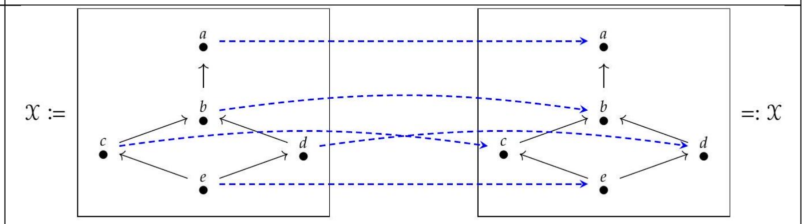
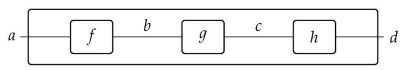
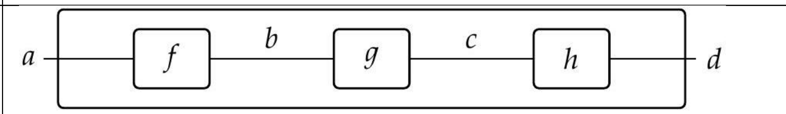
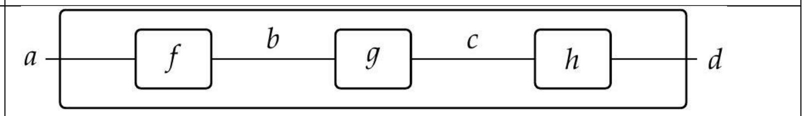
fong-spivak-seven-sketches_DIA_0120	106	—	*(dup)*	7f1e85c6-4350-401f-a960-eb8a0176e786-106_226_181_1966_733.jpg	1. $\begin{tikzcd} \underset{z}{\bullet} \arrow[loop above, "s"] \end{tikzcd}$ $s \fatsemi s = z$	1. $\begin{tikzcd} \underset{z}{\bullet} \arrow[loop above, "s"] \end{tikzcd}$ $s \fatsemi s = z$	1. $\begin{tikzcd} \underset{z}{\bullet} \arrow[loop above, "s"] \end{tikzcd}$ $s \fatsemi s = z$
fong-spivak-seven-sketches_DIA_0121	106	—	*(dup)*	7f1e85c6-4350-401f-a960-eb8a0176e786-106_93_324_1998_1217.jpg	$\begin{tikzcd} \overset{a}{\bullet} \arrow[r, "f"] & \overset{b}{\bullet} \arrow[r, shift left, "g"] \arrow[r, shift right, "h"'] & \overset{c}{\bullet} \end{tikzcd}$	$\begin{tikzcd} \overset{a}{\bullet} \arrow[r, "f"] & \overset{b}{\bullet} \arrow[r, shift left, "g"] \arrow[r, shift right, "h"'] & \overset{c}{\bullet} \end{tikzcd}$	$\begin{tikzcd} \overset{a}{\bullet} \arrow[r, "f"] & \overset{b}{\bullet} \arrow[r, shift left, "g"] \arrow[r, shift right, "h"'] & \overset{c}{\bullet} \end{tikzcd}$
fong-spivak-seven-sketches_DIA_0122	106	—	*(dup)*	7f1e85c6-4350-401f-a960-eb8a0176e786-106_43_172_2124_1294.jpg	$f \fatsemi g = f \fatsemi h$	$f \fatsemi g = f \fatsemi h$	$f \fatsemi g = f \fatsemi h$
fong-spivak-seven-sketches_DIA_0123	107	—	*(dup)*	7f1e85c6-4350-401f-a960-eb8a0176e786-107_152_244_527_937.jpg	$\begin{tikzcd} \mathcal{C} \arrow[r, bend left, "F"] \arrow[r, bend right, "G"'] \arrow[r, phantom, "\Downarrow\alpha"] & \mathcal{D}. \end{tikzcd}$	$\begin{tikzcd} \mathcal{C} \arrow[r, bend left, "F"] \arrow[r, bend right, "G"'] \arrow[r, phantom, "\Downarrow\alpha"] & \mathcal{D}. \end{tikzcd}$	$\begin{tikzcd} \mathcal{C} \arrow[r, bend left, "F"] \arrow[r, bend right, "G"'] \arrow[r, phantom, "\Downarrow\alpha"] & \mathcal{D}. \end{tikzcd}$
fong-spivak-seven-sketches_DIA_0124	107	—	*(dup)*	7f1e85c6-4350-401f-a960-eb8a0176e786-107_199_339_1638_889.jpg	$\begin{tikzcd} F(c) \arrow[r, "\alpha_c"] \arrow[d, "F(f)"'] & G(c) \arrow[d, "G(f)"] \\ F(d) \arrow[r, "\alpha_d"'] & G(d) \end{tikzcd}$	$\begin{tikzcd} F(c) \arrow[r, "\alpha_c"] \arrow[d, "F(f)"'] & G(c) \arrow[d, "G(f)"] \\ F(d) \arrow[r, "\alpha_d"'] & G(d) \end{tikzcd}$	$\begin{tikzcd} F(c) \arrow[r, "\alpha_c"] \arrow[d, "F(f)"'] & G(c) \arrow[d, "G(f)"] \\ F(d) \arrow[r, "\alpha_d"'] & G(d) \end{tikzcd}$
fong-spivak-seven-sketches_DIA_0125	108	—	*(dup)*	7f1e85c6-4350-401f-a960-eb8a0176e786-108_276_1195_380_462.jpg			
fong-spivak-seven-sketches_DIA_0126	109	—	*(dup)*	7f1e85c6-4350-401f-a960-eb8a0176e786-109_190_529_2330_792.jpg	**Gr** := $\begin{tikzcd} \overset{\text{Arrow}}{\bullet} \arrow[r, shift left, "\text{source}"] \arrow[r, shift right, "\text{target}"'] & \overset{\text{Vertex}}{\bullet} \end{tikzcd}$ *no equations*	**Gr** := $\begin{tikzcd} \overset{\text{Arrow}}{\bullet} \arrow[r, shift left, "\text{source}"] \arrow[r, shift right, "\text{target}"'] & \overset{\text{Vertex}}{\bullet} \end{tikzcd}$ *no equations*	**Gr** := $\begin{tikzcd} \overset{\text{Arrow}}{\bullet} \arrow[r, shift left, "\text{source}"] \arrow[r, shift right, "\text{target}"'] & \overset{\text{Vertex}}{\bullet} \end{tikzcd}$ *no equations*
fong-spivak-seven-sketches_DIA_0127	110	—	*(dup)*	7f1e85c6-4350-401f-a960-eb8a0176e786-110_206_611_2172_353.jpg	$\begin{tikzcd} G(\text{Arrow}) \arrow[r, "\alpha_{\text{Arrow}}"] \arrow[d, "G(\text{source})"'] & H(\text{Arrow}) \arrow[d, "H(\text{source})"] \\ G(\text{Vertex}) \arrow[r, "\alpha_{\text{Vertex}}"'] & H(\text{Vertex}) \end{tikzcd}$	$\begin{tikzcd} G(\text{Arrow}) \arrow[r, "\alpha_{\text{Arrow}}"] \arrow[d, "G(\text{source})"'] & H(\text{Arrow}) \arrow[d, "H(\text{source})"] \\ G(\text{Vertex}) \arrow[r, "\alpha_{\text{Vertex}}"'] & H(\text{Vertex}) \end{tikzcd}$	$\begin{tikzcd} G(\text{Arrow}) \arrow[r, "\alpha_{\text{Arrow}}"] \arrow[d, "G(\text{source})"'] & H(\text{Arrow}) \arrow[d, "H(\text{source})"] \\ G(\text{Vertex}) \arrow[r, "\alpha_{\text{Vertex}}"'] & H(\text{Vertex}) \end{tikzcd}$
fong-spivak-seven-sketches_DIA_0128	110	—	*(dup)*	7f1e85c6-4350-401f-a960-eb8a0176e786-110_206_591_2172_1172.jpg	$\begin{tikzcd} G(\text{Arrow}) \arrow[r, "\alpha_{\text{Arrow}}"] \arrow[d, "G(\text{target})"'] & H(\text{Arrow}) \arrow[d, "H(\text{target})"] \\ G(\text{Vertex}) \arrow[r, "\alpha_{\text{Vertex}}"'] & H(\text{Vertex}) \end{tikzcd}$	$\begin{tikzcd} G(\text{Arrow}) \arrow[r, "\alpha_{\text{Arrow}}"] \arrow[d, "G(\text{target})"'] & H(\text{Arrow}) \arrow[d, "H(\text{target})"] \\ G(\text{Vertex}) \arrow[r, "\alpha_{\text{Vertex}}"'] & H(\text{Vertex}) \end{tikzcd}$	$\begin{tikzcd} G(\text{Arrow}) \arrow[r, "\alpha_{\text{Arrow}}"] \arrow[d, "G(\text{target})"'] & H(\text{Arrow}) \arrow[d, "H(\text{target})"] \\ G(\text{Vertex}) \arrow[r, "\alpha_{\text{Vertex}}"'] & H(\text{Vertex}) \end{tikzcd}$

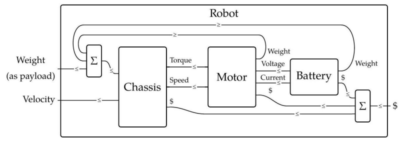
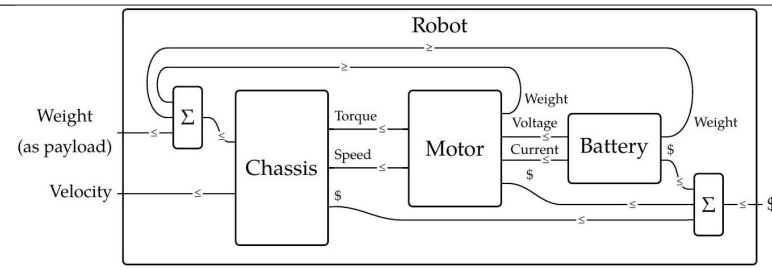
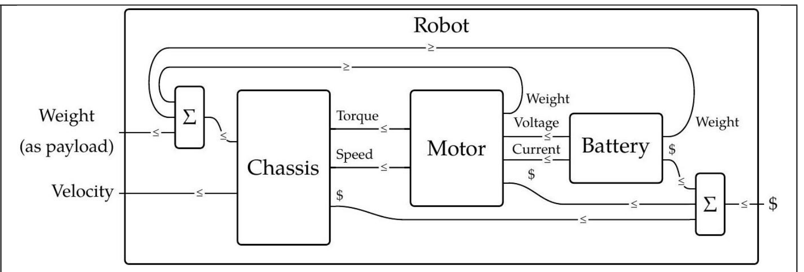
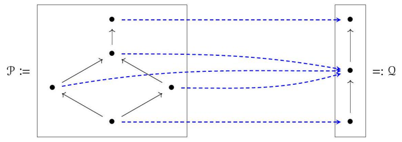
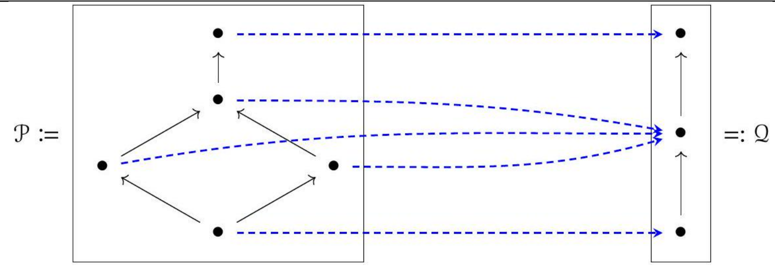
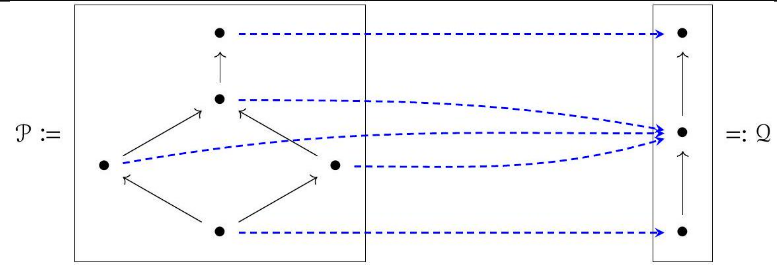
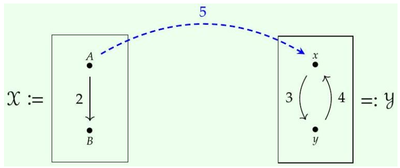
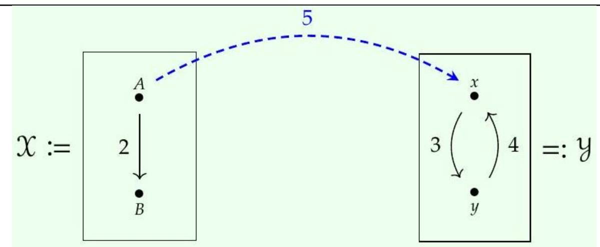
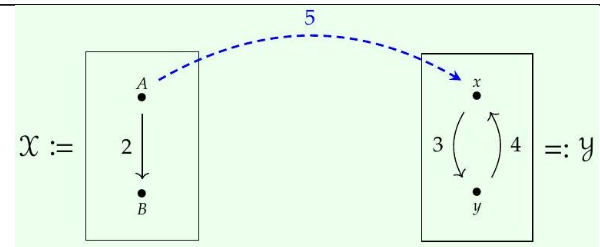
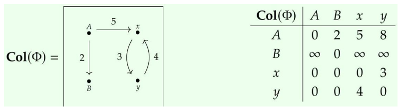
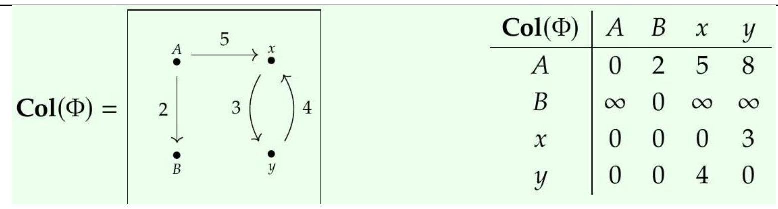
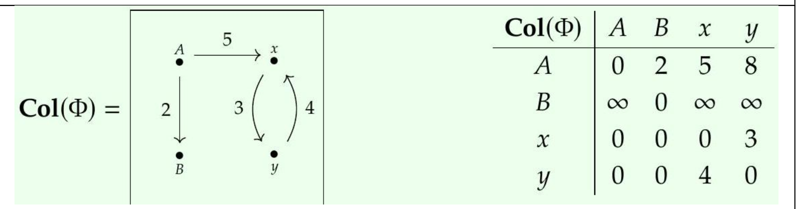
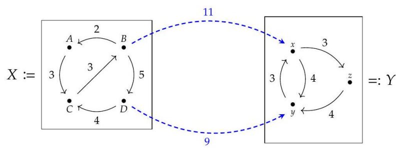
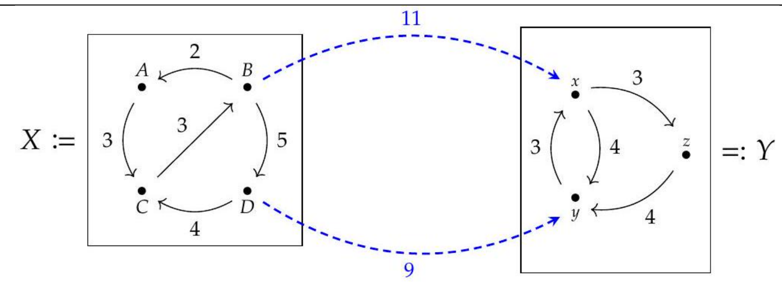
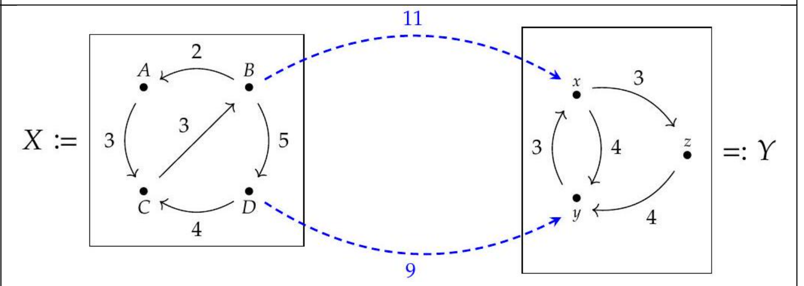
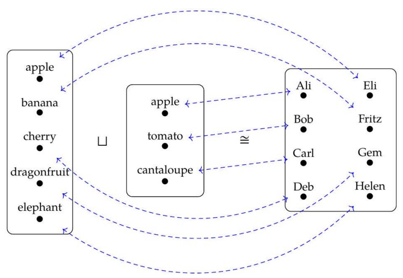
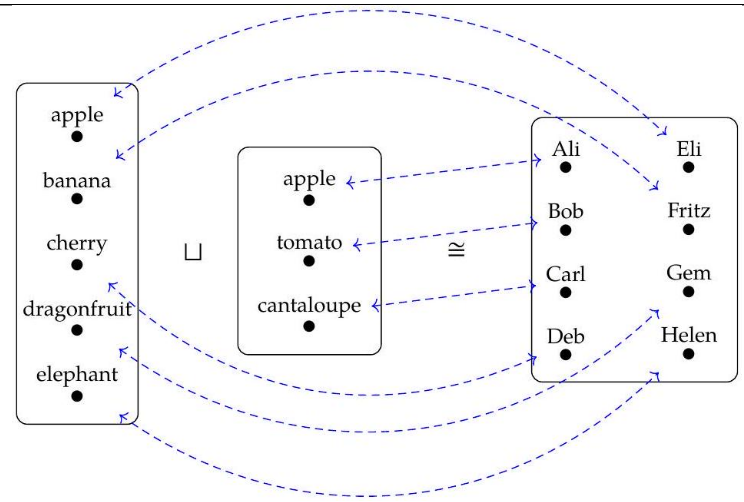
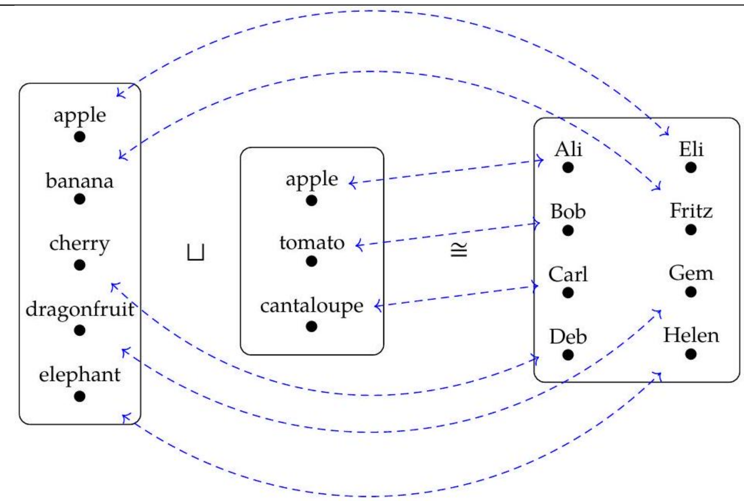
fong-spivak-seven-sketches_112_0129	112	— (dup)	7f1e85c6-4350-401f-a960-eb8a0176e786-112_330_543_1955_787.jpg			
fong-spivak-seven-sketches_113_0130	113	— (dup)	7f1e85c6-4350-401f-a960-eb8a0176e786-113_373_767_1600_676.jpg			
fong-spivak-seven-sketches_114_0131	114	— (dup)	7f1e85c6-4350-401f-a960-eb8a0176e786-114_163_1007_568_557.jpg			
fong-spivak-seven-sketches_115_0132	115	— (dup)	7f1e85c6-4350-401f-a960-eb8a0176e786-115_206_480_2258_819.jpg			
fong-spivak-seven-sketches_116_0133	116	— (dup)	7f1e85c6-4350-401f-a960-eb8a0176e786-116_172_498_2342_810.jpg			
fong-spivak-seven-sketches_117_0134	117	— (dup)	7f1e85c6-4350-401f-a960-eb8a0176e786-117_389_1296_1389_410.jpg			
fong-spivak-seven-sketches_121_0135	121	— (dup)	7f1e85c6-4350-401f-a960-eb8a0176e786-121_219_416_353_851.jpg			

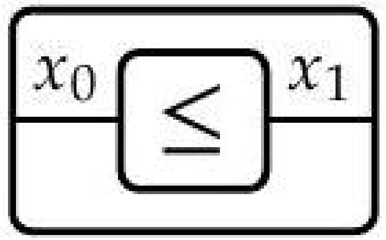
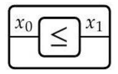
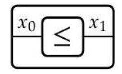
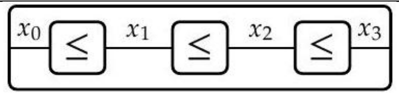
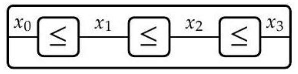
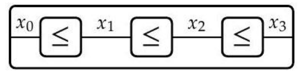
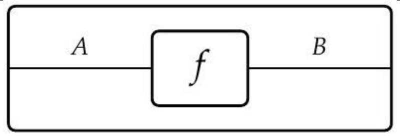
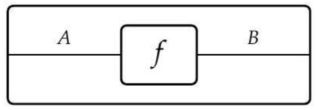
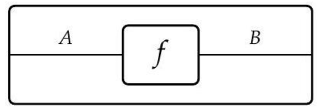
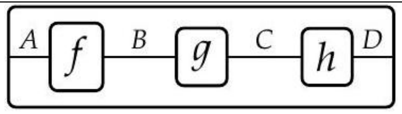
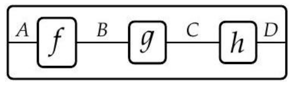
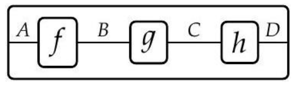
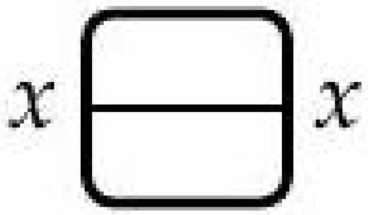
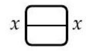
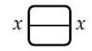
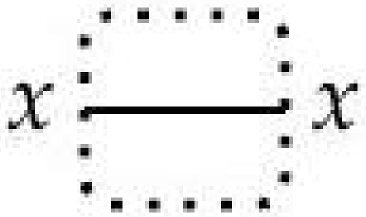
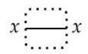
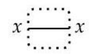
fong-spivak-seven-sketches_DIA_0136	121	— *(dup)*	7f1e85c6-4350-401f-a960-eb8a0176e786-121_710_955_1529_575.jpg			
fong-spivak-seven-sketches_DIA_0137	122	— *(dup)*	7f1e85c6-4350-401f-a960-eb8a0176e786-122_222_355_1837_882.jpg			
fong-spivak-seven-sketches_DIA_0138	123	— *(dup)*	7f1e85c6-4350-401f-a960-eb8a0176e786-123_176_475_1095_824.jpg			
fong-spivak-seven-sketches_DIA_0139	123	— *(dup)*	7f1e85c6-4350-401f-a960-eb8a0176e786-123_317_613_1441_351.jpg			
fong-spivak-seven-sketches_DIA_0140	123	— *(dup)*	7f1e85c6-4350-401f-a960-eb8a0176e786-123_314_615_1443_1154.jpg			
fong-spivak-seven-sketches_DIA_0141	125	— *(dup)*	7f1e85c6-4350-401f-a960-eb8a0176e786-125_208_229_330_663.jpg			

fong-spivak-seven-sketches_DIA_0142	125	— *(dup)*	7f1e85c6-4350-401f-a960-eb8a0176e786-125_213_296_328_1156.jpg			
fong-spivak-seven-sketches_DIA_0143	125	— *(dup)*	7f1e85c6-4350-401f-a960-eb8a0176e786-125_509_787_876_665.jpg			
fong-spivak-seven-sketches_DIA_0144	130	— *(dup)*	7f1e85c6-4350-401f-a960-eb8a0176e786-130_461_1345_1308_387.jpg			
fong-spivak-seven-sketches_DIA_0145	132	— *(dup)*	7f1e85c6-4350-401f-a960-eb8a0176e786-132_233_1136_1649_498.jpg			
fong-spivak-seven-sketches_DIA_0146	134	— *(dup)*	7f1e85c6-4350-401f-a960-eb8a0176e786-134_486_1041_760_538.jpg			
fong-spivak-seven-sketches_DIA_0147	135	— *(dup)*	7f1e85c6-4350-401f-a960-eb8a0176e786-135_367_1014_559_552.jpg			

fong-spivak-seven-sketches_DIA_0148	136	— *(dup)*	7f1e85c6-4350-401f-a960-eb8a0176e786-136_217_407_611_855.jpg			
fong-spivak-seven-sketches_DIA_0149	136	— *(dup)*	7f1e85c6-4350-401f-a960-eb8a0176e786-136_165_285_1860_916.jpg			
fong-spivak-seven-sketches_DIA_0150	136	— *(dup)*	7f1e85c6-4350-401f-a960-eb8a0176e786-136_296_319_2172_575.jpg			
fong-spivak-seven-sketches_DIA_0151	136	— *(dup)*	7f1e85c6-4350-401f-a960-eb8a0176e786-136_292_226_2176_973.jpg			

fong-spivak-seven-sketches_DIA_0152	136	— *(dup)*	7f1e85c6-4350-401f-a960-eb8a0176e786-136_357_215_2145_1339.jpg			
fong-spivak-seven-sketches_DIA_0153	137	— *(dup)*	7f1e85c6-4350-401f-a960-eb8a0176e786-137_423_993_369_563.jpg			
fong-spivak-seven-sketches_DIA_0154	138	— *(dup)*	7f1e85c6-4350-401f-a960-eb8a0176e786-138_452_1213_346_407.jpg			
fong-spivak-seven-sketches_DIA_0155	139	— *(dup)*	7f1e85c6-4350-401f-a960-eb8a0176e786-139_351_1118_692_500.jpg			
fong-spivak-seven-sketches_DIA_0156	140	— *(dup)*	7f1e85c6-4350-401f-a960-eb8a0176e786-140_376_1335_665_391.jpg			
fong-spivak-seven-sketches_DIA_0157	141	— *(dup)*	7f1e85c6-4350-401f-a960-eb8a0176e786-141_136_919_1009_600.jpg			

fong-spivak-seven-sketche	141	_0158 — *(dup)*	7f1e85c6-4350-401f-a960-eb8a0176e786-141_459_1346_1373_387.jpg			
fong-spivak-seven-sketche	142	_0159 — *(dup)*	7f1e85c6-4350-401f-a960-eb8a0176e786-142_398_1156_1602_480.jpg			
fong-spivak-seven-sketche	144	_0160 — *(dup)*	7f1e85c6-4350-401f-a960-eb8a0176e786-144_314_760_523_679.jpg			
fong-spivak-seven-sketche	144	_0161 — *(dup)*	7f1e85c6-4350-401f-a960-eb8a0176e786-144_294_1131_1416_500.jpg			
fong-spivak-seven-sketche	144	_0162 — *(dup)*	7f1e85c6-4350-401f-a960-eb8a0176e786-144_367_1014_1860_552.jpg			
fong-spivak-seven-sketche	145	_0163 — *(dup)*	7f1e85c6-4350-401f-a960-eb8a0176e786-145_645_950_1335_586.jpg			

fong-spivak-seven-sketches_146_0164	146	— *(dup)*	7f1e85c6-4350-401f-a960-eb8a0176e786-146_90_143_1000_989.jpg			
fong-spivak-seven-sketches_146_0165	146	— *(dup)*	7f1e85c6-4350-401f-a960-eb8a0176e786-146_88_380_1242_869.jpg			
fong-spivak-seven-sketches_146_0166	146	— *(dup)*	7f1e85c6-4350-401f-a960-eb8a0176e786-146_138_405_1719_857.jpg			
fong-spivak-seven-sketches_146_0167	146	— *(dup)*	7f1e85c6-4350-401f-a960-eb8a0176e786-146_102_371_2054_873.jpg			
fong-spivak-seven-sketches_147_0168	147	— *(dup)*	7f1e85c6-4350-401f-a960-eb8a0176e786-147_68_104_387_636.jpg			
fong-spivak-seven-sketches_147_0169	147	— *(dup)*	7f1e85c6-4350-401f-a960-eb8a0176e786-147_68_104_387_1007.jpg			
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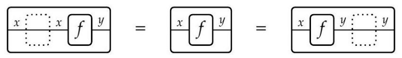
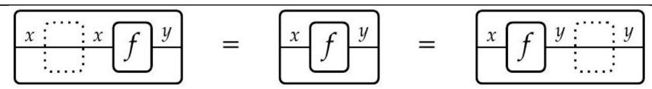
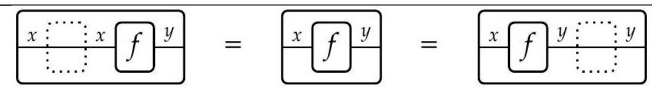
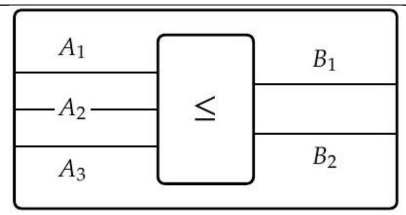
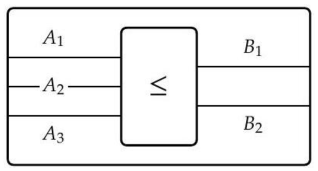
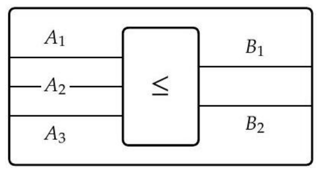
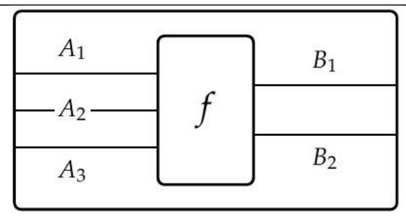
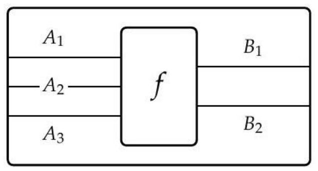
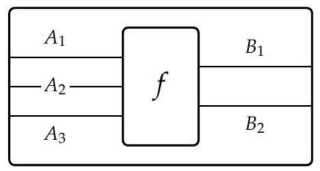
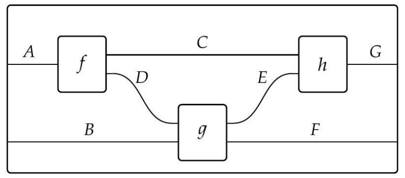
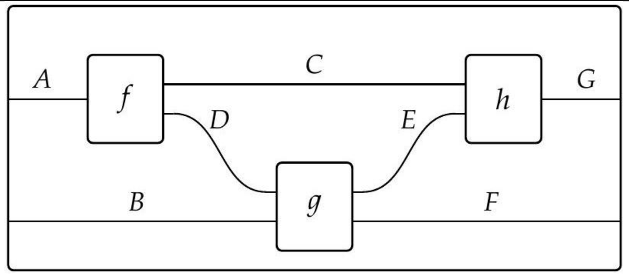
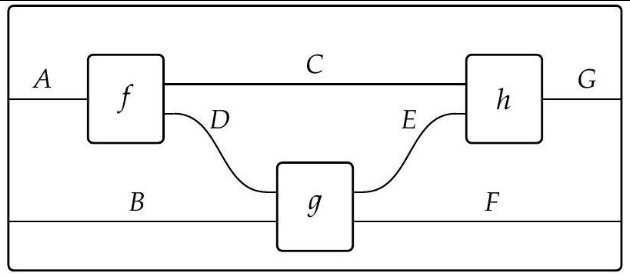
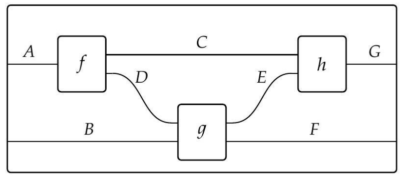
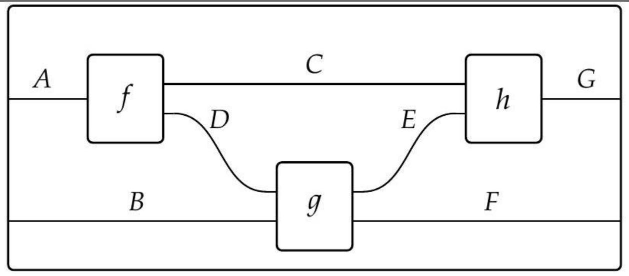
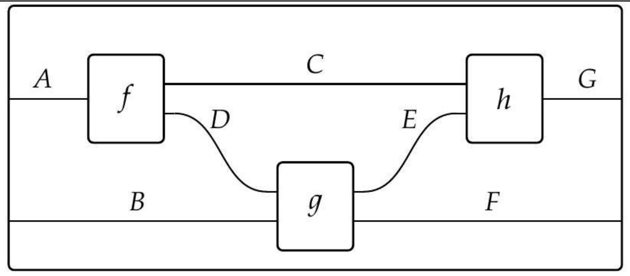
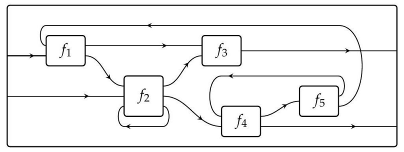
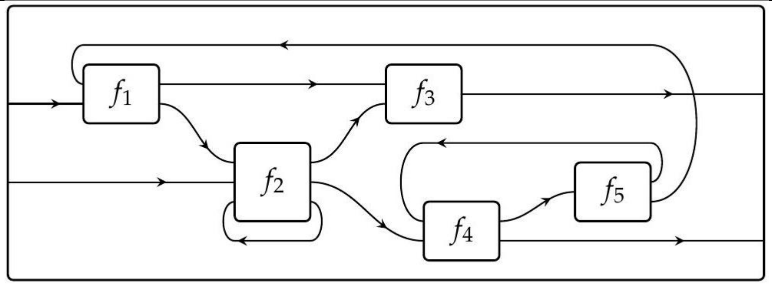
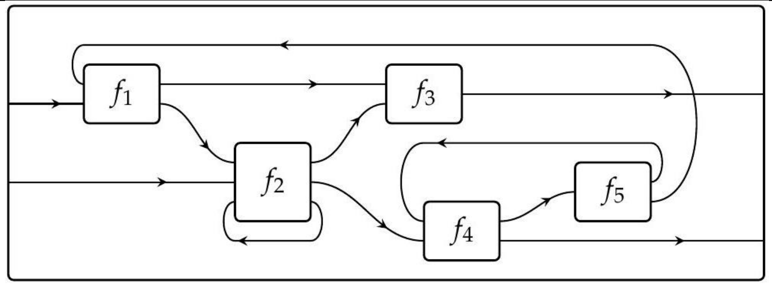
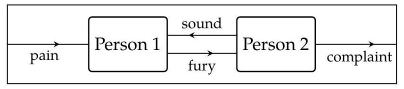
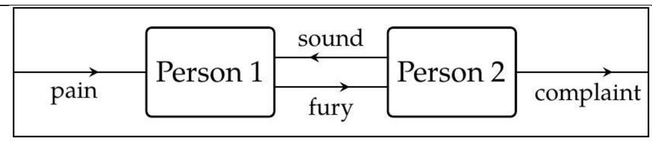
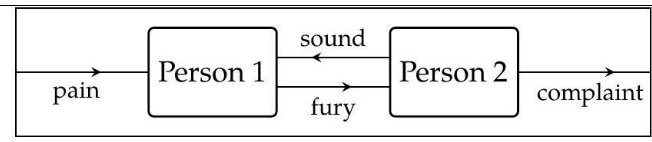
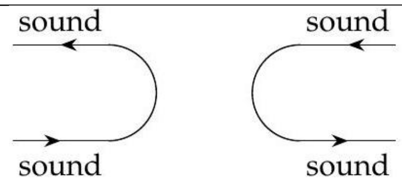
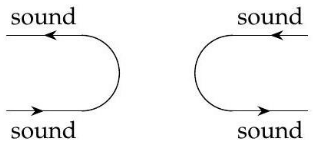
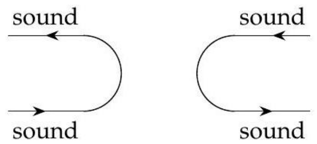
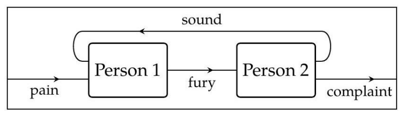
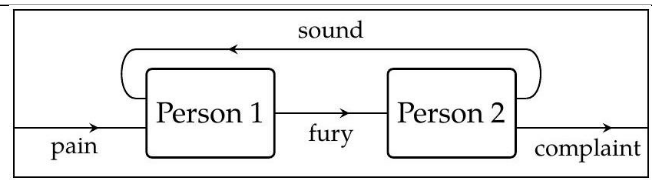
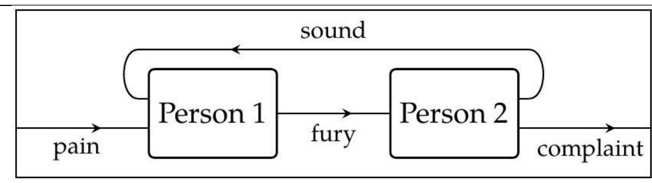
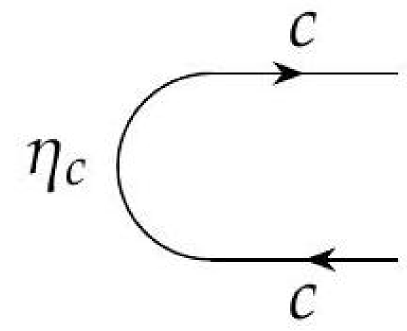
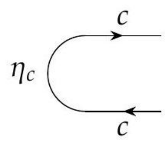
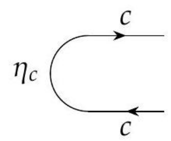
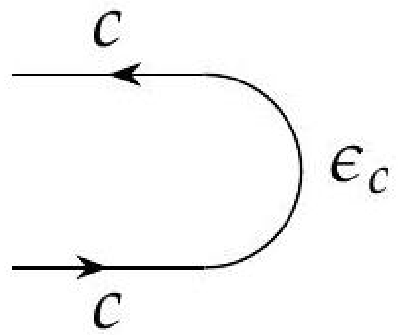
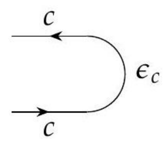
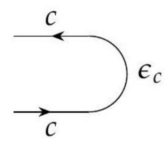
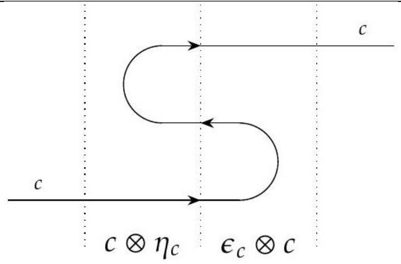
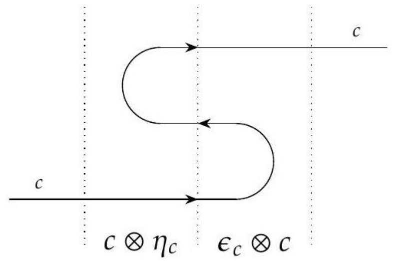
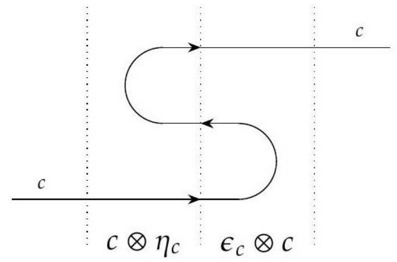
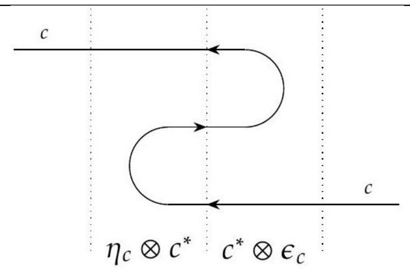
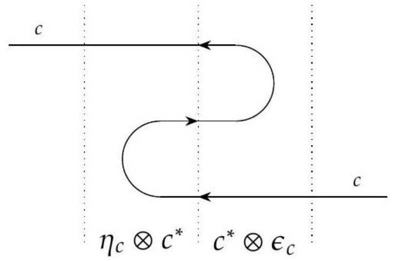
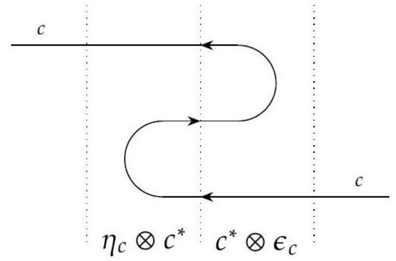
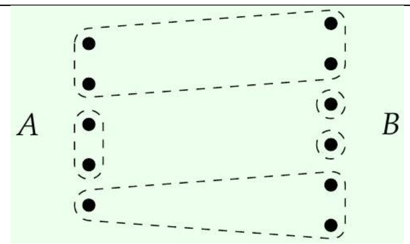
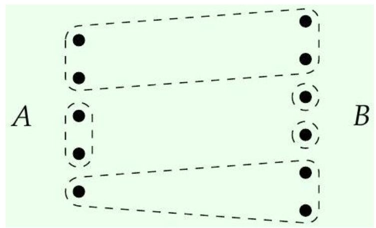
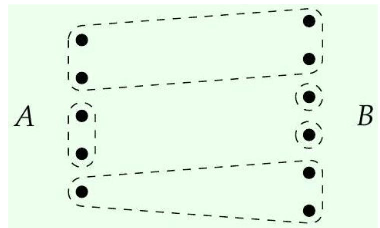
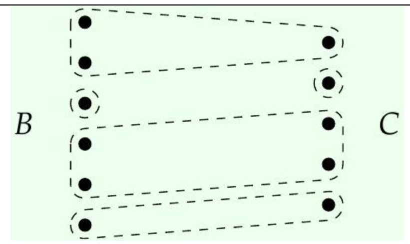
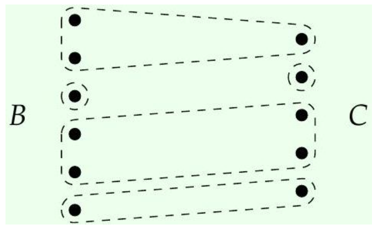
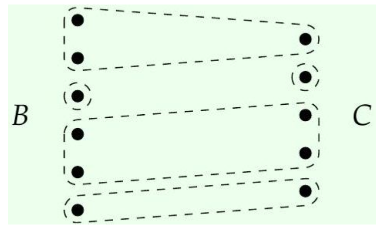
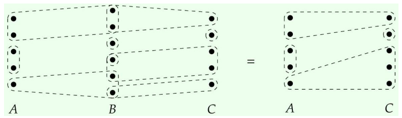
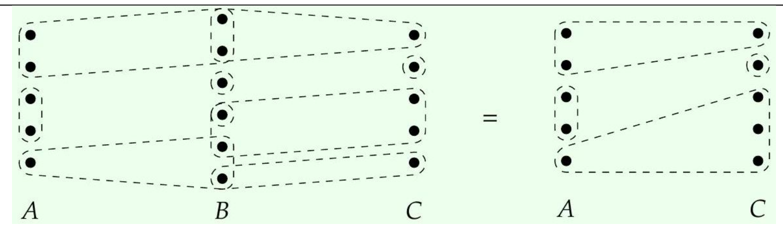
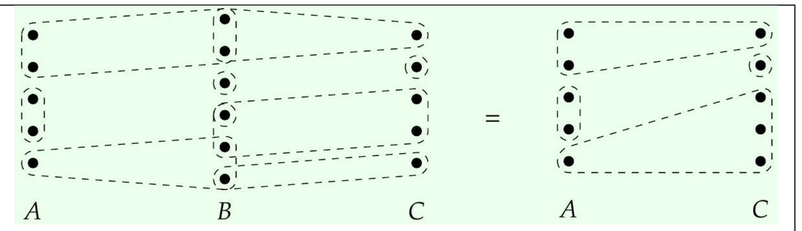
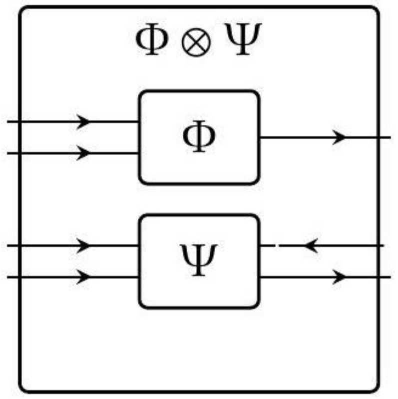
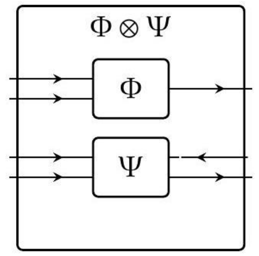
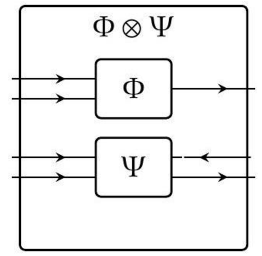
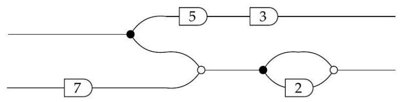
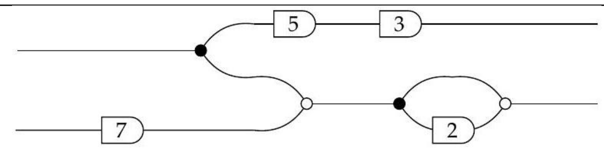
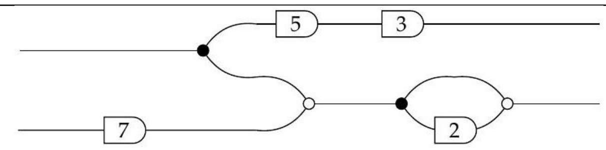
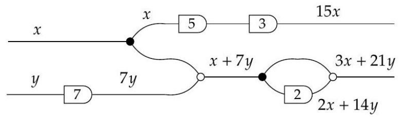
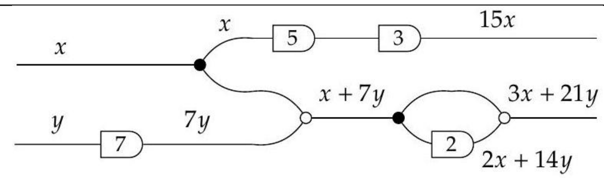
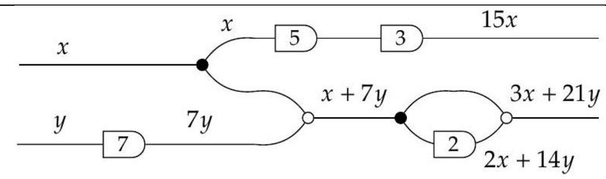
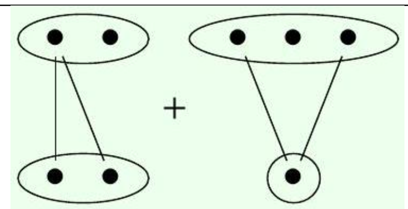
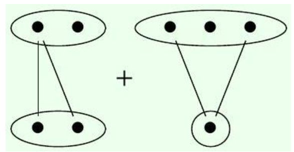
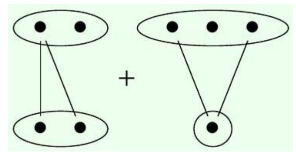
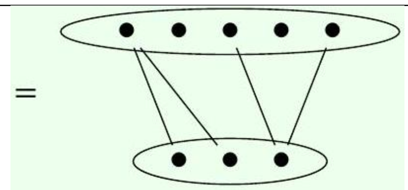
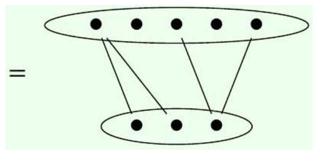
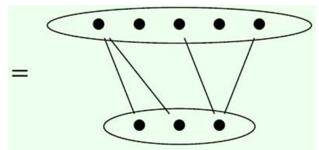
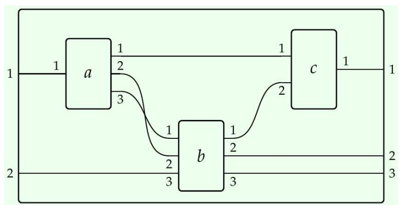
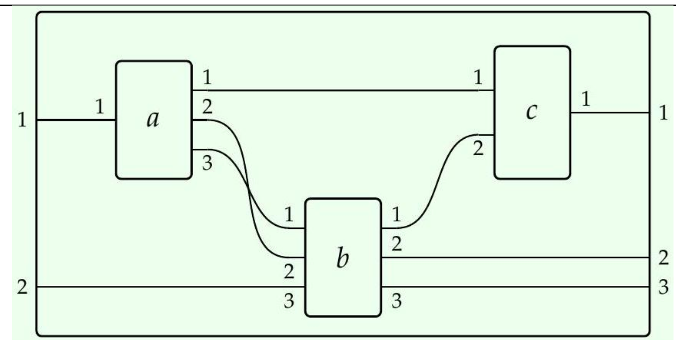
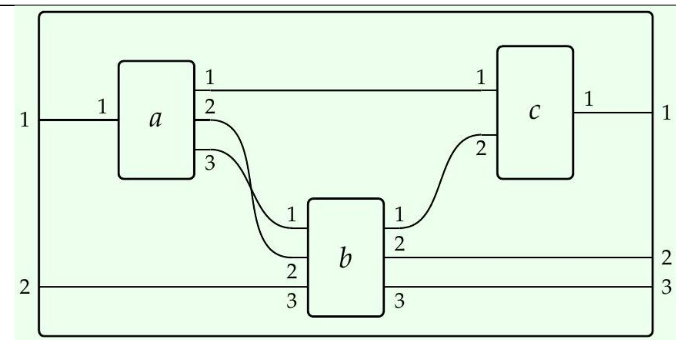
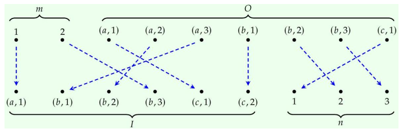
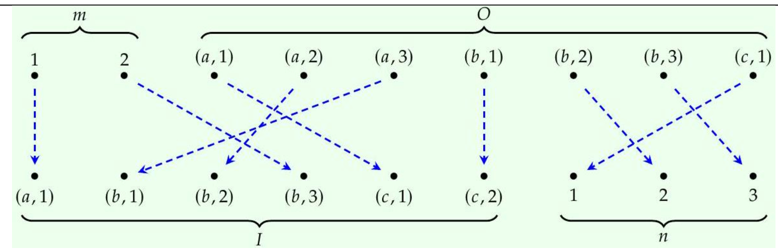
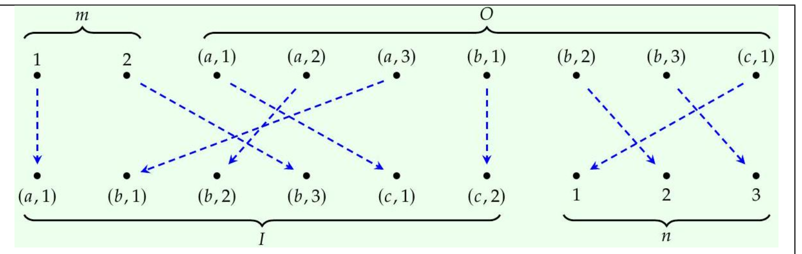
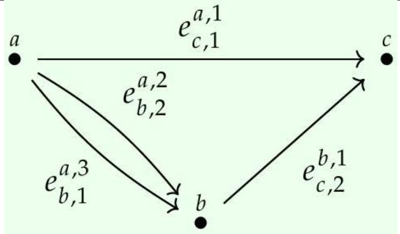
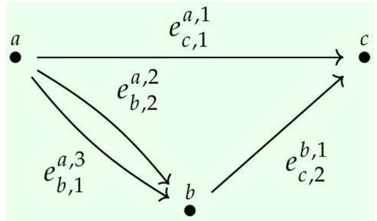
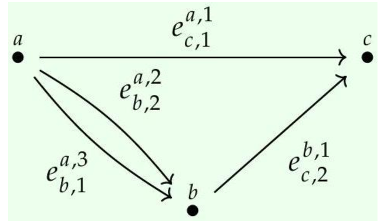
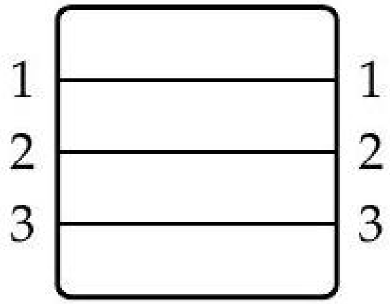
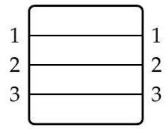
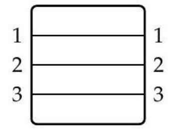
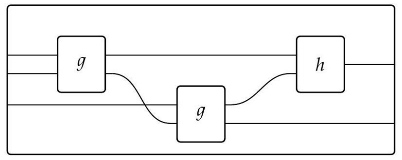
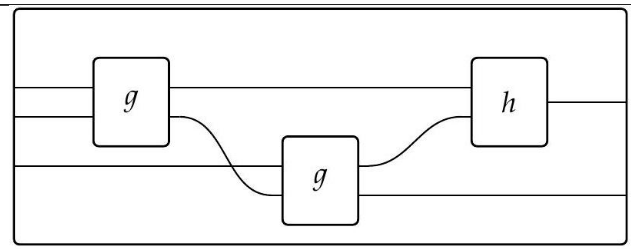
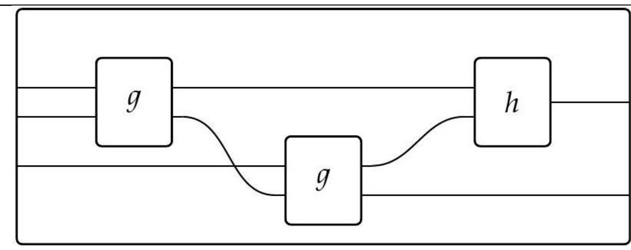
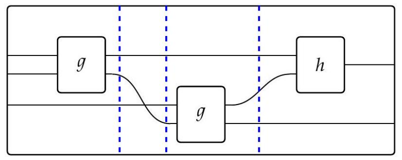
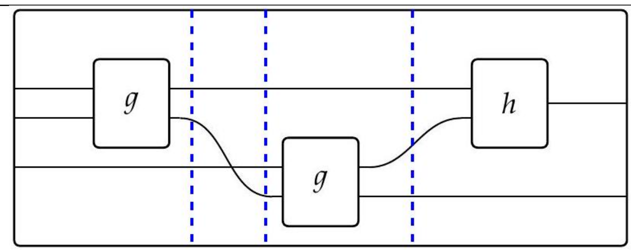
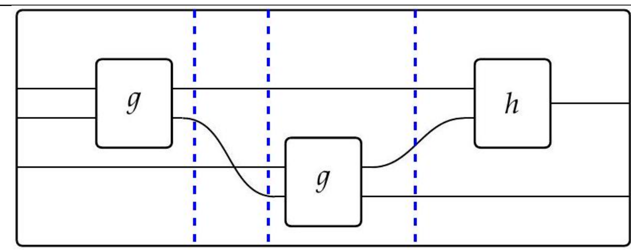
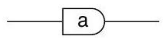
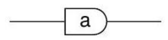
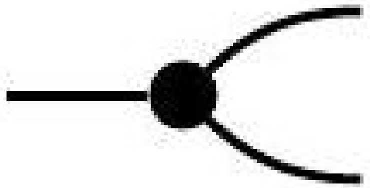
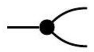
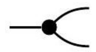
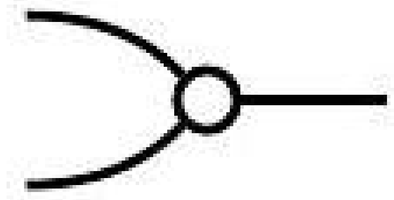
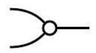
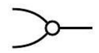
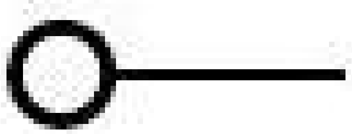
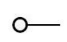
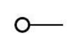
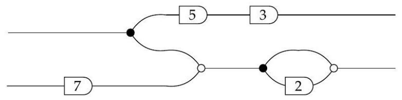
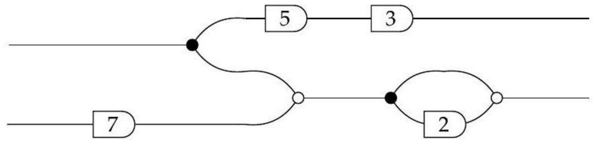
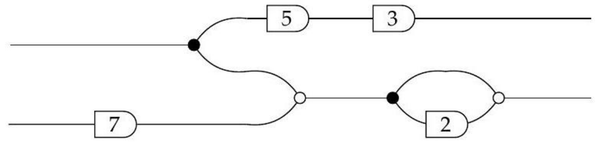
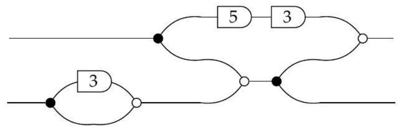
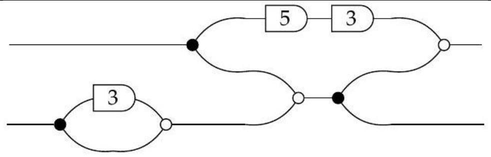
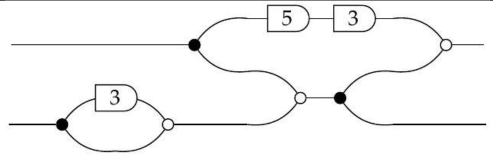
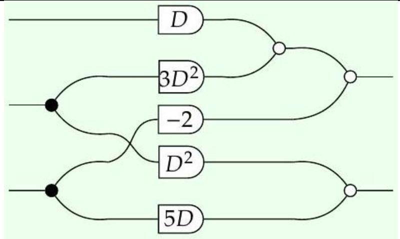
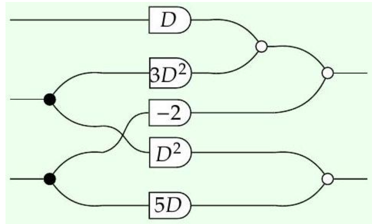
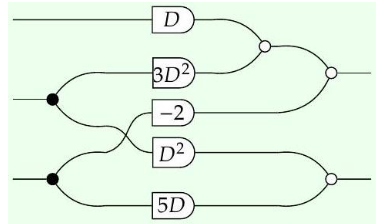
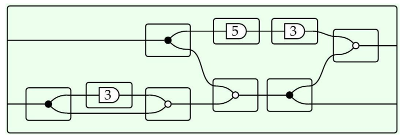
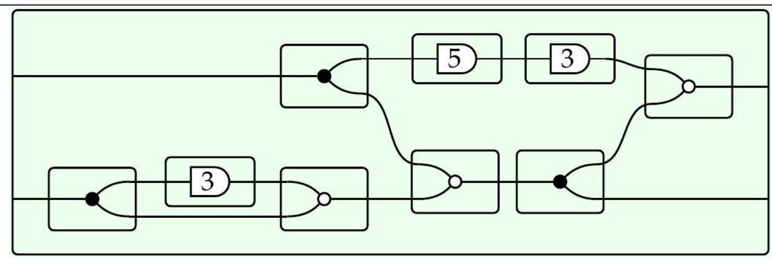
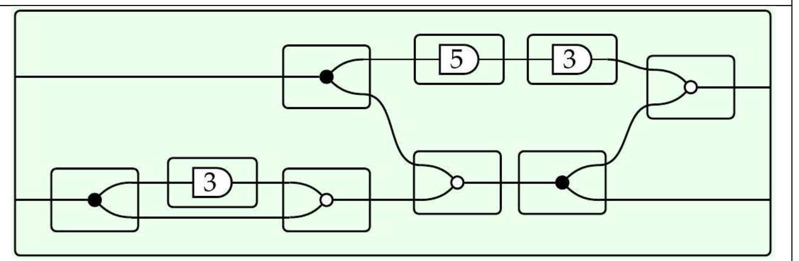
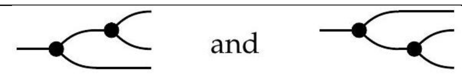
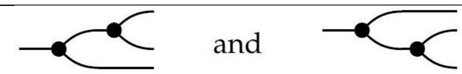
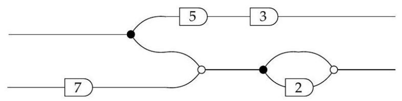
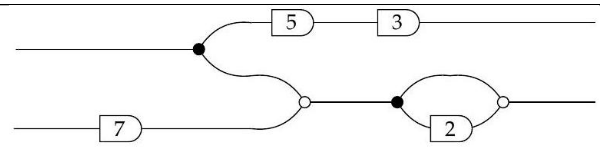
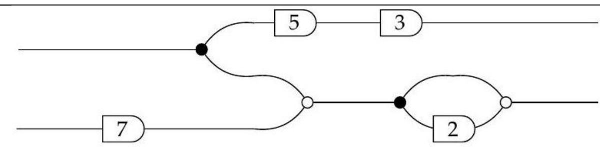
fong-spivak-seven-sketches_147_0171	147	— *(dup)*	7f1e85c6-4350-401f-a960-eb8a0176e786-147_106_833_701_643.jpg			
fong-spivak-seven-sketches_147_0172	147	— *(dup)*	7f1e85c6-4350-401f-a960-eb8a0176e786-147_213_405_1274_857.jpg			
fong-spivak-seven-sketches_147_0173	147	— *(dup)*	7f1e85c6-4350-401f-a960-eb8a0176e786-147_213_405_1753_857.jpg			

fong-spivak-seven-sketches_DIA_0174	147	— *(dup)*	7f1e85c6-4350-401f-a960-eb8a0176e786-147_355_805_2147_656.jpg			
fong-spivak-seven-sketches_DIA_0175	150	— *(dup)*	7f1e85c6-4350-401f-a960-eb8a0176e786-150_355_808_529_656.jpg			
fong-spivak-seven-sketches_DIA_0176	152	— *(dup)*	7f1e85c6-4350-401f-a960-eb8a0176e786-152_370_995_336_563.jpg			
fong-spivak-seven-sketches_DIA_0177	152	— *(dup)*	7f1e85c6-4350-401f-a960-eb8a0176e786-152_176_836_1365_640.jpg			
fong-spivak-seven-sketches_DIA_0178	152	— *(dup)*	7f1e85c6-4350-401f-a960-eb8a0176e786-152_182_404_1712_862.jpg			
fong-spivak-seven-sketches_DIA_0179	152	— *(dup)*	7f1e85c6-4350-401f-a960-eb8a0176e786-152_227_836_2019_640.jpg			
fong-spivak-seven-sketches_DIA_0180	153	— *(dup)*	7f1e85c6-4350-401f-a960-eb8a0176e786-153_301_502_903_387.jpg	$\begin{tikzcd} c \arrow[rr, equal] \arrow[d, "\cong"'] & & c \\ c \otimes I \arrow[d, "c\otimes\eta_c"'] & & I \otimes c \arrow[u, "\cong"'] \\ c \otimes (c^* \otimes c) \arrow[rr, "\cong"'] & & (c \otimes c^*) \otimes c \arrow[u, "\epsilon_c \otimes c"'] \end{tikzcd}$	$\begin{tikzcd} c \arrow[rr, equal] \arrow[d, "\cong"'] & & c \\ c \otimes I \arrow[d, "c\otimes\eta_c"'] & & I \otimes c \arrow[u, "\cong"'] \\ c \otimes (c^* \otimes c) \arrow[rr, "\cong"'] & & (c \otimes c^*) \otimes c \arrow[u, "\epsilon_c \otimes c"'] \end{tikzcd}$	$\begin{tikzcd} c \arrow[rr, equal] \arrow[d, "\cong"'] & & c \\ c \otimes I \arrow[d, "c\otimes\eta_c"'] & & I \otimes c \arrow[u, "\cong"'] \\ c \otimes (c^* \otimes c) \arrow[rr, "\cong"'] & & (c \otimes c^*) \otimes c \arrow[u, "\epsilon_c \otimes c"'] \end{tikzcd}$
fong-spivak-seven-sketches_DIA_0181	153	— *(dup)*	7f1e85c6-4350-401f-a960-eb8a0176e786-153_305_525_900_1113.jpg	$\begin{tikzcd} c^* \arrow[rr, equal] \arrow[d, "\cong"'] & & c^* \\ I \otimes c^* \arrow[d, "\eta_c\otimes c^*"'] & & c^* \otimes I \arrow[u, "\cong"'] \\ (c^* \otimes c) \otimes c^* \arrow[rr, "\cong"'] & & c^* \otimes (c \otimes c^*) \arrow[u, "c^* \otimes \epsilon_c"'] \end{tikzcd}$	$\begin{tikzcd} c^* \arrow[rr, equal] \arrow[d, "\cong"'] & & c^* \\ I \otimes c^* \arrow[d, "\eta_c\otimes c^*"'] & & c^* \otimes I \arrow[u, "\cong"'] \\ (c^* \otimes c) \otimes c^* \arrow[rr, "\cong"'] & & c^* \otimes (c \otimes c^*) \arrow[u, "c^* \otimes \epsilon_c"'] \end{tikzcd}$	$\begin{tikzcd} c^* \arrow[rr, equal] \arrow[d, "\cong"'] & & c^* \\ I \otimes c^* \arrow[d, "\eta_c\otimes c^*"'] & & c^* \otimes I \arrow[u, "\cong"'] \\ (c^* \otimes c) \otimes c^* \arrow[rr, "\cong"'] & & c^* \otimes (c \otimes c^*) \arrow[u, "c^* \otimes \epsilon_c"'] \end{tikzcd}$

fong-spivak-seven-sketches_DIA_0182	153	— *(dup)*	7f1e85c6-4350-401f-a960-eb8a0176e786-153_176_208_1814_710.jpg			
fong-spivak-seven-sketches_DIA_0183	153	— *(dup)*	7f1e85c6-4350-401f-a960-eb8a0176e786-153_174_201_1814_1204.jpg			
fong-spivak-seven-sketches_DIA_0184	153	— *(dup)*	7f1e85c6-4350-401f-a960-eb8a0176e786-153_333_502_2158_437.jpg			
fong-spivak-seven-sketches_DIA_0185	153	— *(dup)*	7f1e85c6-4350-401f-a960-eb8a0176e786-153_335_502_2156_1181.jpg			
fong-spivak-seven-sketches_DIA_0186	154	— *(dup)*	7f1e85c6-4350-401f-a960-eb8a0176e786-154_290_480_1570_817.jpg			
fong-spivak-seven-sketches_DIA_0187	154	— *(dup)*	7f1e85c6-4350-401f-a960-eb8a0176e786-154_285_477_2145_821.jpg			

fong-spivak-seven-sketches_DIA_0188	155	— *(dup)*	7f1e85c6-4350-401f-a960-eb8a0176e786-155_337_1176_317_471.jpg			
fong-spivak-seven-sketches_DIA_0189	156	— *(dup)*	7f1e85c6-4350-401f-a960-eb8a0176e786-156_328_324_810_880.jpg			
fong-spivak-seven-sketches_DIA_0190	160	— *(dup)*	7f1e85c6-4350-401f-a960-eb8a0176e786-160_186_767_337_676.jpg			
fong-spivak-seven-sketches_DIA_0191	160	— *(dup)*	7f1e85c6-4350-401f-a960-eb8a0176e786-160_224_769_1100_670.jpg			
fong-spivak-seven-sketches_DIA_0192	162	— *(dup)*	7f1e85c6-4350-401f-a960-eb8a0176e786-162_192_371_2294_656.jpg			
fong-spivak-seven-sketches_DIA_0193	162	— *(dup)*	7f1e85c6-4350-401f-a960-eb8a0176e786-162_188_400_2299_1061.jpg			
fong-spivak-seven-sketches_DIA_0194	164	— *(dup)*	7f1e85c6-4350-401f-a960-eb8a0176e786-164_434_860_330_629.jpg			

fong-spivak-seven-sketches_DIA_0195	164	— (dup)	7f1e85c6-4350-401f-a960-eb8a0176e786-164_376_1186_1109_466.jpg			
fong-spivak-seven-sketches_DIA_0196	164	— (dup)	7f1e85c6-4350-401f-a960-eb8a0176e786-164_285_480_1640_819.jpg			
fong-spivak-seven-sketches_DIA_0197	165	— (dup)	7f1e85c6-4350-401f-a960-eb8a0176e786-165_167_208_905_955.jpg			
fong-spivak-seven-sketches_DIA_0198	168	— (dup)	7f1e85c6-4350-401f-a960-eb8a0176e786-168_317_810_751_654.jpg			
fong-spivak-seven-sketches_DIA_0199	170	— (dup)	7f1e85c6-4350-401f-a960-eb8a0176e786-170_317_810_491_654.jpg			
fong-spivak-seven-sketches_DIA_0200	172	— (dup)	7f1e85c6-4350-401f-a960-eb8a0176e786-172_43_208_2197_955.jpg			
fong-spivak-seven-sketches_DIA_0201	173	— (dup)	7f1e85c6-4350-401f-a960-eb8a0176e786-173_63_769_385_674.jpg			
fong-spivak-seven-sketches_DIA_0202	173	— (dup)	7f1e85c6-4350-401f-a960-eb8a0176e786-173_66_115_593_1002.jpg			
fong-spivak-seven-sketches_DIA_0203	173	— (dup)	7f1e85c6-4350-401f-a960-eb8a0176e786-173_34_72_943_995.jpg			

fong-spivak-seven-sketches_DIA_0204	173	— *(dup)*	7f1e85c6-4350-401f-a960-eb8a0176e786-173_61_113_1217_1002.jpg			
fong-spivak-seven-sketches_DIA_0205	173	— *(dup)*	7f1e85c6-4350-401f-a960-eb8a0176e786-173_27_72_1439_1050.jpg			
fong-spivak-seven-sketches_DIA_0206	173	— *(dup)*	7f1e85c6-4350-401f-a960-eb8a0176e786-173_188_767_1855_676.jpg			
fong-spivak-seven-sketches_DIA_0207	174	— *(dup)*	7f1e85c6-4350-401f-a960-eb8a0176e786-174_204_631_729_744.jpg			
fong-spivak-seven-sketches_DIA_0208	174	— *(dup)*	7f1e85c6-4350-401f-a960-eb8a0176e786-174_290_477_1955_821.jpg			
fong-spivak-seven-sketches_DIA_0209	175	— *(dup)*	7f1e85c6-4350-401f-a960-eb8a0176e786-175_355_1075_1072_523.jpg			
fong-spivak-seven-sketches_DIA_0210	175	— *(dup)*	7f1e85c6-4350-401f-a960-eb8a0176e786-175_90_584_2005_769.jpg			
fong-spivak-seven-sketches_DIA_0211	177	— *(dup)*	7f1e85c6-4350-401f-a960-eb8a0176e786-177_186_769_452_674.jpg			
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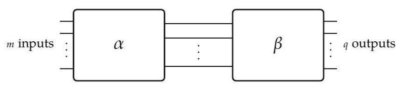
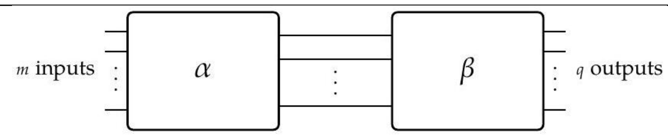
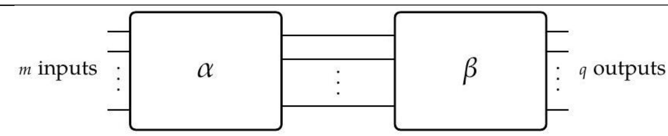
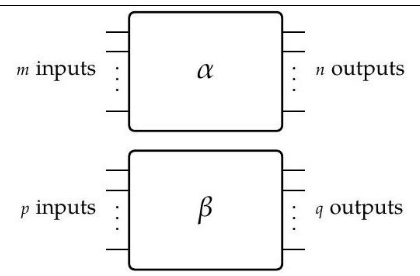
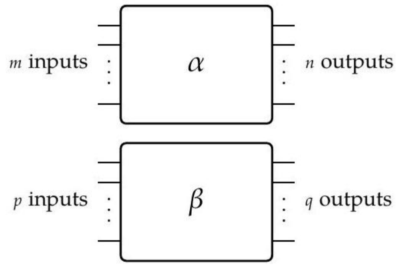
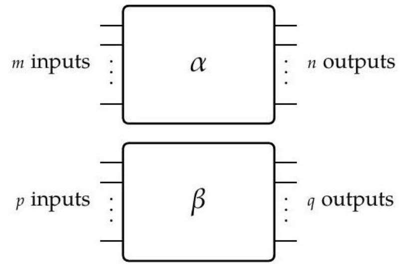
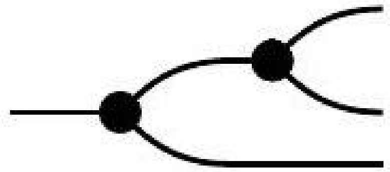
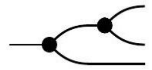
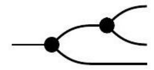
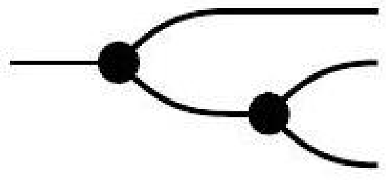
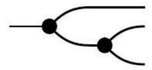
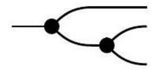
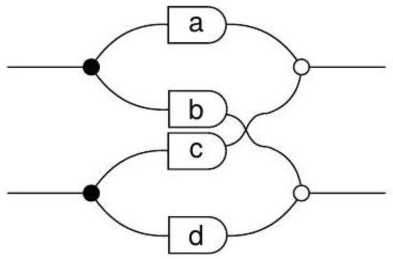
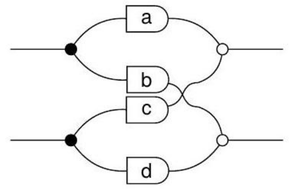
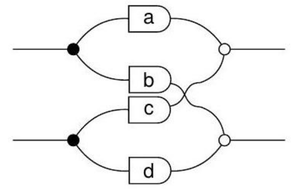
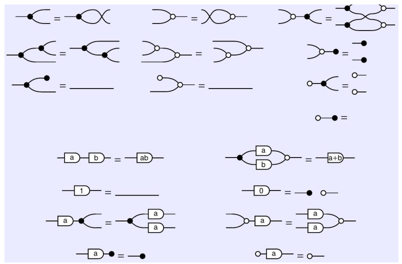
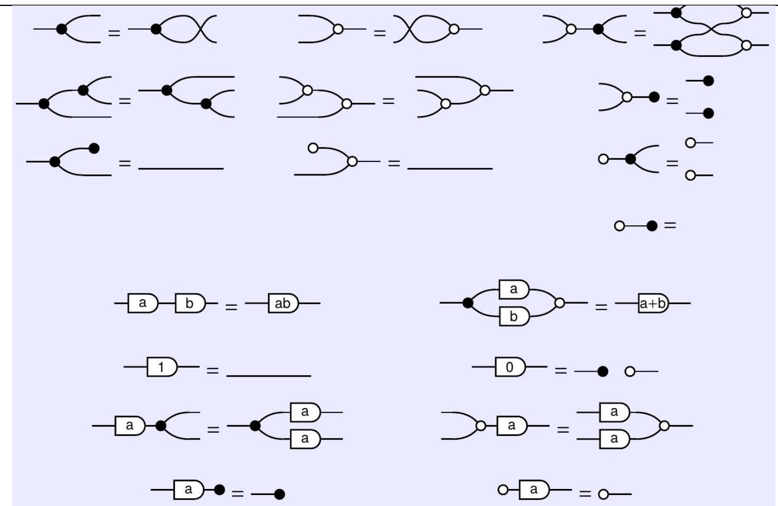
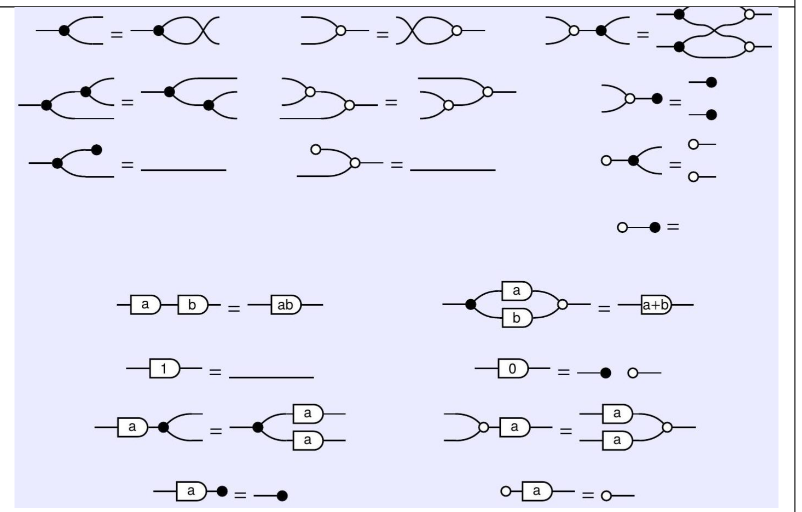
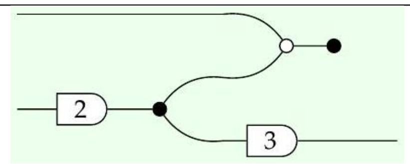
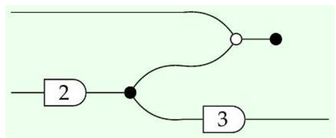
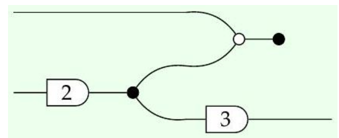
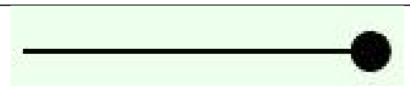
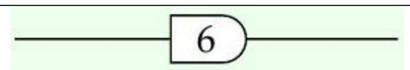
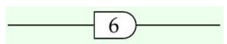
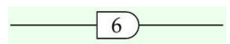
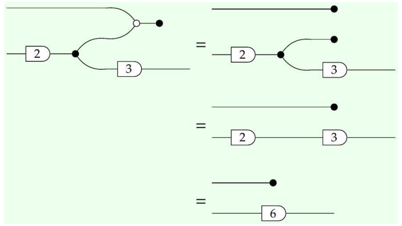
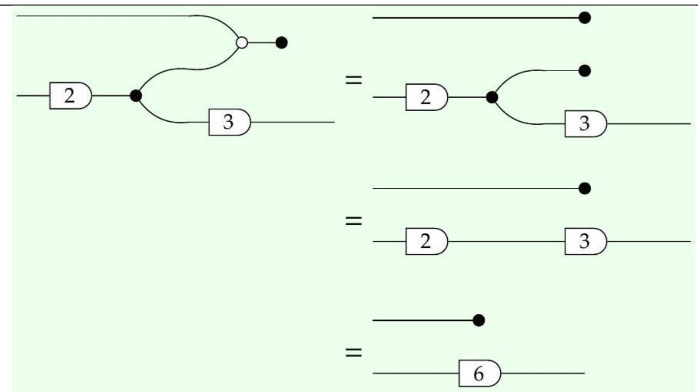
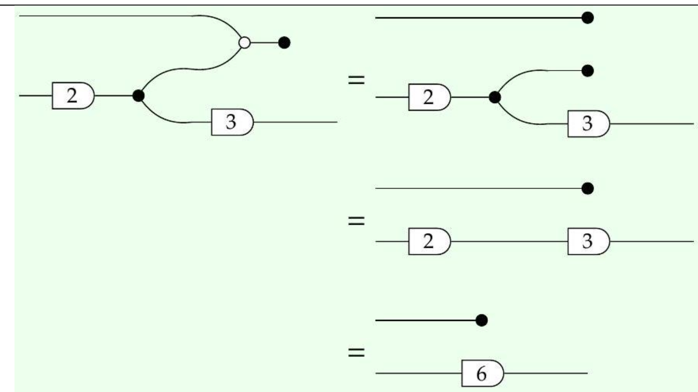
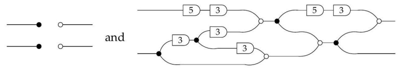
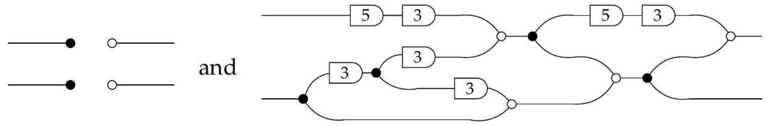
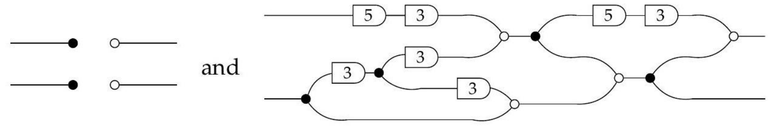
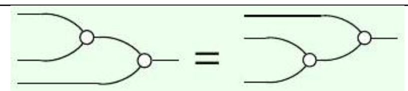
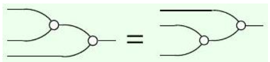
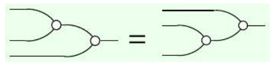
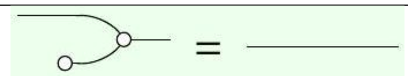
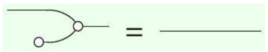
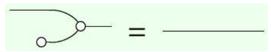
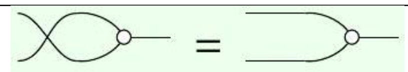
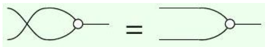
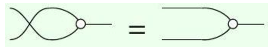
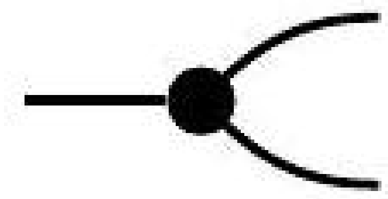
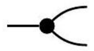
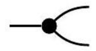
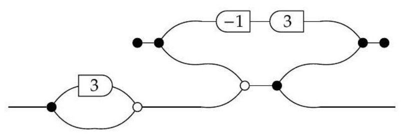
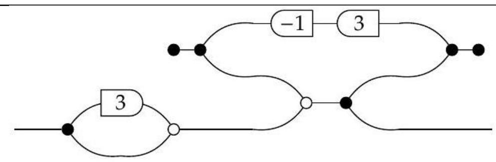
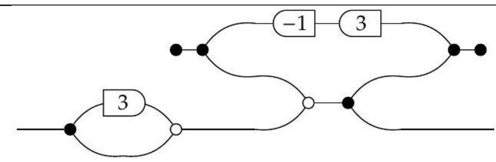
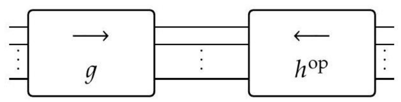
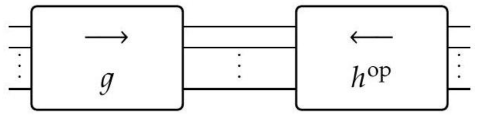
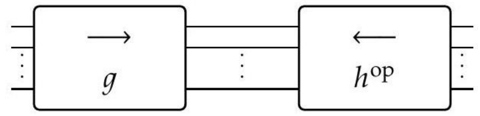
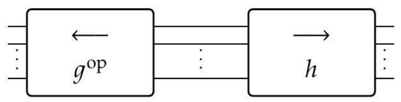
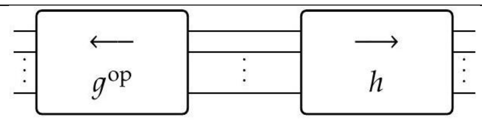
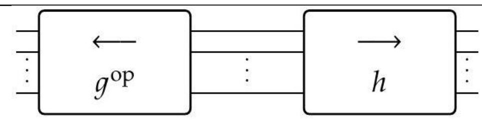
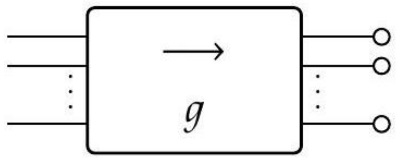
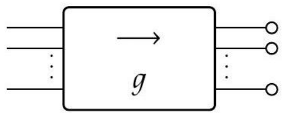
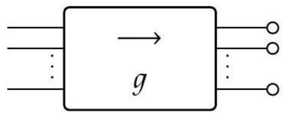
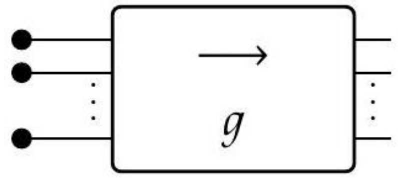
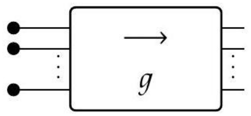
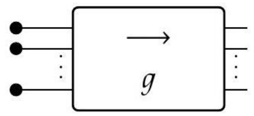
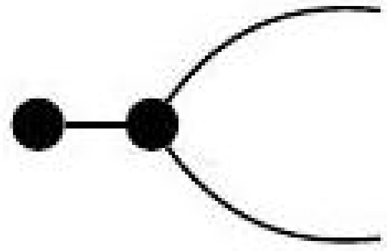
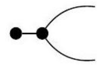
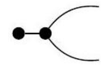
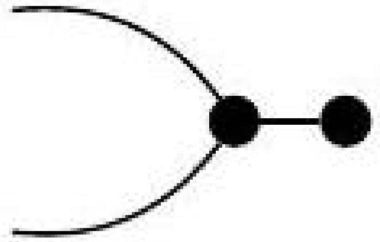
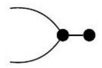
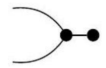
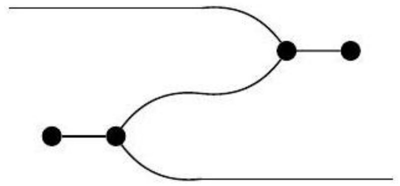
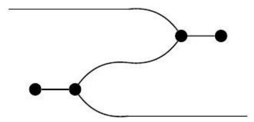
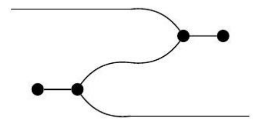
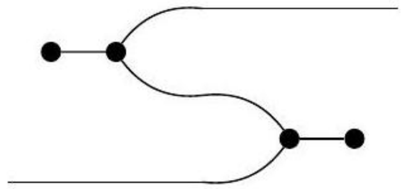
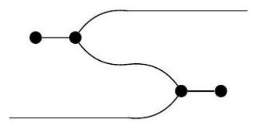
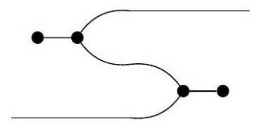
fong-spivak-seven-sketches_DIA_0213	179	— *(dup)*	7f1e85c6-4350-401f-a960-eb8a0176e786-179_170_853_337_633.jpg			

fong-spivak-seven-sketches_DIA_0214	179	— *(dup)*	7f1e85c6-4350-401f-a960-eb8a0176e786-179_344_507_1434_805.jpg			
fong-spivak-seven-sketches_DIA_0215	179	— *(dup)*	7f1e85c6-4350-401f-a960-eb8a0176e786-179_88_186_2328_1014.jpg			
fong-spivak-seven-sketches_DIA_0216	180	— *(dup)*	7f1e85c6-4350-401f-a960-eb8a0176e786-180_88_188_337_1014.jpg			
fong-spivak-seven-sketches_DIA_0217	181	— *(dup)*	7f1e85c6-4350-401f-a960-eb8a0176e786-181_244_369_1138_876.jpg			
fong-spivak-seven-sketches_DIA_0218	182	— *(dup)*	7f1e85c6-4350-401f-a960-eb8a0176e786-182_916_1396_663_364.jpg			
fong-spivak-seven-sketches_DIA_0219	183	— *(dup)*	7f1e85c6-4350-401f-a960-eb8a0176e786-183_172_428_520_636.jpg			
fong-spivak-seven-sketches_DIA_0220	183	— *(dup)*	7f1e85c6-4350-401f-a960-eb8a0176e786-183_32_156_548_1195.jpg			
fong-spivak-seven-sketches_DIA_0221	183	— *(dup)*	7f1e85c6-4350-401f-a960-eb8a0176e786-183_48_283_611_1197.jpg			

fong-spivak-seven-sketches_DIA_0222	183	— (dup)	7f1e85c6-4350-401f-a960-eb8a0176e786-183_502_889_864_615.jpg			
fong-spivak-seven-sketches_DIA_0223	183	— (dup)	7f1e85c6-4350-401f-a960-eb8a0176e786-183_208_1262_1873_382.jpg			
fong-spivak-seven-sketches_DIA_0224	184	— (dup)	7f1e85c6-4350-401f-a960-eb8a0176e786-184_75_344_2115_441.jpg			
fong-spivak-seven-sketches_DIA_0225	184	— (dup)	7f1e85c6-4350-401f-a960-eb8a0176e786-184_61_342_2217_441.jpg			
fong-spivak-seven-sketches_DIA_0226	184	— (dup)	7f1e85c6-4350-401f-a960-eb8a0176e786-184_57_342_2303_441.jpg			
fong-spivak-seven-sketches_DIA_0227	186	— (dup)	7f1e85c6-4350-401f-a960-eb8a0176e786-186_66_115_1061_1002.jpg			
fong-spivak-seven-sketches_DIA_0228	187	— (dup)	7f1e85c6-4350-401f-a960-eb8a0176e786-187_206_633_351_742.jpg			
fong-spivak-seven-sketches_DIA_0229	188	— (dup)	7f1e85c6-4350-401f-a960-eb8a0176e786-188_149_615_389_751.jpg			
fong-spivak-seven-sketches_DIA_0230	188	— (dup)	7f1e85c6-4350-401f-a960-eb8a0176e786-188_147_613_919_753.jpg			
fong-spivak-seven-sketches_DIA_0231	188	— (dup)	7f1e85c6-4350-401f-a960-eb8a0176e786-188_145_362_1805_928.jpg			
fong-spivak-seven-sketches_DIA_0232	188	— (dup)	7f1e85c6-4350-401f-a960-eb8a0176e786-188_145_321_2145_946.jpg			

fong-spivak-seven-sketches_OIA_0233	189	—	*(dup)*	7f1e85c6-4350-401f-a960-eb8a0176e786-189_82_119_1700_756.jpg			
fong-spivak-seven-sketches_OIA_0234	189	—	*(dup)*	7f1e85c6-4350-401f-a960-eb8a0176e786-189_82_122_1700_1240.jpg			
fong-spivak-seven-sketches_OIA_0235	189	—	*(dup)*	7f1e85c6-4350-401f-a960-eb8a0176e786-189_154_318_2209_478.jpg			
fong-spivak-seven-sketches_OIA_0236	189	—	*(dup)*	7f1e85c6-4350-401f-a960-eb8a0176e786-189_154_313_2212_938.jpg			
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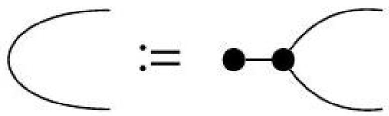
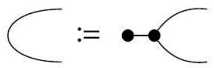
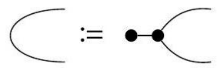
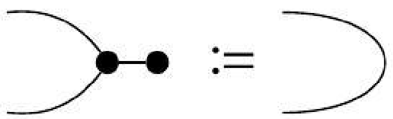
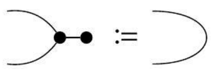
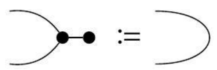
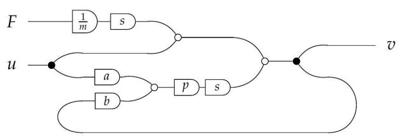
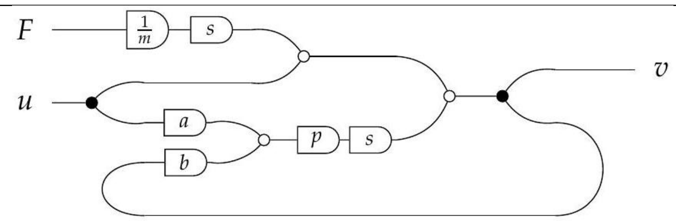
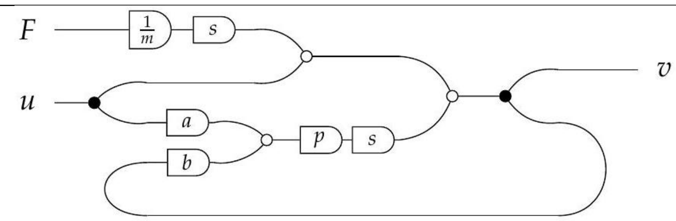
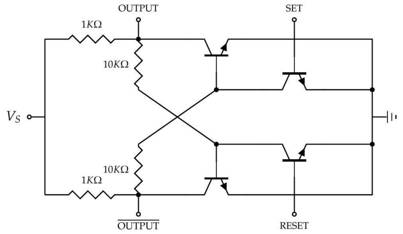
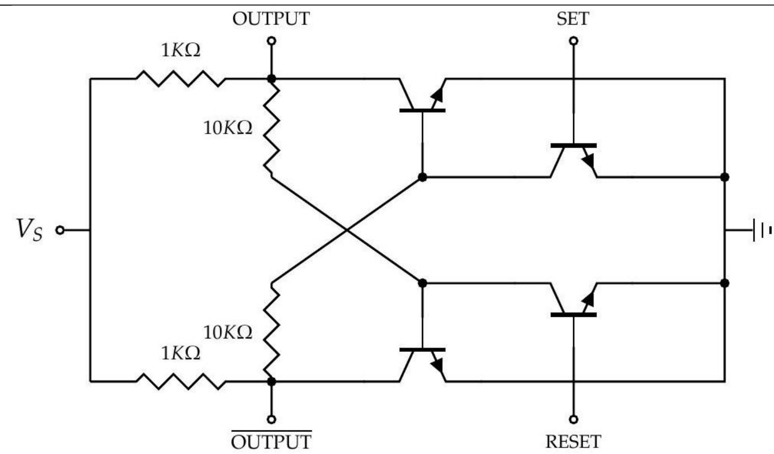
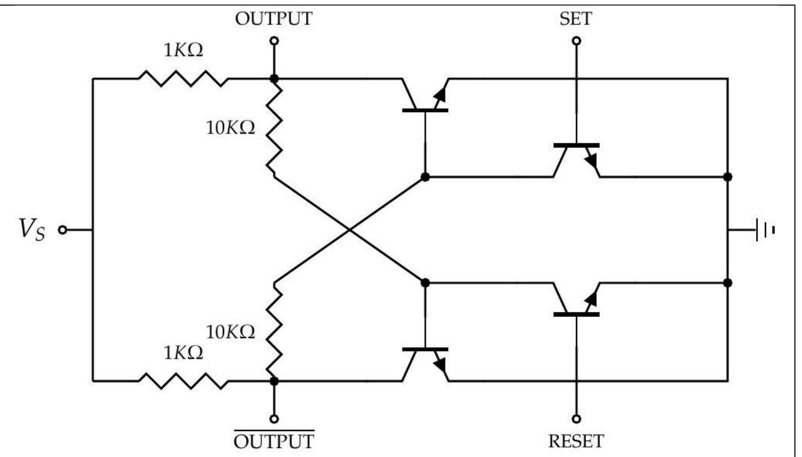
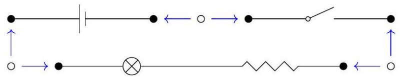
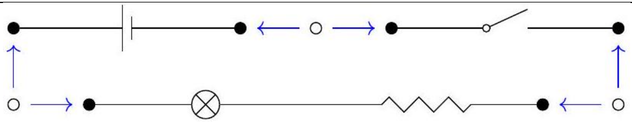
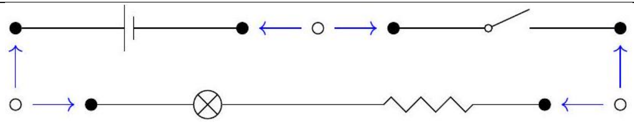
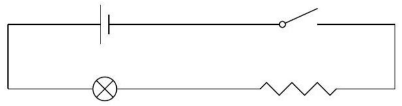
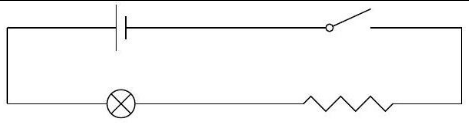
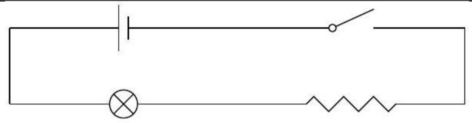
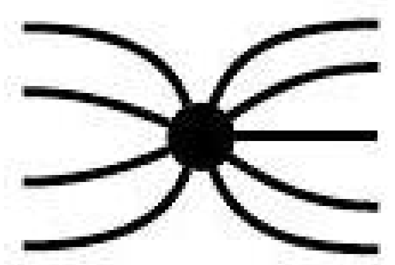
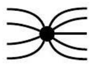
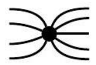
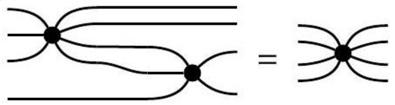
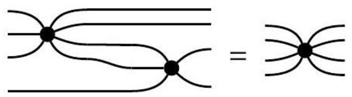
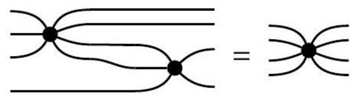
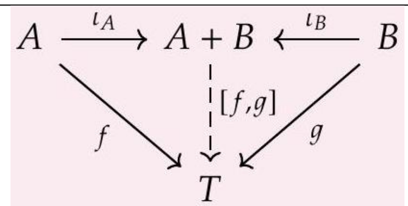
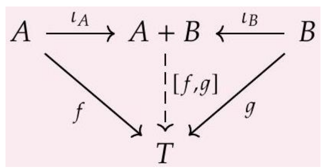
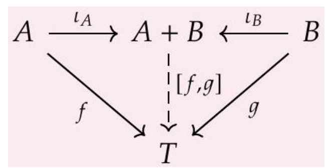
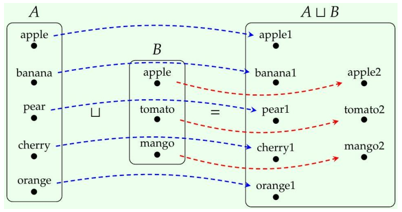
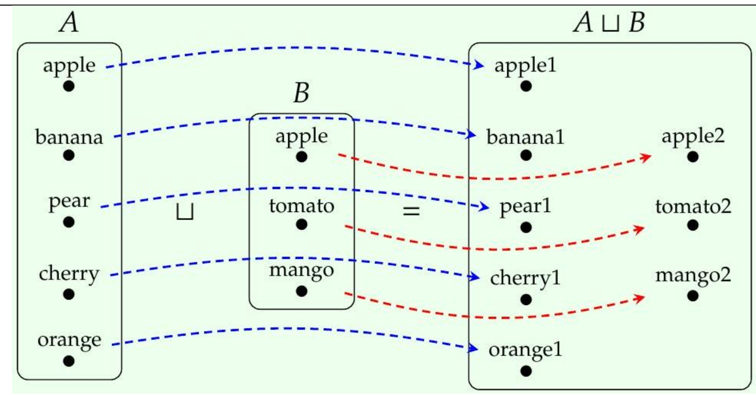
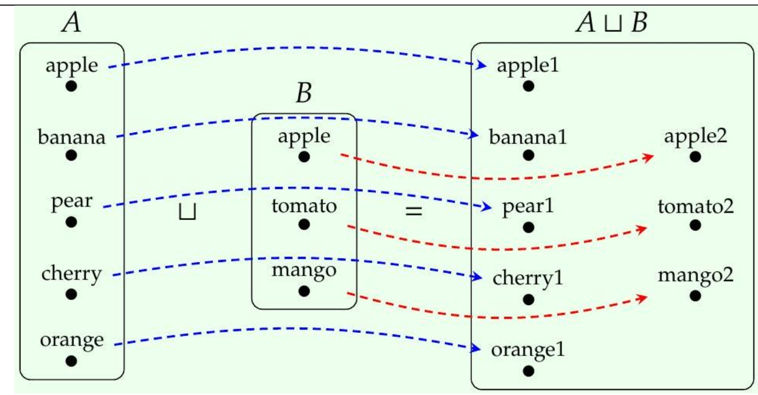
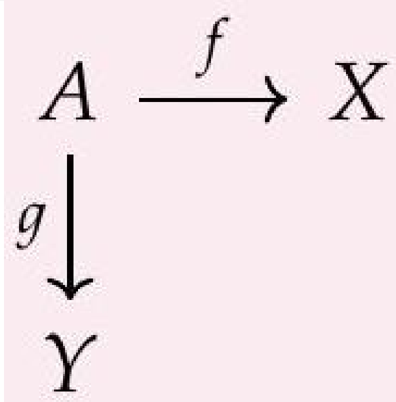
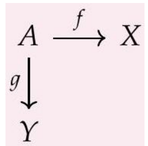
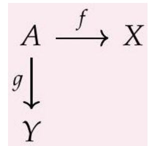
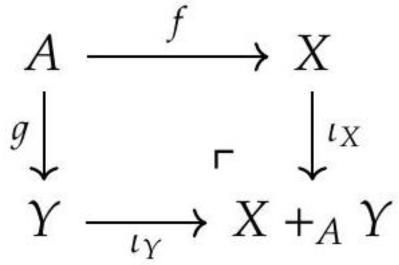
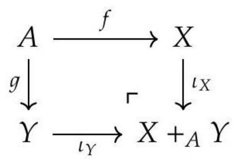
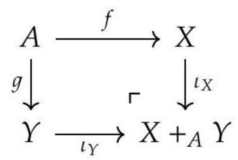
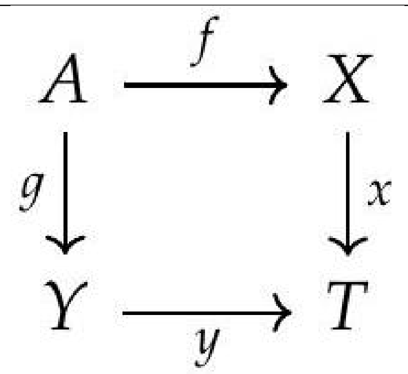
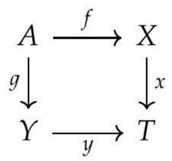
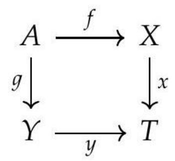
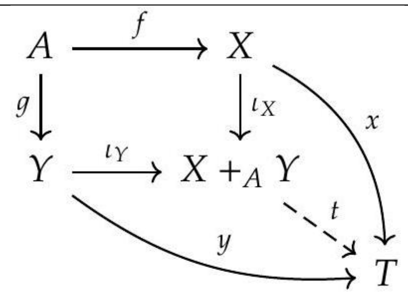
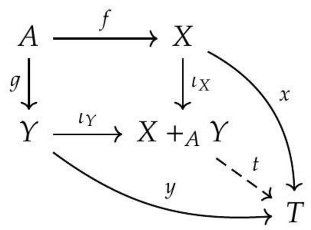
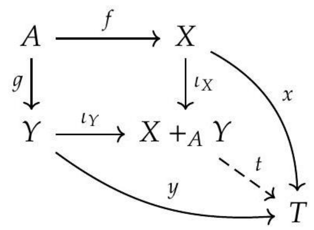
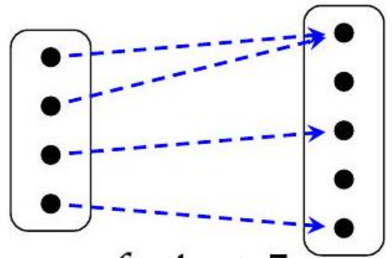
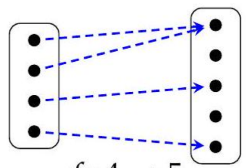
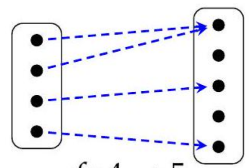
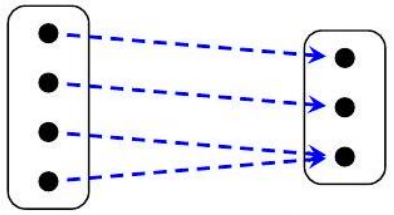
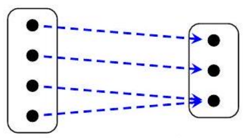
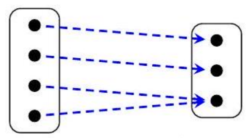
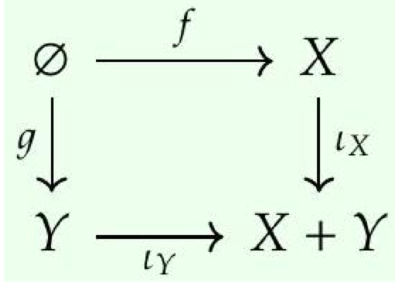
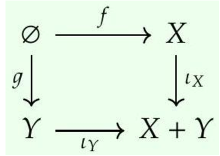
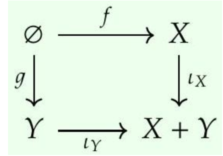
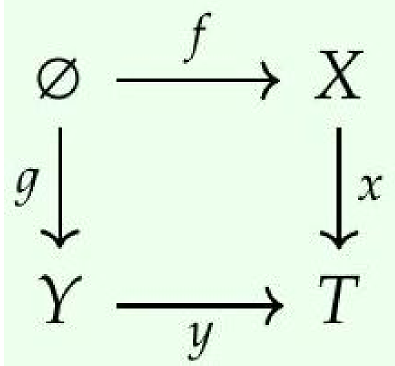
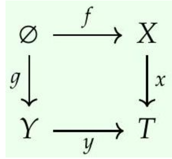
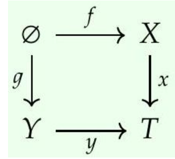
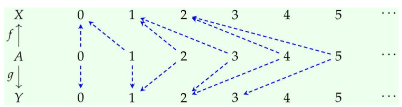
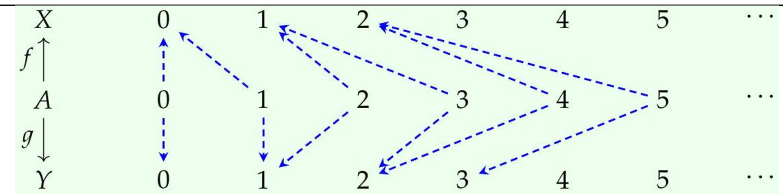
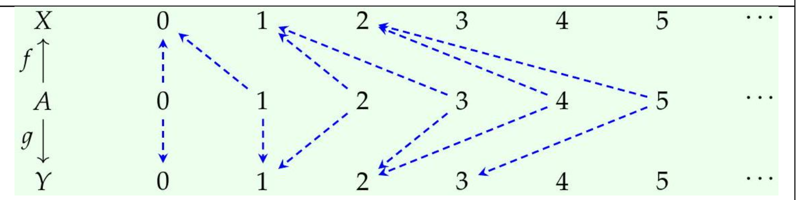
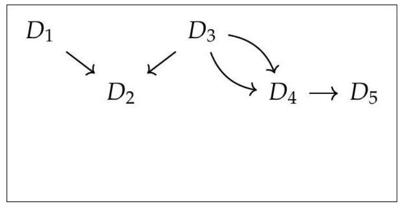
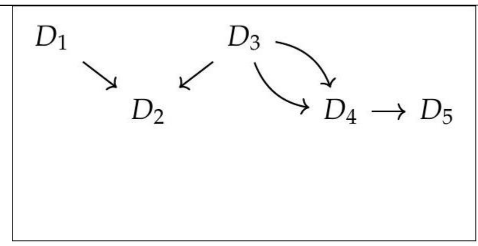
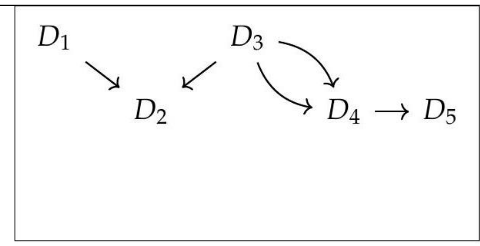
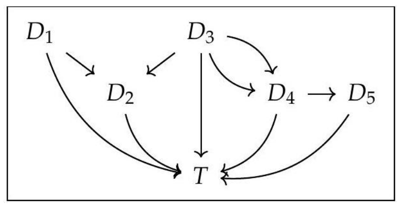
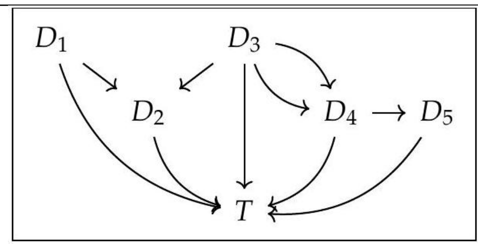
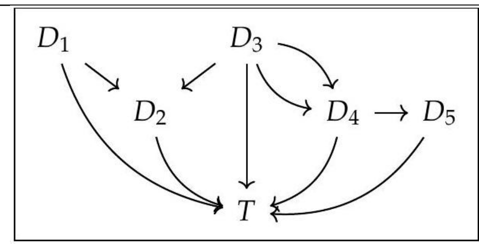
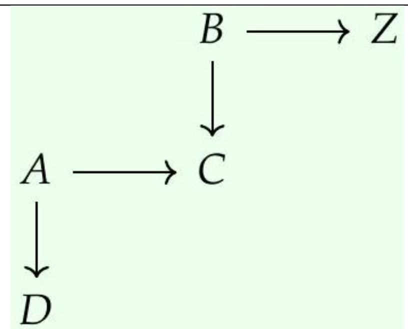
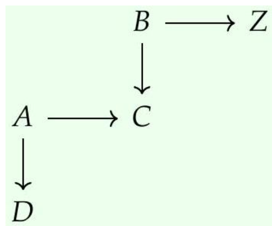
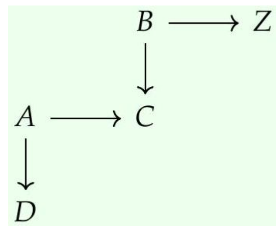
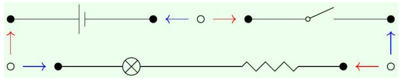
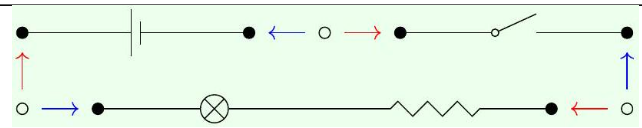
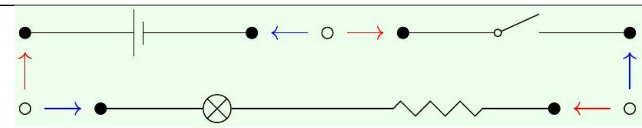
fong-spivak-seven-sketches_OIA_0238	190	—	*(dup)*	7f1e85c6-4350-401f-a960-eb8a0176e786-190_88_274_520_606.jpg			
fong-spivak-seven-sketches_OIA_0239	190	—	*(dup)*	7f1e85c6-4350-401f-a960-eb8a0176e786-190_86_271_520_1240.jpg			
fong-spivak-seven-sketches_OIA_0240	190	—	*(dup)*	7f1e85c6-4350-401f-a960-eb8a0176e786-190_281_860_1231_629.jpg			

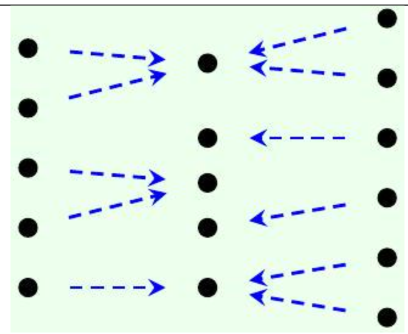
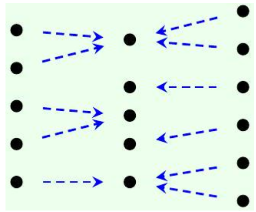
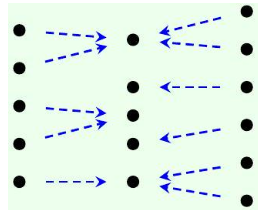
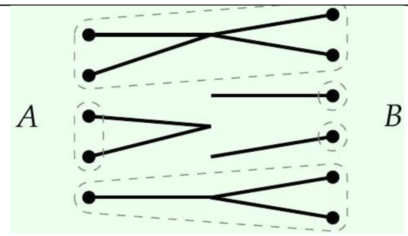
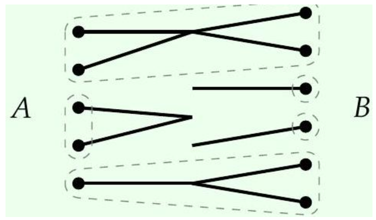
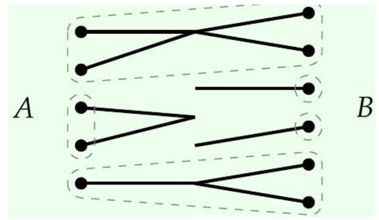
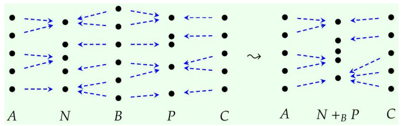
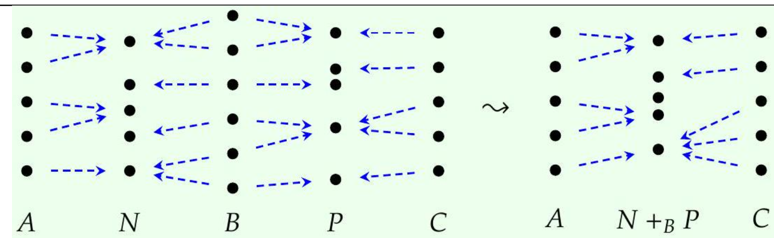
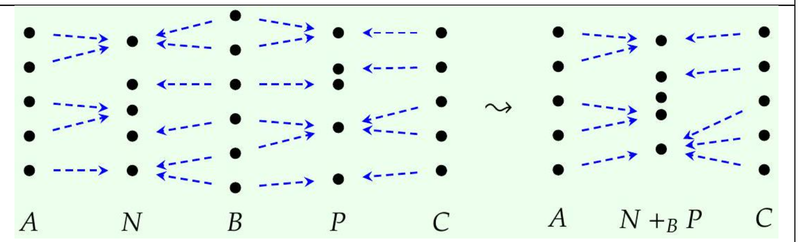
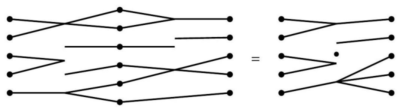
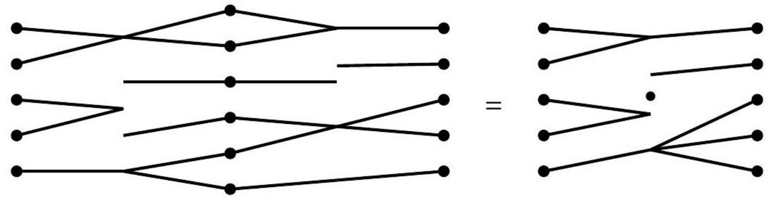
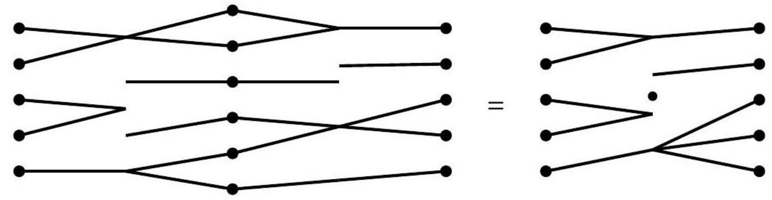
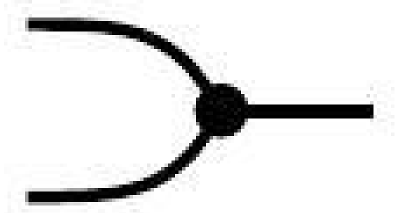
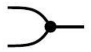
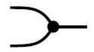
fong-spivak-seven-sketches_DIA_0241	193	— (dup)	7f1e85c6-4350-401f-a960-eb8a0176e786-193_583_995_1442_569.jpg			
fong-spivak-seven-sketches_DIA_0242	194	— (dup)	7f1e85c6-4350-401f-a960-eb8a0176e786-194_163_771_993_674.jpg			
fong-spivak-seven-sketches_DIA_0243	194	— (dup)	7f1e85c6-4350-401f-a960-eb8a0176e786-194_158_814_1769_654.jpg			
fong-spivak-seven-sketches_DIA_0244	194	— (dup)	7f1e85c6-4350-401f-a960-eb8a0176e786-194_154_817_2213_652.jpg			
fong-spivak-seven-sketches_DIA_0245	195	— (dup)	7f1e85c6-4350-401f-a960-eb8a0176e786-195_158_602_385_758.jpg			
fong-spivak-seven-sketches_DIA_0246	195	— (dup)	7f1e85c6-4350-401f-a960-eb8a0176e786-195_81_115_2176_1002.jpg			
fong-spivak-seven-sketches_DIA_0247	196	— (dup)	7f1e85c6-4350-401f-a960-eb8a0176e786-196_122_455_387_833.jpg			
fong-spivak-seven-sketches_DIA_0248	198	— (dup)	7f1e85c6-4350-401f-a960-eb8a0176e786-198_208_410_1509_853.jpg			

fong-spivak-seven-sketches_DIA_0249	199	—	*(dup)*	7f1e85c6-4350-401f-a960-eb8a0176e786-199_505_968_337_575.jpg		 
fong-spivak-seven-sketches_DIA_0250	200	—	*(dup)*	7f1e85c6-4350-401f-a960-eb8a0176e786-200_186_181_1014_964.jpg		 
fong-spivak-seven-sketches_DIA_0251	200	—	*(dup)*	7f1e85c6-4350-401f-a960-eb8a0176e786-200_199_294_1439_955.jpg		 
fong-spivak-seven-sketches_DIA_0252	200	—	*(dup)*	7f1e85c6-4350-401f-a960-eb8a0176e786-200_204_215_1805_1000.jpg		 
fong-spivak-seven-sketches_DIA_0253	200	—	*(dup)*	7f1e85c6-4350-401f-a960-eb8a0176e786-200_292_396_2136_905.jpg		 

fong-spivak-seven-sketches_DIA_0254	201	— *(dup)*	7f1e85c6-4350-401f-a960-eb8a0176e786-201_210_217_878_950.jpg	$\begin{tikzcd} A \arrow[r, "f"] \arrow[d, "i"'] & X \arrow[d, "j"] \\ A' \arrow[r, "f'"'] \arrow[ur, phantom, "\ulcorner" very near end] & X' \end{tikzcd}$	$\begin{tikzcd} A \arrow[r, "f"] \arrow[d, "i"'] & X \arrow[d, "j"] \\ A' \arrow[r, "f'"'] \arrow[ur, phantom, "\ulcorner" very near end] & X' \end{tikzcd}$	$\begin{tikzcd} A \arrow[r, "f"] \arrow[d, "i"'] & X \arrow[d, "j"] \\ A' \arrow[r, "f'"'] \arrow[ur, phantom, "\ulcorner" very near end] & X' \end{tikzcd}$
fong-spivak-seven-sketches_DIA_0255	201	— *(dup)*	7f1e85c6-4350-401f-a960-eb8a0176e786-201_197_217_1222_955.jpg	$\begin{tikzcd} A \arrow[r, "f"] \arrow[d, "i"'] & X \arrow[d, "x"] \\ A' \arrow[r, "a"'] & T \end{tikzcd}$	$\begin{tikzcd} A \arrow[r, "f"] \arrow[d, "i"'] & X \arrow[d, "x"] \\ A' \arrow[r, "a"'] & T \end{tikzcd}$	$\begin{tikzcd} A \arrow[r, "f"] \arrow[d, "i"'] & X \arrow[d, "x"] \\ A' \arrow[r, "a"'] & T \end{tikzcd}$
fong-spivak-seven-sketches_DIA_0256	201	— *(dup)*	7f1e85c6-4350-401f-a960-eb8a0176e786-201_287_314_1588_903.jpg	$\begin{tikzcd} A \arrow[r, "f"] \arrow[d, "i"'] & X \arrow[d, "j"] \arrow[ddr, bend left, "x"] & \\ A' \arrow[r, "f'"'] \arrow[drr, bend right, "a"'] & X' \arrow[dr, dashed, "x'"] & \\ & & T \end{tikzcd}$	$\begin{tikzcd} A \arrow[r, "f"] \arrow[d, "i"'] & X \arrow[d, "j"] \arrow[ddr, bend left, "x"] & \\ A' \arrow[r, "f'"'] \arrow[drr, bend right, "a"'] & X' \arrow[dr, dashed, "x'"] & \\ & & T \end{tikzcd}$	$\begin{tikzcd} A \arrow[r, "f"] \arrow[d, "i"'] & X \arrow[d, "j"] \arrow[ddr, bend left, "x"] & \\ A' \arrow[r, "f'"'] \arrow[drr, bend right, "a"'] & X' \arrow[dr, dashed, "x'"] & \\ & & T \end{tikzcd}$
fong-spivak-seven-sketches_DIA_0257	202	— *(dup)*	7f1e85c6-4350-401f-a960-eb8a0176e786-202_251_317_588_570.jpg	 $f\colon \underline{4} \to \underline{5}$	 $f\colon \underline{4} \to \underline{5}$	 $f\colon \underline{4} \to \underline{5}$
fong-spivak-seven-sketches_DIA_0258	202	— *(dup)*	7f1e85c6-4350-401f-a960-eb8a0176e786-202_224_314_609_1233.jpg	 $g\colon \underline{4} \to \underline{3}$	 $g\colon \underline{4} \to \underline{3}$	 $g\colon \underline{4} \to \underline{3}$

fong-spivak-seven-sketch	202	0259 — *(dup)*	7f1e85c6-4350-401f-a960-eb8a0176e786-202_201_278_1348_679.jpg			
fong-spivak-seven-sketch	202	0260 — *(dup)*	7f1e85c6-4350-401f-a960-eb8a0176e786-202_204_217_1348_1222.jpg			
fong-spivak-seven-sketch	203	0261 — *(dup)*	7f1e85c6-4350-401f-a960-eb8a0176e786-203_278_1129_342_491.jpg			
fong-spivak-seven-sketch	203	0262 — *(dup)*	7f1e85c6-4350-401f-a960-eb8a0176e786-203_308_609_2000_355.jpg			
fong-spivak-seven-sketch	203	0263 — *(dup)*	7f1e85c6-4350-401f-a960-eb8a0176e786-203_310_613_1998_1154.jpg			
fong-spivak-seven-sketch	204	0264 — *(dup)*	7f1e85c6-4350-401f-a960-eb8a0176e786-204_287_347_1666_884.jpg			

fong-spivak-seven-sketches_DIA_0265	204	—	*(dup)*	7f1e85c6-4350-401f-a960-eb8a0176e786-204_293_350_2161_572.jpg	$\begin{tikzcd} & B \arrow[r] \arrow[d] & Z \arrow[d] \\ A \arrow[r] \arrow[d] & Y \arrow[r] \arrow[d] \arrow[ur, phantom, "\ulcorner", very near start] & R \\ X \arrow[r] & Q \arrow[ul, phantom, "\ulcorner", very near start] & \end{tikzcd}$	$\begin{tikzcd} & B \arrow[r] \arrow[d] & Z \arrow[d] \\ A \arrow[r] \arrow[d] & Y \arrow[r] \arrow[d] \arrow[ur, phantom, "\ulcorner", very near start] & R \\ X \arrow[r] & Q \arrow[ul, phantom, "\ulcorner", very near start] & \end{tikzcd}$	$\begin{tikzcd} & B \arrow[r] \arrow[d] & Z \arrow[d] \\ A \arrow[r] \arrow[d] & Y \arrow[r] \arrow[d] \arrow[ur, phantom, "\ulcorner", very near start] & R \\ X \arrow[r] & Q \arrow[ul, phantom, "\ulcorner", very near start] & \end{tikzcd}$
fong-spivak-seven-sketches_DIA_0266	204	—	*(dup)*	7f1e85c6-4350-401f-a960-eb8a0176e786-204_293_353_2161_1194.jpg	$\begin{tikzcd} & B \arrow[r] \arrow[d] & Z \arrow[d] \\ A \arrow[r] \arrow[d] & Y \arrow[r] \arrow[d] & R \arrow[d] \\ X \arrow[r] & Q \arrow[r] & S \end{tikzcd}$	$\begin{tikzcd} & B \arrow[r] \arrow[d] & Z \arrow[d] \\ A \arrow[r] \arrow[d] & Y \arrow[r] \arrow[d] & R \arrow[d] \\ X \arrow[r] & Q \arrow[r] & S \end{tikzcd}$	$\begin{tikzcd} & B \arrow[r] \arrow[d] & Z \arrow[d] \\ A \arrow[r] \arrow[d] & Y \arrow[r] \arrow[d] & R \arrow[d] \\ X \arrow[r] & Q \arrow[r] & S \end{tikzcd}$
fong-spivak-seven-sketches_DIA_0267	206	—	*(dup)*	7f1e85c6-4350-401f-a960-eb8a0176e786-206_158_817_1321_652.jpg			
fong-spivak-seven-sketches_DIA_0268	207	—	*(dup)*	7f1e85c6-4350-401f-a960-eb8a0176e786-207_131_351_394_593.jpg	$\begin{tikzcd} & N & \\ A \arrow[ur, "f"] & & B \arrow[ul, "g"'] \end{tikzcd}$	$\begin{tikzcd} & N & \\ A \arrow[ur, "f"] & & B \arrow[ul, "g"'] \end{tikzcd}$	$\begin{tikzcd} & N & \\ A \arrow[ur, "f"] & & B \arrow[ul, "g"'] \end{tikzcd}$
fong-spivak-seven-sketches_DIA_0269	207	—	*(dup)*	7f1e85c6-4350-401f-a960-eb8a0176e786-207_131_337_394_1183.jpg	$\begin{tikzcd} & P & \\ B \arrow[ur, "h"] & & C \arrow[ul, "k"'] \end{tikzcd}$	$\begin{tikzcd} & P & \\ B \arrow[ur, "h"] & & C \arrow[ul, "k"'] \end{tikzcd}$	$\begin{tikzcd} & P & \\ B \arrow[ur, "h"] & & C \arrow[ul, "k"'] \end{tikzcd}$
fong-spivak-seven-sketches_DIA_0270	207	—	*(dup)*	7f1e85c6-4350-401f-a960-eb8a0176e786-207_133_652_699_733.jpg	$\begin{tikzcd} & N & & P & \\ A \arrow[ur, "f"] & & B \arrow[ul, "g"'] \arrow[ur, "h"] & & C \arrow[ul, "k"'] \end{tikzcd}$	$\begin{tikzcd} & N & & P & \\ A \arrow[ur, "f"] & & B \arrow[ul, "g"'] \arrow[ur, "h"] & & C \arrow[ul, "k"'] \end{tikzcd}$	$\begin{tikzcd} & N & & P & \\ A \arrow[ur, "f"] & & B \arrow[ul, "g"'] \arrow[ur, "h"] & & C \arrow[ul, "k"'] \end{tikzcd}$
fong-spivak-seven-sketches_DIA_0271	207	—	*(dup)*	7f1e85c6-4350-401f-a960-eb8a0176e786-207_231_1276_1027_439.jpg	$\begin{tikzcd} & & N +_B P & & \\ & N \arrow[ur, "\iota_N"] & & P \arrow[ul, "\iota_P"'] & \\ A \arrow[ur, "f"] & & B \arrow[ul, "g"'] \arrow[ur, "h"] \arrow[uu, phantom, "\vee", near end] & & C \arrow[ul, "k"'] \end{tikzcd} \rightsquigarrow \begin{tikzcd} & N +_B P & \\ A \arrow[ur, "f ; \iota_N"] & & C \arrow[ul, "k ; \iota_P"'] \end{tikzcd}$	$\begin{tikzcd} & & N +_B P & & \\ & N \arrow[ur, "\iota_N"] & & P \arrow[ul, "\iota_P"'] & \\ A \arrow[ur, "f"] & & B \arrow[ul, "g"'] \arrow[ur, "h"] \arrow[uu, phantom, "\vee", near end] & & C \arrow[ul, "k"'] \end{tikzcd} \rightsquigarrow \begin{tikzcd} & N +_B P & \\ A \arrow[ur, "f ; \iota_N"] & & C \arrow[ul, "k ; \iota_P"'] \end{tikzcd}$	$\begin{tikzcd} & & N +_B P & & \\ & N \arrow[ur, "\iota_N"] & & P \arrow[ul, "\iota_P"'] & \\ A \arrow[ur, "f"] & & B \arrow[ul, "g"'] \arrow[ur, "h"] \arrow[uu, phantom, "\vee", near end] & & C \arrow[ul, "k"'] \end{tikzcd} \rightsquigarrow \begin{tikzcd} & N +_B P & \\ A \arrow[ur, "f ; \iota_N"] & & C \arrow[ul, "k ; \iota_P"'] \end{tikzcd}$
fong-spivak-seven-sketches_DIA_0272	207	—	*(dup)*	7f1e85c6-4350-401f-a960-eb8a0176e786-207_156_260_1993_661.jpg	$\begin{tikzcd} & P & \\ A \arrow[ur] \arrow[dr] & & B \arrow[ul] \arrow[dl] \\ & P' & \end{tikzcd}$	$\begin{tikzcd} & P & \\ A \arrow[ur] \arrow[dr] & & B \arrow[ul] \arrow[dl] \\ & P' & \end{tikzcd}$	$\begin{tikzcd} & P & \\ A \arrow[ur] \arrow[dr] & & B \arrow[ul] \arrow[dl] \\ & P' & \end{tikzcd}$
fong-spivak-seven-sketches_DIA_0273	207	—	*(dup)*	7f1e85c6-4350-401f-a960-eb8a0176e786-207_152_258_1995_1195.jpg	$\begin{tikzcd} & P \arrow[dd, "\cong"] & \\ A \arrow[ur] \arrow[dr] & & B \arrow[ul] \arrow[dl] \\ & P' & \end{tikzcd}$	$\begin{tikzcd} & P \arrow[dd, "\cong"] & \\ A \arrow[ur] \arrow[dr] & & B \arrow[ul] \arrow[dl] \\ & P' & \end{tikzcd}$	$\begin{tikzcd} & P \arrow[dd, "\cong"] & \\ A \arrow[ur] \arrow[dr] & & B \arrow[ul] \arrow[dl] \\ & P' & \end{tikzcd}$

fong-spivak-seven-sketches_208_0274	208	— *(dup)*	7f1e85c6-4350-401f-a960-eb8a0176e786-208_269_324_1328_520.jpg			
fong-spivak-seven-sketches_208_0275	208	— *(dup)*	7f1e85c6-4350-401f-a960-eb8a0176e786-208_276_477_1355_1122.jpg			
fong-spivak-seven-sketches_209_0276	209	— *(dup)*	7f1e85c6-4350-401f-a960-eb8a0176e786-209_333_1095_376_466.jpg			
fong-spivak-seven-sketches_209_0277	209	— *(dup)*	7f1e85c6-4350-401f-a960-eb8a0176e786-209_276_1057_1133_532.jpg			
fong-spivak-seven-sketches_210_0278	210	— *(dup)*	7f1e85c6-4350-401f-a960-eb8a0176e786-210_63_113_536_824.jpg			
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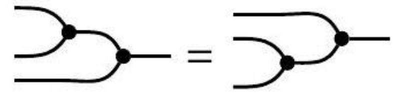
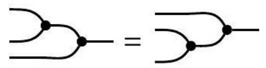
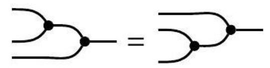
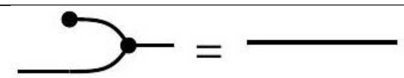
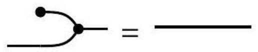
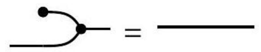
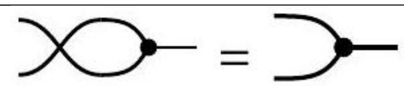
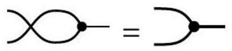
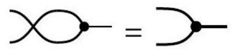
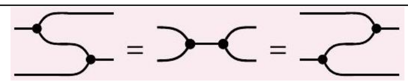
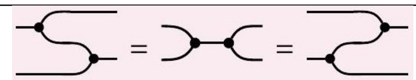
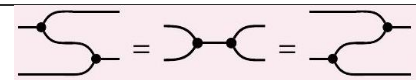
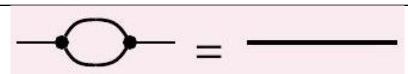
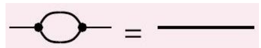
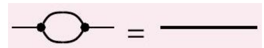
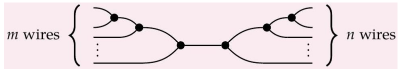
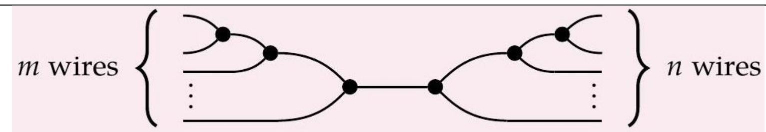
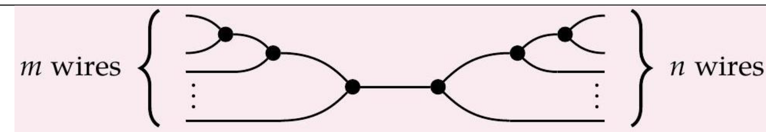
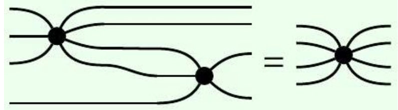
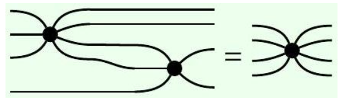
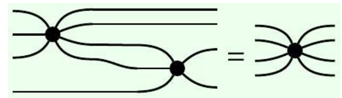
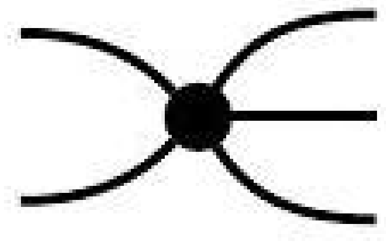
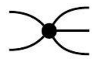
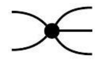
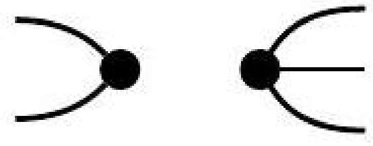
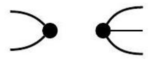
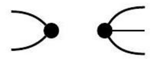
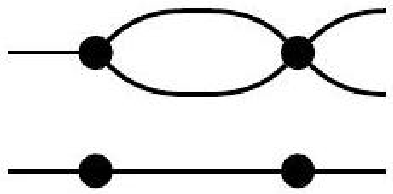
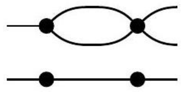
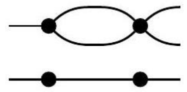
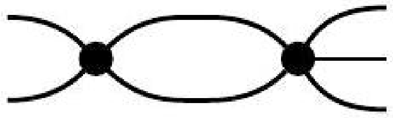
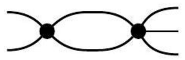
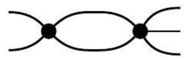
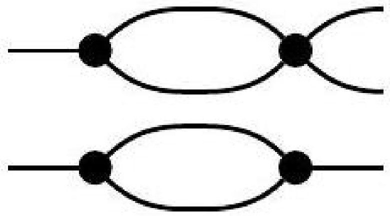
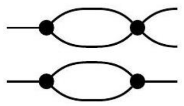
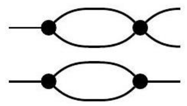
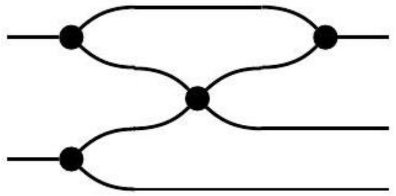
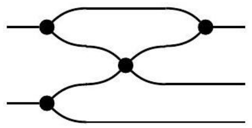
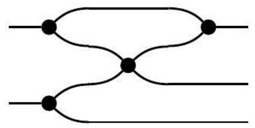
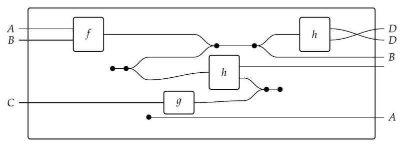
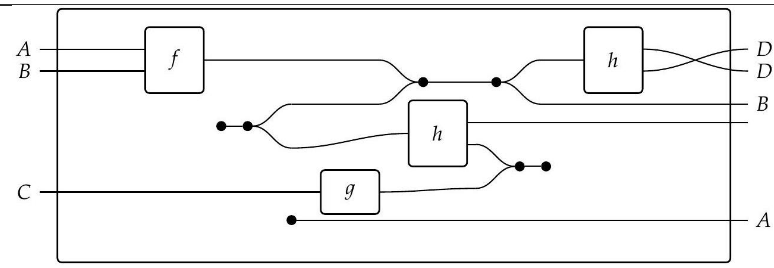
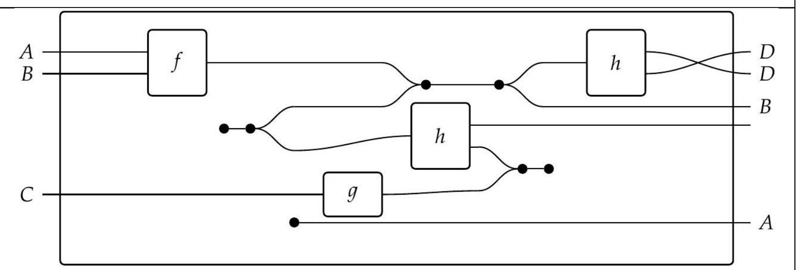
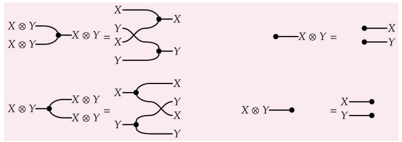
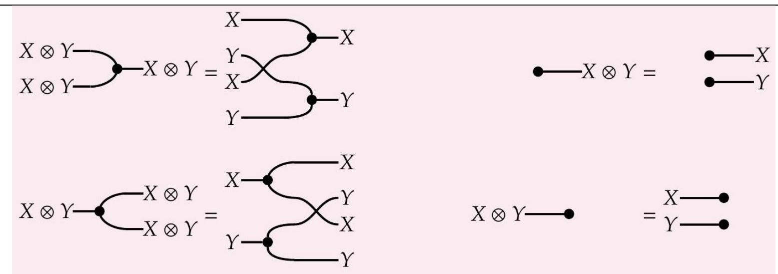
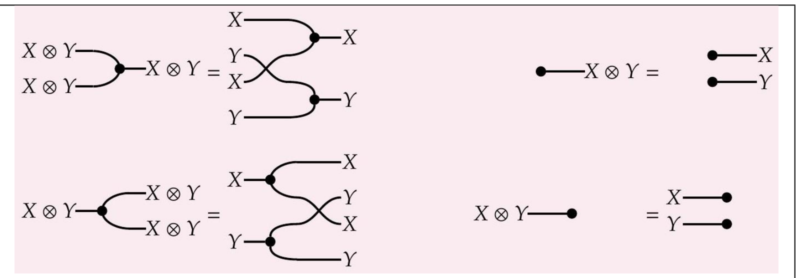
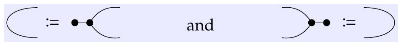
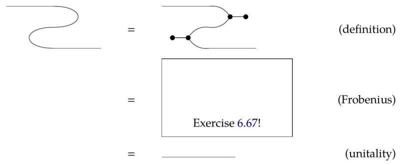
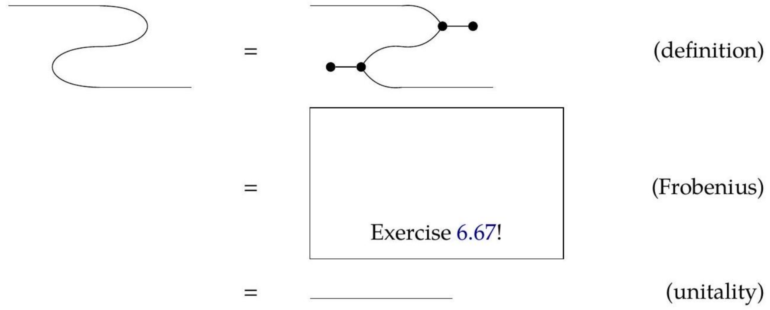
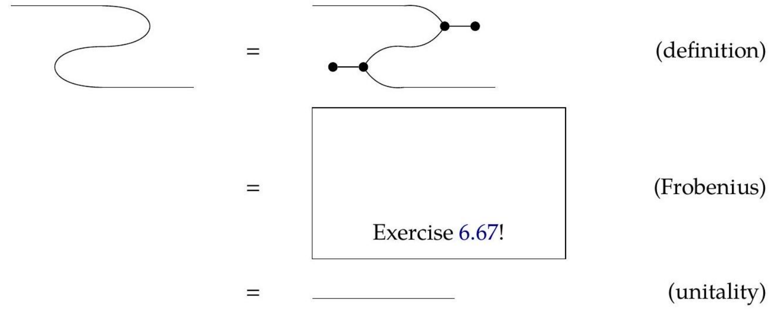
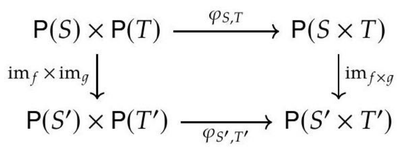
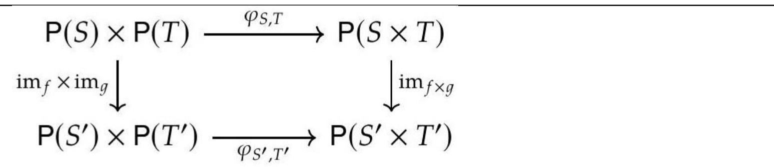
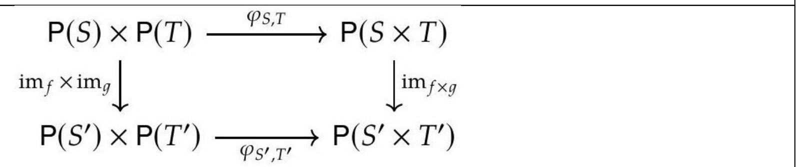
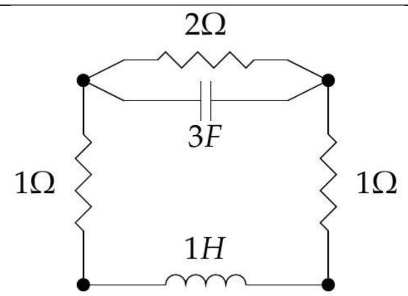
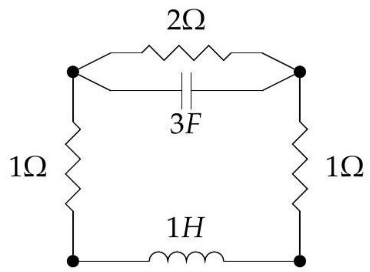
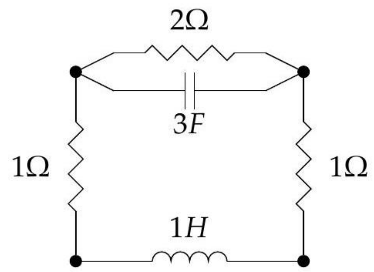
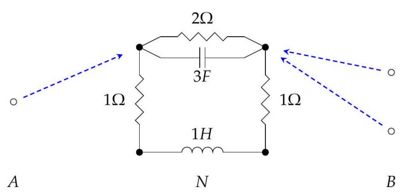
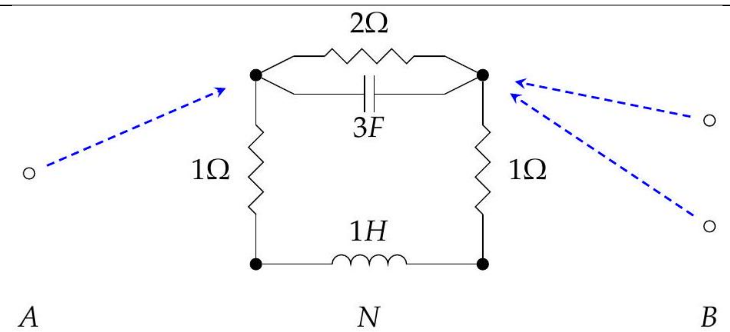
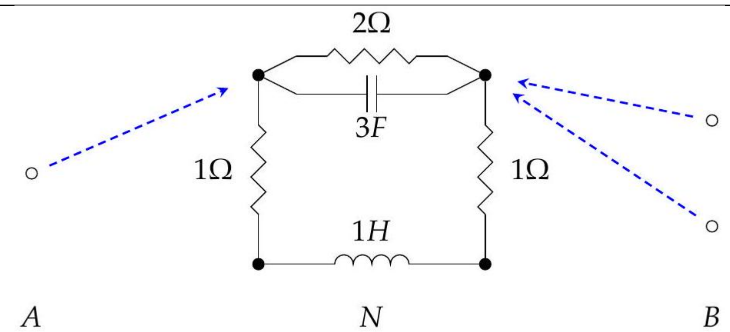
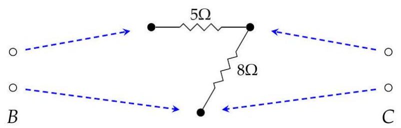
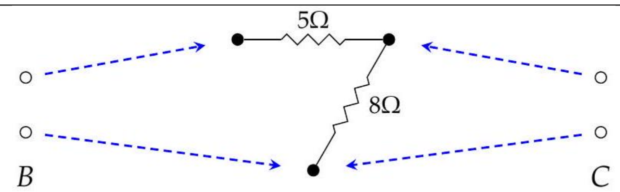
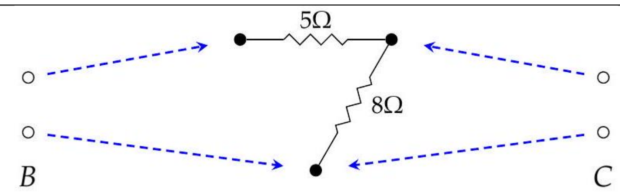
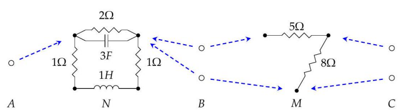
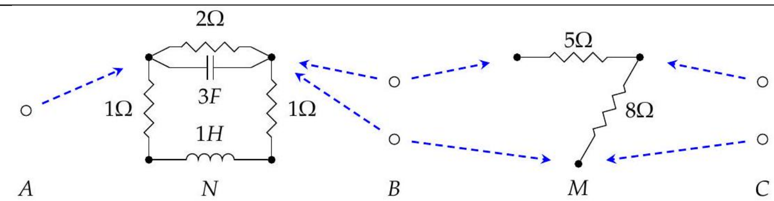
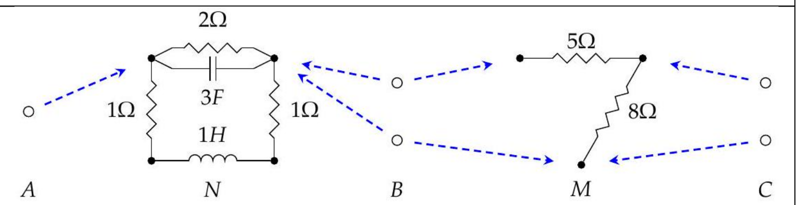
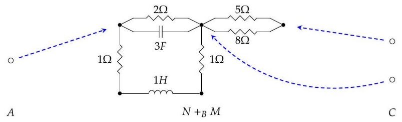
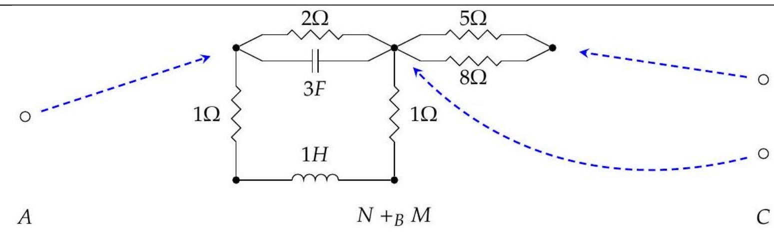
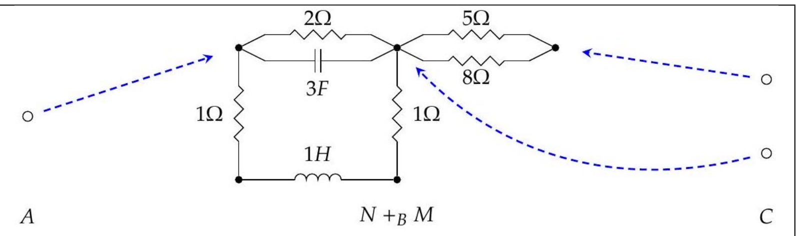
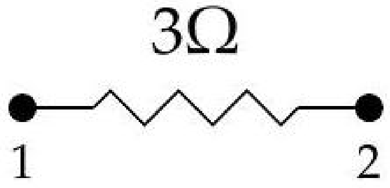
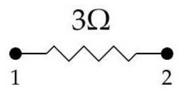
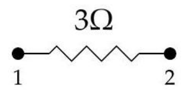
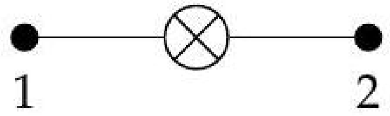
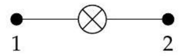
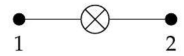
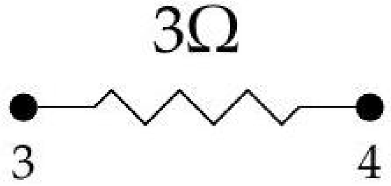
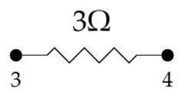
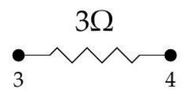
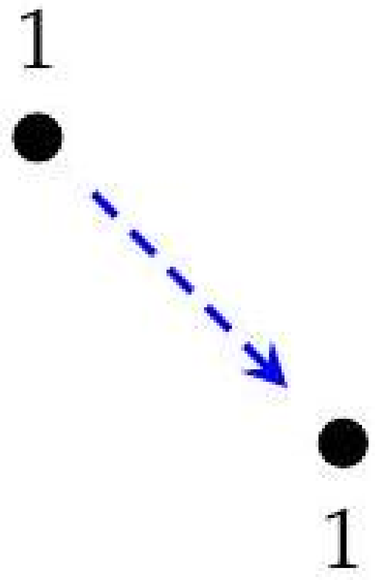
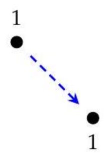
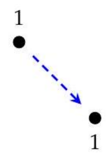
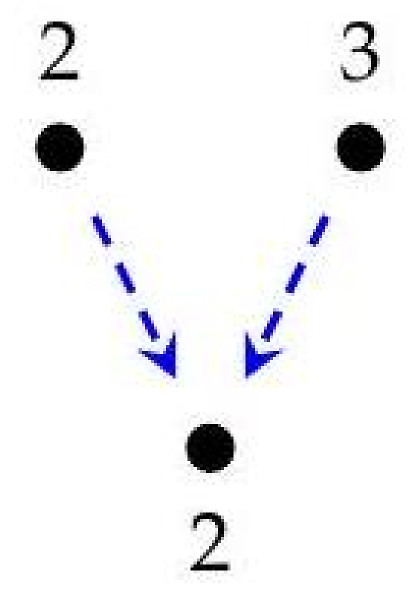
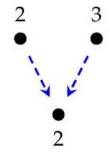
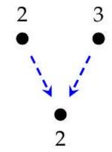
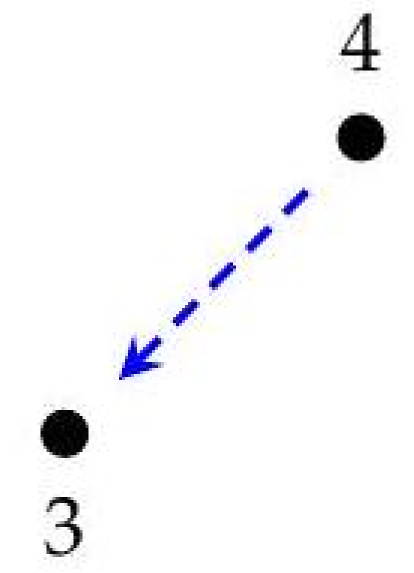
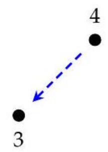
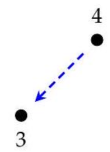
fong-spivak-seven-sketches_210_0280	210	— *(dup)*	7f1e85c6-4350-401f-a960-eb8a0176e786-210_84_342_817_425.jpg			
fong-spivak-seven-sketches_210_0281	210	— *(dup)*	7f1e85c6-4350-401f-a960-eb8a0176e786-210_66_330_828_869.jpg			
fong-spivak-seven-sketches_210_0282	210	— *(dup)*	7f1e85c6-4350-401f-a960-eb8a0176e786-210_61_294_828_1308.jpg			
fong-spivak-seven-sketches_210_0283	210	— *(dup)*	7f1e85c6-4350-401f-a960-eb8a0176e786-210_97_523_2163_579.jpg			
fong-spivak-seven-sketches_210_0284	210	— *(dup)*	7f1e85c6-4350-401f-a960-eb8a0176e786-210_59_330_2181_1208.jpg			

fong-spivak-seven-sketches_DIA_0285	211	— *(dup)*	7f1e85c6-4350-401f-a960-eb8a0176e786-211_165_980_636_568.jpg			
fong-spivak-seven-sketches_DIA_0286	211	— *(dup)*	7f1e85c6-4350-401f-a960-eb8a0176e786-211_122_432_2095_844.jpg			
fong-spivak-seven-sketches_DIA_0287	211	— *(dup)*	7f1e85c6-4350-401f-a960-eb8a0176e786-211_72_115_2432_1050.jpg			
fong-spivak-seven-sketches_DIA_0288	212	— *(dup)*	7f1e85c6-4350-401f-a960-eb8a0176e786-212_75_195_287_1005.jpg			
fong-spivak-seven-sketches_DIA_0289	212	— *(dup)*	7f1e85c6-4350-401f-a960-eb8a0176e786-212_120_231_428_991.jpg			
fong-spivak-seven-sketches_DIA_0290	212	— *(dup)*	7f1e85c6-4350-401f-a960-eb8a0176e786-212_72_231_615_991.jpg			
fong-spivak-seven-sketches_DIA_0291	212	— *(dup)*	7f1e85c6-4350-401f-a960-eb8a0176e786-212_133_233_758_991.jpg			
fong-spivak-seven-sketches_DIA_0292	212	— *(dup)*	7f1e85c6-4350-401f-a960-eb8a0176e786-212_161_321_959_946.jpg			
fong-spivak-seven-sketches_DIA_0293	213	— *(dup)*	7f1e85c6-4350-401f-a960-eb8a0176e786-213_438_1285_660_415.jpg			

fong-spivak-seven-sketches_DIA_0294	214	— (dup)	7f1e85c6-4350-401f-a960-eb8a0176e786-214_484_1382_324_348.jpg			
fong-spivak-seven-sketches_DIA_0295	215	— (dup)	7f1e85c6-4350-401f-a960-eb8a0176e786-215_88_907_925_606.jpg			
fong-spivak-seven-sketches_DIA_0296	215	— (dup)	7f1e85c6-4350-401f-a960-eb8a0176e786-215_525_1294_1181_489.jpg			
fong-spivak-seven-sketches_DIA_0297	216	— (dup)	7f1e85c6-4350-401f-a960-eb8a0176e786-216_208_581_1851_765.jpg			
fong-spivak-seven-sketches_DIA_0298	217	— (dup)	7f1e85c6-4350-401f-a960-eb8a0176e786-217_351_475_378_824.jpg			
fong-spivak-seven-sketches_DIA_0299	217	— (dup)	7f1e85c6-4350-401f-a960-eb8a0176e786-217_425_925_1158_595.jpg			
fong-spivak-seven-sketches_DIA_0300	217	— (dup)	7f1e85c6-4350-401f-a960-eb8a0176e786-217_242_785_2104_667.jpg			
fong-spivak-seven-sketches_DIA_0301	218	— (dup)	7f1e85c6-4350-401f-a960-eb8a0176e786-218_283_1102_342_462.jpg			

fong-spivak-seven-sketches_DIA_0302	218	— *(dup)*	7f1e85c6-4350-401f-a960-eb8a0176e786-218_335_1140_819_443.jpg			
fong-spivak-seven-sketches_DIA_0303	219	— *(dup)*	7f1e85c6-4350-401f-a960-eb8a0176e786-219_111_224_2382_948.jpg			
fong-spivak-seven-sketches_DIA_0304	220	— *(dup)*	7f1e85c6-4350-401f-a960-eb8a0176e786-220_109_226_1335_799.jpg			
fong-spivak-seven-sketches_DIA_0305	220	— *(dup)*	7f1e85c6-4350-401f-a960-eb8a0176e786-220_109_224_1335_1095.jpg			
fong-spivak-seven-sketches_DIA_0306	220	— *(dup)*	7f1e85c6-4350-401f-a960-eb8a0176e786-220_195_127_1561_799.jpg			

fong-spivak-seven-sketches_DIA_0307	220	— (dup)	7f1e85c6-4350-401f-a960-eb8a0176e786-220_195_129_1561_995.jpg			
fong-spivak-seven-sketches_DIA_0308	220	— (dup)	7f1e85c6-4350-401f-a960-eb8a0176e786-220_192_131_1563_1190.jpg			
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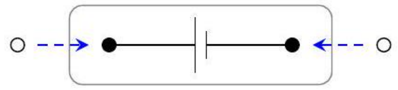
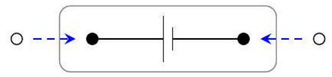
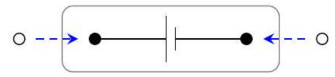
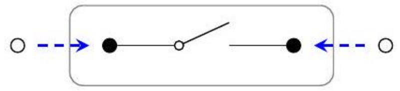
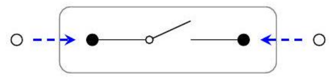
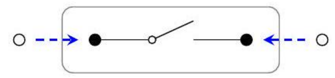
fong-spivak-seven-sketches_DIA_0311	221	— (dup)	7f1e85c6-4350-401f-a960-eb8a0176e786-221_97_423_1186_848.jpg			
fong-spivak-seven-sketches_DIA_0312	221	— (dup)	7f1e85c6-4350-401f-a960-eb8a0176e786-221_97_421_1878_848.jpg			
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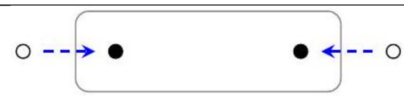
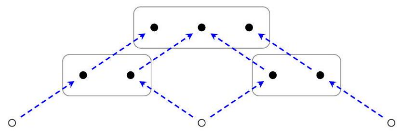
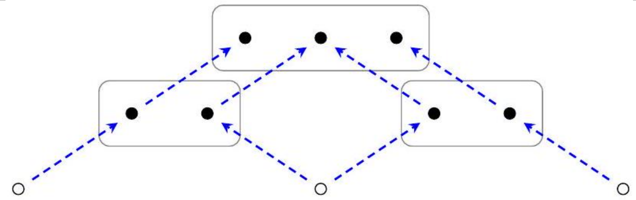
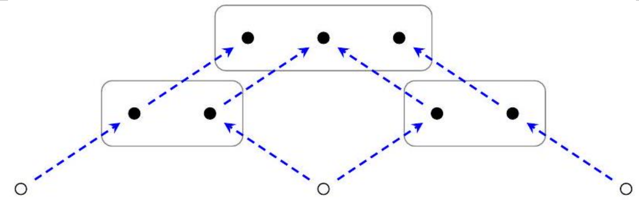
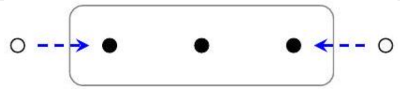
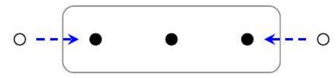
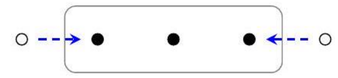
fong-spivak-seven-sketches_DIA_0314	221	— (dup)	7f1e85c6-4350-401f-a960-eb8a0176e786-221_97_419_2410_1283.jpg			

fong-spivak-seven-sketches_DIA_0315	222	— (dup)	7f1e85c6-4350-401f-a960-eb8a0176e786-222_262_817_378_649.jpg			
fong-spivak-seven-sketches_DIA_0316	222	— (dup)	7f1e85c6-4350-401f-a960-eb8a0176e786-222_95_421_751_848.jpg			
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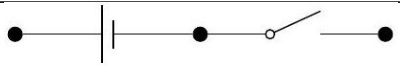
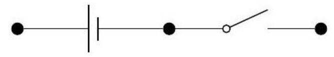
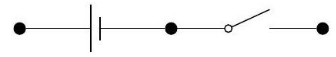
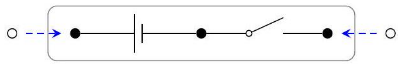
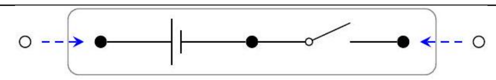
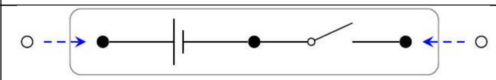
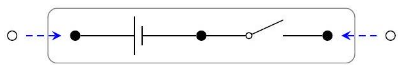
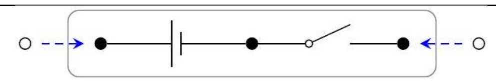
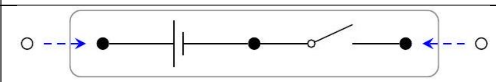
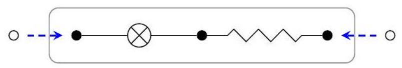
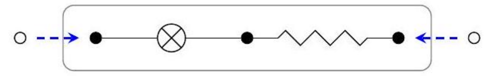
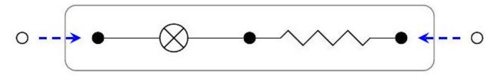
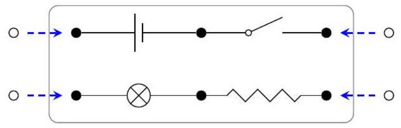
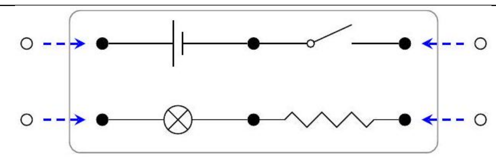
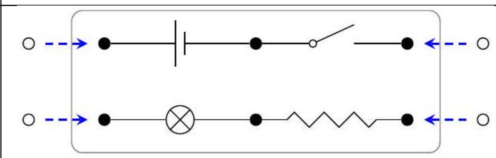
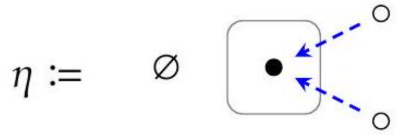
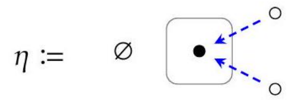
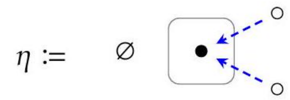
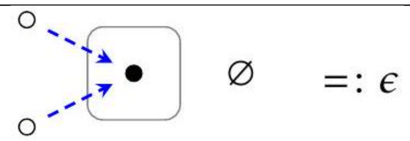
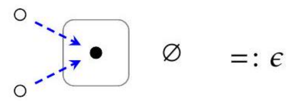
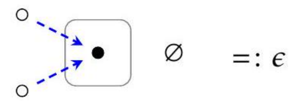
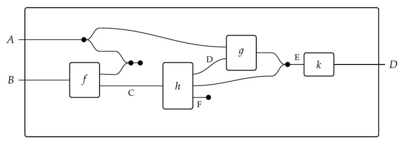
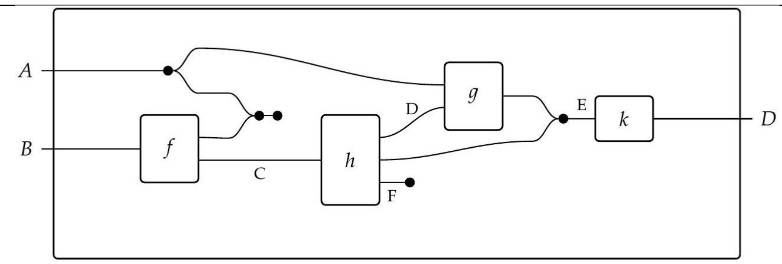
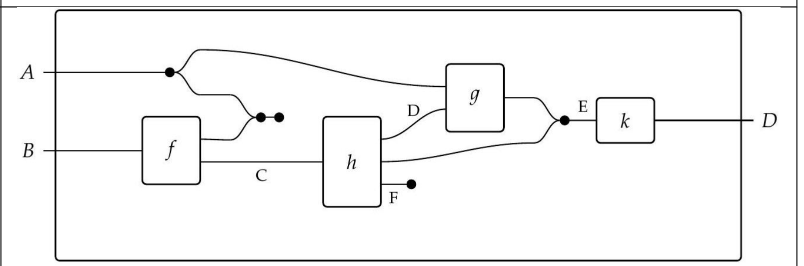
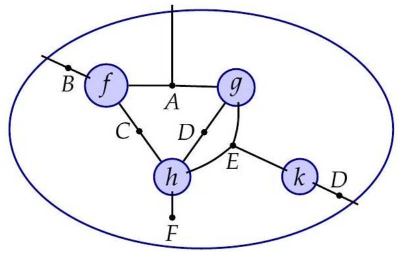
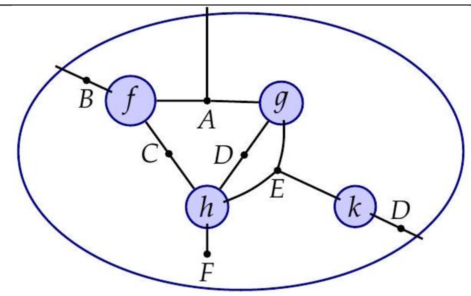
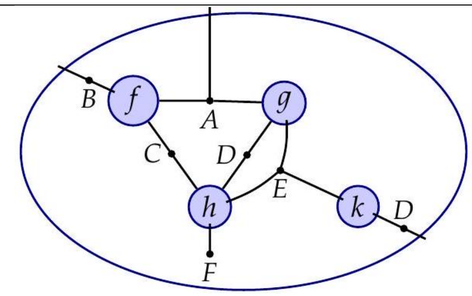
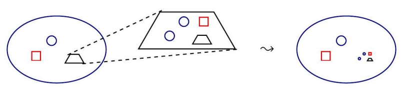
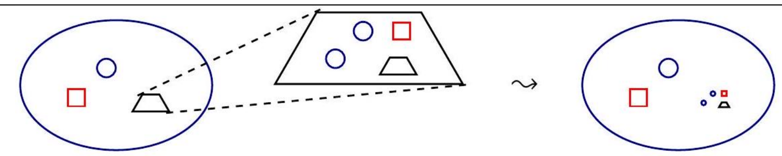
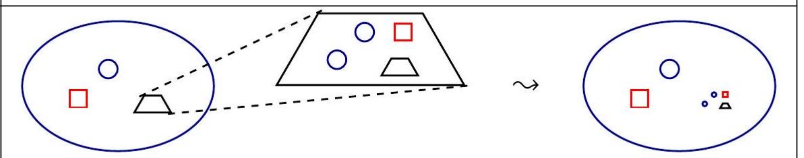
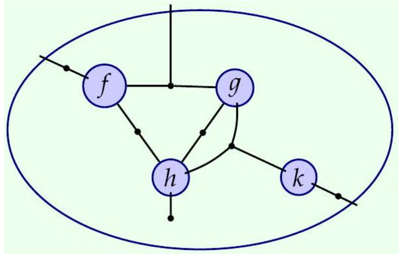
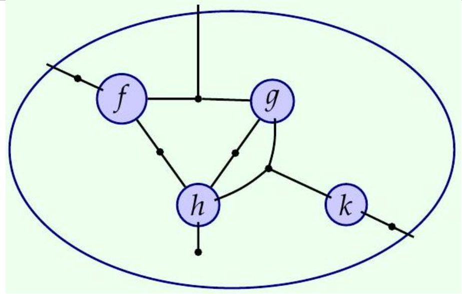
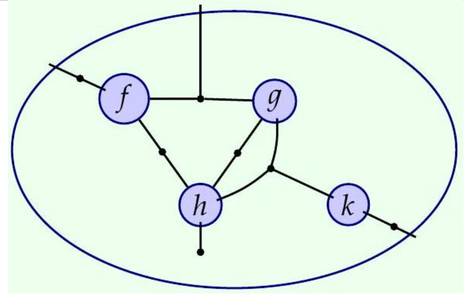
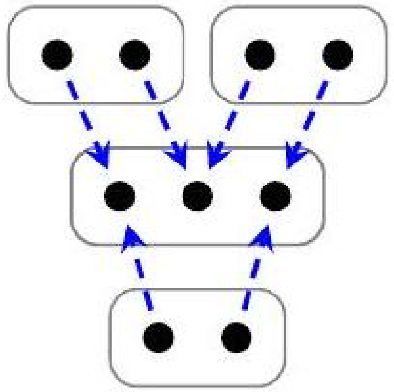
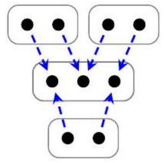
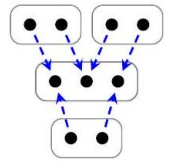
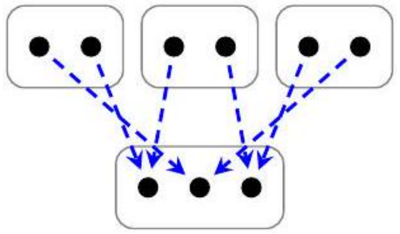
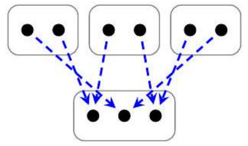
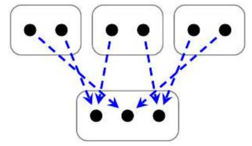
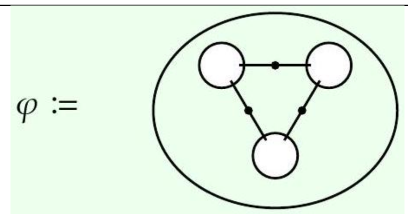
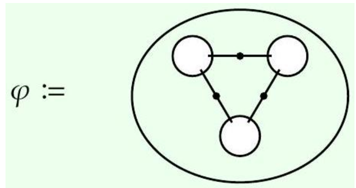
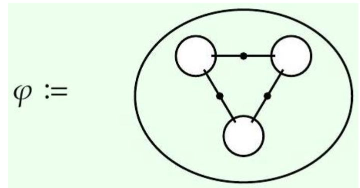
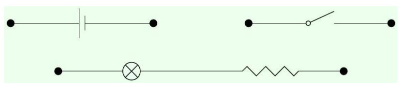
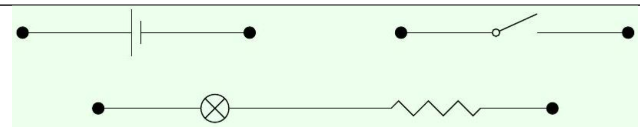
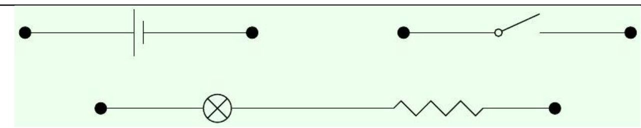
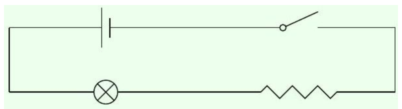
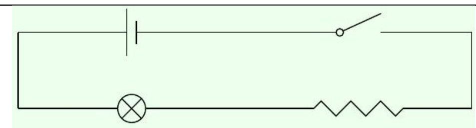
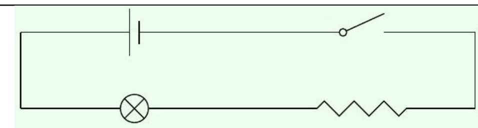
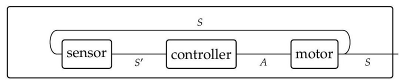
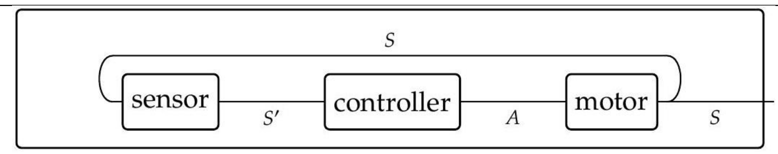
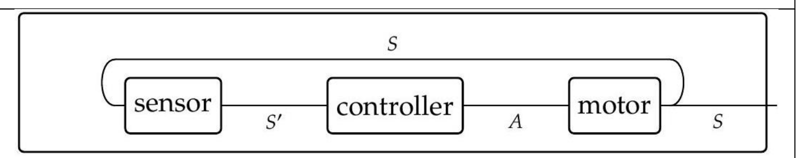
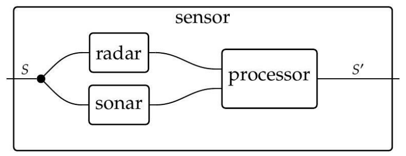
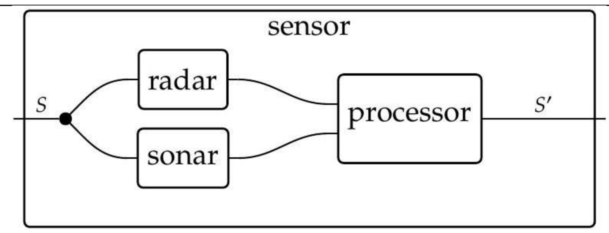
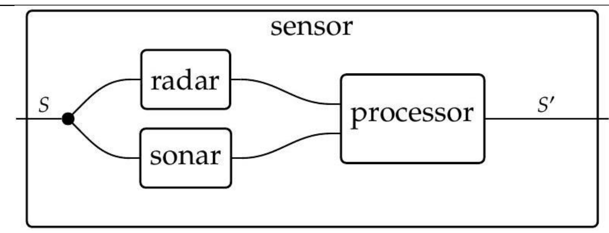
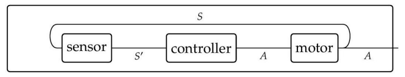
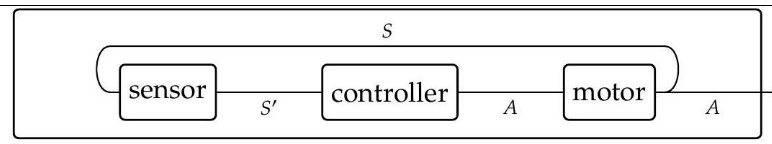
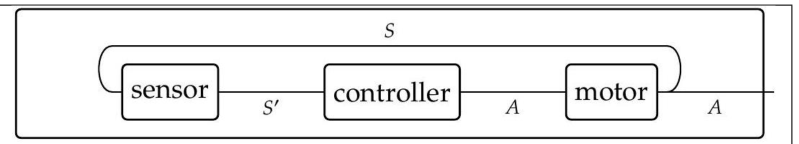
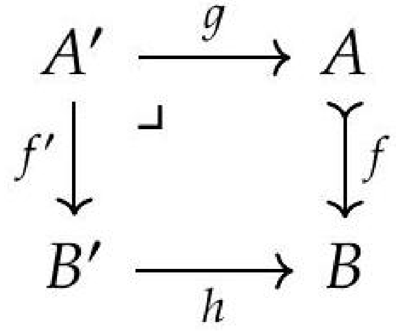
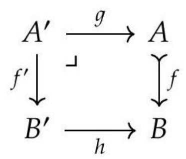
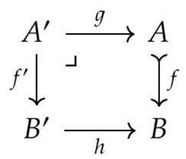
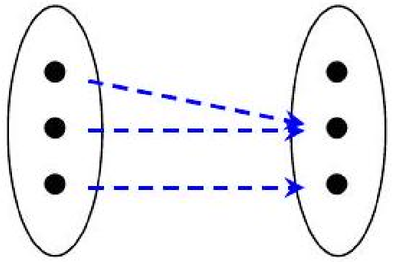
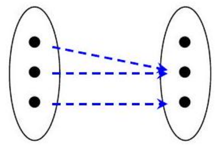
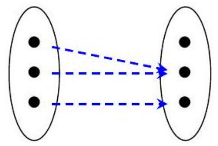
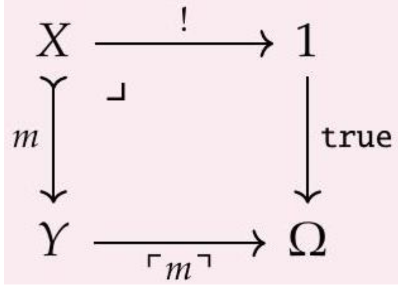
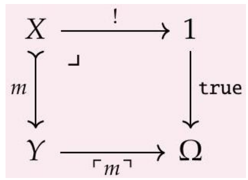
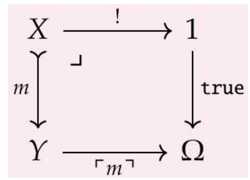
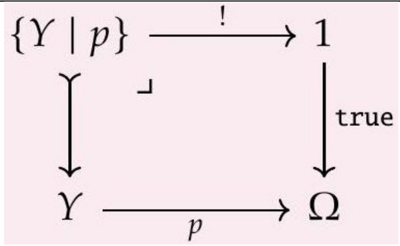
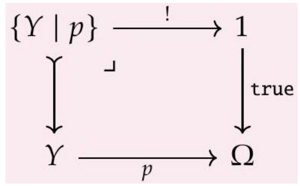
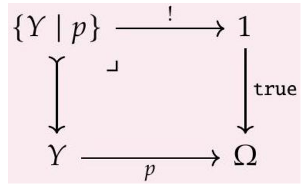
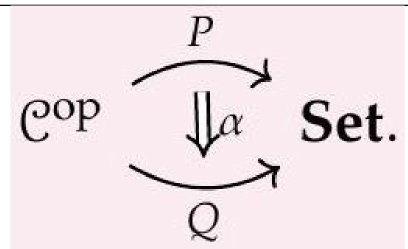
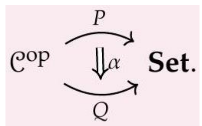
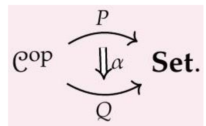
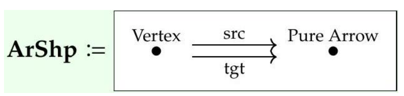
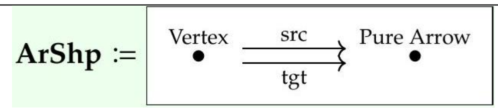
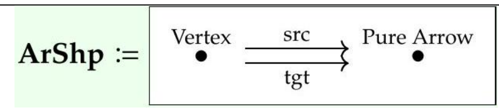
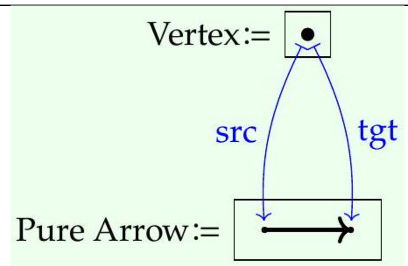
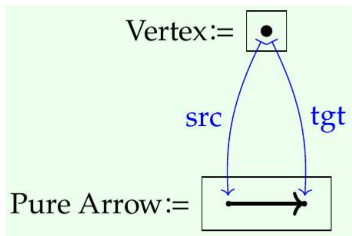
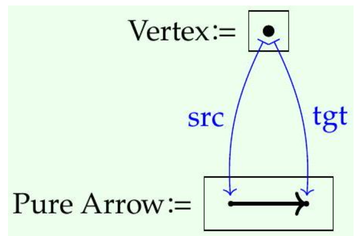
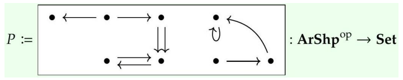
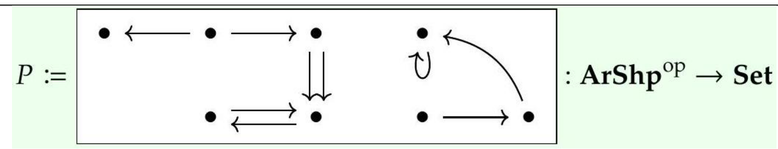
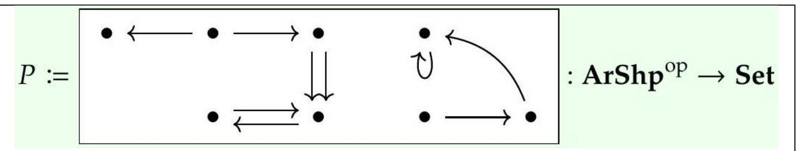
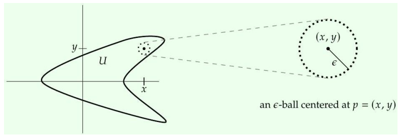
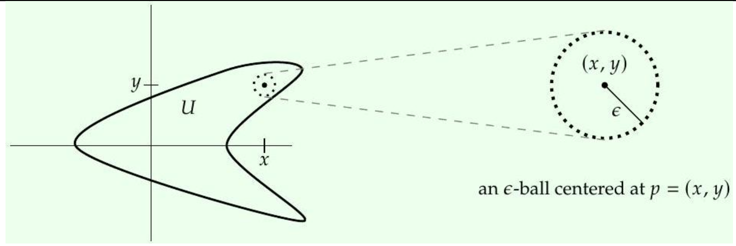
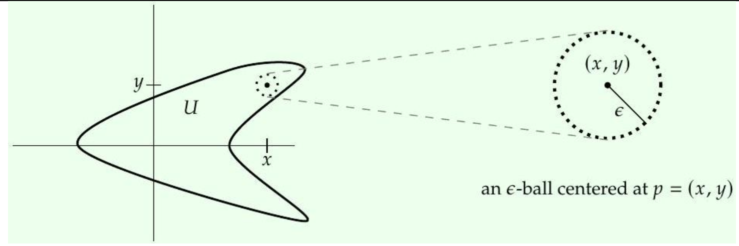
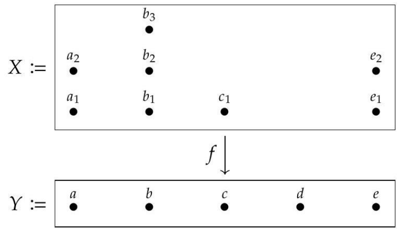
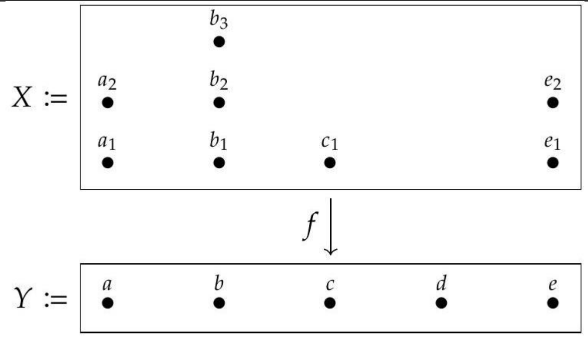
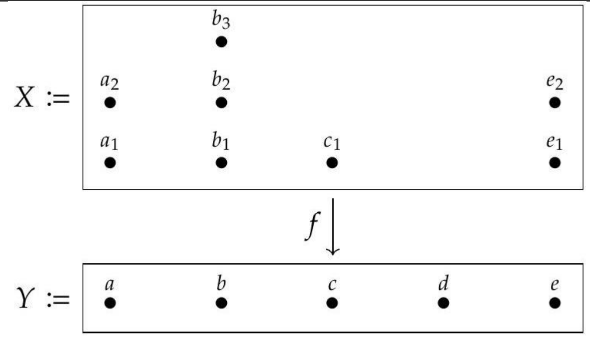
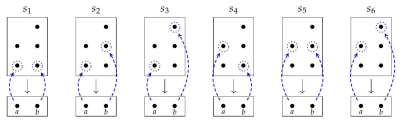
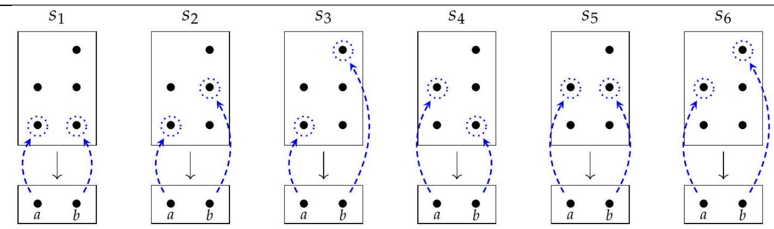
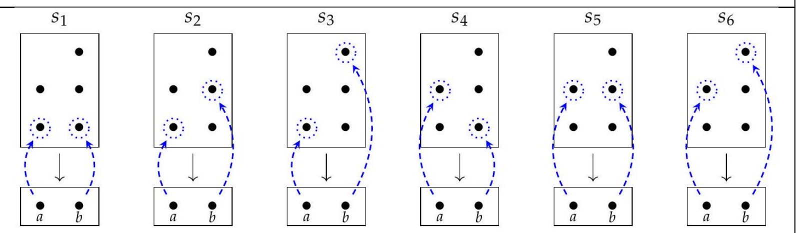
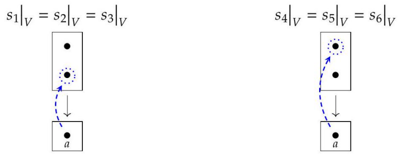
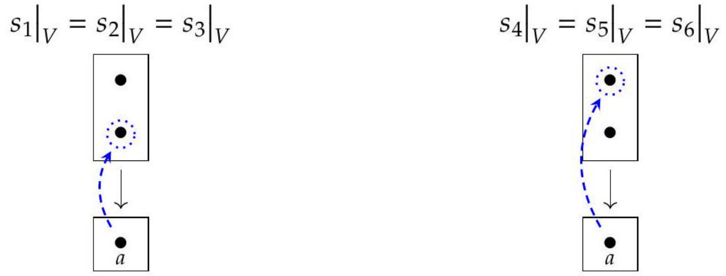
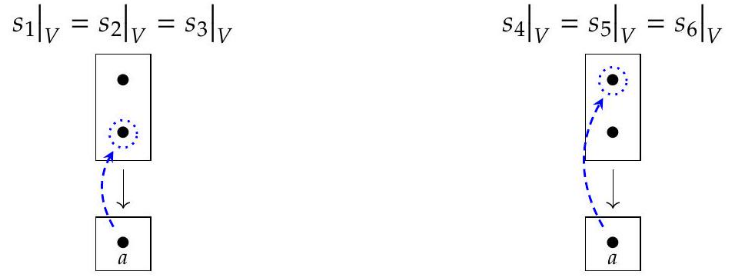
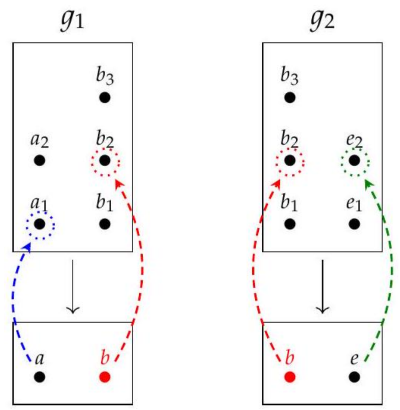
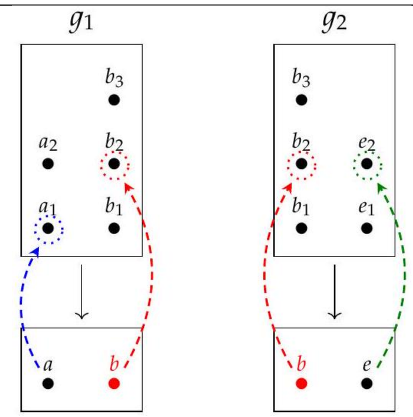
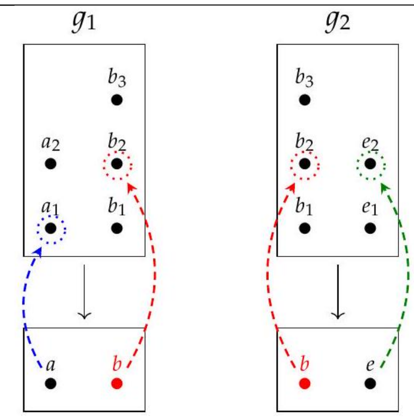
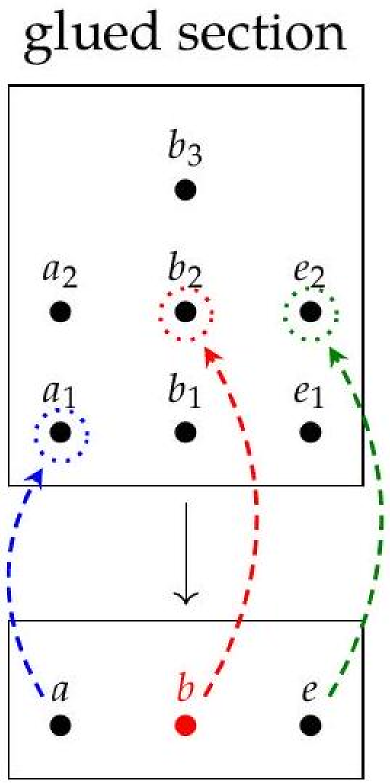
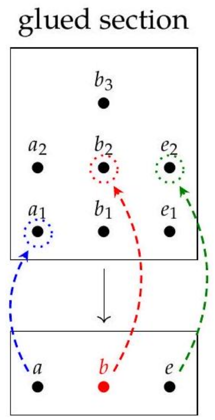
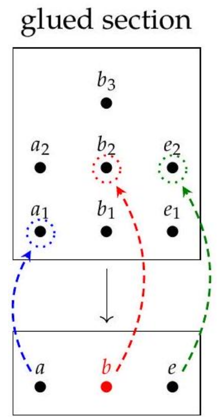
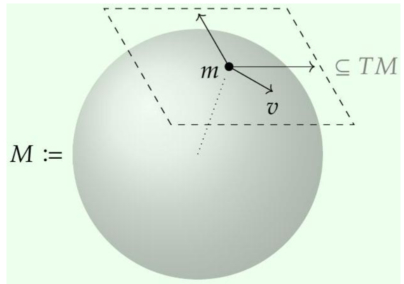
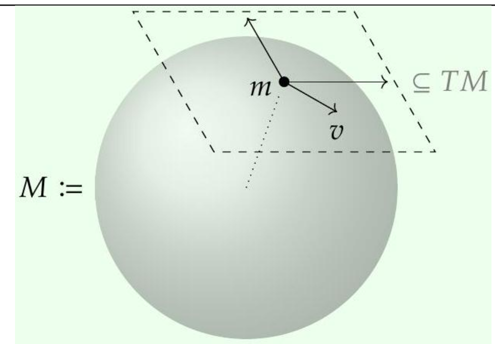
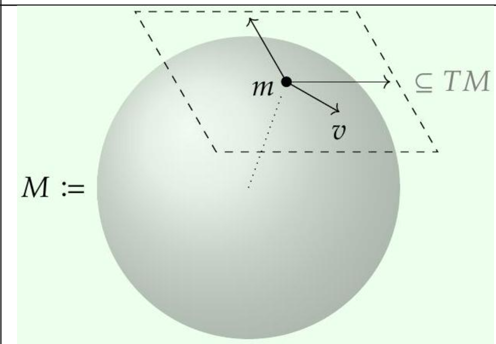
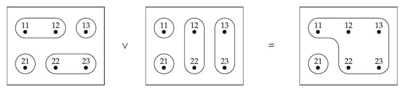
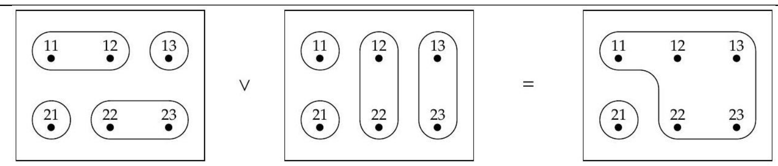
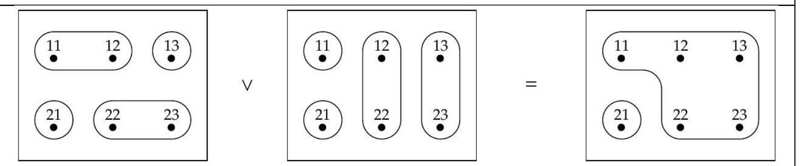
fong-spivak-seven-sketches_DIA_0318	222	— (dup)	7f1e85c6-4350-401f-a960-eb8a0176e786-222_68_419_1421_851.jpg			
fong-spivak-seven-sketches_DIA_0319	222	— (dup)	7f1e85c6-4350-401f-a960-eb8a0176e786-222_95_615_1606_751.jpg			
fong-spivak-seven-sketches_DIA_0320	222	— (dup)	7f1e85c6-4350-401f-a960-eb8a0176e786-222_95_615_2407_751.jpg			
fong-spivak-seven-sketches_DIA_0321	223	— (dup)	7f1e85c6-4350-401f-a960-eb8a0176e786-223_100_618_285_749.jpg			
fong-spivak-seven-sketches_DIA_0322	223	— (dup)	7f1e85c6-4350-401f-a960-eb8a0176e786-223_197_620_486_749.jpg			
fong-spivak-seven-sketches_DIA_0323	223	— (dup)	7f1e85c6-4350-401f-a960-eb8a0176e786-223_129_362_962_579.jpg			
fong-spivak-seven-sketches_DIA_0324	223	— (dup)	7f1e85c6-4350-401f-a960-eb8a0176e786-223_124_362_964_1176.jpg			
fong-spivak-seven-sketches_DIA_0325	224	— (dup)	7f1e85c6-4350-401f-a960-eb8a0176e786-224_438_1299_1237_409.jpg			
fong-spivak-seven-sketches_DIA_0326	224	— (dup)	7f1e85c6-4350-401f-a960-eb8a0176e786-224_375_594_2127_762.jpg			
fong-spivak-seven-sketches_DIA_0327	226	— (dup)	7f1e85c6-4350-401f-a960-eb8a0176e786-226_224_1133_495_493.jpg			

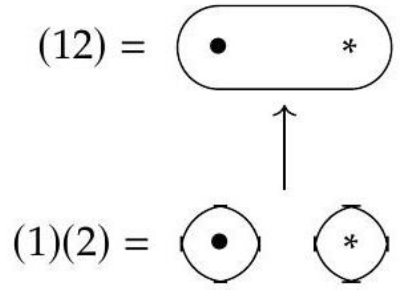
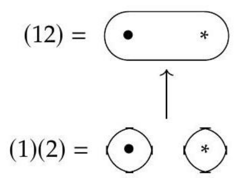
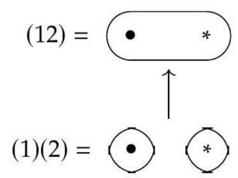
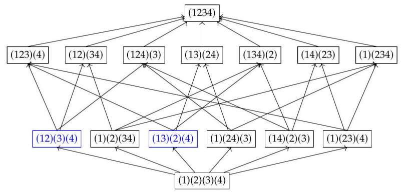
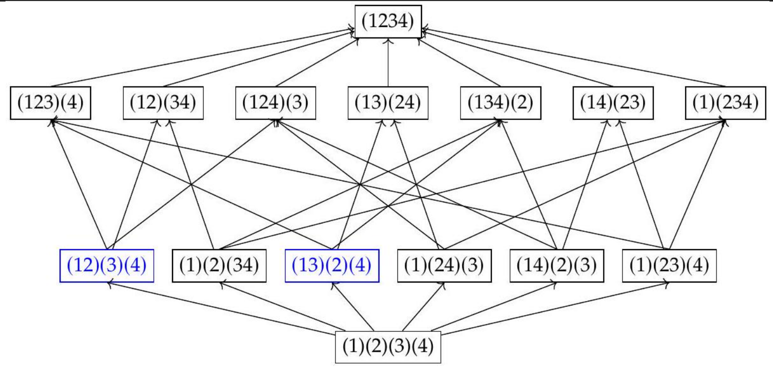
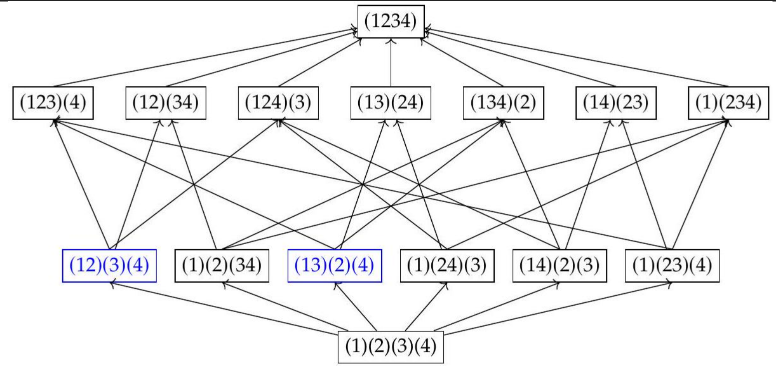
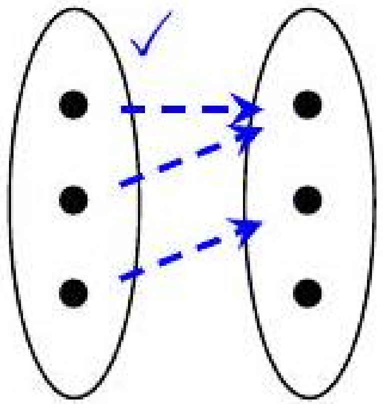
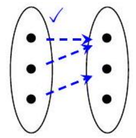
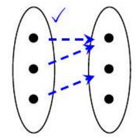
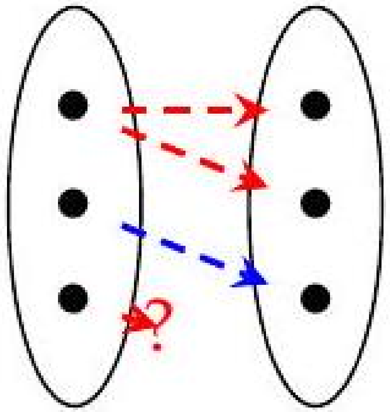
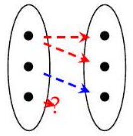
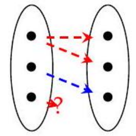
fong-spivak-seven-sketches_DIA_0328	227	— *(dup)*	7f1e85c6-4350-401f-a960-eb8a0176e786-227_382_593_1466_765.jpg			
fong-spivak-seven-sketches_DIA_0329	228	— *(dup)*	7f1e85c6-4350-401f-a960-eb8a0176e786-228_210_206_527_1005.jpg			
fong-spivak-seven-sketches_DIA_0330	228	— *(dup)*	7f1e85c6-4350-401f-a960-eb8a0176e786-228_186_310_980_952.jpg			
fong-spivak-seven-sketches_DIA_0331	229	— *(dup)*	7f1e85c6-4350-401f-a960-eb8a0176e786-229_242_455_1699_839.jpg			
fong-spivak-seven-sketches_DIA_0332	229	— *(dup)*	7f1e85c6-4350-401f-a960-eb8a0176e786-229_158_814_2312_652.jpg			
fong-spivak-seven-sketches_DIA_0333	230	— *(dup)*	7f1e85c6-4350-401f-a960-eb8a0176e786-230_159_603_333_756.jpg			
fong-spivak-seven-sketches_DIA_0334	234	— *(dup)*	7f1e85c6-4350-401f-a960-eb8a0176e786-234_204_1061_446_527.jpg			
fong-spivak-seven-sketches_DIA_0335	234	— *(dup)*	7f1e85c6-4350-401f-a960-eb8a0176e786-234_292_774_1305_672.jpg			

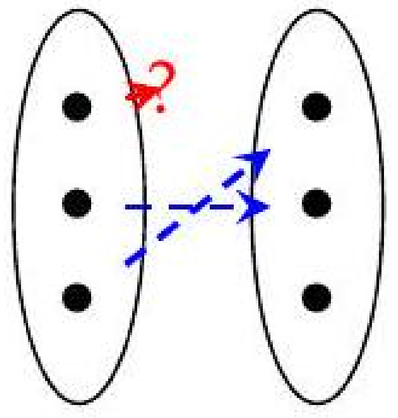
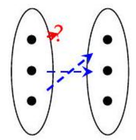
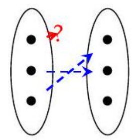
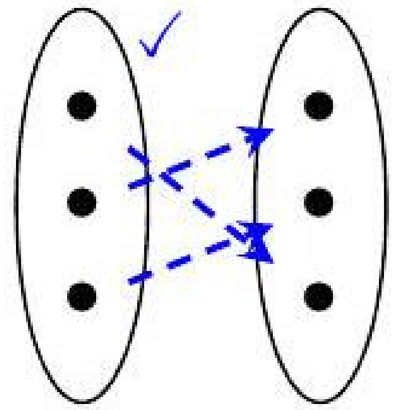
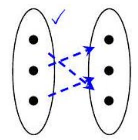
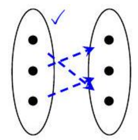
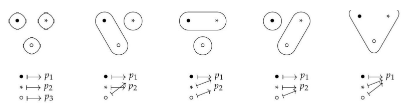
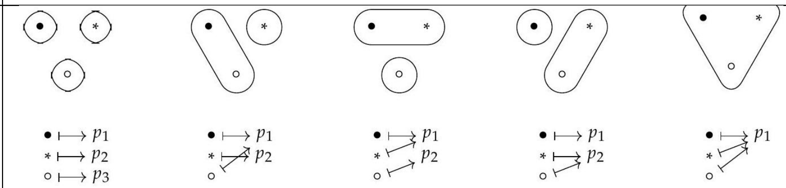
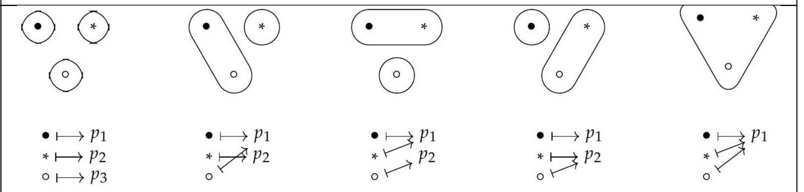
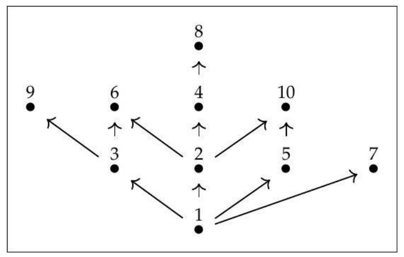
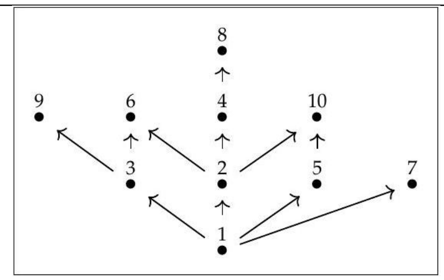
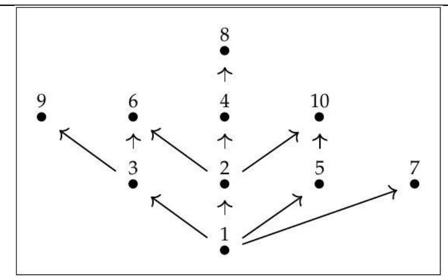
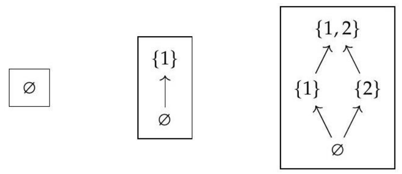
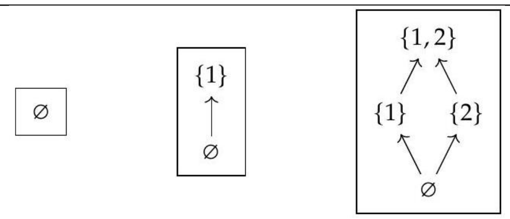
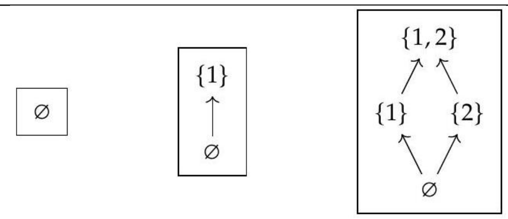
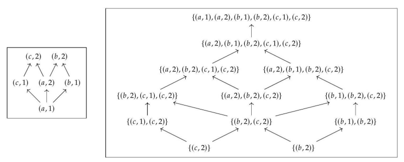
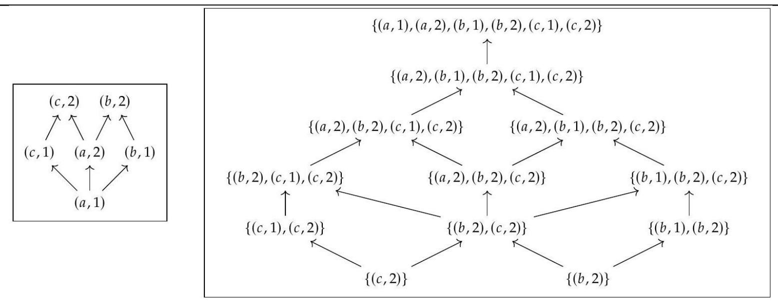
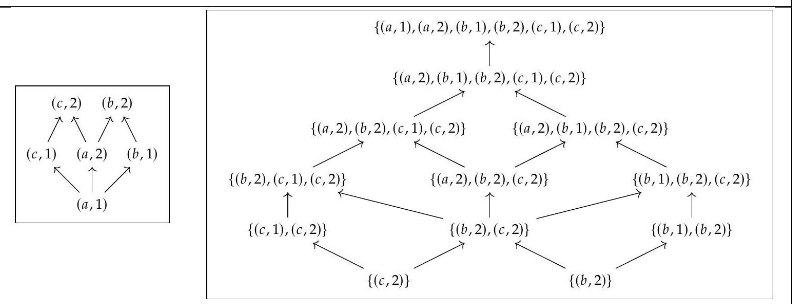
fong-spivak-seven-sketches_236_0336	236	— (dup)	7f1e85c6-4350-401f-a960-eb8a0176e786-236_190_1061_301_527.jpg			
fong-spivak-seven-sketches_237_0337	237	— (dup)	7f1e85c6-4350-401f-a960-eb8a0176e786-237_165_344_1532_602.jpg	$\begin{tikzcd} A \arrow[r] \arrow[d] & B \arrow[r] \arrow[d] \arrow[dr, phantom, "\lrcorner"] & C \arrow[d] \\ D \arrow[r] & E \arrow[r] & F \end{tikzcd}$	$\begin{tikzcd} A \arrow[r] \arrow[d] & B \arrow[r] \arrow[d] \arrow[dr, phantom, "\lrcorner"] & C \arrow[d] \\ D \arrow[r] & E \arrow[r] & F \end{tikzcd}$	$\begin{tikzcd} A \arrow[r] \arrow[d] & B \arrow[r] \arrow[d] \arrow[dr, phantom, "\lrcorner"] & C \arrow[d] \\ D \arrow[r] & E \arrow[r] & F \end{tikzcd}$
fong-spivak-seven-sketches_237_0338	237	— (dup)	7f1e85c6-4350-401f-a960-eb8a0176e786-237_163_344_1534_1172.jpg	$\begin{tikzcd} A \arrow[r] \arrow[d] \arrow[dr, phantom, "\lrcorner"] & B \arrow[r] \arrow[d] & C \arrow[d] \\ D \arrow[r] & E \arrow[r] & F \end{tikzcd}$	$\begin{tikzcd} A \arrow[r] \arrow[d] \arrow[dr, phantom, "\lrcorner"] & B \arrow[r] \arrow[d] & C \arrow[d] \\ D \arrow[r] & E \arrow[r] & F \end{tikzcd}$	$\begin{tikzcd} A \arrow[r] \arrow[d] \arrow[dr, phantom, "\lrcorner"] & B \arrow[r] \arrow[d] & C \arrow[d] \\ D \arrow[r] & E \arrow[r] & F \end{tikzcd}$
fong-spivak-seven-sketches_237_0339	237	— (dup)	7f1e85c6-4350-401f-a960-eb8a0176e786-237_167_371_2131_873.jpg	$\begin{tikzcd} A \arrow[r] \arrow[d] \arrow[dr, phantom, "\lrcorner"] & B \arrow[r] \arrow[d] \arrow[dr, phantom, "\lrcorner"] & C \arrow[d] \\ A' \arrow[r] & B' \arrow[r] & C' \end{tikzcd}$	$\begin{tikzcd} A \arrow[r] \arrow[d] \arrow[dr, phantom, "\lrcorner"] & B \arrow[r] \arrow[d] \arrow[dr, phantom, "\lrcorner"] & C \arrow[d] \\ A' \arrow[r] & B' \arrow[r] & C' \end{tikzcd}$	$\begin{tikzcd} A \arrow[r] \arrow[d] \arrow[dr, phantom, "\lrcorner"] & B \arrow[r] \arrow[d] \arrow[dr, phantom, "\lrcorner"] & C \arrow[d] \\ A' \arrow[r] & B' \arrow[r] & C' \end{tikzcd}$
fong-spivak-seven-sketches_238_0340	238	— (dup)	7f1e85c6-4350-401f-a960-eb8a0176e786-238_210_606_701_753.jpg	$\begin{tikzcd} A \arrow[r, "\mathrm{id}_A"] \arrow[d, "\mathrm{id}_A"'] \arrow[dr, phantom, "\lrcorner"] & A \arrow[d, "f"] \\ A \arrow[r, "f"'] & B \end{tikzcd} \qquad \begin{tikzcd} A \arrow[r, "f"] \arrow[d, "f"'] & B \arrow[d, "\mathrm{id}_B"] \\ B \arrow[r, "\mathrm{id}_B"'] \arrow[ur, phantom, "\ulcorner"] & B \end{tikzcd}$	$\begin{tikzcd} A \arrow[r, "\mathrm{id}_A"] \arrow[d, "\mathrm{id}_A"'] \arrow[dr, phantom, "\lrcorner"] & A \arrow[d, "f"] \\ A \arrow[r, "f"'] & B \end{tikzcd} \qquad \begin{tikzcd} A \arrow[r, "f"] \arrow[d, "f"'] & B \arrow[d, "\mathrm{id}_B"] \\ B \arrow[r, "\mathrm{id}_B"'] \arrow[ur, phantom, "\ulcorner"] & B \end{tikzcd}$	$\begin{tikzcd} A \arrow[r, "\mathrm{id}_A"] \arrow[d, "\mathrm{id}_A"'] \arrow[dr, phantom, "\lrcorner"] & A \arrow[d, "f"] \\ A \arrow[r, "f"'] & B \end{tikzcd} \qquad \begin{tikzcd} A \arrow[r, "f"] \arrow[d, "f"'] & B \arrow[d, "\mathrm{id}_B"] \\ B \arrow[r, "\mathrm{id}_B"'] \arrow[ur, phantom, "\ulcorner"] & B \end{tikzcd}$
fong-spivak-seven-sketches_238_0341	238	— (dup)	7f1e85c6-4350-401f-a960-eb8a0176e786-238_210_217_1419_995.jpg	$\begin{tikzcd} A' \arrow[r, "f'"] \arrow[d, "i'"', "\cong"] \arrow[dr, phantom, "\lrcorner"] & B' \arrow[d, "i", "\cong"'] \\ A \arrow[r, "f"'] & B \end{tikzcd}$	$\begin{tikzcd} A' \arrow[r, "f'"] \arrow[d, "i'"', "\cong"] \arrow[dr, phantom, "\lrcorner"] & B' \arrow[d, "i", "\cong"'] \\ A \arrow[r, "f"'] & B \end{tikzcd}$	$\begin{tikzcd} A' \arrow[r, "f'"] \arrow[d, "i'"', "\cong"] \arrow[dr, phantom, "\lrcorner"] & B' \arrow[d, "i", "\cong"'] \\ A \arrow[r, "f"'] & B \end{tikzcd}$
fong-spivak-seven-sketches_238_0342	238	— (dup)	7f1e85c6-4350-401f-a960-eb8a0176e786-238_208_190_1747_1009.jpg	$\begin{tikzcd} A \arrow[r, "f"] \arrow[d, equal] \arrow[dr, phantom, "\lrcorner"] & B \arrow[d, equal] \\ A \arrow[r, "f"'] & B \end{tikzcd}$	$\begin{tikzcd} A \arrow[r, "f"] \arrow[d, equal] \arrow[dr, phantom, "\lrcorner"] & B \arrow[d, equal] \\ A \arrow[r, "f"'] & B \end{tikzcd}$	$\begin{tikzcd} A \arrow[r, "f"] \arrow[d, equal] \arrow[dr, phantom, "\lrcorner"] & B \arrow[d, equal] \\ A \arrow[r, "f"'] & B \end{tikzcd}$

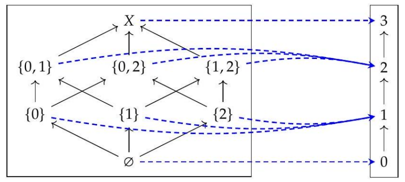
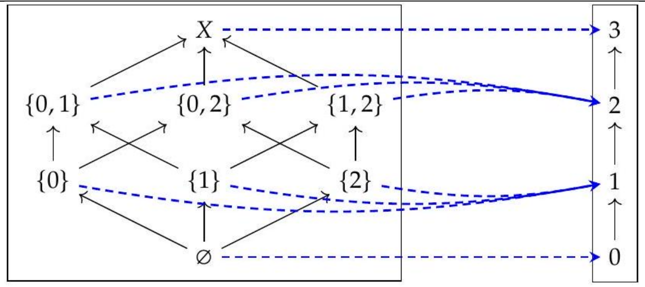
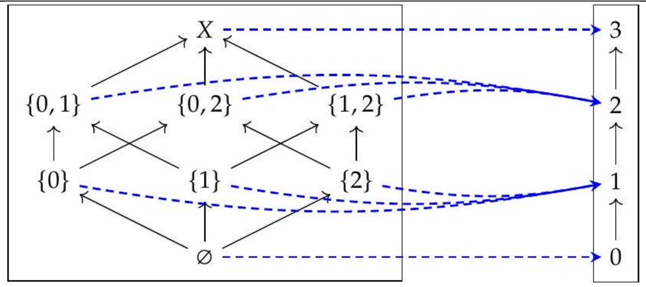
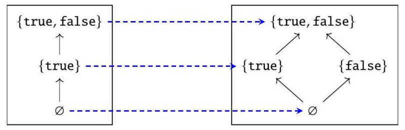
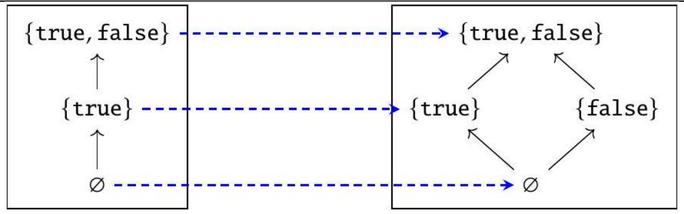
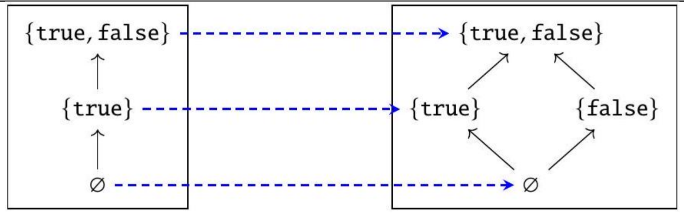
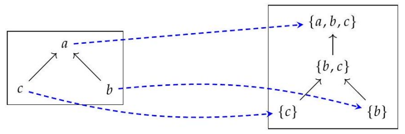
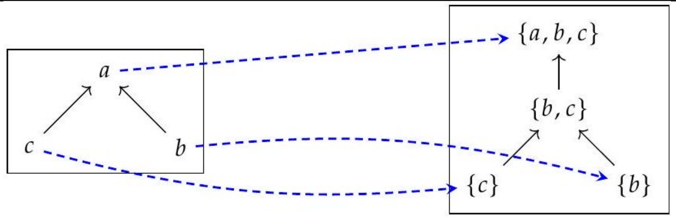
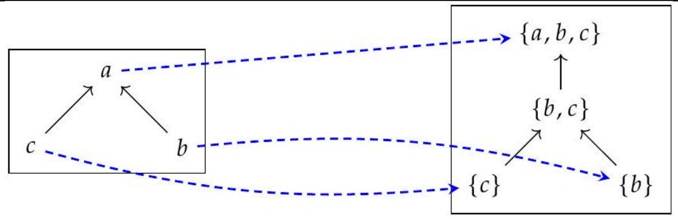
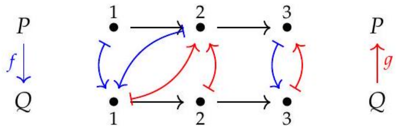
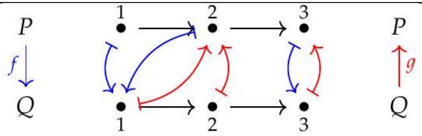
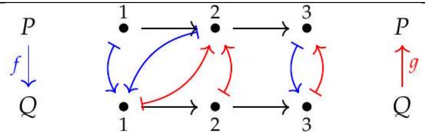
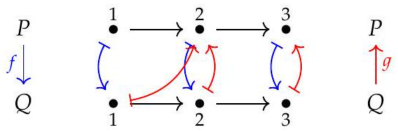
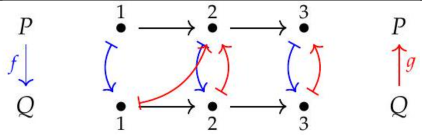
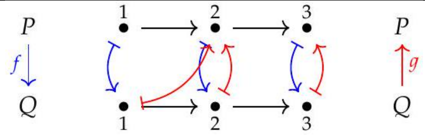
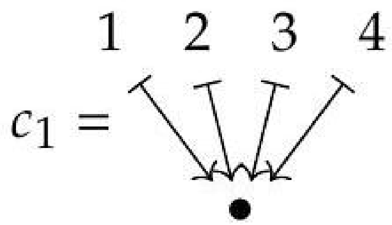
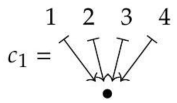
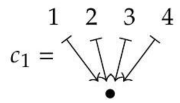
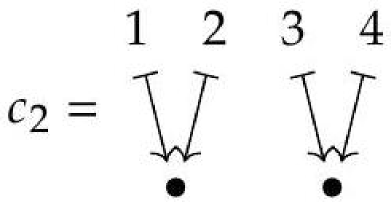
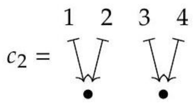
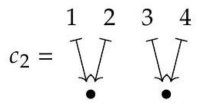
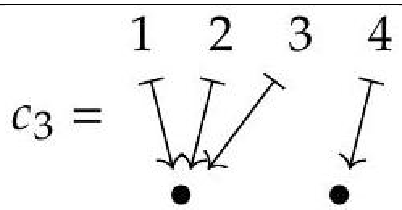
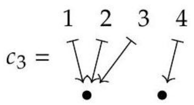
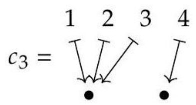
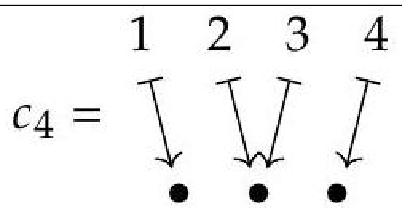
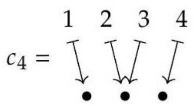
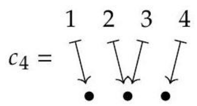
fong-spivak-seven-sketches_DIA_0343	238	— *(dup)*	7f1e85c6-4350-401f-a960-eb8a0176e786-238_195_231_2077_943.jpg			
fong-spivak-seven-sketches_DIA_0344	239	— *(dup)*	7f1e85c6-4350-401f-a960-eb8a0176e786-239_186_274_796_923.jpg			
fong-spivak-seven-sketches_DIA_0345	240	— *(dup)*	7f1e85c6-4350-401f-a960-eb8a0176e786-240_226_312_1448_579.jpg			
fong-spivak-seven-sketches_DIA_0346	240	— *(dup)*	7f1e85c6-4350-401f-a960-eb8a0176e786-240_235_380_1446_1158.jpg			
fong-spivak-seven-sketches_DIA_0347	245	— *(dup)*	7f1e85c6-4350-401f-a960-eb8a0176e786-245_158_253_344_932.jpg			
fong-spivak-seven-sketches_DIA_0348	245	— *(dup)*	7f1e85c6-4350-401f-a960-eb8a0176e786-245_133_629_683_744.jpg			
fong-spivak-seven-sketches_DIA_0349	245	— *(dup)*	7f1e85c6-4350-401f-a960-eb8a0176e786-245_299_450_928_835.jpg			

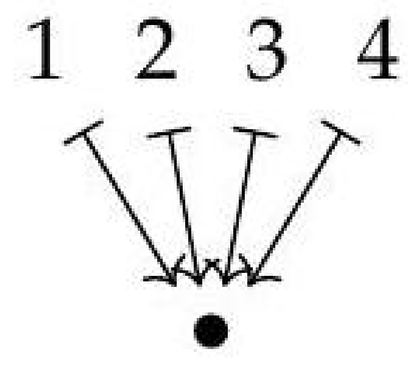
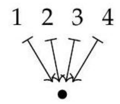
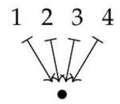
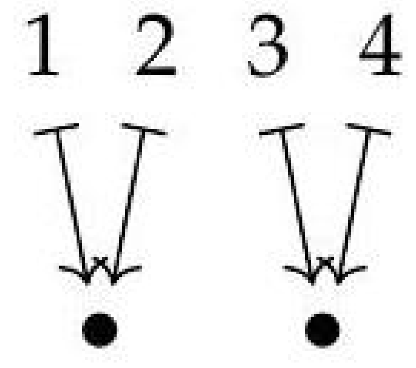
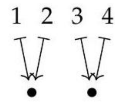
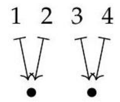
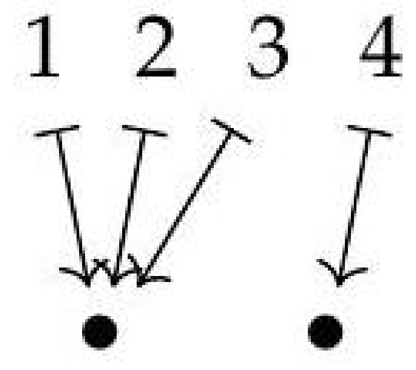
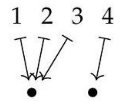
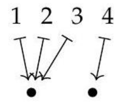
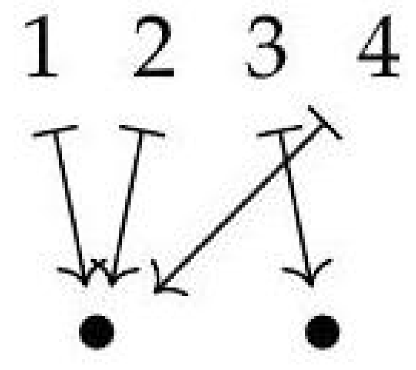
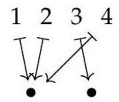
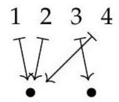
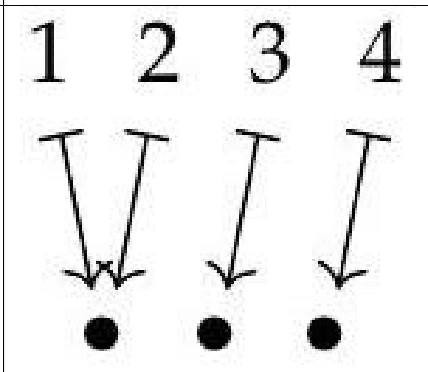
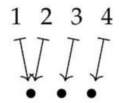
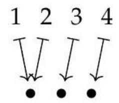
fong-spivak-seven-sketches_[illegible]_0350	245	— *(dup)*	7f1e85c6-4350-401f-a960-eb8a0176e786-245_195_1043_1484_538.jpg			
fong-spivak-seven-sketches_[illegible]_0351	247	— *(dup)*	7f1e85c6-4350-401f-a960-eb8a0176e786-247_316_947_466_591.jpg			
fong-spivak-seven-sketches_[illegible]_0352	250	— *(dup)*	7f1e85c6-4350-401f-a960-eb8a0176e786-250_434_753_466_688.jpg			
fong-spivak-seven-sketches_[illegible]_0353	251	— *(dup)*	7f1e85c6-4350-401f-a960-eb8a0176e786-251_407_1394_367_357.jpg			
fong-spivak-seven-sketches_[illegible]_0354	251	— *(dup)*	7f1e85c6-4350-401f-a960-eb8a0176e786-251_362_934_1602_604.jpg			
fong-spivak-seven-sketches_[illegible]_0355	252	— *(dup)*	7f1e85c6-4350-401f-a960-eb8a0176e786-252_534_520_339_796.jpg			

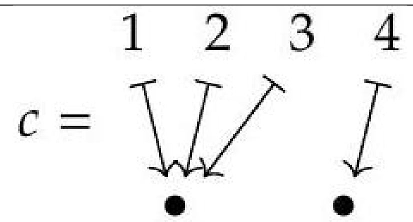
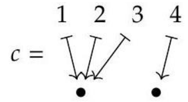
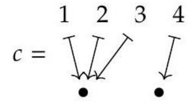
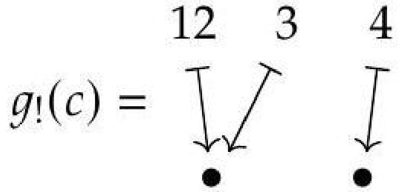
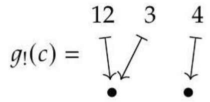
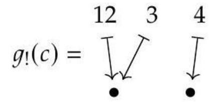
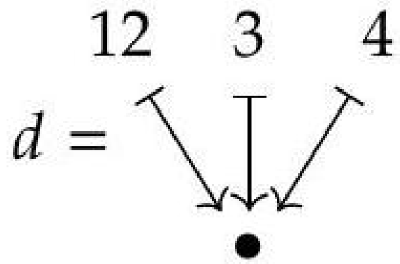
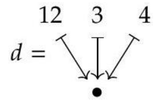
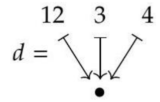
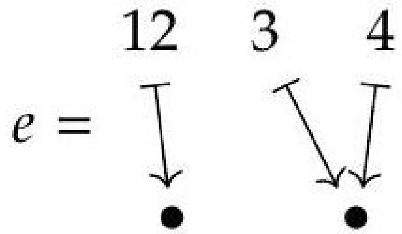
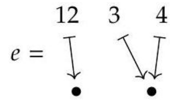
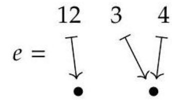
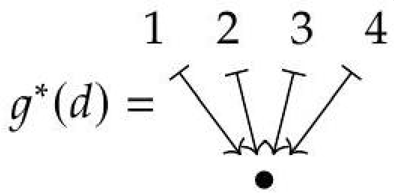
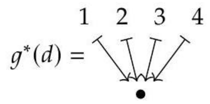
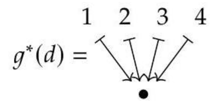
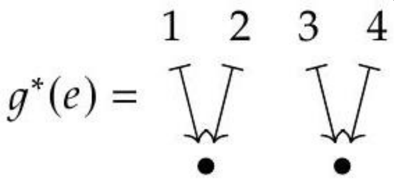
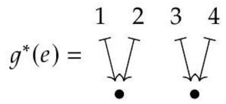
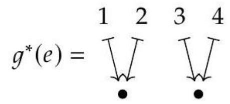
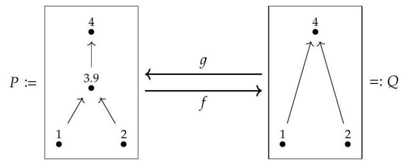
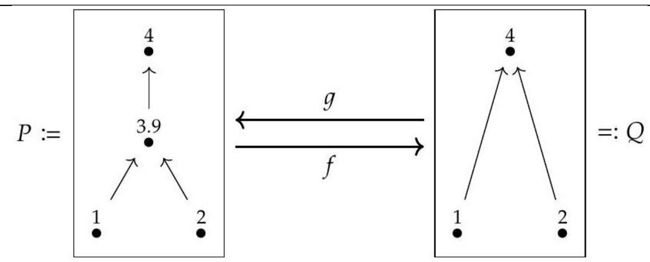
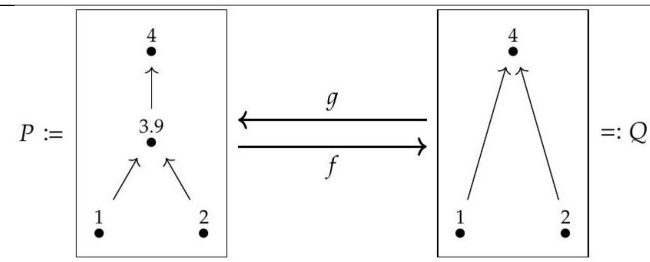
fong-spivak-seven-sketches_DIA_0356	252	— *(dup)*	7f1e85c6-4350-401f-a960-eb8a0176e786-252_541_271_1029_923.jpg			
fong-spivak-seven-sketches_DIA_0357	253	— *(dup)*	7f1e85c6-4350-401f-a960-eb8a0176e786-253_441_620_1191_748.jpg			
fong-spivak-seven-sketches_DIA_0358	261	— *(dup)*	7f1e85c6-4350-401f-a960-eb8a0176e786-261_206_301_1722_912.jpg	$\begin{array}{ccc} \forall_t p & \longrightarrow & 1 \\ \downarrow & \lrcorner & \downarrow \mathtt{true}^T \\ S & \xrightarrow[p']{} & \Omega^T \end{array}$	$\begin{array}{ccc} \forall_t p & \longrightarrow & 1 \\ \downarrow & \lrcorner & \downarrow \mathtt{true}^T \\ S & \xrightarrow[p']{} & \Omega^T \end{array}$	$\begin{array}{ccc} \forall_t p & \longrightarrow & 1 \\ \downarrow & \lrcorner & \downarrow \mathtt{true}^T \\ S & \xrightarrow[p']{} & \Omega^T \end{array}$
fong-spivak-seven-sketches_DIA_0359	262	— *(dup)*	7f1e85c6-4350-401f-a960-eb8a0176e786-262_186_403_860_860.jpg	$\begin{array}{ccc} \{S \times T \mid p\} & \rightarrowtail & S \times T \\ \twoheaddownarrow & & \downarrow \pi_S \\ \exists_t p & \rightarrowtail & S \end{array}$	$\begin{array}{ccc} \{S \times T \mid p\} & \rightarrowtail & S \times T \\ \twoheaddownarrow & & \downarrow \pi_S \\ \exists_t p & \rightarrowtail & S \end{array}$	$\begin{array}{ccc} \{S \times T \mid p\} & \rightarrowtail & S \times T \\ \twoheaddownarrow & & \downarrow \pi_S \\ \exists_t p & \rightarrowtail & S \end{array}$
fong-spivak-seven-sketches_DIA_0360	271	— *(dup)*	7f1e85c6-4350-401f-a960-eb8a0176e786-271_262_1253_1774_450.jpg			

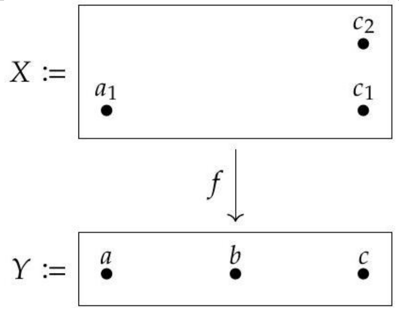
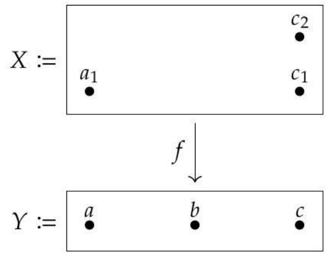
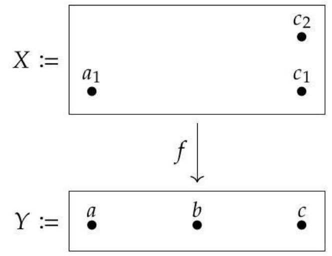
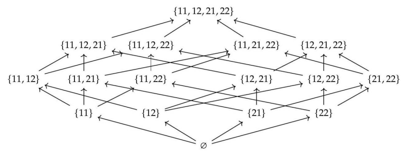
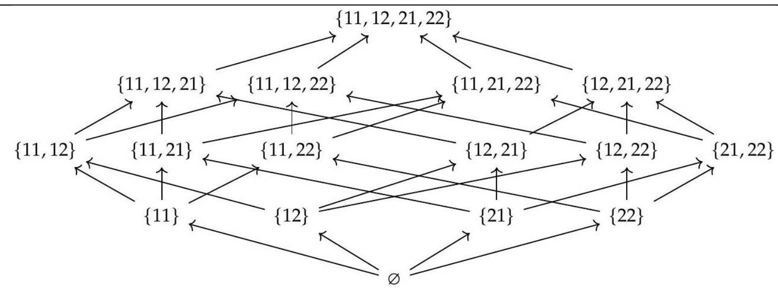
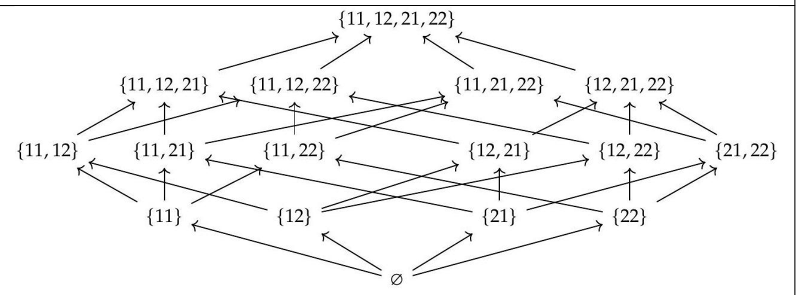
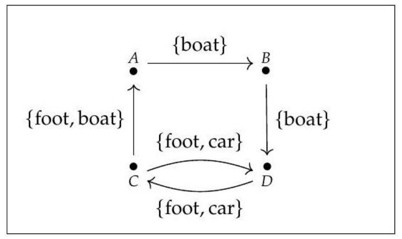
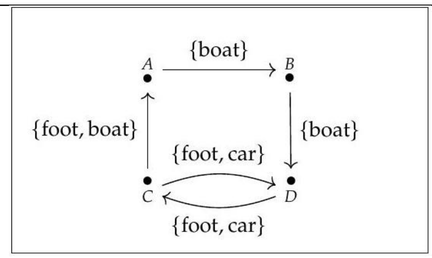
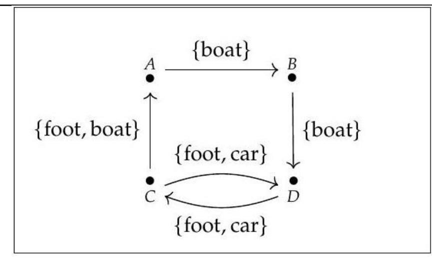
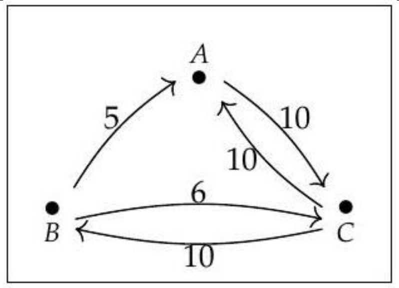
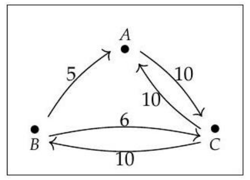
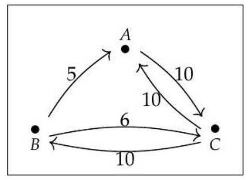
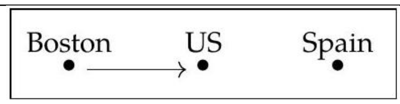
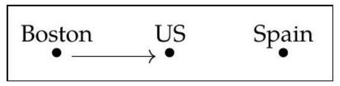
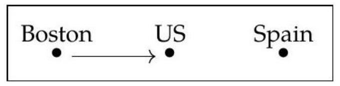
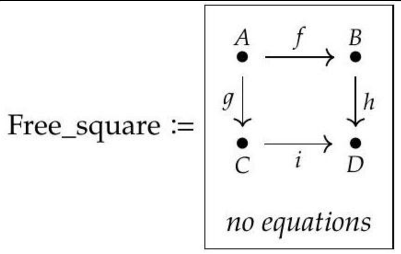
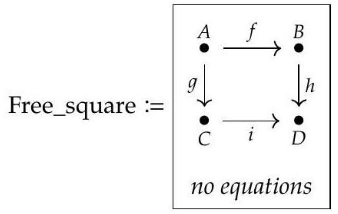
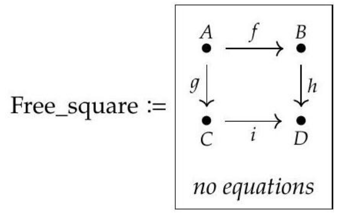
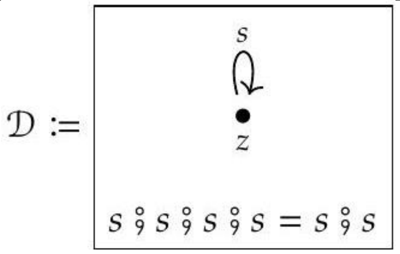
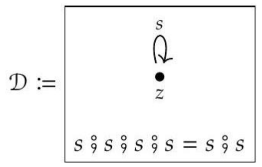
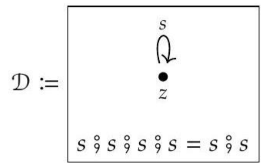
fong-spivak-seven-sketches_0361	271	— *(dup)*	7f1e85c6-4350-401f-a960-eb8a0176e786-271_224_296_2272_971.jpg			
fong-spivak-seven-sketches_0362	272	— *(dup)*	7f1e85c6-4350-401f-a960-eb8a0176e786-272_574_1203_322_516.jpg			
fong-spivak-seven-sketches_0363	273	— *(dup)*	7f1e85c6-4350-401f-a960-eb8a0176e786-273_174_165_1946_593.jpg			
fong-spivak-seven-sketches_0364	273	— *(dup)*	7f1e85c6-4350-401f-a960-eb8a0176e786-273_176_161_1946_891.jpg			

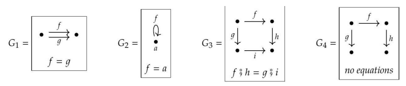
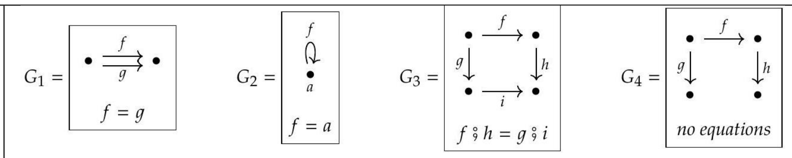
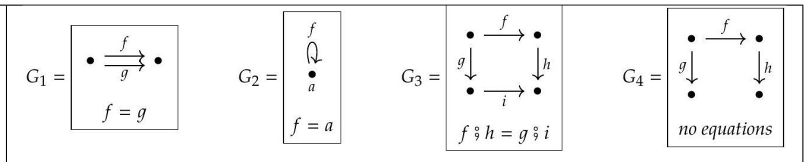
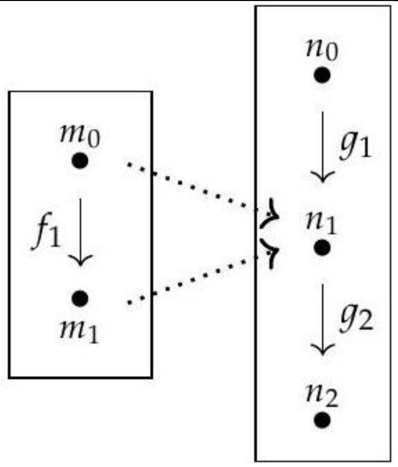
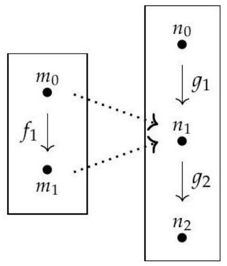
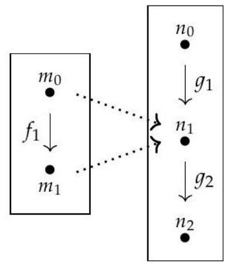
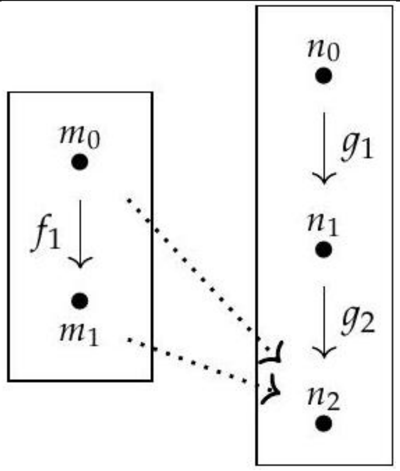
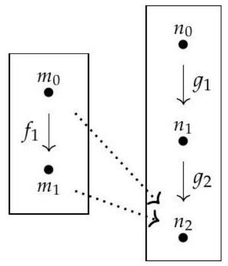
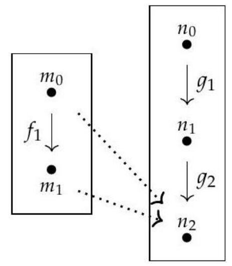
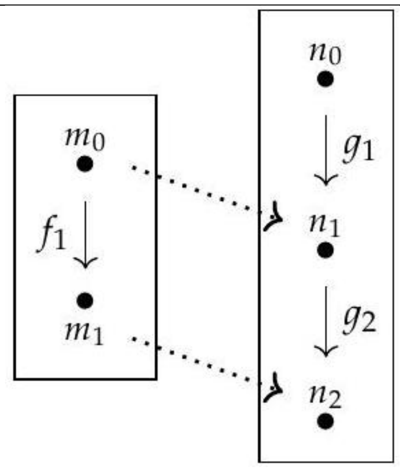
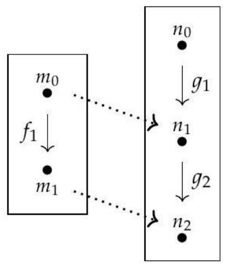
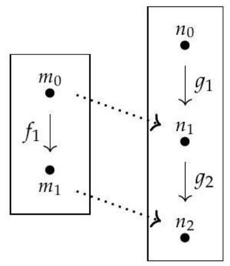
fong-spivak-seven-sketches_DIA_0365	273	—	*(dup)*	7f1e85c6-4350-401f-a960-eb8a0176e786-273_176_163_1946_1186.jpg			
fong-spivak-seven-sketches_DIA_0366	273	—	*(dup)*	7f1e85c6-4350-401f-a960-eb8a0176e786-273_176_163_1946_1480.jpg			
fong-spivak-seven-sketches_DIA_0367	274	—	*(dup)*	7f1e85c6-4350-401f-a960-eb8a0176e786-274_321_1360_398_389.jpg			
fong-spivak-seven-sketches_DIA_0368	275	—	*(dup)*	7f1e85c6-4350-401f-a960-eb8a0176e786-275_355_557_337_799.jpg			
fong-spivak-seven-sketches_DIA_0369	275	—	*(dup)*	7f1e85c6-4350-401f-a960-eb8a0176e786-275_276_645_1050_753.jpg			
fong-spivak-seven-sketches_DIA_0370	275	—	*(dup)*	7f1e85c6-4350-401f-a960-eb8a0176e786-275_570_1489_1937_371.jpg			

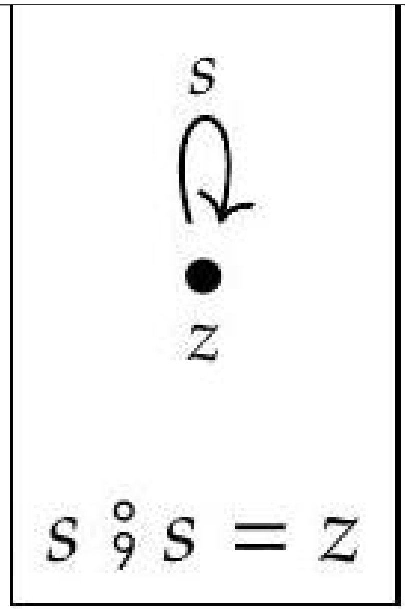
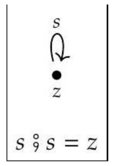
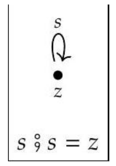
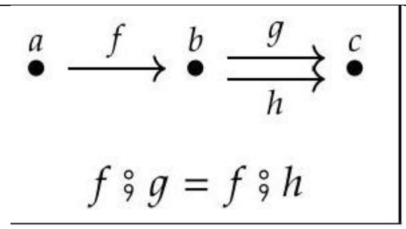
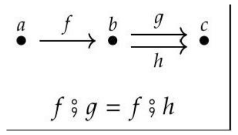
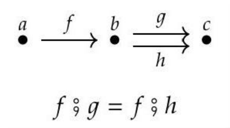
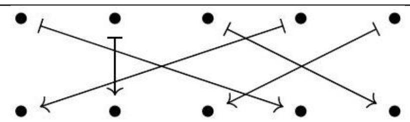
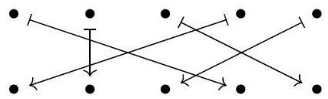
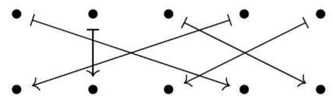
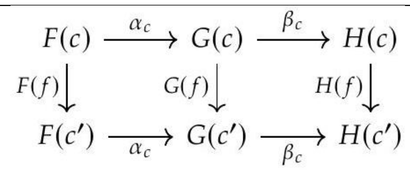
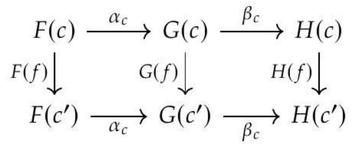
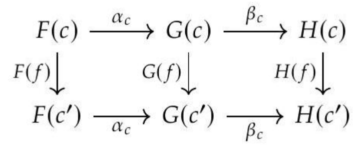
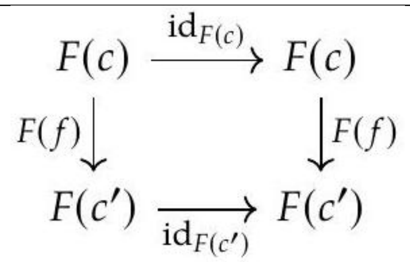
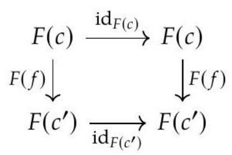
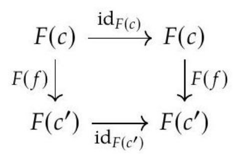
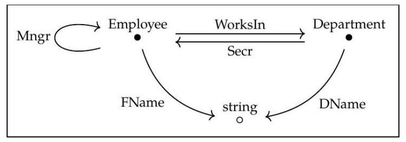
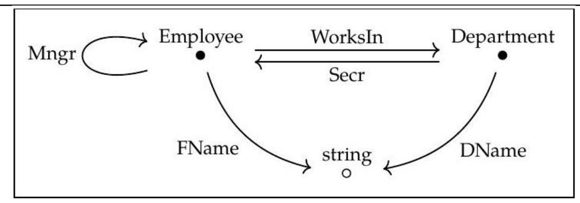
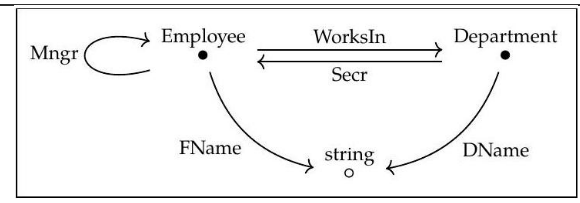
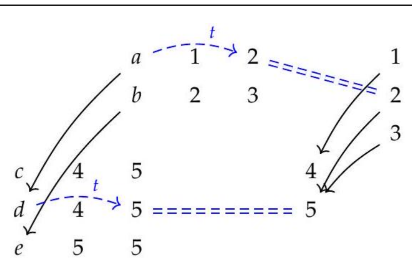
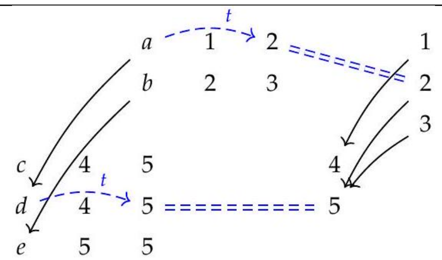
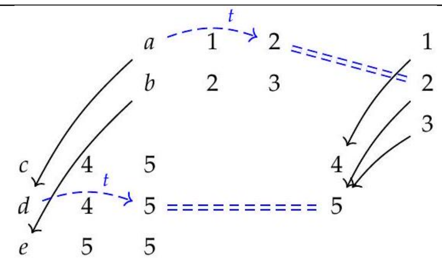
fong-spivak-seven-sketches_DIA_0371	276	— *(dup)*	7f1e85c6-4350-401f-a960-eb8a0176e786-276_369_826_391_663.jpg			
fong-spivak-seven-sketches_DIA_0372	276	— *(dup)*	7f1e85c6-4350-401f-a960-eb8a0176e786-276_274_878_887_638.jpg			
fong-spivak-seven-sketches_DIA_0373	276	— *(dup)*	7f1e85c6-4350-401f-a960-eb8a0176e786-276_281_869_1609_683.jpg			
fong-spivak-seven-sketches_DIA_0374	278	— *(dup)*	7f1e85c6-4350-401f-a960-eb8a0176e786-278_172_541_1041_846.jpg			
fong-spivak-seven-sketches_DIA_0375	278	— *(dup)*	7f1e85c6-4350-401f-a960-eb8a0176e786-278_174_543_1520_846.jpg			
fong-spivak-seven-sketches_DIA_0376	278	— *(dup)*	7f1e85c6-4350-401f-a960-eb8a0176e786-278_131_222_2353_448.jpg			
fong-spivak-seven-sketches_DIA_0377	278	— *(dup)*	7f1e85c6-4350-401f-a960-eb8a0176e786-278_129_247_2355_769.jpg			
fong-spivak-seven-sketches_DIA_0378	278	— *(dup)*	7f1e85c6-4350-401f-a960-eb8a0176e786-278_129_244_2355_1115.jpg			
fong-spivak-seven-sketches_DIA_0379	278	— *(dup)*	7f1e85c6-4350-401f-a960-eb8a0176e786-278_129_247_2355_1459.jpg			

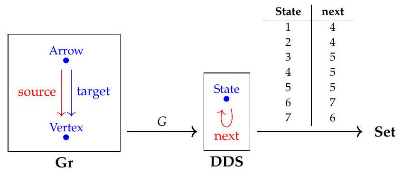
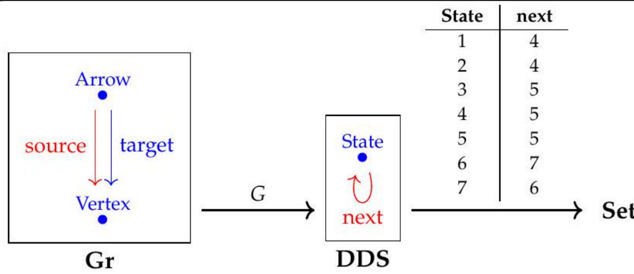
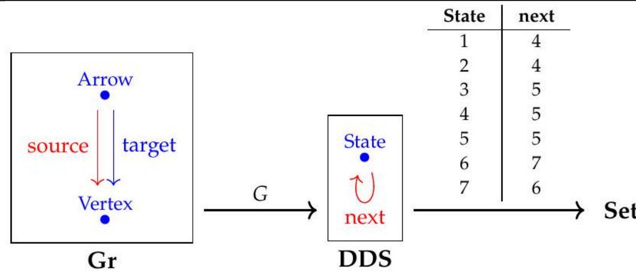
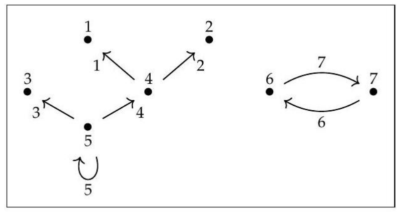
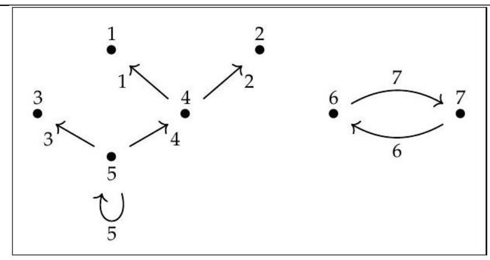
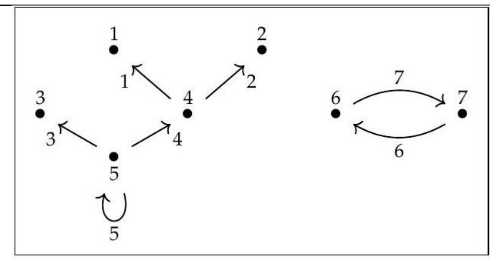
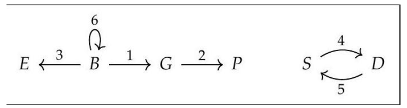
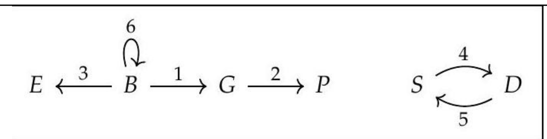
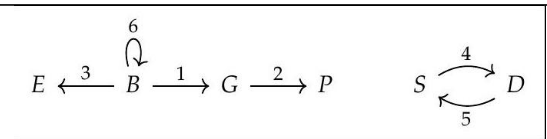
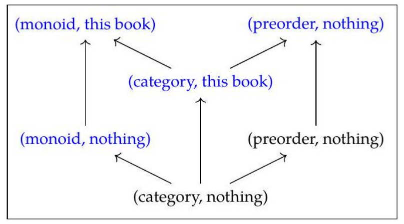
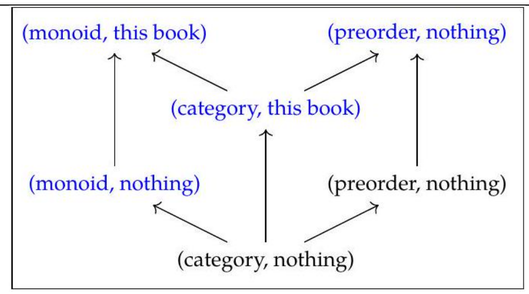
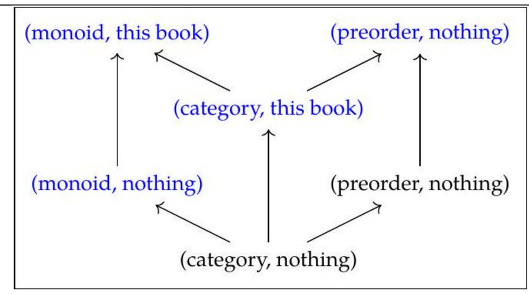
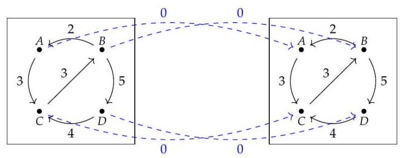
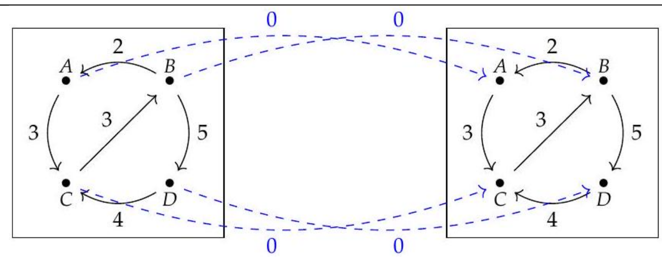
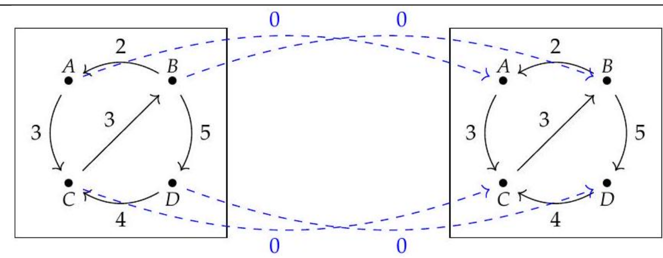
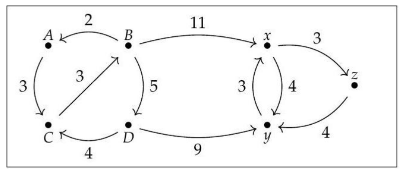
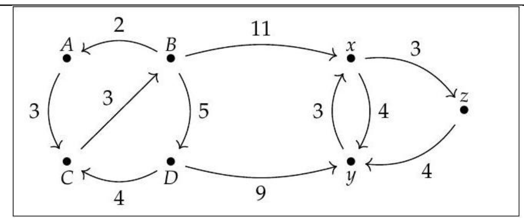
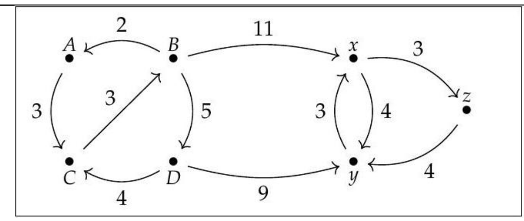
fong-spivak-seven-sketches_DIA_0380	279	— *(dup)*	7f1e85c6-4350-401f-a960-eb8a0176e786-279_131_140_674_534.jpg			
fong-spivak-seven-sketches_DIA_0381	279	— *(dup)*	7f1e85c6-4350-401f-a960-eb8a0176e786-279_129_138_676_771.jpg			
fong-spivak-seven-sketches_DIA_0382	279	— *(dup)*	7f1e85c6-4350-401f-a960-eb8a0176e786-279_129_138_676_1007.jpg			
fong-spivak-seven-sketches_DIA_0383	279	— *(dup)*	7f1e85c6-4350-401f-a960-eb8a0176e786-279_129_138_676_1244.jpg			
fong-spivak-seven-sketches_DIA_0384	279	— *(dup)*	7f1e85c6-4350-401f-a960-eb8a0176e786-279_129_140_676_1480.jpg			

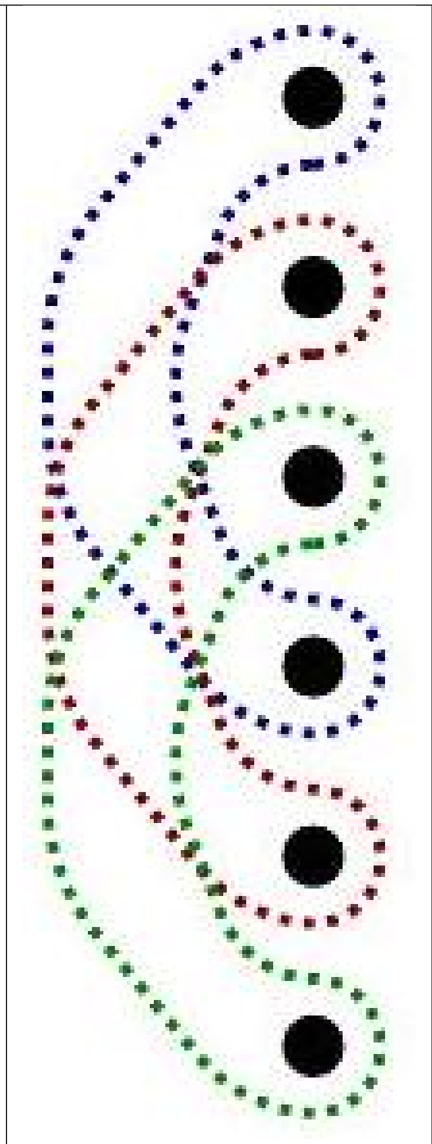
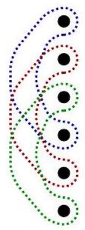
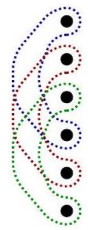
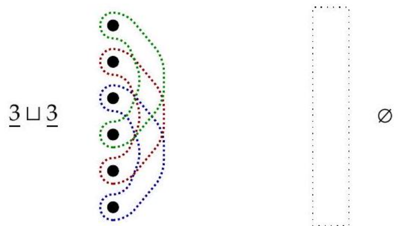
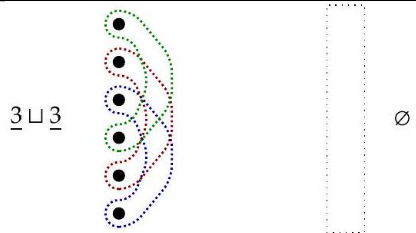
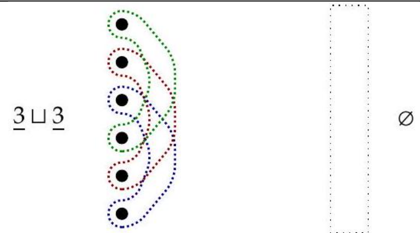
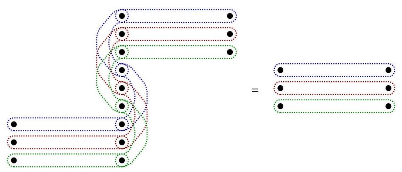
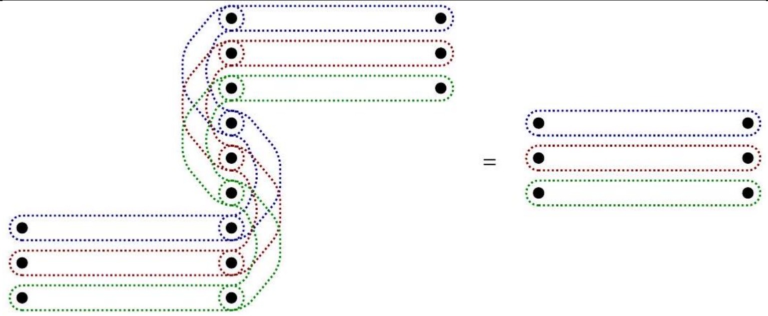
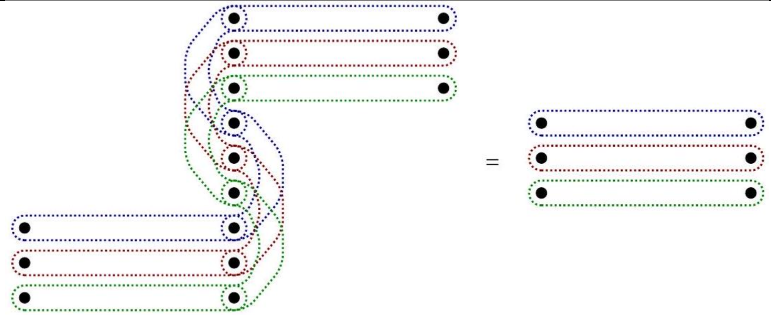
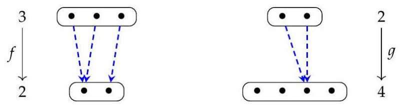
fong-spivak-seven-sketches_DIA_0385	279	—	*(dup)*	7f1e85c6-4350-401f-a960-eb8a0176e786-279_129_229_1025_765.jpg			
fong-spivak-seven-sketches_DIA_0386	279	—	*(dup)*	7f1e85c6-4350-401f-a960-eb8a0176e786-279_129_258_1025_1217.jpg			
fong-spivak-seven-sketches_DIA_0387	279	—	*(dup)*	7f1e85c6-4350-401f-a960-eb8a0176e786-279_127_188_1281_1025.jpg			
fong-spivak-seven-sketches_DIA_0388	279	—	*(dup)*	7f1e85c6-4350-401f-a960-eb8a0176e786-279_127_210_1538_1014.jpg			
fong-spivak-seven-sketches_DIA_0389	279	—	*(dup)*	7f1e85c6-4350-401f-a960-eb8a0176e786-279_131_262_1792_790.jpg			
fong-spivak-seven-sketches_DIA_0390	279	—	*(dup)*	7f1e85c6-4350-401f-a960-eb8a0176e786-279_131_283_1792_1163.jpg			
fong-spivak-seven-sketches_DIA_0391	280	—	*(dup)*	7f1e85c6-4350-401f-a960-eb8a0176e786-280_333_828_1005_661.jpg			

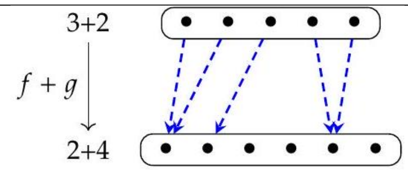
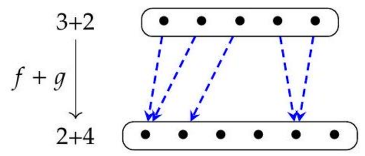
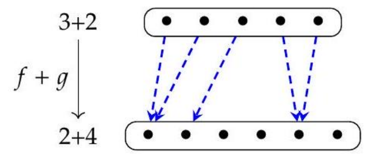
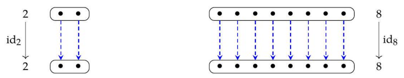
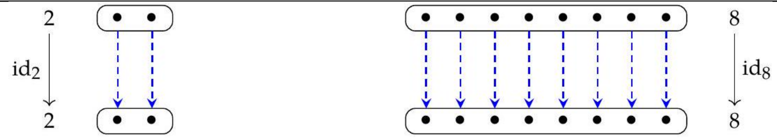
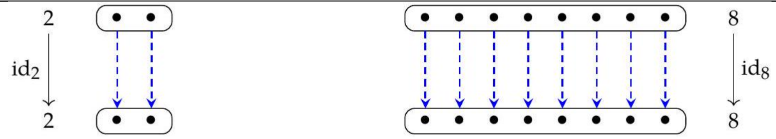
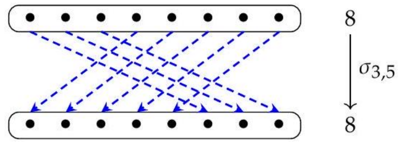
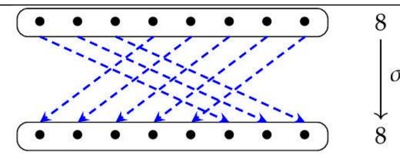
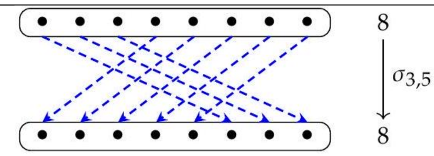
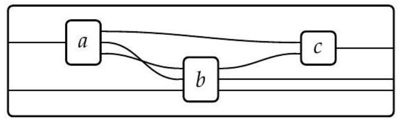
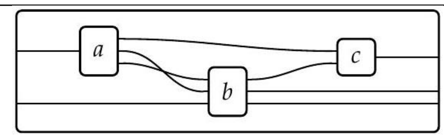
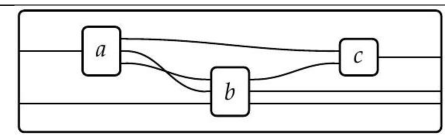
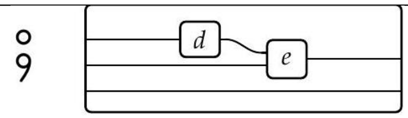
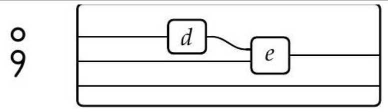
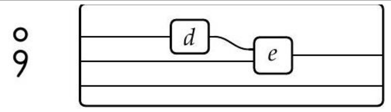
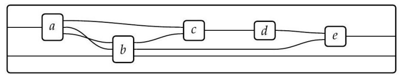
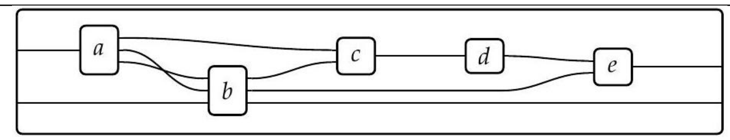
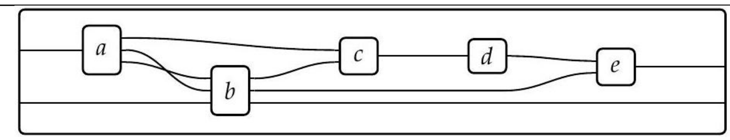
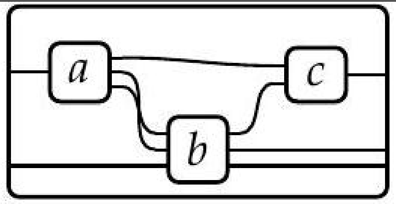
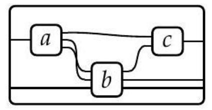
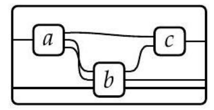
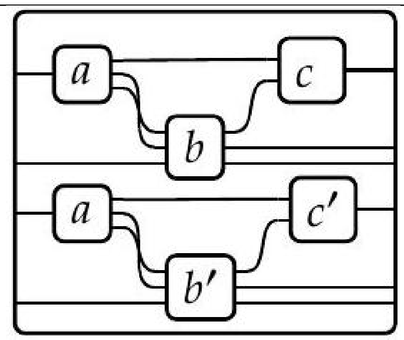
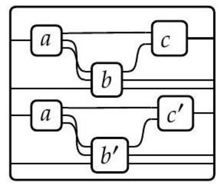
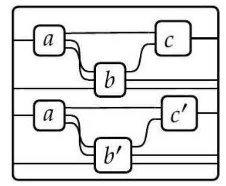
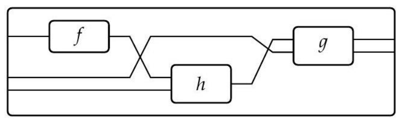
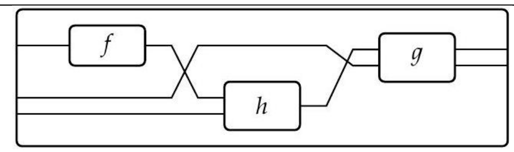
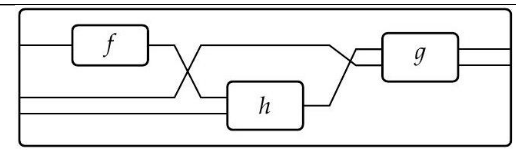
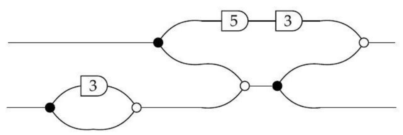
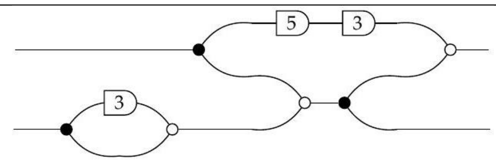
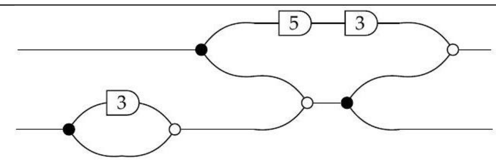
fong-spivak-seven-sketches_0392 — *(dup)*	280		7f1e85c6-4350-401f-a960-eb8a0176e786-280_328_419_2192_873.jpg			
fong-spivak-seven-sketches_0393 — *(dup)*	281		7f1e85c6-4350-401f-a960-eb8a0176e786-281_473_1278_982_437.jpg			
fong-spivak-seven-sketches_0394 — *(dup)*	285		7f1e85c6-4350-401f-a960-eb8a0176e786-285_326_550_290_844.jpg			
fong-spivak-seven-sketches_0395 — *(dup)*	285		7f1e85c6-4350-401f-a960-eb8a0176e786-285_229_314_1367_962.jpg			
fong-spivak-seven-sketches_0396 — *(dup)*	285		7f1e85c6-4350-401f-a960-eb8a0176e786-285_104_432_2396_860.jpg			
fong-spivak-seven-sketches_0397 — *(dup)*	290		7f1e85c6-4350-401f-a960-eb8a0176e786-290_274_428_523_860.jpg			
fong-spivak-seven-sketches_0398 — *(dup)*	290		7f1e85c6-4350-401f-a960-eb8a0176e786-290_208_328_1557_912.jpg			

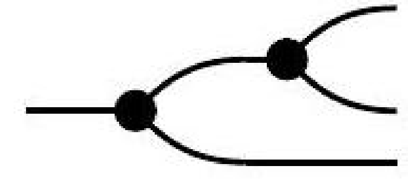
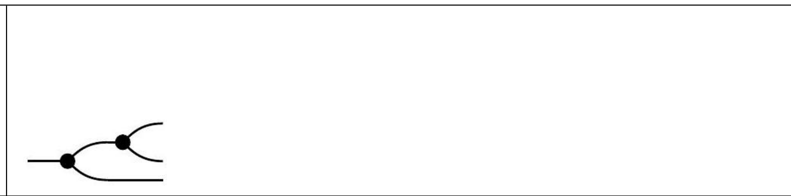
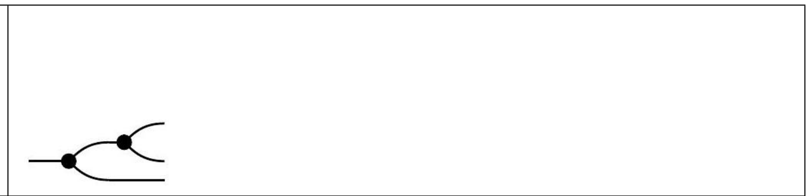
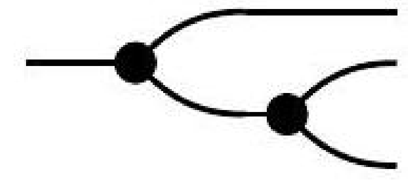
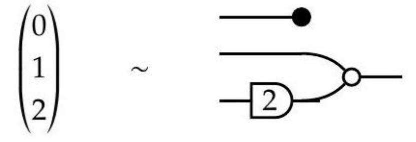
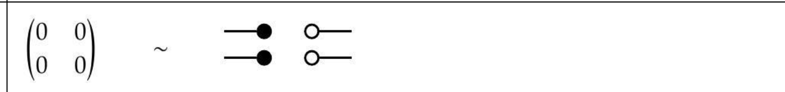
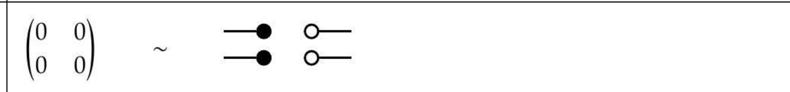
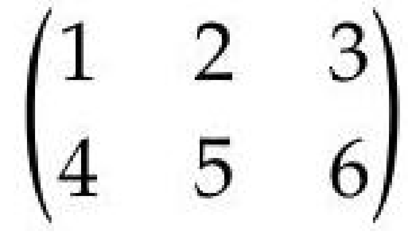
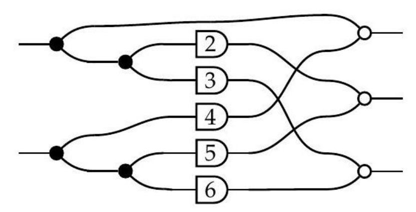
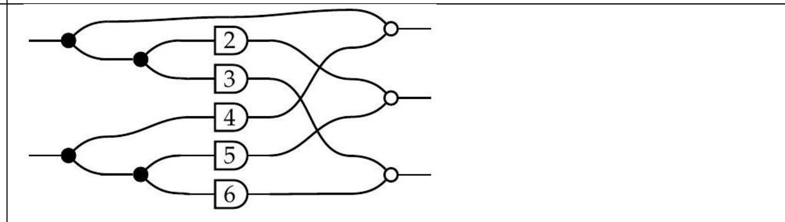
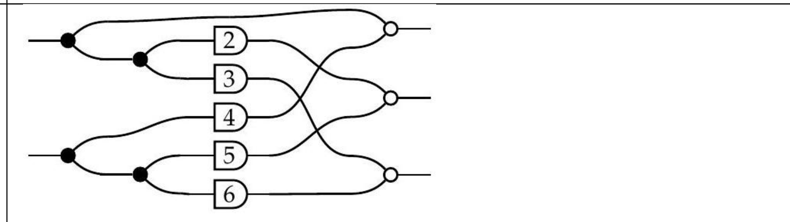
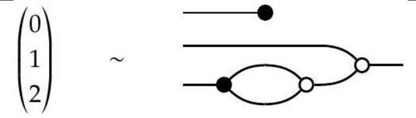
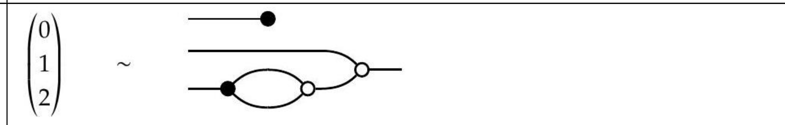
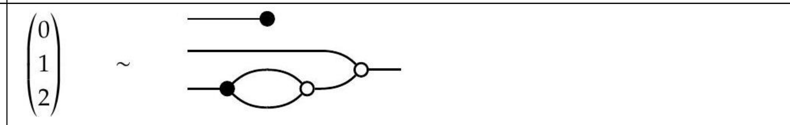
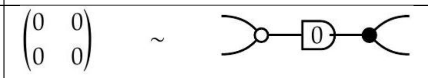
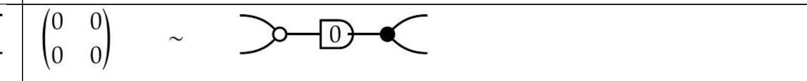
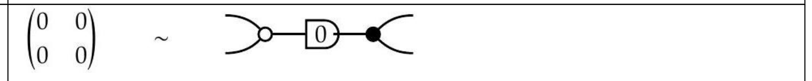
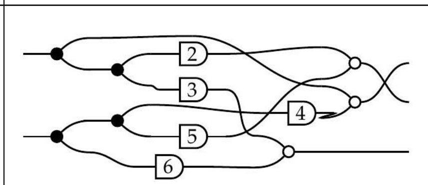
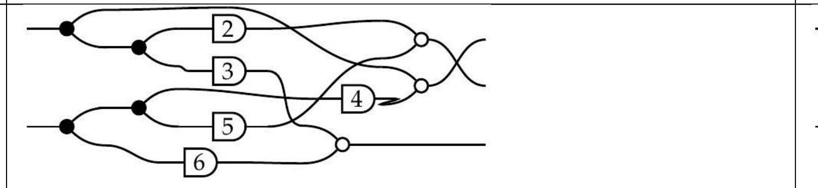
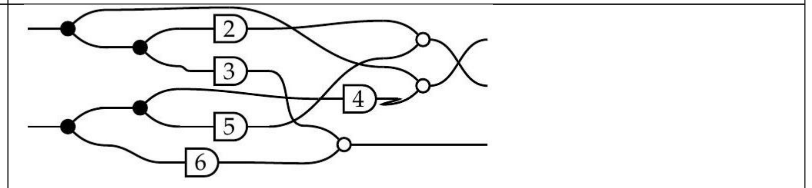
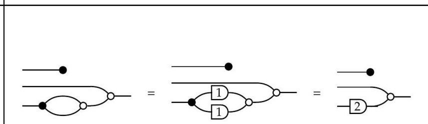
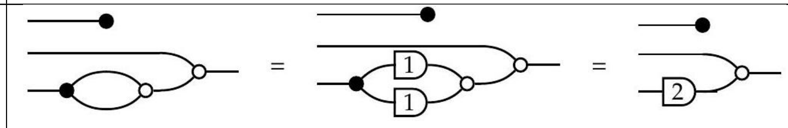
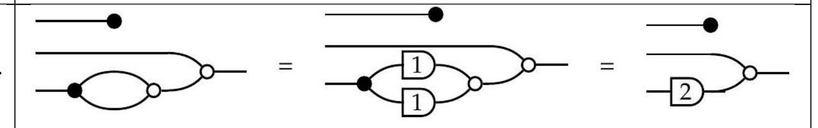
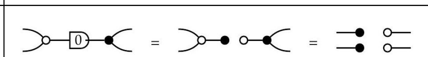
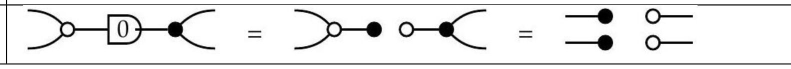
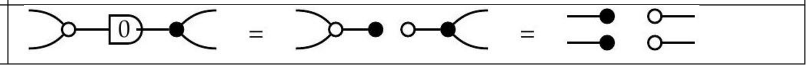
fong-spivak-seven-sketches_290_0399	290	— *(dup)*	7f1e85c6-4350-401f-a960-eb8a0176e786-290_247_1285_1968_434.jpg			
fong-spivak-seven-sketches_291_0400	291	— *(dup)*	7f1e85c6-4350-401f-a960-eb8a0176e786-291_335_285_1774_405.jpg			
fong-spivak-seven-sketches_291_0401	291	— *(dup)*	7f1e85c6-4350-401f-a960-eb8a0176e786-291_337_283_1774_932.jpg			
fong-spivak-seven-sketches_291_0402	291	— *(dup)*	7f1e85c6-4350-401f-a960-eb8a0176e786-291_337_287_1774_1455.jpg			

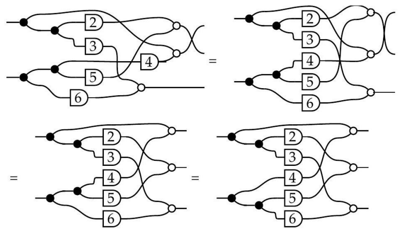
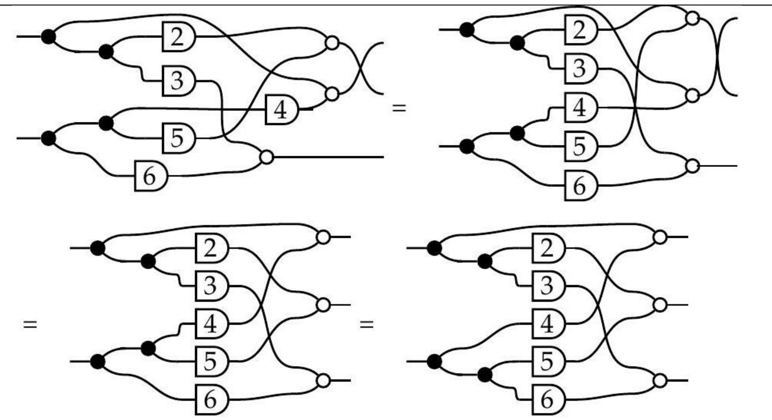
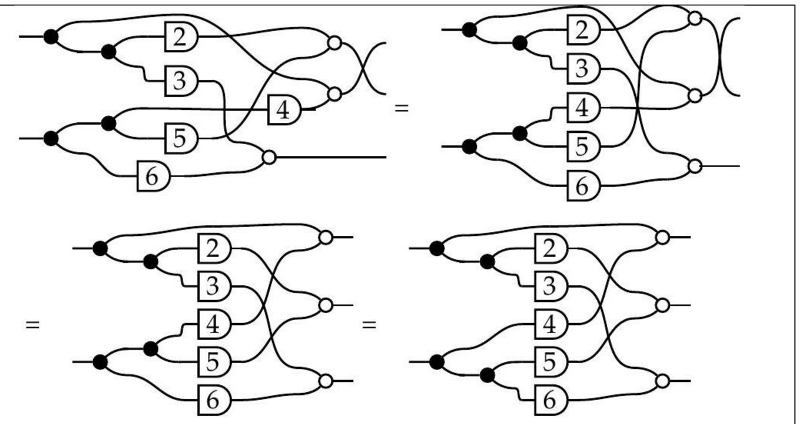
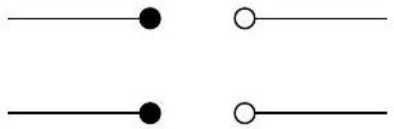
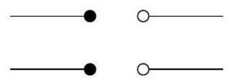
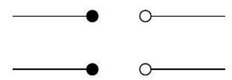
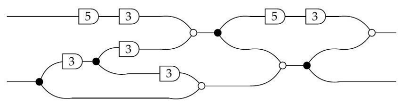
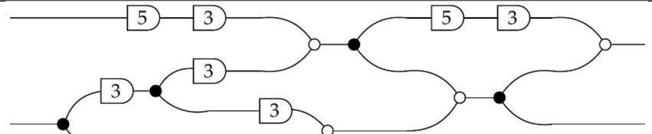
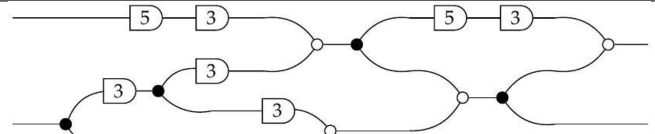
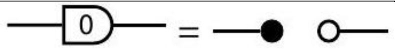
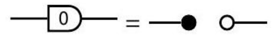
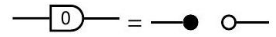
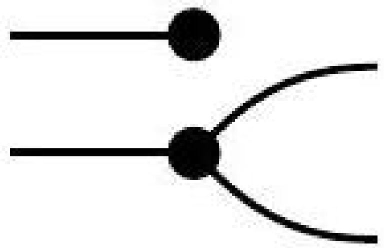
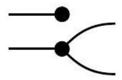
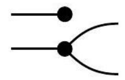
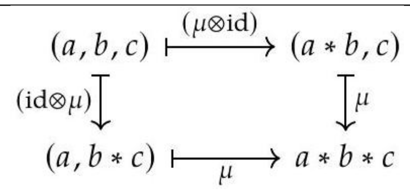
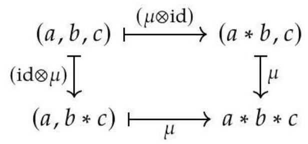
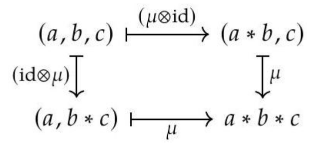
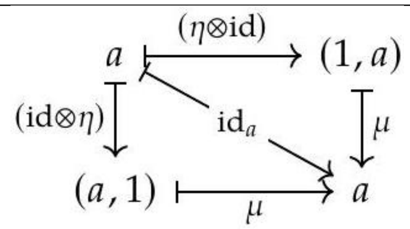
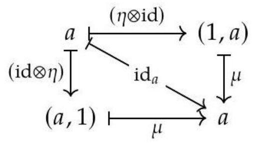
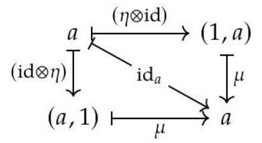
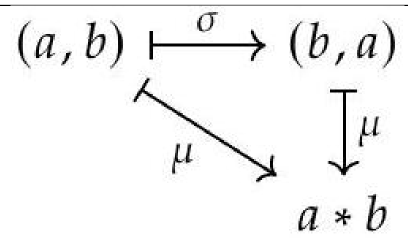
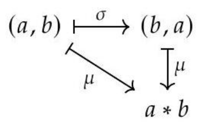
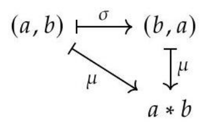
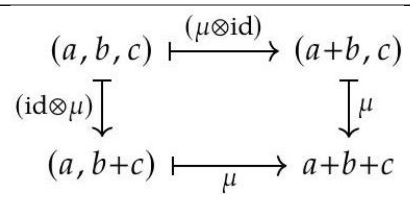
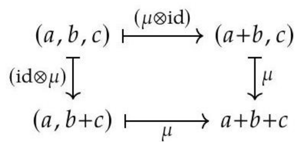
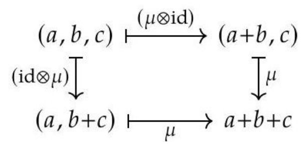
fong-spivak-seven-sketches_292_0403	292	— (dup)	7f1e85c6-4350-401f-a960-eb8a0176e786-292_204_133_1991_410.jpg			
fong-spivak-seven-sketches_292_0404	292	— (dup)	7f1e85c6-4350-401f-a960-eb8a0176e786-292_165_294_2016_837.jpg			
fong-spivak-seven-sketches_292_0405	292	— (dup)	7f1e85c6-4350-401f-a960-eb8a0176e786-292_124_416_2376_910.jpg			
fong-spivak-seven-sketches_293_0406	293	— (dup)	7f1e85c6-4350-401f-a960-eb8a0176e786-293_186_448_1271_889.jpg			
fong-spivak-seven-sketches_293_0407	293	— (dup)	7f1e85c6-4350-401f-a960-eb8a0176e786-293_190_292_1638_973.jpg			
fong-spivak-seven-sketches_294_0408	294	— (dup)	7f1e85c6-4350-401f-a960-eb8a0176e786-294_251_733_373_710.jpg			
fong-spivak-seven-sketches_294_0409	294	— (dup)	7f1e85c6-4350-401f-a960-eb8a0176e786-294_330_550_1529_844.jpg			

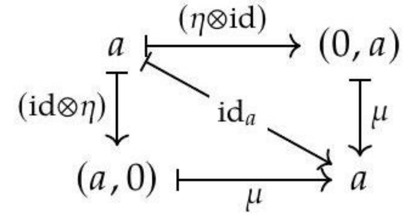
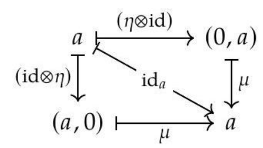
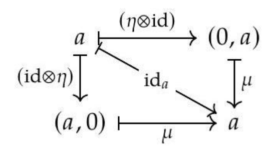
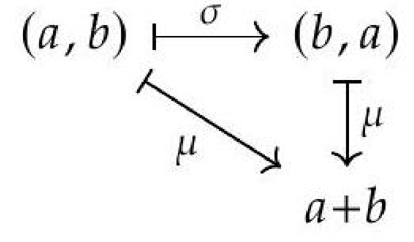
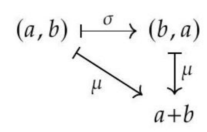
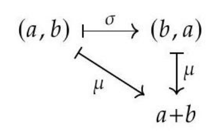
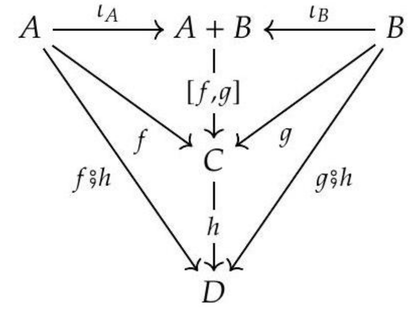
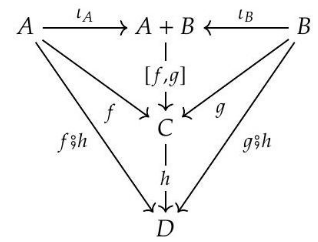
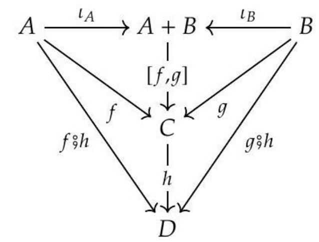
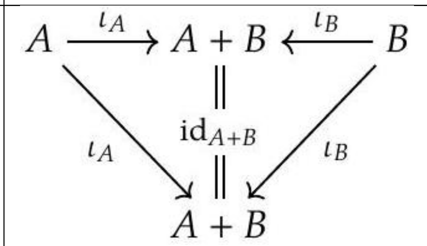
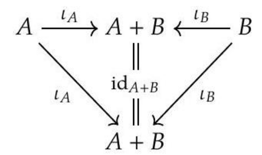
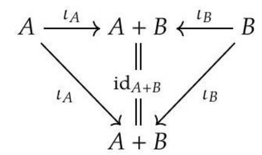
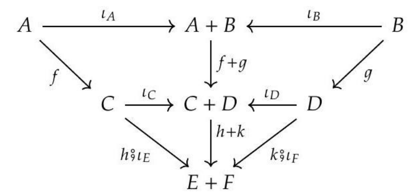
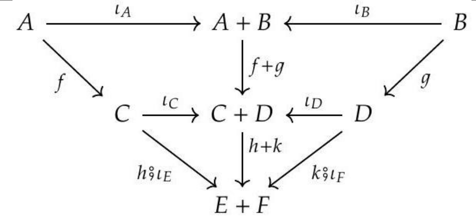
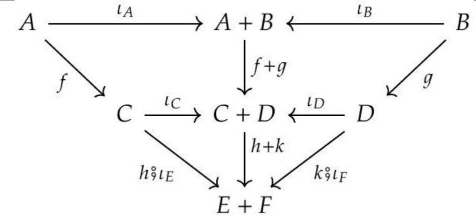
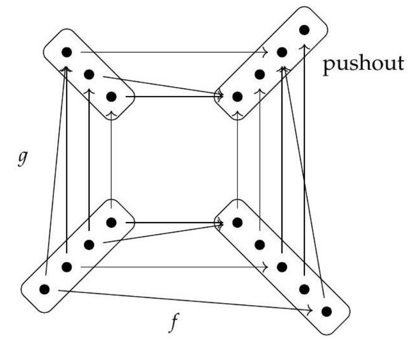
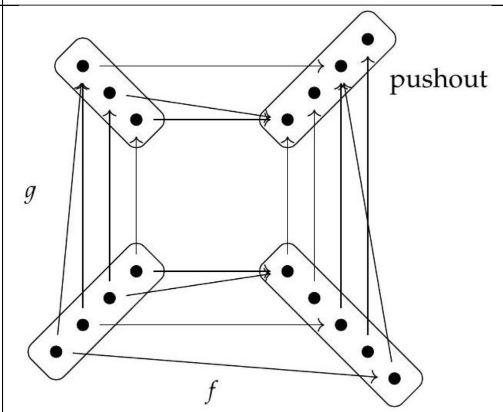
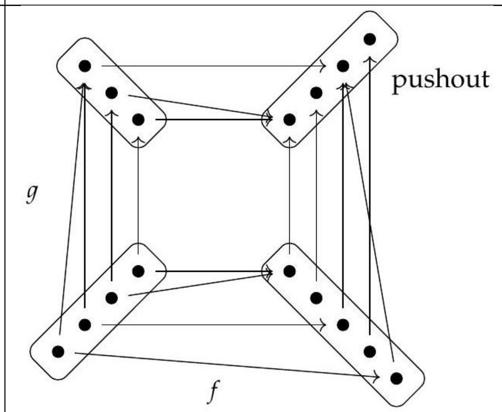
fong-spivak-seven-sketches_DIA_0410	294	— *(dup)*	7f1e85c6-4350-401f-a960-eb8a0176e786-294_353_830_2140_656.jpg			
fong-spivak-seven-sketches_DIA_0411	295	— *(dup)*	7f1e85c6-4350-401f-a960-eb8a0176e786-295_333_627_333_1138.jpg			
fong-spivak-seven-sketches_DIA_0412	295	— *(dup)*	7f1e85c6-4350-401f-a960-eb8a0176e786-295_176_692_2156_733.jpg			
fong-spivak-seven-sketches_DIA_0413	298	— *(dup)*	7f1e85c6-4350-401f-a960-eb8a0176e786-298_372_672_438_782.jpg			
fong-spivak-seven-sketches_DIA_0414	300	— *(dup)*	7f1e85c6-4350-401f-a960-eb8a0176e786-300_326_851_1357_652.jpg			
fong-spivak-seven-sketches_DIA_0415	301	— *(dup)*	7f1e85c6-4350-401f-a960-eb8a0176e786-301_276_663_1767_747.jpg			

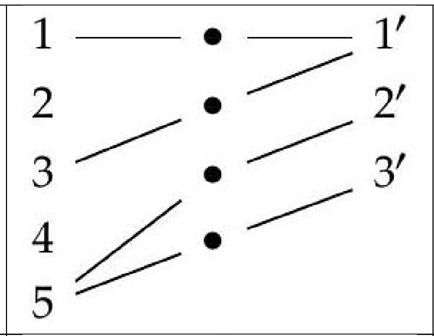
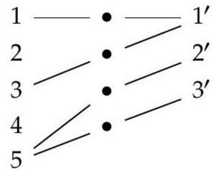
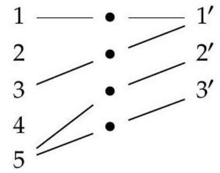
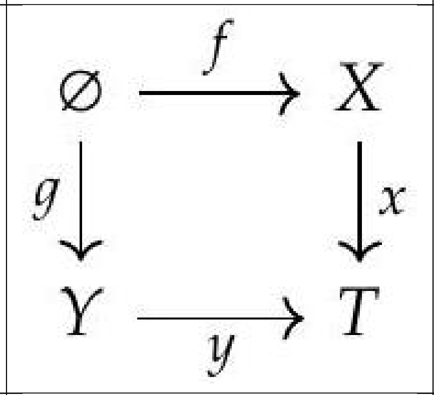
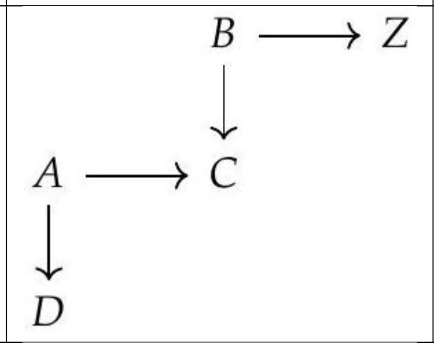
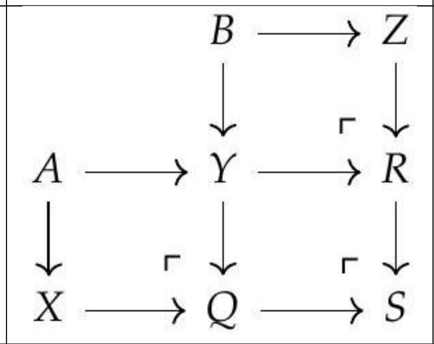
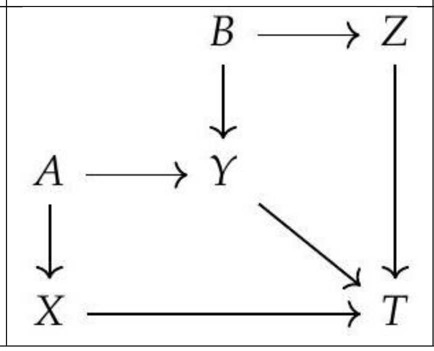
fong-spivak-seven-sketches_DIA_0417	302	— *(dup)*	7f1e85c6-4350-401f-a960-eb8a0176e786-302_294_102_2204_1163.jpg			
fong-spivak-seven-sketches_DIA_0418	303	— *(dup)*	7f1e85c6-4350-401f-a960-eb8a0176e786-303_308_534_324_851.jpg			
fong-spivak-seven-sketches_DIA_0419	303	— *(dup)*	7f1e85c6-4350-401f-a960-eb8a0176e786-303_443_1077_729_579.jpg			
fong-spivak-seven-sketches_DIA_0420	305	— *(dup)*	7f1e85c6-4350-401f-a960-eb8a0176e786-305_195_783_452_726.jpg			

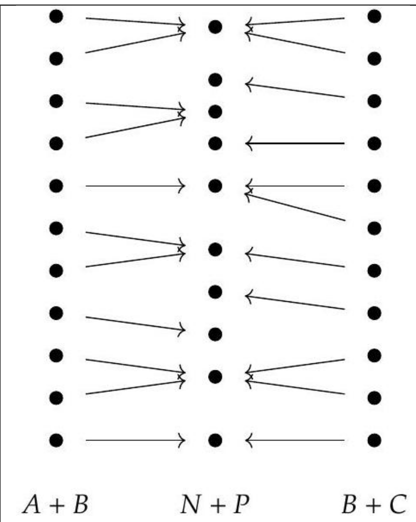
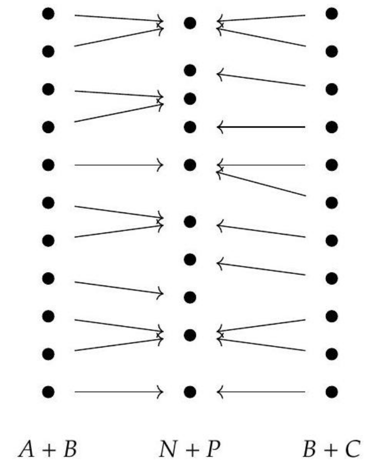
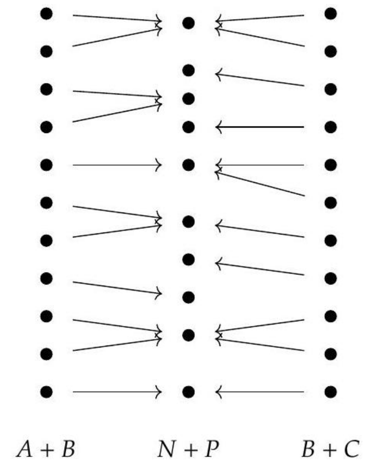
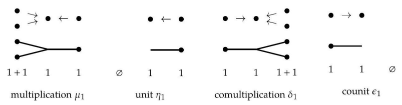
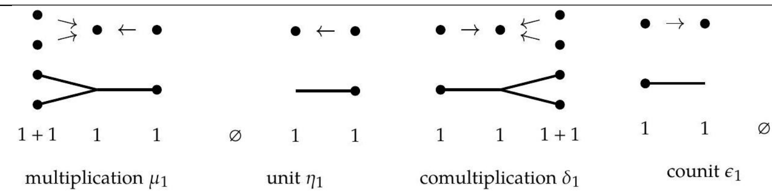
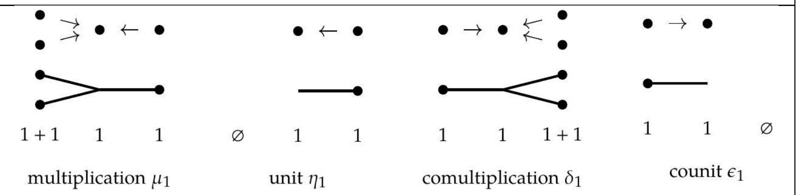
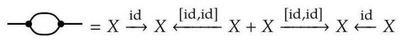
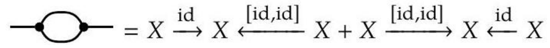
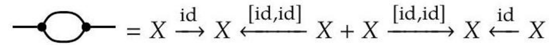
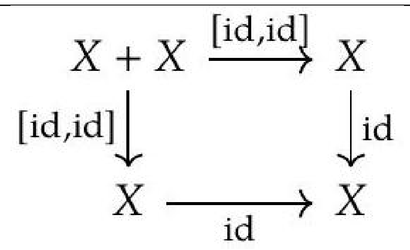
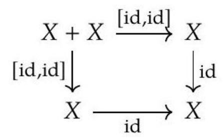
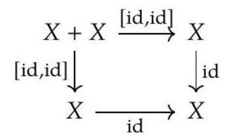
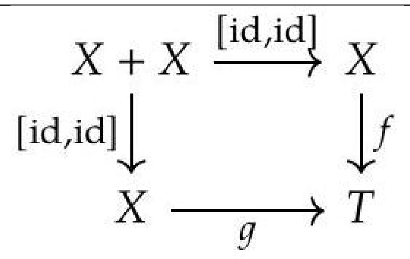
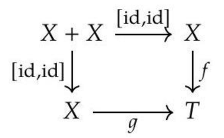
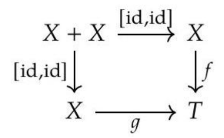
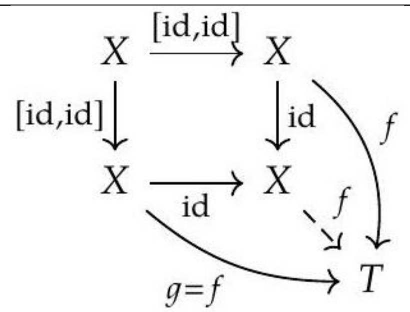
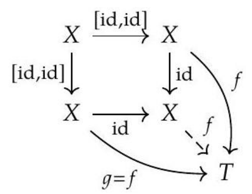
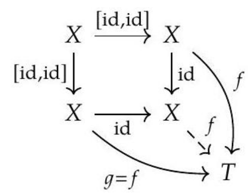
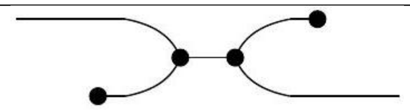
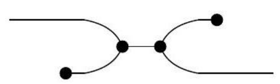
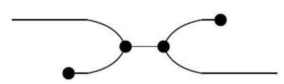
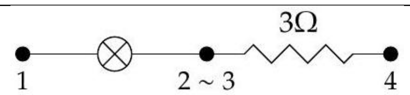
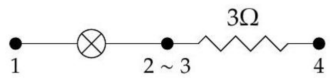
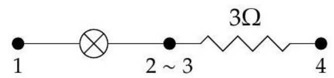
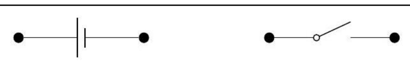
fong-spivak-seven-sketches_DIA_0421	305	— *(dup)*	7f1e85c6-4350-401f-a960-eb8a0176e786-305_192_462_729_891.jpg			
fong-spivak-seven-sketches_DIA_0422	305	— *(dup)*	7f1e85c6-4350-401f-a960-eb8a0176e786-305_195_1097_1102_568.jpg			
fong-spivak-seven-sketches_DIA_0423	305	— *(dup)*	7f1e85c6-4350-401f-a960-eb8a0176e786-305_192_545_1407_839.jpg			
fong-spivak-seven-sketches_DIA_0424	306	— *(dup)*	7f1e85c6-4350-401f-a960-eb8a0176e786-306_167_561_2353_523.jpg			
fong-spivak-seven-sketches_DIA_0425	306	— *(dup)*	7f1e85c6-4350-401f-a960-eb8a0176e786-306_138_491_2378_1138.jpg			
fong-spivak-seven-sketches_DIA_0426	307	— *(dup)*	7f1e85c6-4350-401f-a960-eb8a0176e786-307_170_932_317_611.jpg			
fong-spivak-seven-sketches_DIA_0427	307	— *(dup)*	7f1e85c6-4350-401f-a960-eb8a0176e786-307_136_262_808_686.jpg			
fong-spivak-seven-sketches_DIA_0428	307	— *(dup)*	7f1e85c6-4350-401f-a960-eb8a0176e786-307_235_276_708_1190.jpg			
fong-spivak-seven-sketches_DIA_0429	309	— *(dup)*	7f1e85c6-4350-401f-a960-eb8a0176e786-309_188_652_389_751.jpg			
fong-spivak-seven-sketches_DIA_0430	309	— *(dup)*	7f1e85c6-4350-401f-a960-eb8a0176e786-309_201_627_1880_765.jpg			

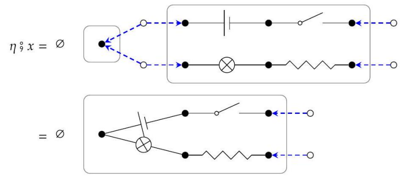
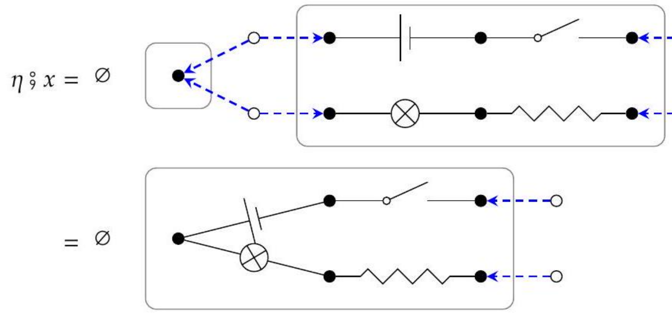
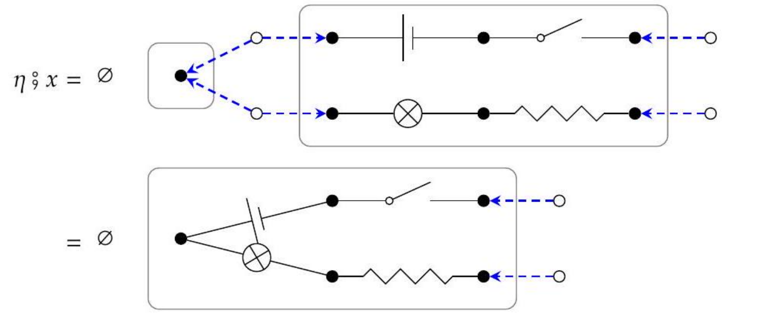
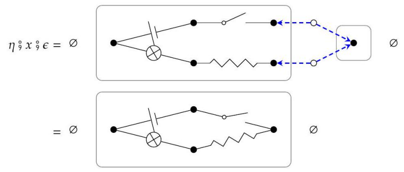
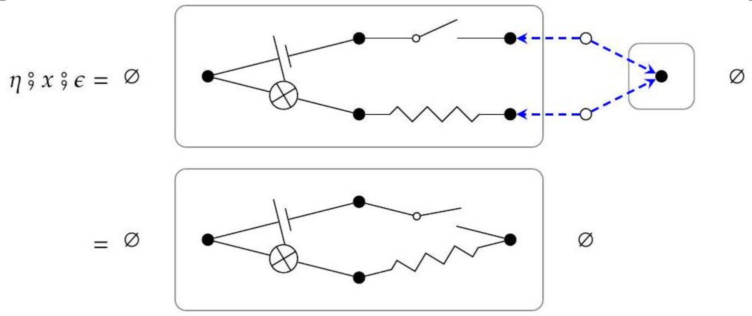
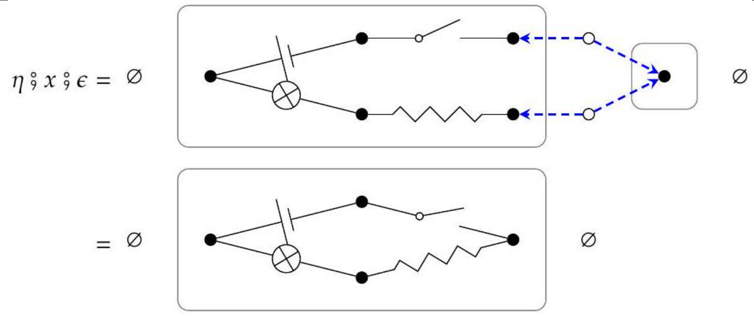
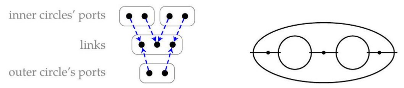
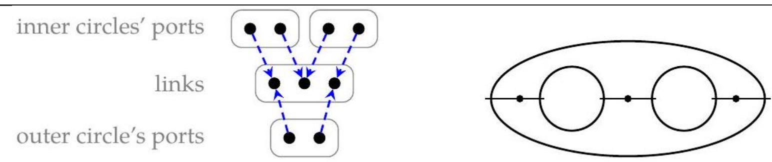
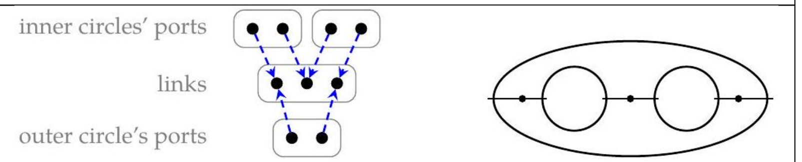
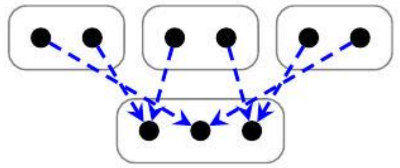
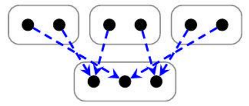
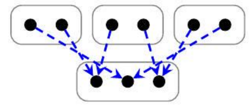
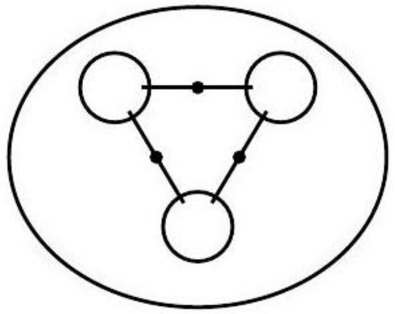
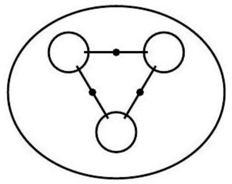
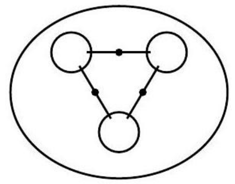
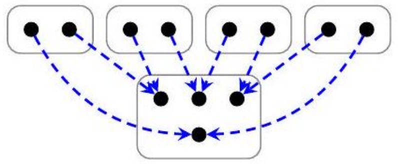
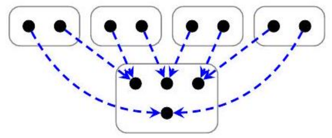
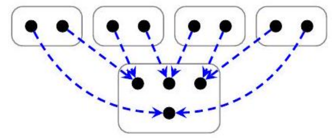
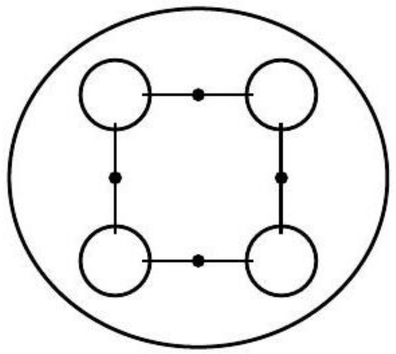
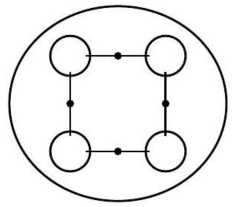
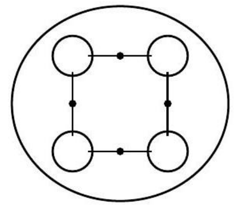
fong-spivak-seven-sketches_310_0431	310	— *(dup)*	7f1e85c6-4350-401f-a960-eb8a0176e786-310_86_187_370_739.jpg			
fong-spivak-seven-sketches_310_0432	310	— *(dup)*	7f1e85c6-4350-401f-a960-eb8a0176e786-310_86_187_554_739.jpg			
fong-spivak-seven-sketches_310_0433	310	— *(dup)*	7f1e85c6-4350-401f-a960-eb8a0176e786-310_142_408_834_915.jpg			
fong-spivak-seven-sketches_310_0434	310	— *(dup)*	7f1e85c6-4350-401f-a960-eb8a0176e786-310_88_432_1016_904.jpg			
fong-spivak-seven-sketches_310_0435	310	— *(dup)*	7f1e85c6-4350-401f-a960-eb8a0176e786-310_86_146_1242_700.jpg			
fong-spivak-seven-sketches_310_0436	310	— *(dup)*	7f1e85c6-4350-401f-a960-eb8a0176e786-310_271_537_1143_1005.jpg			
fong-spivak-seven-sketches_311_0437	311	— *(dup)*	7f1e85c6-4350-401f-a960-eb8a0176e786-311_143_498_611_905.jpg			
fong-spivak-seven-sketches_311_0438	311	— *(dup)*	7f1e85c6-4350-401f-a960-eb8a0176e786-311_88_511_799_900.jpg			
fong-spivak-seven-sketches_311_0439	311	— *(dup)*	7f1e85c6-4350-401f-a960-eb8a0176e786-311_229_609_928_1005.jpg			
fong-spivak-seven-sketches_311_0440	311	— *(dup)*	7f1e85c6-4350-401f-a960-eb8a0176e786-311_152_1005_1294_652.jpg			
fong-spivak-seven-sketches_311_0441	311	— *(dup)*	7f1e85c6-4350-401f-a960-eb8a0176e786-311_63_878_1493_715.jpg			

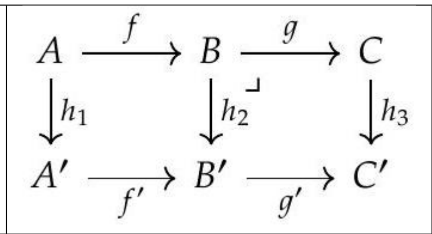
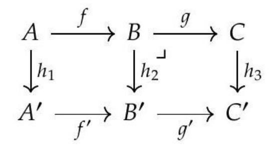
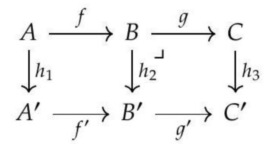
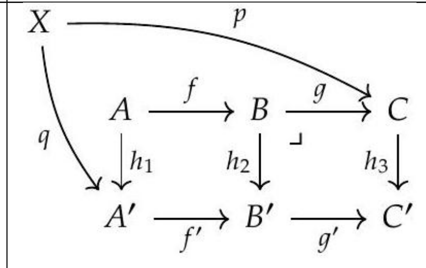
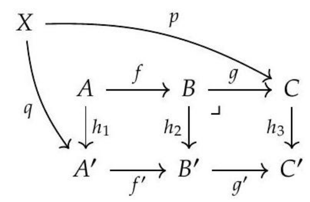
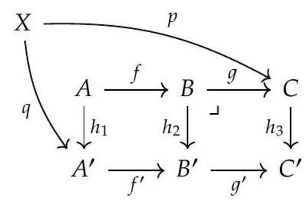
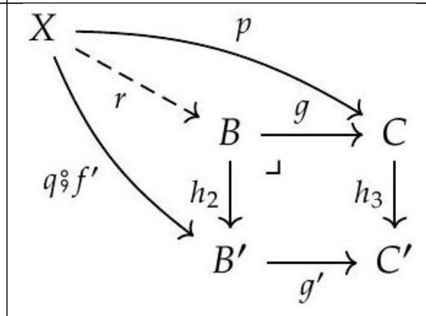
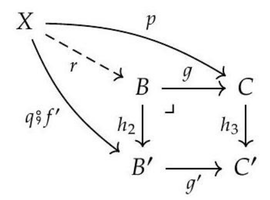
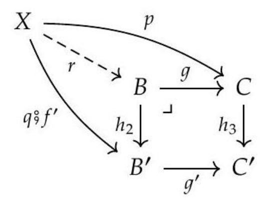
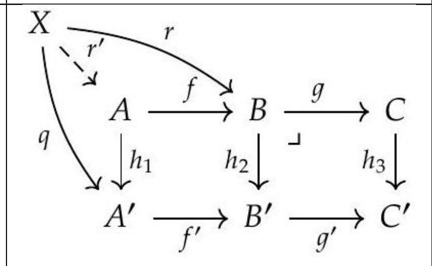
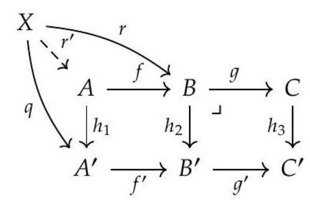
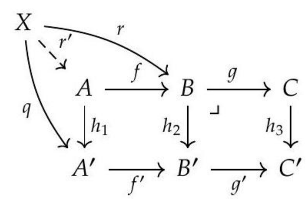
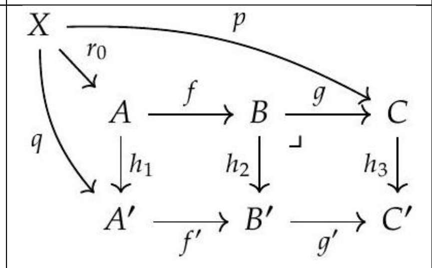
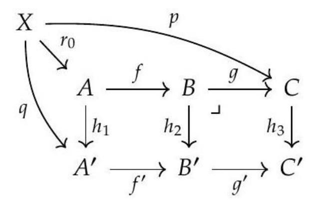
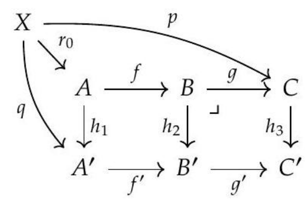
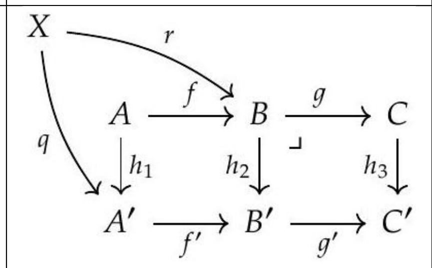
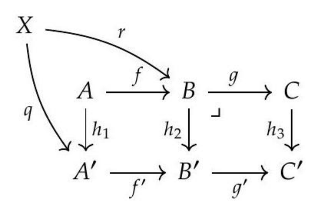
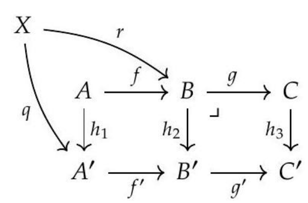
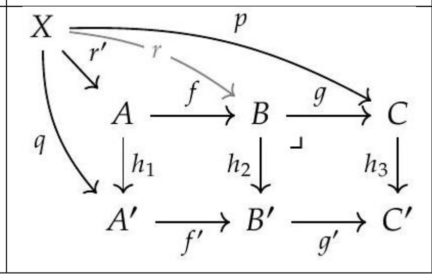
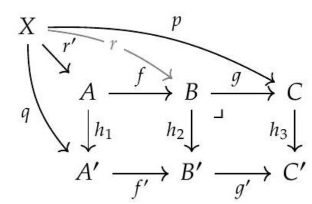
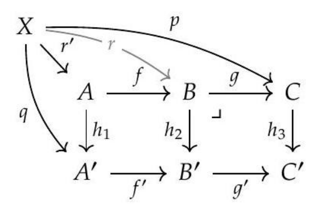
fong-spivak-seven-sketches_311_0442	311	— (dup)	7f1e85c6-4350-401f-a960-eb8a0176e786-311_538_950_1624_710.jpg			
fong-spivak-seven-sketches_311_0443	311	— (dup)	7f1e85c6-4350-401f-a960-eb8a0176e786-311_97_287_2357_502.jpg			
fong-spivak-seven-sketches_311_0444	311	— (dup)	7f1e85c6-4350-401f-a960-eb8a0176e786-311_199_837_2305_896.jpg			
fong-spivak-seven-sketches_312_0445	312	— (dup)	7f1e85c6-4350-401f-a960-eb8a0176e786-312_43_344_410_946.jpg			
fong-spivak-seven-sketches_312_0446	312	— (dup)	7f1e85c6-4350-401f-a960-eb8a0176e786-312_100_149_679_1045.jpg			
fong-spivak-seven-sketches_312_0447	312	— (dup)	7f1e85c6-4350-401f-a960-eb8a0176e786-312_181_389_1271_554.jpg			
fong-spivak-seven-sketches_312_0448	312	— (dup)	7f1e85c6-4350-401f-a960-eb8a0176e786-312_181_319_1271_1020.jpg			
fong-spivak-seven-sketches_312_0449	312	— (dup)	7f1e85c6-4350-401f-a960-eb8a0176e786-312_154_249_1290_1425.jpg			
fong-spivak-seven-sketches_312_0450	312	— (dup)	7f1e85c6-4350-401f-a960-eb8a0176e786-312_181_378_1588_559.jpg			

fong-spivak-seven-sketch[illegible]0451	312	— *(dup)*	7f1e85c6-4350-401f-a960-eb8a0176e786-312_179_324_1593_1016.jpg			
fong-spivak-seven-sketch[illegible]0452	312	— *(dup)*	7f1e85c6-4350-401f-a960-eb8a0176e786-312_154_249_1611_1421.jpg			
fong-spivak-seven-sketch[illegible]0453	316	— *(dup)*	*no image for this region (498 in zip)*	*(no tex.zip crop for this row)*	*(image not on disk — pass –texzip or download the CDN crop)*	*(image not on disk — pass –texzip or download the CDN crop)*
fong-spivak-seven-sketch[illegible]0454	316	— *(dup)*	7f1e85c6-4350-401f-a960-eb8a0176e786-316_308_391_1785_921.jpg			
fong-spivak-seven-sketch[illegible]0455	316	— *(dup)*	7f1e85c6-4350-401f-a960-eb8a0176e786-316_192_317_2258_959.jpg			
fong-spivak-seven-sketch[illegible]0456	317	— *(dup)*	7f1e85c6-4350-401f-a960-eb8a0176e786-317_283_597_1088_819.jpg			
fong-spivak-seven-sketch[illegible]0457	318	— *(dup)*	7f1e85c6-4350-401f-a960-eb8a0176e786-318_527_615_387_769.jpg			

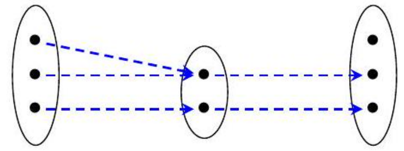
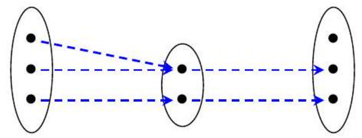
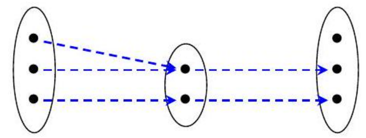
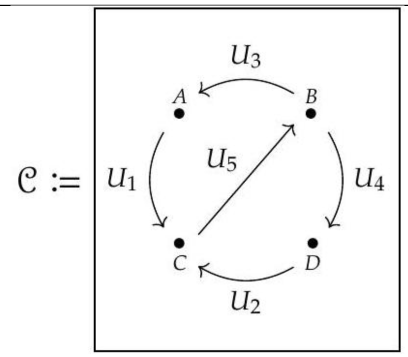
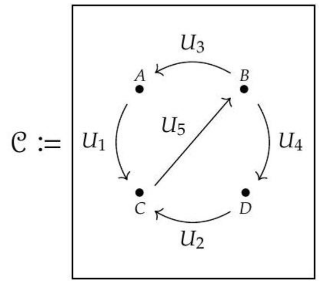
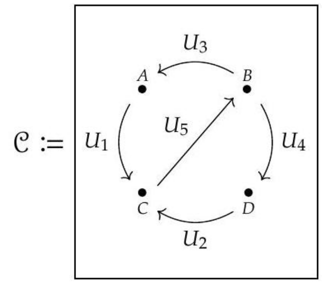
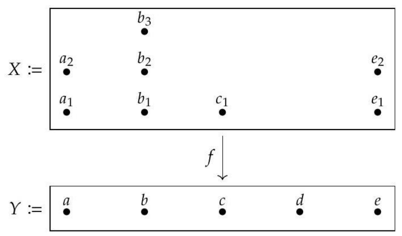
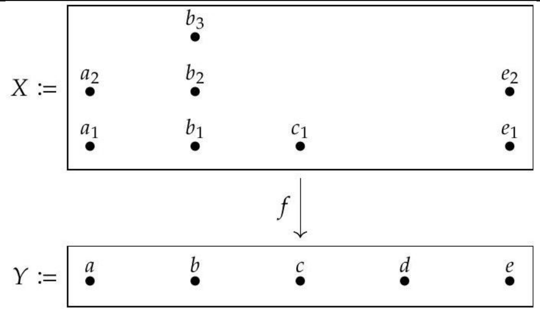
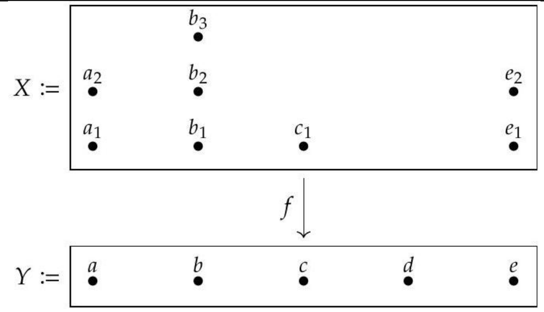
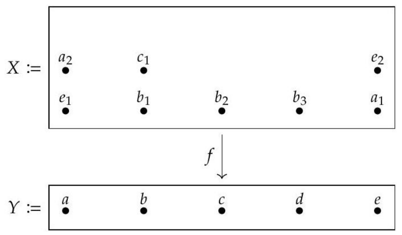
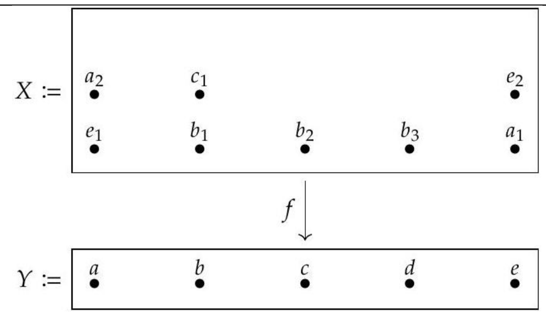
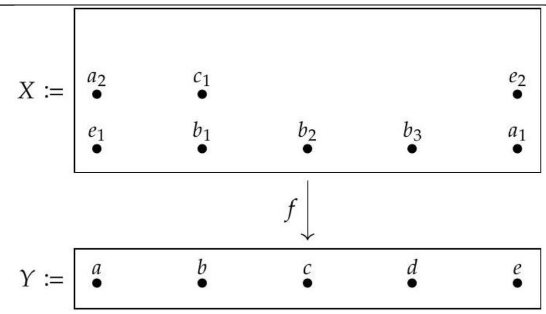
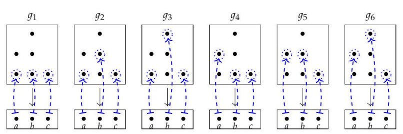
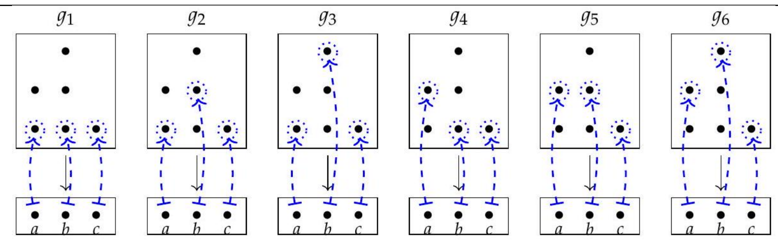
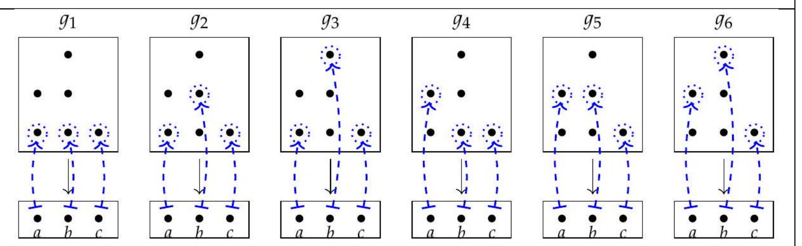
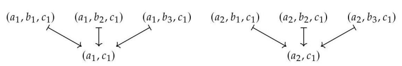
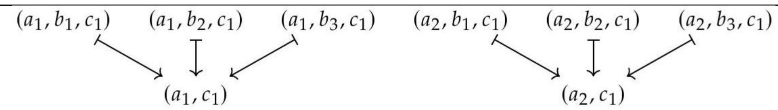
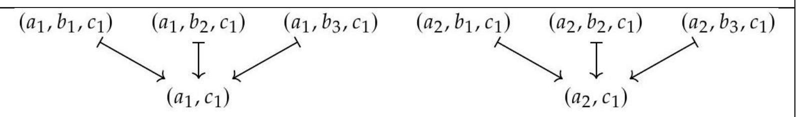
fong-spivak-seven-sketches_DIA_0458	318	— *(dup)*	7f1e85c6-4350-401f-a960-eb8a0176e786-318_222_274_1100_939.jpg			
fong-spivak-seven-sketches_DIA_0459	318	— *(dup)*	7f1e85c6-4350-401f-a960-eb8a0176e786-318_172_179_1824_1029.jpg		$\begin{tikzcd} \varnothing \arrow[r, "f"] \arrow[d, "g"'] & X \arrow[d, "x"] \\ Y \arrow[r, "y"'] & T \end{tikzcd}$	$\begin{tikzcd} \varnothing \arrow[r, "f"] \arrow[d, "g"'] & X \arrow[d, "x"] \\ Y \arrow[r, "y"'] & T \end{tikzcd}$
fong-spivak-seven-sketches_DIA_0460	318	— *(dup)*	7f1e85c6-4350-401f-a960-eb8a0176e786-318_238_287_2269_654.jpg		$\begin{tikzcd} & B \arrow[r] \arrow[d] & Z \\ A \arrow[r] \arrow[d] & C \\ D \end{tikzcd}$	$\begin{tikzcd} & B \arrow[r] \arrow[d] & Z \\ A \arrow[r] \arrow[d] & C \\ D \end{tikzcd}$
fong-spivak-seven-sketches_DIA_0461	318	— *(dup)*	7f1e85c6-4350-401f-a960-eb8a0176e786-318_242_290_2267_1208.jpg		$\begin{tikzcd} & B \arrow[r] \arrow[d] & Z \arrow[d] \\ A \arrow[r] \arrow[d] & Y \arrow[r] \arrow[d] & R \arrow[d] \arrow[ul, phantom, "\ulcorner", very near start] \\ X \arrow[r] & Q \arrow[r] \arrow[ul, phantom, "\ulcorner", very near start] & S \arrow[ul, phantom, "\ulcorner", very near start] \end{tikzcd}$	$\begin{tikzcd} & B \arrow[r] \arrow[d] & Z \arrow[d] \\ A \arrow[r] \arrow[d] & Y \arrow[r] \arrow[d] & R \arrow[d] \arrow[ul, phantom, "\ulcorner", very near start] \\ X \arrow[r] & Q \arrow[r] \arrow[ul, phantom, "\ulcorner", very near start] & S \arrow[ul, phantom, "\ulcorner", very near start] \end{tikzcd}$
fong-spivak-seven-sketches_DIA_0462	319	— *(dup)*	7f1e85c6-4350-401f-a960-eb8a0176e786-319_240_287_376_932.jpg		$\begin{tikzcd} & B \arrow[r] \arrow[d] & Z \arrow[dd] \\ A \arrow[r] \arrow[d] & Y \arrow[dr] \\ X \arrow[rr] & & T \end{tikzcd}$	$\begin{tikzcd} & B \arrow[r] \arrow[d] & Z \arrow[dd] \\ A \arrow[r] \arrow[d] & Y \arrow[dr] \\ X \arrow[rr] & & T \end{tikzcd}$

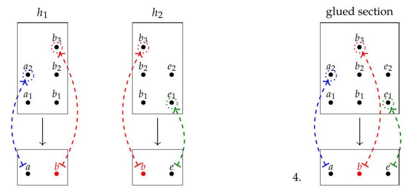
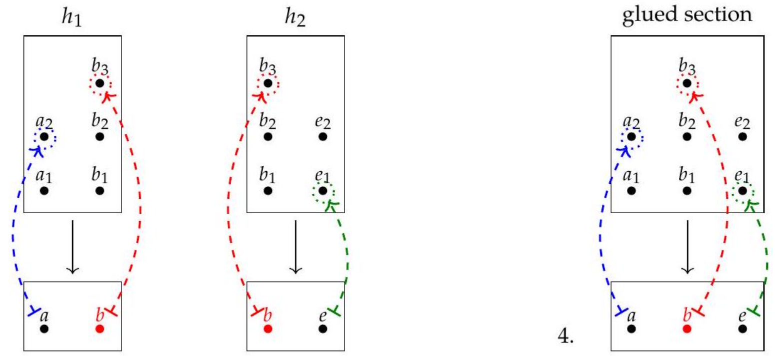
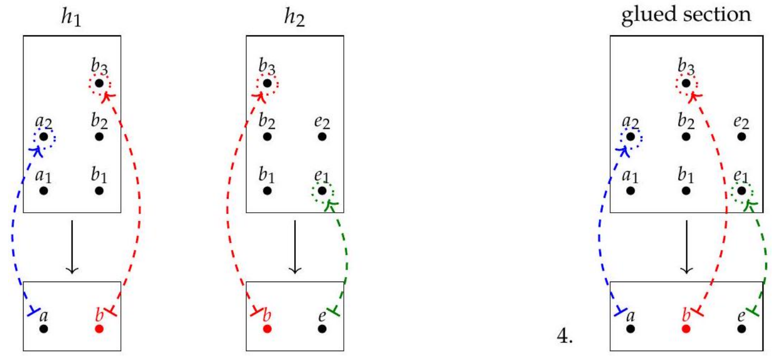
fong-spivak-seven-sketches_DIA_0463	319	—	*(dup)*	7f1e85c6-4350-401f-a960-eb8a0176e786-319_597_457_1290_846.jpg			
fong-spivak-seven-sketches_DIA_0464	320	—	*(dup)*	7f1e85c6-4350-401f-a960-eb8a0176e786-320_310_1260_609_446.jpg			
fong-spivak-seven-sketches_DIA_0465	320	—	*(dup)*	7f1e85c6-4350-401f-a960-eb8a0176e786-320_59_704_1131_726.jpg			
fong-spivak-seven-sketches_DIA_0466	320	—	*(dup)*	7f1e85c6-4350-401f-a960-eb8a0176e786-320_172_278_1308_939.jpg			
fong-spivak-seven-sketches_DIA_0467	320	—	*(dup)*	7f1e85c6-4350-401f-a960-eb8a0176e786-320_172_269_1602_943.jpg			
fong-spivak-seven-sketches_DIA_0468	320	—	*(dup)*	7f1e85c6-4350-401f-a960-eb8a0176e786-320_240_308_1964_923.jpg			
fong-spivak-seven-sketches_DIA_0469	320	—	*(dup)*	7f1e85c6-4350-401f-a960-eb8a0176e786-320_93_353_2416_900.jpg			
fong-spivak-seven-sketches_DIA_0470	321	—	*(dup)*	7f1e85c6-4350-401f-a960-eb8a0176e786-321_95_421_1910_867.jpg			
fong-spivak-seven-sketches_DIA_0471	321	—	*(dup)*	7f1e85c6-4350-401f-a960-eb8a0176e786-321_70_618_2213_769.jpg			

fong-spivak-seven-sketches_DIA_0472	322	— (dup)	7f1e85c6-4350-401f-a960-eb8a0176e786-322_410_930_1256_613.jpg			
fong-spivak-seven-sketches_DIA_0473	322	— (dup)	7f1e85c6-4350-401f-a960-eb8a0176e786-322_410_968_1738_588.jpg			
fong-spivak-seven-sketches_DIA_0474	322	— (dup)	7f1e85c6-4350-401f-a960-eb8a0176e786-322_215_1032_2294_611.jpg			
fong-spivak-seven-sketches_DIA_0475	323	— (dup)	7f1e85c6-4350-401f-a960-eb8a0176e786-323_136_312_480_896.jpg			
fong-spivak-seven-sketches_DIA_0476	323	— (dup)	7f1e85c6-4350-401f-a960-eb8a0176e786-323_240_294_452_1348.jpg			
fong-spivak-seven-sketches_DIA_0477	323	— (dup)	7f1e85c6-4350-401f-a960-eb8a0176e786-323_181_421_805_842.jpg			
fong-spivak-seven-sketches_DIA_0478	323	— (dup)	7f1e85c6-4350-401f-a960-eb8a0176e786-323_267_294_792_1403.jpg			

fong-spivak-seven-sketches_DIA_0479	324	— *(dup)*	7f1e85c6-4350-401f-a960-eb8a0176e786-324_183_330_448_914.jpg			
fong-spivak-seven-sketches_DIA_0480	324	— *(dup)*	7f1e85c6-4350-401f-a960-eb8a0176e786-324_249_382_765_885.jpg			
fong-spivak-seven-sketches_DIA_0481	324	— *(dup)*	7f1e85c6-4350-401f-a960-eb8a0176e786-324_249_321_1147_588.jpg			
fong-spivak-seven-sketches_DIA_0482	324	— *(dup)*	7f1e85c6-4350-401f-a960-eb8a0176e786-324_247_385_1149_1181.jpg			
fong-spivak-seven-sketches_DIA_0483	324	— *(dup)*	7f1e85c6-4350-401f-a960-eb8a0176e786-324_249_382_1618_885.jpg			
fong-spivak-seven-sketches_DIA_0484	324	— *(dup)*	7f1e85c6-4350-401f-a960-eb8a0176e786-324_249_382_2131_885.jpg			
fong-spivak-seven-sketches_DIA_0485	325	— *(dup)*	7f1e85c6-4350-401f-a960-eb8a0176e786-325_251_385_367_882.jpg			

fong-spivak-seven-sketches_DIA_0486	325	— *(dup)*	7f1e85c6-4350-401f-a960-eb8a0176e786-325_233_265_998_948.jpg	$\begin{tikzcd} X \arrow[dr, dashed] \arrow[drr, bend left, "g_2"] \arrow[ddr, bend right, "g_1"'] & & \\ & A \arrow[r, "\mathrm{id}_A"] \arrow[d, "\mathrm{id}_A"'] \arrow[dr, phantom, "\lrcorner", very near start] & A \arrow[d, "f"] \\ & A \arrow[r, "f"'] & B \end{tikzcd}$	$\begin{tikzcd} X \arrow[dr, dashed] \arrow[drr, bend left, "g_2"] \arrow[ddr, bend right, "g_1"'] & & \\ & A \arrow[r, "\mathrm{id}_A"] \arrow[d, "\mathrm{id}_A"'] \arrow[dr, phantom, "\lrcorner", very near start] & A \arrow[d, "f"] \\ & A \arrow[r, "f"'] & B \end{tikzcd}$	$\begin{tikzcd} X \arrow[dr, dashed] \arrow[drr, bend left, "g_2"] \arrow[ddr, bend right, "g_1"'] & & \\ & A \arrow[r, "\mathrm{id}_A"] \arrow[d, "\mathrm{id}_A"'] \arrow[dr, phantom, "\lrcorner", very near start] & A \arrow[d, "f"] \\ & A \arrow[r, "f"'] & B \end{tikzcd}$
fong-spivak-seven-sketches_DIA_0487	325	— *(dup)*	7f1e85c6-4350-401f-a960-eb8a0176e786-325_179_183_1697_1025.jpg	$\begin{tikzcd} A' \arrow[r, "f'"] \arrow[d, "i'"'] \arrow[dr, phantom, "\lrcorner", very near start] & B' \arrow[d, "i", "\cong"'] \\ A \arrow[r, "f"'] & B \end{tikzcd}$	$\begin{tikzcd} A' \arrow[r, "f'"] \arrow[d, "i'"'] \arrow[dr, phantom, "\lrcorner", very near start] & B' \arrow[d, "i", "\cong"'] \\ A \arrow[r, "f"'] & B \end{tikzcd}$	$\begin{tikzcd} A' \arrow[r, "f'"] \arrow[d, "i'"'] \arrow[dr, phantom, "\lrcorner", very near start] & B' \arrow[d, "i", "\cong"'] \\ A \arrow[r, "f"'] & B \end{tikzcd}$
fong-spivak-seven-sketches_DIA_0488	325	— *(dup)*	7f1e85c6-4350-401f-a960-eb8a0176e786-325_179_156_2183_1038.jpg	$\begin{tikzcd} A \arrow[r, "f"] \arrow[d, equal] \arrow[dr, phantom, "\lrcorner", very near start] & B \arrow[d, equal] \\ A \arrow[r, "f"'] & B \end{tikzcd}$	$\begin{tikzcd} A \arrow[r, "f"] \arrow[d, equal] \arrow[dr, phantom, "\lrcorner", very near start] & B \arrow[d, equal] \\ A \arrow[r, "f"'] & B \end{tikzcd}$	$\begin{tikzcd} A \arrow[r, "f"] \arrow[d, equal] \arrow[dr, phantom, "\lrcorner", very near start] & B \arrow[d, equal] \\ A \arrow[r, "f"'] & B \end{tikzcd}$
fong-spivak-seven-sketches_DIA_0489	326	— *(dup)*	7f1e85c6-4350-401f-a960-eb8a0176e786-326_283_335_376_615.jpg	$\begin{tikzcd} & & A \arrow[dr, equal] \arrow[dd, equal] & \\ & A' \arrow[rr, "g"] \arrow[dd, "f'"'] & & A \arrow[dd, "f"] \\ A' \arrow[rr, "g"'] \arrow[dr, "f'"'] & & A \arrow[dr, "f"] & \\ & B' \arrow[rr, "h"'] & & B \end{tikzcd}$	$\begin{tikzcd} & & A \arrow[dr, equal] \arrow[dd, equal] & \\ & A' \arrow[rr, "g"] \arrow[dd, "f'"'] & & A \arrow[dd, "f"] \\ A' \arrow[rr, "g"'] \arrow[dr, "f'"'] & & A \arrow[dr, "f"] & \\ & B' \arrow[rr, "h"'] & & B \end{tikzcd}$	$\begin{tikzcd} & & A \arrow[dr, equal] \arrow[dd, equal] & \\ & A' \arrow[rr, "g"] \arrow[dd, "f'"'] & & A \arrow[dd, "f"] \\ A' \arrow[rr, "g"'] \arrow[dr, "f'"'] & & A \arrow[dr, "f"] & \\ & B' \arrow[rr, "h"'] & & B \end{tikzcd}$
fong-spivak-seven-sketches_DIA_0490	326	— *(dup)*	7f1e85c6-4350-401f-a960-eb8a0176e786-326_296_333_364_1208.jpg	$\begin{tikzcd} A' \arrow[rr, "g"] \arrow[dr, equal] \arrow[dd, equal] \arrow[drrr, dotted] & & A \arrow[dr, equal] \arrow[dd, equal] & \\ & A' \arrow[rr, "g"] \arrow[dd, "f'"'] & & A \arrow[dd, "f"] \\ A' \arrow[rr, "g"'] \arrow[dr, "f'"'] \arrow[drrr, dotted] & & A \arrow[dr, "f"] & \\ & B' \arrow[rr, "h"'] & & B \end{tikzcd}$	$\begin{tikzcd} A' \arrow[rr, "g"] \arrow[dr, equal] \arrow[dd, equal] \arrow[drrr, dotted] & & A \arrow[dr, equal] \arrow[dd, equal] & \\ & A' \arrow[rr, "g"] \arrow[dd, "f'"'] & & A \arrow[dd, "f"] \\ A' \arrow[rr, "g"'] \arrow[dr, "f'"'] \arrow[drrr, dotted] & & A \arrow[dr, "f"] & \\ & B' \arrow[rr, "h"'] & & B \end{tikzcd}$	$\begin{tikzcd} A' \arrow[rr, "g"] \arrow[dr, equal] \arrow[dd, equal] \arrow[drrr, dotted] & & A \arrow[dr, equal] \arrow[dd, equal] & \\ & A' \arrow[rr, "g"] \arrow[dd, "f'"'] & & A \arrow[dd, "f"] \\ A' \arrow[rr, "g"'] \arrow[dr, "f'"'] \arrow[drrr, dotted] & & A \arrow[dr, "f"] & \\ & B' \arrow[rr, "h"'] & & B \end{tikzcd}$

fong-spivak-seven-sketches_326_0491	326	— (dup)	7f1e85c6-4350-401f-a960-eb8a0176e786-326_176_459_1122_846.jpg			
fong-spivak-seven-sketches_326_0492	326	— (dup)	7f1e85c6-4350-401f-a960-eb8a0176e786-326_344_559_1914_833.jpg	$\begin{array}{ccccccc}(0,0)&\leftarrow&(0,1)&\leftarrow&(0,2)&\leftarrow&(0,3)\cdots\\ \uparrow&&\uparrow&&\uparrow&&\\ (1,0)&\leftarrow&(1,1)&\leftarrow&(1,2)\cdots&&\\ \uparrow&&\uparrow&&&&\\ (2,0)&\leftarrow&(1,1)\cdots&&&&\\ \uparrow&&&&&&\\ (3,0)\cdots&&&&&&\\ \vdots&&&&&&\end{array}$	$\begin{array}{ccccccc}(0,0)&\leftarrow&(0,1)&\leftarrow&(0,2)&\leftarrow&(0,3)\cdots\\ \uparrow&&\uparrow&&\uparrow&&\\ (1,0)&\leftarrow&(1,1)&\leftarrow&(1,2)\cdots&&\\ \uparrow&&\uparrow&&&&\\ (2,0)&\leftarrow&(1,1)\cdots&&&&\\ \uparrow&&&&&&\\ (3,0)\cdots&&&&&&\\ \vdots&&&&&&\end{array}$	$\begin{array}{ccccccc}(0,0)&\leftarrow&(0,1)&\leftarrow&(0,2)&\leftarrow&(0,3)\cdots\\ \uparrow&&\uparrow&&\uparrow&&\\ (1,0)&\leftarrow&(1,1)&\leftarrow&(1,2)\cdots&&\\ \uparrow&&\uparrow&&&&\\ (2,0)&\leftarrow&(1,1)\cdots&&&&\\ \uparrow&&&&&&\\ (3,0)\cdots&&&&&&\\ \vdots&&&&&&\end{array}$
fong-spivak-seven-sketches_328_0493	328	— (dup)	7f1e85c6-4350-401f-a960-eb8a0176e786-328_362_407_2034_878.jpg			
fong-spivak-seven-sketches_329_0494	329	— (dup)	7f1e85c6-4350-401f-a960-eb8a0176e786-329_400_692_697_735.jpg			
fong-spivak-seven-sketches_329_0495	329	— (dup)	7f1e85c6-4350-401f-a960-eb8a0176e786-329_398_690_1301_738.jpg			
fong-spivak-seven-sketches_329_0496	329	— (dup)	7f1e85c6-4350-401f-a960-eb8a0176e786-329_367_1199_1982_505.jpg			
fong-spivak-seven-sketches_330_0497	330	— (dup)	7f1e85c6-4350-401f-a960-eb8a0176e786-330_156_1129_412_511.jpg			

fong-spivak-seven-sketches_000_0498	330	— *(dup)*	7f1e85c6-4350-401f-a960-eb8a0176e786-330_500_1072_787_572.jpg			
fong-spivak-seven-sketches_000_0499	331	— *(dup)*	7f1e85c6-4350-401f-a960-eb8a0176e786-331_229_1118_579_516.jpg